

THE SULTANATE OF BORNU

TRANSLATED FROM THE GERMAN
OF DR. A. SCHULTZE
FIRST LIEUTENANT (RETIRED) IMPERIAL GERMAN ARMY

WITH
ADDITIONS AND APPENDICES

BY
P. ASKELL BENTON, B.A., F.R.G.S.
C. C. C. OXON.

AND A THIRD CLASS RESIDENT, BORNU PROVINCE, NORTHERN NIGERIA

Σπάρταν ἔλαχες, Σπάρταν κόσμει
Spartam nactus es, hanc exorna

HUMPHREY MILFORD
OXFORD UNIVERSITY PRESS
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BY THE SAME AUTHOR

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HUMPHREY MILFORD

TRANSLATOR'S PREFACE

THIS translation of Dr. Schultze's monograph on Bornu has been undertaken in the hope that it may be of use and interest to those who, like the translator, cannot read German 'with their feet on the fender'. The translator's own knowledge of German is, in fact, almost *nil*, and the translation has been painfully worried out at odd times with the constant aid of a dictionary. In the work of revision, however, he has had the assistance of Alexander Siemens, Esq., and of various members of his family, who have an intimate knowledge of both English and German; to them the translator wishes to express his most cordial gratitude.^[1] The translation is published by permission of Herr G. W. Baedeker, of Essen, owner of the German copyright.

It is not generally realized, even in Nigeria, how ancient and powerful the old Bornu Empire was. In the fifteenth century it was the greatest power in Central Africa and its boundaries extended to Fezzan, the Niger, and practically to the Benue. The Hausas, of whom one hears so much nowadays, were then a *congrégie* of obscure semi-pagan tribes,^[2] while the Fulani are, of course, politically speaking, mere upstarts of a century ago. Bornu was never conquered by the Fulani—the only state of any size in Nigeria that was not—and, as regards the Hausas, the Kanuri have always despised them as an inferior race.^[3] According to the article on the Azhar Mosque, the great Mohammedan University in Cairo, by K. Vollers, in the *Encyclopaedia of Islam* now in course of publication, the Kanuri are given as one of the twenty-six great 'Arwika', into which the Azhar students are divided, and which seem to correspond roughly to the 'nations' in the older German Universities: the Hausas are not even mentioned among the 'also rans'.

These facts are emphasized because there is a tendency among those who have no close knowledge of Bornu towards the attitude that outside the Hausas or Fulani there is no salvation. It is commonly believed—utterly erroneously—and the statement has even appeared in print, that the Hausa

language is generally understood in Bornu. As a matter of fact not 1 per cent. of the Kanuri proper either speak or understand it, and it has made little, if any, progress during the ten years of the British occupation of their country.

No one who knows the Kanuri of the present day would pretend that he is a very admirable or a very amiable creature, or that he is the equal of the Hausa either morally or mentally. He has, however, a history and a social system of his own, as will be seen in the following pages.

The [maps](#) in this translation have been printed from the same plates as those in the German edition in order to save expense. They, therefore, do not show the exchange of the small piece of territory on the Shari, known as the 'Duck's Bill', between France and Germany by the recent international agreement. A map of the country involved appears in Macleod, *Chiefs and Cities of Central Africa*, p. 37.

The references in the foot-notes to the English edition of Barth have been verified for me by Mr. John Arthur. He informs me that the English and German editions vary considerably, and whole paragraphs are sometimes omitted.

The notes in square brackets—thus []—and all the Appendices except No. I have been added by the translator. They naturally refer chiefly to British Bornu, whereas the original author was more interested in German Bornu.

For assistance in giving the English names of the Fauna mentioned in Chapter VI, I have to thank the kindness of various officials of the South Kensington Museum: Mammalia by Mr. J. G. Dollman; Birds by Mr. Ogilvie Grant; Reptilia by Mr. Boulenger, F.R.S.; Fish by Mr. Tate Regan; Insects by Mr. Austin and Mr. F. W. Edwards; Arachnids by Mr. S. Hurst; Worms by Mr. Bell; Mollusca by Mr. E. A. Smith and Mr. G. C. Robson.

I have much pleasure in acknowledging the generous kindness which allows me to print the translation of extracts from the *Documents scientifiques de la Mission Tilho* appearing in the Appendices. The information contained therein was mostly collected by MM. Tilho and Landeroin.

For information about Denham portraits and relics in Appendix XVII I am indebted to the present representatives of the family, H. M. Denham, Esq., and Captains H. A. and L. S. Denham, grandsons of Admiral H. M. Denham, the explorer's first cousin.

I have to thank W. P. Hewby, Esq., C.M.G., Resident of Bornu, for several corrections and suggestions, and my brother, S. Landor Benton, for assisting me to read the proof sheets.

P. A. BENTON.

ROYAL SOCIETIES CLUB,
63 ST. JAMES ST.,

October 15, 1913.

Also to D. S. K. Crosbie, Esq., 93rd Highlanders and 2nd Batt. Northern Nigeria Regt., for assistance in revising the latter half of the translation.

Bornu was converted to Islam in the eleventh century, the Hausa States not till the sixteenth.

Cf. Benton, *Kanuri Readings*, p. 25.

AUTHOR'S PREFACE

I BECAME partly acquainted with the districts of which the present volume treats as a member of the Anglo-German Yola-Chad Boundary Commission in 1903-4.^[4] In endeavouring to inform myself with regard to the region then traversed, I found it a great inconvenience that the standard works, which contain descriptions of the former Sultanate of Bornu—works which are comprehensively ponderous—suffer from being very unhandy, and that, owing to the fact of their being published a generation ago, cannot include the latest investigations. Moreover, the literature of the last ten or twenty years on this subject being very scattered, the study of a highly interesting district is rendered unusually difficult. No monograph on it has hitherto existed.

I have now attempted in the present book to comprehend in such a monograph the essentials of the literature already known and of my own observations, which can correct particular errors in zoology and botany and will also deal generally with all lines of inquiry.

I have been most kindly assisted in my work by a whole number of gentlemen, who have helped me with information in the kindred sciences of zoology and botany, who have enabled me to examine the comprehensively ponderous literature, who have answered various questions, or who have simply given me advice and suggestions. They are Drs. Marquardsen and Moisel, Professor Reichenow, Professor Volken of Berlin, Professor Rein of Bonn, Herr G. Seligmann of Coblenz, Professor Boettger of Frankfurt a/M., Professor Lacroix of Paris, and Professor Aurivilius of Stockholm. It is my pleasant duty to render them herewith my heartiest thanks.

BONN,

March, 1910.

[Since writing this book Herr Schultze was a member of the African Expedition (1910-11) of Duke Adolf Friedrich zu Mecklenburg, and contributed a chapter on 'German Congo and South

Kameruns' to the account of the expedition published under the title of *From the Congo to the Niger and the Nile.*

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I

INTRODUCTION

ON more than one ground the country which embraces the districts of the once mighty Central African Sultanate of Bornu, deserves our quite particular interest. Enclosed in the centre of the Sudan and lying on the shores of an inland sea, this highly fruitful and thickly populated district was the first of all the great African water-basins to be known in Europe. For centuries, over the central plateau, two important articles of trade, ivory and human flesh, were up to the middle of the nineteenth century almost the exclusive aim of all expeditions which concerned themselves with the exploration of the dark continent. It was German travellers above all, and amongst them the most scientifically important and authoritative of all African explorers, Heinrich Barth,^[5] who brought news to Europe of these remarkable countries, first reported the unheard-of atrocities of the slave-trade in the heart of Africa, and thus gave a fresh impetus to the anti-slavery movement. Scarcely were the main avenues of exploration closed at the end of the nineteenth century than events happened such as these parts of Africa had never seen hitherto—events which the Great Powers of Europe made necessary by their policy of annexation, and the land of Bornu was pushed once more into the foreground of public interest. The end of these events was the present political understanding, which brought large potential empires under the protection of the three chief European Powers. Thus Germany at the present day again claims a share in those lands in which the self-denying and indefatigable activity of her great sons, the explorers Barth, Overweg, Vogel, von Beurmann, Rohlfs, and Nachtigal, had been so valiantly displayed of old, and in the great scientific and economic problems that have still to be solved in the Sudan.

[Preceded by the Englishmen Denham and Clapperton, and himself in the pay of the British Government.]

II

HISTORY AND EXPLORATION

As in almost all the kingdoms of the Sudan, so also in Bornu, history has exercised the widest influence on the present extent of the country, on the composition of its population, and on their character. The history of Bornu—accounts of which, beginning from the middle of the sixteenth century, remain in native manuscript chronicles—has been made known to us without any very great gaps by the German travellers Heinrich Barth and Gustav Nachtigal, who made use of already existing Arabic sources and as regards modern times wrote from personal observation.^[6] Without their information we should remain to-day in still greater ignorance than as it happens we fortunately are of the recent condition of this Central African empire, owing to the constant shifting of population. This movement had not yet altogether ceased even in very recent times.

The first mention of the name of Bornu is in the Arabic writer Ibn Said,^[7] where it appears as an integral part of the empire of Kanem. Later, as the centre of gravity of its rulers was removed to Bornu, the empire also took the name of this country. So when we speak to-day of a history of Bornu, the history of Kanem is to be understood as regards its earlier phases.

The first dynasty (according to Barth 67, according to Nachtigal 64 kings) which ruled the destinies of Kanem, i.e. Bornu, for nearly 1,200 years, traced their origin back to one Sef of Mecca,^[8] son of the last Himyaritic king. The period of time during which the first kings of Bornu held sway, beginning almost with the Hegira, differs considerably according to the various chronicles;^[9] nevertheless, oral tradition concerning this first dynasty, stretching back for nearly 1,000 years, enables us to distinguish certain fixed epochs, the way in which the Sefs pushed forward their dominion to the southern edge of the middle Sahara, and the period at which the present province of Kanem was included. It seems fairly probable—and several circumstances support this view—that the immigrants halted

a long time in Tubu country before pushing farther south-westwards,^[10] whereby this tribe became a part and even an essential part of the Kanem nation.

A singular expansion of the empire first appears with the acceptance of Islam under King Hume^[11] at the end of the eleventh century. By this step a further advance was secured over the other Central African kingdoms; for it brought with it a proportionately high degree of civilization and culture. Within a hundred years, under King Dunama Dibbalami^[12] (in the thirteenth century), the empire was extended over enormous tracts of country and reached as far as Fezzan in the north, the Nile on the east, and the present district of Dikoa^[13] on the south. Owing to this, however, the problems of administration also increased,^[14] and finally could only be solved by the help of the 'Nokena', a sort of Privy Council, but this again imperilled the unity of government. Under this king also discord broke out between the two chief nations of the empire, the kindred races of the Tubu (or Teda) and the Kanuri, which finally for a time led to an embittered struggle, which likewise did not contribute to the solidity of the empire. It is also during the reign of this Dunama Dibbalami that the name Bornu is mentioned for the first time as that of the most southernly province of the empire, by the chronicler who is called Ibn Said.

But the chief possessions of the Sef dynasty still lay to the north of Chad, including the most important, viz. Kanem on the north shore of the great lake. In the next two centuries, whilst the unfortunate civil wars brought an evident shrinkage of power and pointed to an undeniable decay, the dynasty of the Sefs were gradually expelled from Kanem by the kindred stock of the Bulala, from Lake Fitri, their bitterest opponents, and compelled to remove their head-quarters to the southernly and westernly shores of Lake Chad to the country which gave the empire its final designation. Although Kanem was later temporarily reconquered, still Bornu Proper remains henceforward the centre of gravity of the empire. The tribes which had hitherto possessed the newly conquered districts were absorbed in the conquering nation, being naturally influenced by it both in essence and in externals; or they were completely annihilated, like the Sos, who still live in tradition and who, finally drained of their life-blood by

their long and obstinate resistance, have left behind in Affade the remains of a civilization peculiar to themselves.^[15]

King Ali Ghadjideni ben Dunama,^[16] who reigned towards the end of the fifteenth century, was the first to put an end to the causes of the long-standing civil war, which had nearly led to the downfall of the empire, by restricting within proper bounds the power of the twelve 'Kokenawa',^[17] the members of the 'Nokena', who in the course of time had become almost independent princes. He founded on the lower courses of the River Yo the town of Ghasr Eggomo (Birni),^[18] which remained the capital of Bornu for the next 300 years and saw the empire reach its zenith. It is very probable also that it was in Ali Dunama's reign that the famous Leo Africanus visited Bornu, and undertook a description of the land and people after only a month's stay; his description was evidently coloured by unpleasant personal experiences, and is considered on that account almost worthless; at most the commercial instincts of the Bornu people there depicted may remind us of the reports of later travellers.^[19]

The most important of a line of able princes, whom the country was permitted to enjoy during the succeeding period, was without doubt Edriss Aloma (whose date, according to Barth, was 1571-1603),^[20] who, according to his conscientious native chronicler Iman Ahmed, left a reputation behind him for unusual energy and circumspection together with great philanthropy and clemency. With an army, the flower of whom already carried fire-arms, he undertook successive campaigns for the strengthening of his power, campaigns which extended from Aïr in the north to Kano in the west and the country of the pagan Margis in the south. With the rulers of Kanem, which had been already reconquered by his predecessor, and which remained a province of Bornu till the end of the eighteenth century, Edriss Aloma had already at the beginning of his reign concluded a treaty of friendship. This treaty was even embodied in duplicate written dispatches, which argues a highly developed system of government. It speaks much for this in general, that the country, some of whose kings performed the pilgrimage to Mecca, was and continued to be in close touch with the politics and culture of Tripoli and Egypt, which at that time had reached its economic zenith. Denham was the first to establish

that for the building of the walls of Ghasr Eggomo, the ruins of which he visited, recourse was had to burnt bricks, a material which is now no longer made use of, and he rightly deduced from this a high degree of civilization; on Barth also the remains of the former capital made a similar impression.

[\[21\]](#)

The superior ability which the new rulers of the country possessed over the former inhabitants could not remain unnoticed even by such inhabitants themselves. A rich country, bounded on two sides, the north and east, by stretches of sterility or absolute desert, without being protected by any considerable natural impediments, such as standing water, devious river-courses, or inaccessible mountain features, was necessarily exposed to the constant danger of invasion by foreign tribes from outside its jurisdiction. This danger was bound to increase as soon as the cessation of continuous struggles against the hostile nature of the desert, which kept the qualities of mind and body constantly on the stretch, came into effect, and as soon as increasing prosperity and intermixture with peoples of a lower morale produced an evident weakening and modification of the Bornu nation.

Besides the ever-restless Tuareg, who rendered the border districts unsafe until very recent times, the first advance parties of the Fulani, an Hamitic conquering tribe, also showed themselves in the west of the empire towards the end of the sixteenth century. These Fulani, inspired by the ideas of a strict Mohammedanism, were advancing eastwards from Senegal without a halt, and finally established the powerful empire of Sokoto.[\[22\]](#) When, at the beginning of the nineteenth century, the Fulani on their victorious march arrived at the lower course of the River Yo, they met with but a feeble resistance.[\[23\]](#) The Bornu army was beaten within sight of the palace, and the incapable Sultan Ahmed and his chief men were obliged to fly from the capital, which was to a very great extent destroyed by the victors. This happened in the year 1809.

Yet again was there to arise for the empire at this second period of eclipse a saviour, who preserved it from complete annihilation. But this time he did not come from the royal family itself. A man who stood quite apart from them, Faki Mohammed el Amin el Kanemi from Fezzan, who, however, was connected with Bornu by family ties,[\[24\]](#) dared to withstand

the invaders together with a handful of devoted adherents, and, as he was successful at the very beginning, he was soon joined by so many patriotically-minded followers—to the number of 200 horse and 2,000 foot—that he was able to bring matters to a decisive issue. By the victorious battle of Ngornu at least the eastern part of Bornu was cleared of the invaders, but they were able, however, to retain Katagum on the west of the empire as a stronghold.

With the help of Mohammed el Amin, Ahmed was then enabled to regain the old capital, though only temporarily. But neither he nor his son, who had entirely lost the confidence of their subjects, possessed such qualities as to restore the prestige of the empire. Willy nilly, in spite of their own secret reluctance and the jealousy of the office-holders, the royal family were compelled to seek the help of Mohammed el Amin, who was the favourite of the nation and in whom it reposed the greatest confidence as its deliverer from a foreign yoke. The influence of this man reached such a pitch that he finally became the real ruler, and was able in 1814 to found a capital of his own, which he called Kuka or Kukawa, after a Kuka-tree^[25] which stood on the site of his own house. On this account Barth is quite correct in pointing out that the year 1814 marks the birth of a new dynasty. In spite of being the real ruler and having authority with the people as such, el Amin, with prudent moderation, still only described himself as ‘Sheik’, while leaving all superfluous externals and all pomp to the royal family, now fast becoming ridiculous owing to dwindling power and constantly changing their place of residence in an aimless and helpless fashion. For this reason he sought continually to strengthen his authority with the people by bringing back to their former allegiance the tributary states of Bornu, who had revolted during the period of decay. It was just at this period (1822-4) that the English expedition under Oudney, Clapperton, and Denham^[26] appeared and was favourably received by the Sheik; it is to their inquiries that we owe the first reliable information about the country. Thus Denham was a witness of the battle of Ngala between the troops of el Amin^[27] and the Bagirmis, which resulted in a victory for Bornu. In spite of the multitudinous following of armed men, which the wide extent of the empire made possible, el Amin had done nothing to put an end to the shadowy royalty of the Sefua. This was reserved for Sheik Omar, who

succeeded his father after the latter's death in 1835. Whilst this same Omar and his troops were busy in the west of the empire bringing the unruly Governor of Zinder to account, the Sefua, assisted by the ruler of Wadai, made use of this favourable opportunity to attempt to regain possession, by force of arms, of their earlier authority. This attempt, however, failed, and the course of the struggle was unfavourable to the old dynasty. The puppet King Ibrahim was put on his trial as a traitor to the country by Omar and executed in 1846. His son Ali, the last of the Sefua, died on the field of battle. Omar thus became undisputed ruler of Bornu. Although, according to general opinion in Bornu, he did not inherit his father's indispensable energy, still he possessed many excellent private qualities which were of special advantage to the German travellers, who, during his reign and with his assistance and furtherance, were able to continue in so admirable a manner the exploration of the Central Sudan commenced by Denham.

In the year 1851 there arrived in Bornu, via Tripoli, the expedition under Richardson sent out by the English Government; to it were attached the German explorers, Dr. Heinrich Barth and Dr. Adolf Overweg. After the premature death of the leader, who succumbed to fever^[28] before reaching the capital of Bornu, Dr. Barth undertook its leadership. From Kukawa, which remained the proper base of operations throughout the whole duration of their stay in the Sudan, the two Germans, but especially Barth—Overweg died in the summer of 1852 at Maduari, on the shores of Lake Chad,^[29] which he was the first European to navigate^[30]—made longer or shorter journeys in all directions, which extended as far as the Benue in the heart of Adamawa, and which led to the solution of important geographical questions relative to that river. Owing to a lack of sufficient resources of their own, the travellers, like their predecessor Denham, were practically compelled to join in the 'razzias', or forays, whose chief object was the capture of slaves, unless they were willing to let slip the opportunity of being able to learn about lands hitherto untrodden. As they were compelled to be passive spectators of these slave-raids, taking no part in them yet subsequently describing them, they found themselves in the position of turning the attention of European civilization to the existing traffic in human flesh and to the barbarities it gave rise to, barbarities which were enacted before their eyes and of the culpability of which their black

companions had no perception.^[31] The German astronomer also, Dr. Edward Vogel, who was sent out later by the English Government to join the expedition on Petermann's recommendation, in order to complete their investigations by fixing the position of places astronomically, and who arrived in Kukawa at the beginning of the year 1854, was obliged to make a part of his expedition southwards in the wake of slave-raids. Barth was the only scholar who returned home in safety (1855) from this great Central African expedition, and he was able to leave behind him in a five-volume work the results of his prolonged investigations; his detailed descriptions cannot be superseded in essentials even to-day, so limited is the economic and political revolution occasioned by the events of the end of the nineteenth century. Vogel was less fortunate, being destined to forfeit his life in his attempt to penetrate into Wadai.^[32] He was murdered in 1856^[33] by order of the sultan of that country. For the result of his investigations, which are of special value with regard to the explanation of the flora, we have only the reports published in Petermann's *Mitteilungen* and several of his letters,^[34] but these preserve some important conclusions. The uncertainty of the fate of Edward Vogel induced the German traveller, Maurice von Beurmann, to enter on a further investigation from Bornu with regard to what was still unknown. But the same fate befell him as his unfortunate predecessor. In the year 1863 he was strangled at Mao,^[35] at that time the capital of Kanem, at the instigation of an officer of the Sultan of Wadai, before ever he reached that country. A few years later, in 1866, Gerhard Rohlfs arrived in Bornu on his journey across the dark continent from Tripoli to Lagos; he only stayed a short time however. The last of the German travellers who visited Bornu during the reign of Omar, and enjoyed his hospitality, was Dr. Gustav Nachtigal. Nachtigal was selected to take to Kukawa the presents which King William of Prussia had decided to send to Omar as a recognition of his great services to German travellers. Starting from Tripoli in the year 1869, Nachtigal reached Kukawa in July, 1870, by a little-known route, passing through districts hitherto untrodden by a white man.^[36] After having acquitted himself of his mission he set forth on his travels, choosing, like his predecessor, the capital of Bornu as a starting-point for detailed journeys of inquiry. Thanks to his superior powers of observation and his many-sided interests, which, moreover, lay partly in

different directions from those of his predecessor, he was able to supplement most successfully the fundamental results of Barth's journeys. In order to enlarge his knowledge of the country he had, as his predecessor had pointed out, to join as a spectator in the slave-hunts in the almost inexhaustible slave-reservoir of the districts of the middle Shari and Logone, but he expresses himself in the strongest terms, as did Barth, on the unheard-of atrocities which were committed on these occasions.^[37] Finally, during Omar's reign occurred the visit of the Italians, Matteucci and Massari, who paid a hasty visit to Bornu in 1880, but who were prevented by their untimely end from leaving behind a written account of their discoveries.^[38]

There can be no doubt that, out of all the kings, sultans, or sheiks who had hitherto reigned over Bornu, Omar has been delineated for posterity with the most justice and impartiality, thanks to the German travellers who at such various times enjoyed his hospitality.^[39] All these are in accordance in declaring that Omar possessed a goodness and philanthropy very unusual for a potentate of the Sudan—for he can scarcely be held responsible for the deeply-rooted cruelty of his troops. These qualities were often displayed towards the German travellers in a fatherly providence, a fact which is very eloquent in view of the selfishness natural to the native. But since this goodness was not coupled with the energy which is quite indispensable in the conduct of African affairs, they foreboded danger to the country. Though Omar stood high in his people's regard, this was not increased by the artful intrigues of his nearest associates. His own brother, Abdur Rahman, ventured to openly rebel against him and sought to depose him. Omar was compelled in the end, certainly with reluctance, to resolve to allow his brother to be executed.^[40] It was fortunate for Omar that for a long time he had a counsellor, whom he inherited from Lamino, the first wearer of the dignity of Sheik, and whom Nachtigal describes as 'the only *man* amongst his hangers-on', a race of men whose days were filled with petty gossip.^[41]

The results of Omar's policy of inaction were bound to show themselves fatally in the border districts and in the provinces only loosely attached to Bornu. In spite of a standing army of 7,000 men permanently

mobilized—the old military spirit had long vanished from the rest of the nation—the Tuaregs ventured on the most impudent raids in north-western Bornu; the districts north of the River Yo, which Barth had described as a kind of ‘political quarantine-station’, were thoroughly unsafe.^[42] Moreover, the vassal states of Bornu, such as Zinder, proved themselves little to be depended on during the renewed advance of the Fulani,^[43] and the speedy rise of the neighbouring kingdom of Wadai meant an ever-increasing danger to Bornu. After the death of Omar in 1881 he was succeeded at short intervals by his sons Abubakr, Ibrahim, and Ashimi, who, however, were not in a position to avert the impending ruin. When, in 1892, the French traveller Monteil visited Bornu, whose sultan at that time was Ashimi, Zinder had long belonged to Bornu only in name, while constant robberies and intrigues on the part of Wadai were the order of the day and remained unpunished.^[44]

Such were the relations of Bornu at the end of the eighties of the last century, while far to the east, in the region of the middle Nile, events were preparing which fundamentally revolutionized in the years that followed the whole state of affairs in North Central Africa, and finally delivered the death-blow of the Bornu empire in its old form. When Suliman, son of the notorious slave-raider Zobeir Pasha, after a long and bitter struggle with Gessi Pasha,^[45] surrendered to the latter in 1879, Rabeh, one of his most powerful followers, who had urgently dissuaded him from surrender, declared himself independent of him.^[46] A great part of Suliman’s following joined Rabeh,^[47] and under their new leader soon left the Egyptian Sudan and turned their steps westwards, with Gessi Pasha at first still in pursuit. The flower of Rabeh’s troops consisted of ‘dervishes’, that incomparable material for African soldiery, who gave proofs at the famous battles of Khartoum and Omdurman of such unparalleled courage and contempt of death. With this force, which received constant increases from the army of Slatin Pasha himself, Rabeh conquered by degrees the whole Eastern Sudan; Wadai alone, though it indeed also suffered defeat, was able to maintain independence by means of a compromise concluded with Rabeh. His raids extended southwards as far as Bangasso in the Congo State. Although Rabeh declined all connexion with the Mahdi^[48], then at

the height of his power, he knew how to make skilful use for his own purposes of the wave of fanaticism let loose by the Dervishes, and probably also of the ideas of the religious sect of the Senussi.^[49]

In 1892 began Rabeh's campaign against Bagirmi, which asked for help from Wadai and even from Bornu, but from the latter in vain. After having vanquished the troops of Wadai, Rabeh in 1893 crossed the Shari and laid waste in a very short time the flourishing towns and districts on both banks of this river. At first Sultan Ashimi took some preliminary steps to ward off the formidable danger from his country. But the Bornu troops, unused to war, who were dispatched against Rabeh, suffered a complete defeat; their commander, Malla Kerim, was by order of the conqueror sewn up in a freshly-flayed bullock hide, and so found a horrible death exposed to the rays of the African sun.^[50] Ashimi himself, at the head of his troops, suffered a still more complete defeat not far from Kukawa. He himself fled, but was thereupon killed by his own nephew, Abba Kiari,^[51] with whom before this he had had a quarrel of long standing. Abba Kiari then even succeeded in surprising and beating Rabeh in front of the capital, but this victory of the Bornu troops was changed into crushing defeat, owing to their scattering too early in order to plunder the enemy's camp; their leader was captured and executed. The fate of Bornu was thus sealed. Rabeh's victory was followed immediately by the destruction of Kukawa, at that time one of the largest towns of the Sudan. Such of the royal family as did not fall into captivity were able to save their lives. Omar Sanda, the eldest son of Ashimi, first sought asylum in Mandara, but during Rabeh's campaign against that sultan fled from there to Zinder, where he was dependent on the bounty of his former vassal, who now at last had become quite independent. Rabeh now determined to consolidate his influence in the lately conquered districts before pushing his dominion further. He built up the ancient town of Dikoa into a capital, whither flowed all the riches of the Central Sudan, and made it a place which, with its reported population of 100,000 souls, surpassed all other towns in these latitudes from the Senegal to the Nile. The town was surrounded with a strong wall, inside which the conqueror established an impregnable citadel which contained his palace and the barracks, stabling, armoury, and powder magazine for his troops. Everything was arranged with the greatest precision, probably on

the example of the English in the Egyptian Sudan. The troops were even regularly drilled,^[52] and every Friday a parade of the whole armed force took place, which gave Rabeh an opportunity of assuring himself of the readiness of his forces to take the field.^[53] Estimates vary as to the actual numbers of his army, which was distributed between Dikoa and a border garrison on the Shari, but they certainly amounted to at least 20,000 men.^[54] The provisioning of these masses of troops, who were encumbered with large additions of women and children, weighed heavily on the districts hardly yet incorporated as a result of the recent campaigns. Moreover, the slave-hunts were an oppressive scourge in the populous districts, which provided women and slaves for the soldiery, who were detested for their imperious dealings. To this was added the fact that Rabeh enforced the legal ordinances of the Koran with a strictness that the population, though very largely Mohammedan, were bound to feel as a sweeping limitation of personal liberty. In quite external matters also Rabeh sought to make his influence prevalent. Foureau mentions as 'Rabeh's mark' (a sort of notched cross on the face forming an enormous raised scar) a kind of tattooing or brand, which the people of two communities, whom the traveller met with, bore on their foreheads.^[55] The conqueror's influence extended even to dress, and in this direction, indeed, with very little compulsion; a sort of shirt with three shield-shaped pieces of cloth sewn on, which is very smart and yet practical for all occasions, has partly displaced the Sudan 'tobe', the usual native dress for men.^[56]

Of whatever kind the influence of Rabeh and his followers may have shown itself, this much is certain, that their pitiless cruelty, which spared neither pagan nor Mohammedan, drew upon them the deep-rooted hatred of the subjugated peoples, a circumstance which will be of great weight in the occurrences which now follow. While Rabeh on the one side sought to preserve good relations with the border powers in order to keep open the high roads into his country, on the other, his preparations for extending his rule over Zinder and Sokoto struck a blow at this, and were followed in the years 1893 and 1894 by the partition—at any rate on the map—of the Lake Chad countries between the three colonial powers interested, viz. Germany, England, and France. The French, who had already pushed their advance posts into the region of the upper Shari, were the first of the three powers to

move, and in a very tentative way to further the practical joining up of their newly acquired possessions within their great African colonial empire. They consequently very soon came into conflict with Rabeh.

In 1897 Gentil, a civil officer of the French Congo, undertook a voyage down the Shari from the region of the upper Ubanghi, in order to bring the Sultanate of Bagirmi under French influence. He met with a far-sighted reception from this Sultan, who hoped to be preserved from the yoke of Rabeh by the arrival of the French. The *Léon Blot*,^[57] the little river gunboat which carried the expedition, penetrated right to the Shari delta unmolested by Rabeh's garrisons, and returned unmolested also, being greeted everywhere with a friendly reception by the river people.^[58] Thereupon Gentil without delay betook himself to Paris, in order to set on foot a big expedition from thence. Meanwhile Rabeh revenged himself for Gentil's demonstrative reception there by laying waste the whole of Bagirmi. Moreover, the unfortunate French trader, de Béhagle, who was accidentally mixed up in the economic politics of the Chad region, fell a victim to Rabeh's fury and, by his orders, was hanged in Dikoa.^[59] Gentil's representative Prins only waited for the arrival of the expedition: as soon as it entered the Shari in the summer of 1899 the campaign against Rabeh immediately commenced. Very unfortunately for the French the beginning of the campaign miscarried, for the leader of the advance-guard, Bretonnet, who had greatly under-estimated the enemy's strength, allowed himself to be misled into making a premature attack. In the battle at Togbao on the Shari, Rabeh directed his troops in person and Bretonnet's column was completely annihilated; its leader fell and all the rifles and three guns fell into the hands of the enemy.^[60]

Hereupon Gentil advanced in person and defeated Rabeh in the autumn of the same year in the desperately contested battle of Kuno; but the heavy losses suffered by his own troops determined him to await in Fort Archambault fresh reinforcements from the Congo. By a fortunate accident two further French expeditions arrived almost simultaneously in the Lake Chad region. At the beginning of December the expedition under Joalland and Meynier coming from Zinder by way of Kanem reached the Shari, but it was not till the beginning of the following year that they effected a

junction with Gentil. On the 2nd of November the 'Sahara Mission' under the scientist Foureau arrived in Zinder from Asben. In their presence Omar Sanda, the man who had fled to Zinder, was proclaimed—somewhat prematurely—as Sultan of Bornu by his followers.^[61] Foureau too thereupon marched via Kanem to the Shari, where he joined Joalland. At this point Major Lamy, who had hitherto been in charge of the escort of the 'Sahara Mission', took over chief command of the combined forces, which soon afterwards came into touch with Rabeh's outposts. On the 3rd of March, 1900, the French succeeded in capturing Kusseri, but they could not strike a decisive blow against Rabeh until the arrival of Gentil. On the very next day after his arrival, on the 22nd of April, the decisive battle was fought,^[62] when the French brought into action 800 riflemen armed with modern weapons against the entire fighting force of Rabeh. Whilst the battle was at its height, Rabeh fell;^[63] he had taken up his position at the 'Tata', or earthwork constructed by his troops. His head was brought into the French camp. But the French had also bought their victory dearly: Lamy was mortally wounded and Capt. de Cointet killed. The enemy fled with headlong speed to Dikoa, which however soon afterwards fell into the hands of the French. Yet even this by no means put an end to the struggle, for the remains of the beaten army rallied under Fadel Allah, Niebe, and Mahmud, the still youthful^[64] sons of their fallen leader, of whom Fadel Allah especially gave the French plenty more to do, though the latter had on their side the sympathy and to some extent the assistance of the exasperated natives. Meanwhile Omar was installed as Sultan in Dikoa by Gentil, but was very soon relegated again to banishment for having made attempts on the life of his benefactor. He was then replaced by his brother Garbai.^[65] Fadel Allah had shifted the scene of his activities to the River Yedseram, and thus threatened to be a standing danger to the neighbourhood of Dikoa, although the French remained victors in the hard-fought battles which were contested along the river at Issege and Mubi. In spite of the successes of the French, the energetic son of Rabeh managed to again reconquer Dikoa, after having completely defeated the Sultan. The French punitive expedition, which immediately followed, was so energetically conducted that Fadel Allah was obliged to fly to Gujba, far inside English territory. Here he came into touch with the English,^[66] under whose protection he put himself, and

through whose mediation he hoped to become Sultan of Bornu. The English were favourably disposed towards the suggestions, or rather, the expressed wishes of Fadel Allah on this point,^[67] and would certainly have helped him towards their realization, had not events forestalled them. Fadel Allah was killed in renewed fighting with the French, and the question was thus arranged in the simplest and most satisfactory fashion. Above all, the country recovered the tranquillity which it had lacked for so long.

A short time after these events the English attracted Sultan Garbai and his following to that part of Bornu that belonged to their sphere of influence. Thereupon the French put in Omar Sanda, a nephew of the other Omar deposed by themselves, as an opposition Sultan in Dikoa,^[68] to prevent that town being depopulated. The part of Bornu situated between the Yedseram and the Shari still remained under the administration of the French garrisons in Dikoa and the places on the latter river. It was not till the arrival of the German expedition under Lieut.-Col. Pavel (1902) that the territory, which had hitherto only been German on the map, was actually brought under the protection of the Empire. After various experiments, the part of Bornu which had fallen to Germany was finally formed into a province under the name of the German Chad Territory, whose administrative head-quarters were established at Kusseri. Such was the position taken up by the colonial powers interested towards the economic investigation and development of the country. The English have certainly had more pressing and important problems to solve in their great Niger Protectorate than the development of Bornu, which is its remotest province; so that our knowledge of the British part of the country has not been substantially increased since Rohlfs' and Nachtigal's time.^[69] The exact opposite is true of the German part, which is what will be chiefly described in the following pages; for it has been so thoroughly explored in all directions by Germans as well as French in supplement of already existing materials, that very little remains to be done in the geographical department of inquiry. Besides Commissions, which have laid down in actual practice the political boundaries hitherto only theoretical, the country has been traversed by larger or smaller expeditions, who have accomplished valuable economic discoveries. The French have been especially active in seeking to approach the solution of one of the most interesting of geographical

questions, i.e. the difficult problem of Lake Chad, a task which has not even to-day been finally and incontestably accomplished.

Barth, vol. ii, chap. xxix, and Chronological Table, vol. ii, p. 633; Nachtigal, ii. 392.

Nachtigal, ii. 401. In the same passage he treats of the etymology of the word 'Bornu'.

[Vid. [Appendix III.](#)]

Nachtigal, ii. 394.

Nachtigal, ii. 400.

[Hume ben Abd el Djelil, 1086-97. No. 12 on Vischer's list of Bornu kings, vid. [Appendix II.](#)]

[1221-59. No. 17 *idem.*]

At that time it was even in close relations with Tunis. Ibn Chaldun, ii. 346.

Barth, Chronological Table, vol. ii, p. 647.

Nachtigal, ii. 404. Cf. *Deutsches Kolonialblatt*, xvii, p. 802.

[Does not appear, under that name at any rate, in Vischer's list. Barth gives his date as 1472-1504, Nachtigal as 1465-92, Landeroin as 1437-65.]

[Or Koganawa = grandees. Cf. 'Kogana' in Vocabulary of Benton, *Kanuri Readings.*]

[For derivation vid. note [\[8\]](#).]

Leo Africanus, iii. 308. Still less trustworthy is the account given by Ibn Batuta. Ibn Batuta, iii. 441.

[Vischer, *idem*; Nachtigal, 1563-1614; Landeroin, 1545-96. He is called Aloma because he is said to have been buried in the middle of the little lake of Alo near Maiduguri.]

Denham, i. 211; Barth, iv. 23, vid. also Barth, ii. 658.

Barth, ii. 649.

[St. John, p. 214, relates that the Bornu army mistook a herd of ostriches for the Fulani invaders and fled in panic.]

[Lamino (el Amin) was the son of a Kanembu, Sheikh Langa, and an Arab woman from Zouïla in the Tripolitaine. Like his father, he studied as a young man with the Koyams of Gaserregomo (Kasr Kumo), and accompanied him on a pilgrimage to Mecca. His father having died at Medina, Sheikh Lamino returned to Kanem. Tilho, *Doc. sc.* ii. 358.]

Adansonia digitata. [Tilho, *Doc. sc.* i. 221, gives its astronomical position as lat. 12° 55' 36", long. 11° 13' 34". Kukawa no longer exists except as a mass of ruins and as a farm hamlet. Provincial head-quarters were moved to Maiduguri (Maifoni) in 1906, and the new town of Shehuri (Yeriwa) built for the Shehu and his following.]

[They were accompanied by William Hillman, a naval carpenter, at a salary of £120 per annum (Denham, i. xvii). He returned safely to Europe, as did Denham and Clapperton.]

Dr. Walter Oudney, M.D., died at Murmur, near Katagum, on January 12, 1824 (Denham, ii. 227).

Ensign Toole of the 80th Regt., who proceeded from Malta as a volunteer with stores to revictual the Expedition, and reached Kukawa on December 23, 1823 (Denham, i. 311), died at Ngala, between the Yedseram and the Shari, on February 26, 1824 (Denham, ii. 23). Mr. Tyrwhitt, who arrived in Kukawa on May 20, 1824, with presents for the Shehu and stores for the Expedition (Denham, ii. 41), and who on the departure of Denham and Clapperton was left as British Consul in Kukawa, died there on October 23-4, 1824; the Shehu reported his death to the British Consul at Tripoli and sent a list of his effects (Denham, ii. 385).]

According to Denham's estimate they numbered at that time 30,000 men. Denham, ii. 165.

[He died at Ngurutua (i.e. the place of hippopotami). For Barth's account of his death, vid. Benton, *Notes on Some Languages*, p. 174.]

[For account of Overweg's death and further information about him vid. Benton, *Notes on Some Languages*.]

[For Overweg's boat brought from Malta in pieces, vid. Benton, *Notes on Some Languages*, p. 164. The late Boyd Alexander thought he had discovered the remains of this boat in Lake Chad (*From the Niger to the Nile*, ii. 66). Tilho, in *Documents scientifiques*, i. 10, adduces strong reasons for thinking that it was only the remains of a Kotoko canoe lost by the Lenfant expedition in 1903.]

Barth, iii. 175, 193-5, 236-7.

[For account of Vogel and his death, and of the two English sappers who accompanied the expedition, vid. Benton, *Notes on Some Languages*.]

Barth, iv. 63; Rohlfs, ii. 81.

Cf. also Polko, *Erinnerungen an einen Verschollenen*.

Zeitschrift für allgemeine Erdkunde, vol. xv (1863), p. 538; Rohlfs, ii. 81; Nachtigal, ii. 251 and 264.

[Nachtigal was accompanied by a Piedmontese cook, Giuseppe Valpreda, who became converted to Islam, called himself Mohamed el Mussulmani, and remained in Kuka with the Shehu. He was seen there by Monteil in 1892, and by the Niger Co. mission under M^CIntosh at the end of 1890. He died when fleeing from Rabeh in 1893; cf. Benton, *Notes on Some Languages*, p. 304, quoting from Dujarric, *Vie du Sultan Rabah*, p. 53, and Tilho, *Doc. sc.* i. 15.]

Nachtigal, ii. 626 and 733. He refers here especially to the slave-raids of the sultans of Bagirmi.

Reclus, p. 660; Oppenheim, p. 159. [Cf. note under Massari in [List of Authorities](#).]

[For a less favourable account of Lamino, Omar, and their successors vid. [Appendix IV](#).]

Nachtigal, ii. 413.

Nachtigal, i. 602; ii. 10; vid. also Rohlfs, ii. 103.

Barth, iii. 36 and 116; Nachtigal, i. 564 and 572. Cf. also Monteil, p. 313.

Barth, iii. 171-2; Nachtigal, ii. 501 and 503.

Monteil, p. 344.

[According to Col. Chaillé-Long's *My Life in Four Continents*, Gessi started as Gordon's Italian valet, and used to be unmercifully cuffed and kicked by his master. Later, he was promoted to a more

important position and did not appreciate these attentions, which did not entirely cease. Col. Chaillé-Long was Gordon's American chief of staff. Zobeir died near Khartoum on January 5, 1913.]

Oppenheim, p. 11.

[For account of Rabeh vid. [Appendix V](#), and for a more authoritative account vid. Gentil, *La Chute de l'Empire de Rabah*, and Gaston Dujarric, *La Vie de Rabah*.]

[Vid. Benton, *Notes on Some Languages*, p. 304.]

Oppenheim, pp. 24, 29, and 41.

Oppenheim, p. 44.

[Abba Kiari, or Shehu Kiari, seems to have been almost the only representative of the Kanemy dynasty with a spark of energy or courage. As is mentioned in *Boyd Alexander's Last Journey*, p. 188, he seldom uncovered his face, as his nose had practically been eaten away by a malignant disease, and his speech was so much affected that only one person among his people, a Mallam, could understand him.]

[Rabeh is said always to have personally supervised the musketry instruction of his gun-men.]

Oppenheim, p. 58. This custom is still kept up by Sanda, Shehu of German Bornu. Cf. also Dominik, p. 151.

About 4,000-5,000 men were furnished with fire-arms, some of quite modern pattern, and were well trained in fire-discipline.

Foureau, *D'Alger au Congo*, p. 589. Similar statements were made to me in 1903 at various places in Bornu. I was, however, shown a brand different from Foureau's description as 'Rabeh's mark'. Even the Arab merchants and their servants, settled in Dikoa in Rabeh's time, had to submit to this barbarous tattooing.

Vid. also Foureau, p. 686, and Lenfant, p. 173 (plate).

[This historic little steamer has lately been employed by the French in surveying navigable channels in Lake Chad. There is a picture of her in Gentil, *La Chute de l'Empire de Rabah*.]

Gentil, p. 98.

[Sururu, now headman of one of the 'wards' in Shehuri, the Shehu's quarter of the new capital of British Bornu, was in charge of the execution. It took place on August 15, 1899; vid. Decorse, *Rabah*, &c., p. 36, for native account. For photograph of his monument at Dikoa vid. Tilho, *Doc. sc.*, ii. 376.]

Gentil, p. 125.

Foureau, *D'Alger au Congo*, p. 602.

Gentil, p. 211. [Fadel Allah was not present, vid. [Appendix XVIII](#).]

[The dramatic account of Rabeh's death, reminiscent of *King Solomon's Mines*, as given in Macleod, *Chiefs and Cities of Central Africa*, p. 147, must be apocryphal. It is quite different from that given by Gentil. The Editor of *Boyd Alexander's Last Journey*, p. 186, practically admits that it is incorrect.]

Fadel Allah was then twenty-six years old.

Oppenheim, p. 121. [The present Shehu of British Bornu. There is a photograph of him in Macleod, p. 244. Ex-Shehu Omar Sanda, surnamed Kura (the elder) and Kori (the short), is now Ajia of the British district of Gujba.]

[June, 1901. In the person of the late Major A. M^CClintock, D.S.O., Seaforth Highlanders, formerly commanding the 1st Batt. Northern Nigeria Regt., and until his recent death second senior Political Officer of Bornu Province. His escort was commanded by Sergt. Moman Machena (Auta), now a Preventive Service Agent in Bornu. Cf. Decorse, *Rabah*, &c., p. 18.]

Oppenheim, p. 123; Möckler-Ferryman, p. 142. Fadel Allah was in fact actually recognized as Sultan of Bornu. [This is hardly correct. Sir F. D. Lugard in his *Report on Northern Nigeria for 1901* mentions that Major M^CClintock presented Fadel Allah with a shot gun of his own. This led to a report, believed by the French, that the British authorities were supplying Fadel Allah with rifles. Major M^CClintock formed a high opinion of Fadel Allah, who appears to have been a most gallant soldier and a capable and determined ruler. Major M^CClintock's report was wholly in favour of recognizing Fadel Allah as Emir of Bornu. After the French had killed Fadel Allah at Gujba, they demanded from Shehu Garbai an indemnity of \$80,000. They managed to wring \$73,500 from the exhausted country before the arrival of Col. Morland and a British column. Captain M^CCarthy Morrogh with a company of the West African Frontier Force was left at Maiduguri (Maifoni) as temporary military resident until the arrival of Mr. W. P. Hewby, who was appointed civil Resident of Bornu. Vid. Lugard, *Report on N. Nigeria for 1902*.]

Moisel, p. 184. [Captain M^CCarthy Morrogh escorted Garbai in from Dikoa.]

[Thanks! P. A. B.]

III

SITUATION AND PHYSICAL CONFORMATION

BORNU is a political not a geographical entity. It is difficult, owing to the continuous changes to which the political expansion of Bornu was subjected in the course of centuries, to lay down its outline with even approximate correctness, especially as natural boundaries are conspicuous by their absence. If we leave almost entirely out of consideration the arbitrary divisions which resulted from the partition of the country between the three colonial powers, England, France, and Germany, we cannot do better on the whole even to-day than follow the line left behind by Nachtigal as the result of his conscientious investigations.^[70] The extreme limits of Bornu practically correspond to a rhombus, whose longer diagonal runs from north-west to south-east. The tenth and fourteenth degrees of north latitude and the tenth and sixteenth degrees east of Greenwich denote in general the line beyond which districts belonging to Bornu do not, or do not appreciably, extend.^[71] Bornu is clearly defined only on the east, where Lake Chad and its affluent the Shari form a natural boundary; it is tolerably clear also in part on the south, where indeed the Mandara highlands belonging to Adamawa form a natural boundary-wall. All other boundaries are more or less undetermined. Between the Shari and the highlands, the southern boundary of Bornu is practically coterminous with the southern boundary of the Musgu tribe—whose territory long furnished Bornu with the majority of its slaves—and almost coincides with the tenth degree of north latitude.^[72] East of the Mandara highlands the boundary does not reach quite so far south, it touches here the country of the Marghi pagans between Bornu and Adamawa, and goes from about Kofa on the River Yedseram in a westerly direction till it reaches the River Gongola, follows this upwards for a short distance, and then becomes a western boundary bending round, leaving Katagum to the west, and runs north-west to Zinder.^[73] From the point of intersection of this line with the tenth meridian, the Bornu boundary then runs north of Lake Chad through the country of the

Tuareg, whose forays indeed were the cause of constant alterations of the border. The boundaries given above embrace a territory measuring in round figures 140,000 square kilometres. ^[74]

This great territory was first partitioned among the three colonial powers only on the map; ^[75] and it was not till the year 1903 that the settlement of their boundaries was commenced, a work which has only quite recently been concluded. The northern and smallest part, stretching from the Komadugu Yo and lying north of it, belongs to the French Sahara. The largest and central part consists of British Bornu, the north-easterly province of Northern Nigeria, and faces on the east the German Chad Territory belonging to the German Kameruns, and extends along the River Yedseram and the water-courses which establish a connexion in the rains between this river and Lake Chad. Only to the west of Dikoa does the boundary make a slight deviation over the Yedseram towards the north-west so as to include the arable land belonging to Dikoa.

Bornu is extraordinarily uniform in its physical conformation, perhaps more uniform than any other country of similar size in tropical Africa. It consists of a single vast plain covered with alluvial soil, with an average height above sea-level of between 300 and 450 metres, sloping almost imperceptibly to the lowest level of the Central Sudan, i.e. the shallow depression of Lake Chad, and never reaching a considerable elevation, save in the border districts, if we except the sand-dunes and the Wasa rocks which jut out of the plain. Although the question of mountain formation lies outside Bornu proper, yet it deserves notice, for where the mountains appear they mark the border districts of the otherwise flat country in conspicuous fashion. For not only the mountains in themselves, but also in places the flora which is dependent on them, give the landscape an appearance varying from that of the rest of Bornu. But since the way of life of the ruling tribes of the country is bound up with the plain and its products, these districts coincide naturally with the political boundaries.

The mountains, when they do appear, are of massive granite formation and are often of very peculiar shape; and they are only found in the extreme north-west and in the south of Bornu. The mountains in the north-west include the hill country of Zinder and Munio. They are remarkable on

account of their composition. Immediately east of Zinder stretches north and south a mass of quartz rich in natural cisterns, which is bordered on the west, south, and east by masses of granite, which partly in a continuous chain and partly in the form of a single cone form the connexion with the highlands of Munio.^[76] The granite mountains of Munio, which reach a height of 600 metres, are continued towards the south-south-east by single 'kopjes' of the same rock, which get lower and lower until they finally quite disappear. Peculiar to all this district, which has many of the characteristics of the desert, are the numerous lakes and pools very strongly impregnated with natron, which extend, moreover, as far as Chad and in places attain considerable dimensions, such as the twin lakes of Badamuni or Gadabuni described by Barth.^[77]

Far grander than the above-mentioned mountain districts are the granite masses which form the boundary of Bornu on the south for a long distance. The highest elevation reached is in the highlands of Mandara belonging to Adamawa, which push out their northern spurs, such as the Seledeba Range, which is over 1,300 metres high, like rocky peninsulas far into the Bornu plains. The peculiar wild beauty of these mountains has already been aptly described as follows by Denham: 'Though not to be compared with the higher Alps, the Apennines, the Jura, or even the Sierra Morena in magnitude, yet by none of these are they surpassed in picturesque interest.'^[78] This mountain mass is flanked both east and west by peculiar rock formations, which may best be compared to greater or lesser rocky islets in the ocean, for they jut up quite promiscuously from the alluvial plain. If one stands on an elevated point in the country round Issege and lets one's glance wander southwards, one has a surprising panorama in front of one, provided the weather is clear. The whole plain seems studded with shapes of various sizes, which assume the most bizarre forms and show the most diverse stages of disintegration; one sees hemispherical, cone or needle-shaped hills, even regular mountain ranges, such as that of Uba, some of them of considerable height, all over the plain, which are completely separated from each other and thus give the impression of insularity. The country on the north and east flanks of the Mandara Range is of similar formation, and among them is that of the well-known double cone of the Mendif. In some instances these granite formations hardly rise

above the level of the surrounding country; such masses of rock in the form of giant tables of stone embedded in the soil are found between Uba and Issege on the left bank of the River Yedseram. The already mentioned Wasa^[79] rocks are an offshoot of the extreme limit of this mountain country: they are in about 11° 30' north latitude and their contours are a welcome change in the monotony of the plain.^[80] Otherwise Bornu is entirely free from rock formations, and it is only in the country near Gujba that Rohlfs mentions the appearance of red sandstone.^[81]

The low-lying plain itself consists entirely of a sandy argillaceous soil, the composition of which, however, is not uniform but contains sometimes a larger proportion of loam, sometimes of sand. The appearances which are most characteristic of this plain, viz. sand-dunes and the 'firki' soil peculiar to Bornu, are only intelligible by studying simultaneously its hydrographic conditions, which are indeed of supreme economic importance in this Central African country.

Bornu is, hydrographically considered, one of the most interesting districts in Africa, for its character is essentially influenced by the peculiar lake known as Lake Chad and by its affluents, which form its boundaries for long stretches or flow through it. A study of this important Central African inland lagoon and its water-system, whose investigation is amongst the most disputed of geographical problems, is absolutely necessary for the understanding of many important questions. Lake Chad, whose average water-level according to Tilho is 283 metres above the sea,^[82] fills the tray-shaped depression, to which the Bornu plain slopes gradually from south-west to north-east. While it is itself extraordinarily shallow—in the deepest part the lead only marks fourteen metres—no perceptible effluent can be traced, and for a great part of its surface, owing to the fact that its banks are flat and nowhere sharply defined, and owing to a constantly varying volume of water, it is subject to continuous fluctuations. These fluctuations do not only depend on the seasons, they appear also in the course of long periods during which the shore-line may shift inside a zone of ten or more kilometres broad. A French writer, Lieut. Freydenberg,^[83] relying on native information, calculates shorter periods of twenty years, which bring slight fluctuations in the level of the lake, and longer ones of about seventy years,

during which it is said that there is an alternation between a nearly complete disappearance of water and a very considerable inundation. This theory, however, contradicts the observations of Barth, Rohlfs, and Nachtigal. When Barth on his exploring expedition visited Lake Chad for the second time (in 1855), the places along the western shore of the lake were threatened by an extensive inundation, and the town of Ngornu was actually completely destroyed.^[84] Eleven years later, on Rohlfs' visit, the shore of the lake was quite within normal limits.^[85] According to Freydenberg's theory a steady retrogression of the water-level of the lake ought to have followed until it had nearly completely dried up; instead of which Nachtigal in 1871 found an inundation which probably surpassed that of 1855, because it actually threatened Kukawa, which lies far from the lake.^[86] A similar high-water period has not appeared again since Nachtigal's time, on the contrary a continual shrinkage of the shore of the lake from year to year is to be registered; this shrinkage has been under scientific observation, for since 1900 the lake has been subjected by the French to careful inquiry in every direction.^[87] Freydenberg, who in the summer of 1905 traversed the northern part of Chad, found that the retreat of the water had proceeded so far, that between Barua and Ngigmi on the west bank a broad ridge of completely dried-up country had pushed itself eastwards right into the middle of the lake. As long as the laws which govern the periodically recurring inundations are still unknown, the yearly recurring alterations in the level of the lake can only be attributed to the rise and fall of the rivers, to which the Chad, in addition to its own relatively small rainfall, owes its supply of water. To its most important affluent, the Shari, the lake owes practically two-thirds of the volume of water discharged into it; on this river, as well as on the season, depends its average depth. At the end of December, that is in the middle of the dry season, when all the other affluents are falling and the majority of the river-courses in the southern bay of Chad are already dried up, the lake is actually still rising, for at that time the Shari is still discharging into it a large quantity of water. From the middle of December to the end of January the Chad has reached its high-water mark and then occupies an area of about 20,000 square kilometres. Soon afterwards, in consequence of enormous evaporation, the lake begins to fall rapidly and continues falling even when the rainy season has already

set in. Countless mud- or sand-banks then emerge, which have hitherto been covered with only a few inches of water, and now in some parts put on a green carpet of vegetation: owing to these the lake's navigability, always of a very limited nature, is rendered still more difficult, if not entirely suspended.^[88] In the end, apart from the often canal-like lagoons, which surround the countless islands on the Kanem shore of the lake, there remain only two comparatively small areas of open water, and they lie at the mouths of the two chief rivers, but have so little connexion with each other, that their water shows a different chemical composition.^[89]

The regrettably small proportion of water in the lake during the dry season, which was not inquired into till within the last ten years, has no doubt contributed to render it possible that individual French travellers can express themselves so extremely pessimistically about the lake's future. Audoin even calculates that according to his observations there is in normal years a sinking of about 0·15 of a metre^[90] in the level of the lake. Only the future can teach us whether the retreat of the lake—apart from the loss of water due to natural evaporation—is connected with a subterraneous outlet^[91] (perhaps into the bed of the Bahr-el-Ghazal) which it has constructed for itself in the course of years, with alterations in meteorological conditions at the sources of its chief affluents, or simply with tectonic displacement. Perhaps, as a matter of fact, it will again enter on a high-water period, similar to that observed by Barth and Nachtigal. At any rate, the information now available with regard to the lake itself, comprising only the results of inquiries of the last ten years,^[92] is not sufficient to enable us to form a definite judgement.

Lastly, one must also, with regard to the alterations in the level of the lake, take account of those phenomena recurring daily during the dry season, which certainly have a similarity with the periodical movements of this inland sea, but are attributable solely to the prevailing winds. The north-east trade wind produces, quite mechanically and always according to its strength, a slight rise or fall of the water on the flat south-western shore of the lake. D'Huart's assumption that the greater evaporation caused by the wind also influences this phenomenon seems to rest on slender foundations,

since during the time when the wind drops, i.e. the morning hours, insolation, operating in the same sense but more intensely, sets in.^[93]

Moreover, since evaporation or percolation happens to be exerted over so extraordinarily shallow a basin as that of Lake Chad, alterations are inevitable owing to the deposit of the silt brought down by the rivers. The question is, what becomes of the quantities of water from the Shari and other rivers when the basin of the lake has once been filled up with the silt? This leads to another interesting problem, viz. the connexion between Lake Chad and the Bahr-el-Ghazal, which in a way is a continuation of the lake towards the east. The views on this question, even of recent geographers, differ widely as to whether the Bahr-el-Ghazal is a former affluent or an effluent of the lake. Although Nachtigal tries to produce proof that this valley was formerly a channel of effluence of Chad leading to the depression of Bodele, which lies 100 metres nearer sea-level,^[94] yet some of the travellers who have latterly examined these regions have set forth the opposite assertion.^[95] Freydenberg cites as a proof of this the delta-like passage from the Bahr-el-Ghazal to the lake, as well as the appearance of stones in this channel, which can only have originated from the highlands of Tibesti in the Sahara.^[96] The delta-like formation at the mouth of the Bahr-el-Ghazal is at least not a sufficient argument, for it resembles in many respects the mass of islands stretching north-westwards along the Kanem shore of Chad.^[97] This archipelago is of enormous extent, and the number of its islands and islets is legion. All these islands are of sandy formation and run longitudinally from north-west to south-east, decreasing in height towards the middle of the lake.^[98] Where these rows of islands cut the Kanem shore they are continued by sand-dunes; when the level of the lake is low, those dunes which are nearest to the land assume the form of peninsulas; on the other hand, canals in the form of countless lagoons running between the islands penetrate the mainland and lend an extraordinarily jagged appearance to the coast there.

Even the island, or more correctly the peninsula, of Seyorum on the Bornu side has the same longitudinal direction from north-west to south-east; the same applies to the dune formations which continue along the Bornu shore roughly parallel with it. This remarkable identity of direction

between the dunes and the islands on the Kanem shore on the one hand, and the dunes on the Bornu shore on the other, is at the root of the conjecture in which Marquardsen indulges when he says, 'the proof that the islands belong geologically to the mainland would as a consequence lead us to the conclusion that before the lake was here there existed a desert which created the dunes.'^[99] As a matter of fact, such a conclusion would not be incorrect. How otherwise could the existence of dunes on the Bornu side, in spite of the thickly overgrown bank, be explained, unless in earlier times there was dry land where now is the bed of the lake? And the whole behaviour of the lake speaks in favour of such a theory. At all events, there can to-day be scarcely any doubt that the islands are the ridge-tops of dunes which were cut off from the land by water, but which formerly originated, like the still existing dunes, under the influence of the strong north-east trade wind prevailing in Bornu.^[100] In what manner this collection of water, to which the Chad owes its origin, has found room for itself, whether through the blocking up of the Bahr-el-Ghazal, as Nachtigal assumes,^[101] or through tectonic movement, this indeed has still to be definitely settled. Nachtigal's assumption has in its favour the greater probability; if one accepts his theory, one can also assume as causes of the blocking up, owing to the slightness in depth of the depression, the formation of dunes as well as the copious deposit of sediment by the Shari. Audoin's opinion, that volcanic disturbance has had a say in the matter, would presuppose a high antiquity for the lake, which is scarcely to be assumed, since neither the configuration of the shore nor its vegetation warrants such a conclusion.^[102]

A further point, which has called forth much discussion, concerns the saltiness of the lake. Barth, as well as Nachtigal, lays special stress on the fact that Chad is a fresh-water lake;^[103] Barth does so, although he adds that the soda-impregnated soil on the north shore of the lake 'communicates this quality to the water'. Nay, he even says further on (German edition, iii, p. 237): 'It is very remarkable that, while the water of Chad itself is fresh, the greater part of the water found in the region quite close to the edge of the shore of the lake contains natron.' It was at this place that 'the water was so strongly impregnated with that mineral as to be scarcely drinkable'. Nowadays it can hardly be any longer disputed that the water of Lake Chad is salt—though certainly in a different sense to that of other well-known salt

inland seas—even if the actual quantity of salt present is so small as to be generally not perceptible to the taste, at least at the period of high water.^[104] The saltness of the water varies not only according to the time of year, but also according to its situation. The places which are least salt are those which lie at the mouths of the rivers and which are distinguished by the free flow of water. One may well distinguish the water here at all times of the year as fresh or quite fresh.^[105] According to Tilho's map of Lake Chad,^[106] the saltness is greater to the north of the two stretches of open water than to the south of them. This tallies with the fact that those pools which are partly separated from the lake, and which contain a more or less strong solution of salt according to their volume of water, leaving behind when evaporated a visible layer of this mineral, are also to be found along the banks of the northern half of the lake, just where Denham was struck by the taste of natron in the water.^[107] These salt pools form the continuation of that strip of country already mentioned, dotted with salt lakes, which extends from Munio to the north-west corner of Lake Chad and forms the southern boundary of the Sahara. The soil here contains everywhere a considerable amount of natron, which gradually disappears as one proceeds southwards. It is only natural that the water of Chad should penetrate the natron-impregnated soil and communicate traces of this mineral to the whole basin of the lake, traces which will be perceptible in a greater or less degree according to the hydrographic relations already described.

Of the rivers which mingle their waters in Lake Chad, the Shari with its affluents and side streams is by far the most considerable and also the most important for Bornu, although it is only in its lower reaches that it flows through this country. Rising outside the Sudan proper, with its furthest southerly sources lying somewhat below 6° of north latitude, in a district whose climatic conditions show a strong resemblance to the damp tropical basin of the Congo, it conveys a powerful volume of water, and, including the Logone, which debouches into it, is the only affluent of Lake Chad, which contains continuously flowing water at all times of the year. In its lower reaches, which form the eastern boundary of Bornu, the river has an average breadth of from 2,000 to 4,000 metres, but below Kuseri it has dug itself a bed contracted to 1,000 metres, and its banks have often a sheer drop of as much as ten metres. This bed is, of course, not always full.

Towards the end of the dry season it is beset or constricted by multitudinous sand-banks, and the river itself is generally not more than 400 metres broad and often less, but, nevertheless, even then retains its navigability, and, with the exception of a few fords, cannot be crossed except in canoes.^[108] In June the river begins to rise and reaches its highest point in October, when its level has risen from four to seven metres; it then completely fills its bed, and under certain circumstances even overflows its banks far and wide. Conditions similar to those of the Shari prevail with its most important affluent, the Logone. This river also takes its rise in districts which are lower-lying than Bornu itself, though its sources do not reach quite so far south; it also conveys a constant stream of water, and in the rains shows a still greater inclination to overflow its banks even than the Shari. Besides, in the rains it conveys to its lower reaches, through numerous streams and rivulets, a large part of the heavy rainfall from the Mandara highlands. The Logone, which rises from three to four metres in the rains, shows in places a canal-like character; its course is very tortuous, for with a very small fall it apparently has great difficulty in forcing its way through the clayey soil of the absolutely level plain. To the same reason is attributed the endeavour of the river to build up a system of steep-banked side streams, such as the deep and canal-like Lage-Matia (Laho-Matia) below Musgu—a sort of discharge-valve for the pent-up masses of water. To this also may be traced the origin of the numerous natural canals which unite the Logone and the Shari, and which caused Barth to call this region ‘the African Holland’.^[109] Many of these canals appear to have no current whatever, and are designated in the language of the country ‘ngaldjam’^[110] according to Barth, and are called by the same traveller ‘meadow-water’ or ‘flooded meadows’. This extensive network of natural canals is nevertheless not sufficient to drain off the volume of water brought down by the river, and so from about the end of September to a time when the waters of Lake Chad are still far from being at their highest level, widespread floods occur, which place a large part of Bornu under water. Every year from August to January the whole country between Logone and Matia is one huge lake which, during the time when the water reaches its highest level, may even extend to Mandara, and out of which the towns of the Musgu stick up like islands; the right bank of the Logone till it reaches the Shari is also then far

under water. Communication between village and village is then only possible in canoes, and even the harvest has often to be brought in in this fashion.^[111] At the beginning of the dry season the water in the canals very quickly runs off and only remains in isolated spots—in fact, it often dries up so completely that the canals appear as hollow roads, and can be used as avenues of communication by people on foot.^[112] To the Shari-Logone system belongs also the Ba-Ili, the ‘second great artery of the Musgu country’,^[113] which has just the same characteristics as the Logone, with which it is also connected through the various ‘ngaldjam’. It has this difference, however, from the latter, apart from its size, in that during the dry season it contains no flowing water, but only exhibits larger or smaller stagnant lagoons. Moreover, before the Logone, reinforced by the Ba-Ili, discharges itself into the Shari, it sends towards Lake Chad a regular network of waterways of its own, which are again themselves connected with the Shari, and thus give the impression of a wide-branching delta. The many-armed delta proper at the mouth of the Shari is, however, not in connexion with the network of canals. In their condition during the dry season these water-courses resemble the canals connecting the Shari and Logone; only a part of them then have water communication with the lake. Before the many-limbed river and canal system of the Shari-Logone reaches Lake Chad, it occasions in places the formation of a very remarkable kind of soil especially characteristic of these regions, a formation which is common also to the other rivers flowing into Chad from the south of Bornu. This formation is known in the country under the name of ‘firki’, a peculiar boggy loam—very expressively called ‘terre cassée’ by Lenfant^[114]—which is saturated with water in the rainy season and forms a gigantic sponge, while the humidity thus absorbed is again returned to Chad in the form of little rivulets. Though entirely covered with water in the rainy season, the ‘firki’ dries up within a few days, the deposits disappear, and the rivulets dwindle, so that the soil as far as the eye can reach is split open by gaping cracks over a yard deep, which divide the surface into larger and smaller many-cornered lumps, and render the crossing of the marsh over these stretches extraordinarily arduous both for man and beast. Only isolated spots still remain covered with water, and form during the dry weather perennial pools. Marquardsen’s theory^[115] that the ‘firki’ is a

loamy deposit above a sandy subsoil has the most probability: the sand allows the water to sink through and withdraws it from the upper layer of loam, which, receiving no new accessions of humidity, dries up completely under the intensive insolation and splits asunder into big lumps. It is only where the subsoil also is loamy that it is possible for the water to lie and form those lakes that continue even during the dry weather. The layer of 'firki', quite different in its strength in different localities, is only interrupted by the higher dune formations, it submerges the more insignificant sand outcrops.

In many more striking ways the presence of 'firki' soil on the lower course of the Yedseram, the third largest of the rivers flowing towards the Chad basin, makes itself felt, as also on the smaller rivers Ngadda, Goma, and Ngua, which happen to flow parallel to it. All these do not reach Chad itself, but are in communication with the lake only during the rains by means of larger stretches of flood water, in which the current of the river is still perceptible, whilst the 'firki' soil underneath, probably of considerable thickness here, absorbs vast masses of the water.^[116] The 'firki' is here sometimes piled up into regular dikes—probably built up on sand-dunes underneath—as one can clearly perceive in the dry weather at certain spots between Ulugo and Bornuski. At this time of the year there remain also channels leading from these 'firki' masses to Chad, such as the Mbulu, in the form of stagnant shallow canals consisting of extremely dirty water, whilst the streams ending in the 'firki' district have long ago dwindled away. The Yedseram is remarkable in that its hard gravelly bed contains in its upper courses even in March, i.e. at the end of the dry season, a continuous flow of water; and above all, in that its sources springing from the Mandara Highlands form even then mountain brooks of crystal clearness, and in some cases of considerable size. But below Mutube the water dwindles and becomes more and more exiguous among the pebbles, and then enters the dried-up river-bed consisting of pools more or less deep, whose water becomes more and more dull-coloured as the loamy low-lying plain is reached. At the end of September it has reached its highest level, and is then impassable except at the fords even above Issege, but even by October the running water has quickly fallen again, and recedes further and further towards the source.

Much more considerable than this river, both as regards its length^[117] as well as its volume of water, is the river of Yo or the Komadugu-Yobe. It takes its rise from quite a number of streams, which unite into one river below the ruins of the ancient capital, Ghasr Eggomo, after forming large and numerous fertile islands in a rich low-lying plain. The main stream proper comes from the Hausa States and is known as the river of Katagum, that on the left, which is not much smaller, as the river of Hadeija, and that on the right coming from southernmost Bornu and rising south-east of Gujba is called Ansei. The streams coming from the north-west, such as the river flowing from Yamia, are mere 'wadis', as might be expected from the desert districts where they rise, and seem to contain running water only intermittently. The bed of the Komadugu-Yobe is scarcely fifty metres broad at the mouth, and the river discharges itself into Lake Chad without any delta being formed; during a great part of the year it contains no running water, and at the beginning of the rains has nothing to show but a few larger or smaller pools. At the beginning of July the river begins to contain running water, and by the end of November, when it reaches its highest level and overflows its banks, it can only be crossed by artificial means; but it falls again very quickly, and by the end of January it is again fordable, so that it can only fulfil the duty entrusted to it of forming a bulwark on the border against the hordes of robber Tuaregs in a very imperfect fashion and for a very short time.

Nachtigal, ii. 380.

Denham gives the respective degrees as 10° and 15° N., and 12° and 18° E. (Denham, ii. 138). The difference is accounted for by the fact that Denham regarded Kanem as still forming part of Bornu.

Along the tenth degree of north latitude runs also a part of the present boundary-line between the German Kameruns and the French Congo.

Cf. also the map annexed to Lugard's 'Northern Nigeria' in *Colonial Reports*, 1907. Even this boundary-line is somewhat arbitrary, since the districts in question belonged sometimes to Bornu and sometimes to the Hausa States on its western border. But the spheres of influence of the adjoining Sultanates are not exactly determined.

Nachtigal puts it at 150,000 square kilometres. Nachtigal, ii. 382.

Cf. also below.

Chudeau, p. 334.

Barth, iv. 68. One-half of the lake contains fresh water.

Denham, i. 174.

Barth, iii. 255.

Even immediately to the south of Chad shore, but outside Bornu, in Bagirmi country, there is a similar rock formation, Hadjer el Hamis (called Hadjer Teous by Denham) composed, according to Foureau, of rhyolite. Foureau, *Doc. sc.*, p. 728. Cf. also Denham, ii. 52, and the plate in Lenfant, p. 203. [Cf. [p. 311](#) of the present work.]

Rohlfs, ii. 120.

La Géographie, xiii (1906), p. 203.

La Géographie, xv (1907), p. 169.

Barth, ii. 322; cf. v. 396. Even Denham mentions a considerable rise of Chad (at the beginning of February), owing to which the cotton plants at Wudi were under water. Denham, i. 73.

Marquardsen attributes the high water mentioned by Rohlfs (ii. 87) to the flooding of the River Wobe (*Mitteil. aus d. deutschen Schutzgeb.* xviii. 328).

Nachtigal, ii. 9.

M. Audoin, *La Géographie*, xii. 308.

Freydenberg, *La Géographie*, xv. (1907), 166.

Tilho, *La Géographie*, xiii. 205; cf. also the map annexed to his article.

La Géographie, xii. 310.

Nachtigal, ii, p. 357; Tilho, *La Géographie*, xiii. 204.

Information obtained prior to this only concerns the banks of the lake.

D'Huart, *La Géographie*, ix. 164; M. Audoin, *idem*, xii. 318.

Nachtigal, ii. 116 and 120. Cf. also Denham, ii. 57.

Foureau, *D'Alger au Congo*, p. 670, and *Doc. sc.*, p. 278.

La Géographie, xv. 162.

[Vid. [Appendix IX](#) and note [\[343\]](#).]

Audoin, *La Géographie*, xii. 305.

Mitteilungen a. d. deutsch. Schutzgeb., xviii. 339.

Audoin, *La Géographie*, xii. 307.

Nachtigal, ii. 359.

Mitteilungen a. d. deutsch. Schutzgeb., xviii. 339.

Barth, ii. 325; iii. 53.

Tilho, *La Géographie*, xiii. 204.

In February 1904 I camped at Ulugo on the southern shore of Chad, and for days I had to use the lake water for cooking and drinking. Although the pools, from which I took the water, were cut off from circulation with the open water owing to thick ambach and papyrus vegetation, yet I could not perceive the slightest taste of salt.

La Géographie, xiii, no. 3.

Denham, ii. 92; Nachtigal, ii. 328; Foureau, *Doc. sc.*, ii. 665; Destenave, *La Géographie*, vii. 425.

Cf. Nachtigal, ii. 549; Foureau, *D'Alger au Congo*, p. 712.

Barth, iii. 241.

Kund, as well as Dominik, indeed affirms that 'ngaldjam' means a rhinoceros in the Musgu language, so there may be some misunderstanding here. *Mitteilungen a. d. deutsch. Schutzgeb.*, xix. 18; Dominik, p. 239. ['Ngaljam' is a Kanuri word equivalent to the Hausa 'faddama', meaning a shallow water-course.—P. A. B.]

Stieber, *Kolonialblatt*, xvi. Jahrg., p. 83.

Cf. plate, Dominik, p. 188.

Stieber, *Kolonialblatt*, xvi. Jahrg., p. 117.

Lenfant, *La grande route du Tchad*, chap. vi, viii.

Mitteilungen a. d. deutsch. Schutzgeb., xviii. 342.

Cf. Rohlfs, ii. 66 and 72.

Nachtigal gives the length as about 600 kilometres. Nachtigal, ii. 353.

IV

CLIMATE

CLIMATICALLY, as well as geographically, Bornu belongs completely to the tropic zone; although cut off from the warmth of the Equator, its average temperature throughout the year of 29° to 30° centigrade makes it not only one of the hottest countries in Africa, but even of the whole world. [118] Though the climate of Bornu may on the whole be called regular, considerable fluctuations in temperature nevertheless take place, as is proved by the records of the last sixty years.

Of all the travellers who have spent any length of time in Bornu, the only observations that extend over a detailed period are those of Barth and Nachtigal; moreover, those of later travellers, which embrace a few months at the most, in the main and on the whole, in so far as they are not of small value, confirm the former results. Barth's observations have the advantage of extending over several years, of embracing all parts of the country which he personally visited, and of having been manifestly taken at places whose situation could not influence the result one way or other. But their reliability is somewhat prejudiced by the fact that the necessary thermometrical readings are sometimes discontinued for weeks at a time, and were resumed on odd days at quite different times. These irregularities happen to be of especial importance in the thermometrical readings, which were taken after midday or during the afternoon, when according to experience quite considerable increases of temperature may occur in a short time. Owing to the fact that the traveller, especially at the beginning of his journey, was not always in a position to be able to take observations during the hottest part of the day, many high maximum temperatures no doubt escaped him. Barth himself was well aware of the insufficiency of his data, for he refers to them throughout only as 'Fragments of a Meteorological Register'.

Nachtigal's observations are extraordinarily conscientious and arranged with greater regularity and include also barometrical readings. But they

extend, as far as Bornu is concerned, only over seven months and omit precisely the period when the highest temperatures occur. Moreover, the place where the observations were taken, viz. the court-yard of Nachtigal's house in Kuka, was such that it must have influenced at any rate the thermometrical readings. Inside the walls of the town, daily baking in the sun, the nightly radiation from the soil does not nearly reach the same intensity as in the open plain; this is strikingly apparent in Nachtigal's thermometrical readings. There is an absence throughout the winter months of the low temperatures, shortly before or after sunrise, which have struck every one travelling in Bornu at this season.^[119] Moreover it is not to be forgotten that the shade-tree, under which Nachtigal took his observations, influenced the result both of the maximum and minimum readings.^[120]

The table given below makes it apparent that April, with an average temperature of 34° centigrade, is the hottest month of the year and December the coldest; further, that the chief rainy months, July, August, and September, show the least fluctuations in daily temperature, while the coolest months, November, December, and January, show the greatest. The reasons are not far to seek, they depend on the position of the sun together with the degree of intensity of its rays resulting from its position. At the end of March the sun approaches its zenith for the first time—i.e. as regards Central Bornu—the atmosphere begins to be saturated with moisture and this lessens the fall of temperature at night, which becomes still less marked in April, in which month prevails the most oppressive heat, not yet mitigated by the rains.^[121] The first tornadoes now set in, and become more and more frequent, while their ever-increasing volume lessens the nightly fall of temperature, though it affects the daily temperature in the opposite direction, so that the daily fluctuation of temperature in July and August only reaches 9°. At the beginning of the month of October the tornadoes cease, and consequently there is an ever-increasing rise of temperature to correspond, which, however, very soon drops in the following months.

Month.	Nachtigal's Observations. Degrees in Centigrade.			Barth's Observations. Degrees in Centigrade. (Those given in Fahrenheit reduced to Centigrade.)				Approx. avg.		Mon. avg.
	Avg. snr.	Avg. 2 p.m.	Hst. fl.	Avg. snr.	Avg. 12-1 p.m.	Avg. 1-2.30 p.m.	Hst. fl.	Avg. snr.	Avg. 1- 2.30 p.m.	
January	18·9	30	16·2	15	35	33 (?)	24	17	32	24·5
February	19	30·2	16·1	15	31	35	19	17	33	25
March				21	37	36	16	21	36	28·5

April				25	37	43	18	25	43	34
May				23	40		17	23	43	33
June				24	37		20 (?)	24	40	32
July				24	31		9	24	34	29
August	23·3	29·3	9·2	22	30			23	31	27
September	24·1	31·4	10·8	22	31	36	11	23	34	28·5
October	24·5	35·5	16·1	24		36	15	24	36	30
November	20	33	16·5	16		32	27	18	32	25
December	18·5	30·4	16·3	12		31	26	15	31	23
Yearly average of temperature measured at sunrise and at 2 p.m.								21	35	
Total yearly average										28°

Naturally the conditions for the northern and southern parts of Bornu vary somewhat, and local circumstances may also cause quite considerable fluctuations of temperature. Thus I noticed at Idjege in Southern Bornu, at the beginning of October, that the granite masses thickly studded with trees, on the slopes of which the place lies, had the result of causing an unusual increase in heat, which close to the surface of the soil measured 51° centigrade in the shade. While the nights here were even oppressively hot, a few days later on the plain at Dissa, not far from Idjege, the thermometer only showed 10° centigrade in the morning before sunrise, and by so doing registered the lowest degree of temperature, whose fluctuations far exceeded 30°. In general the heat reaches its highest point about 2 p.m. and then only declines very slowly, so that even in the evening abnormally high temperatures may be registered,^[122] in fact the interesting phenomenon may occasionally be observed of a higher temperature at sunset than at midday.^[123] If one takes the yearly average temperature according to Barth's and Nachtigal's tables, one cannot but notice that Nachtigal's tables leave out of consideration just the hottest months. Having regard to this circumstance, one may reckon without much inaccuracy the yearly average temperature at from 29° to 30° centigrade.^[124]

With regard to the second most important factor in the climate, viz. the winds, there exist no such detailed observations as with regard to the temperature; still the facts communicated on this subject by Nachtigal and recent French explorers, especially Foureau,^[125] are sufficient to show that just as the atmospheric pressure is extremely regular so likewise are the winds. During the greatest part of the year, under the influence of the north-east trade-wind, north-easterly or easterly winds predominate, but south of 12° north latitude, where the wind comes mostly from the south with a decided inclination towards the south-east,^[126] it is less constant, a fact

which Foureau attributes to the counter-influence of the south-east trade-wind which reaches as far as here.^[127] According to the universal verdict of all observers, easterly and north-easterly winds predominate especially in the dry weather, while in the rains, especially in July and August, westerly winds gain the upper hand, only to surrender their predominance to the easterly winds in October.

Views as to the daily fluctuation in the strength and direction of the wind appear to vary. Unfortunately Nachtigal's tables^[128] do not inform us whether observations with regard to the wind were taken also at night-time. The French naval officer Audoin, who personally took observations of this sort on Lake Chad, where the conduct of the watery plain is bound to give indications conformable to every movement of the wind, lays stress on the predominance of the north-east wind, especially from 11 p.m. to 9 a.m., while it is replaced by the south-west wind for the rest of the day.^[129] The north-east trade-wind brings with it from the Sahara a great volume of dust, which often renders the atmosphere so opaque that the sun only appears through it like a dark-red disk. This phenomenon, which causes a peculiarly oppressive sensation,^[130] is known throughout the whole of Western Tropical Africa under the name of Harmattan, but is especially characteristic of the Central Sudan.

Apart from the violent atmospheric disturbances which the tropical thunderstorms bring with them, the cyclones or tornadoes, coming sometimes from the south-east, sometimes from the north-east or south-west, especially during the times of the change in the wind, affect Bornu as well as the whole tropical district of the Western and Central Sudan.

In Bornu, where over wide stretches of country the weakening effect of dense forests is lacking, they attain an unparalleled violence. They give notice from afar of their arrival by thick clouds and whirling dust, and sweep everything off the earth that has not been made trim and taut. Sometimes cornfields which have been overtaken by them look as if a roller had passed over them. During the autumn of 1903 the roofs of the houses at Dikoa were blown off by a violent tornado. In general the cyclones are not preceded by discharges of electricity, but there are exceptions to this rule.

The tornadoes may be matched in violence by the thunderstorms which occur in Bornu. As is often the case in the tropics, they are heralded beforehand by the discharge of a large amount of electricity, a phenomenon which it is especially fine to observe at night, when the lightning flashes following each other uninterruptedly render it possible to read a book. [\[131\]](#)

Whilst on the subject of hydrometry, the question of cloud formation has next to be considered. The sky in Bornu, quite apart from the Harmattan, is very seldom quite clear; it is only in winter that there may be several successive days free from clouds; at other times 'cirrus' clouds at any rate are perceptible. The most wonderful cloud formations can be observed at the transition period shortly before the bursting of a thunderstorm. The results are analogous to those prevailing in Europe under similar circumstances; only in the Central Sudan the 'cumulus' formations differ in being grander, and they discharge themselves more quickly.

Fogs occur, but they are chiefly confined to the cooler parts of the year, and usually, as in northern latitudes, are connected with the presence of swampy districts, such as the damp low-lying shores of Chad. Besides it would be difficult, especially in the cooler morning hours, to distinguish sharply between fog and Harmattan.

Bornu lies exactly in the zone of the tropical summer rains, whose northerly boundary, naturally not to be sharply defined, coincides roughly with the fifteenth parallel of latitude, and thus includes also the most southerly part of the Sahara. [\[132\]](#) The moisture that falls there is only in the form of dew and rain. The mountains which border the country are not high enough to favour the formation of hail, which, however, is often enough to be seen in the neighbouring highlands of Central Adamawa and the grasslands of the Kameruns that lie to the south of them. Dew appears—though only seldom—chiefly in September, October, and March; but then usually in profuse volume. Thus Denham in March and Barth in September experienced a quite extraordinarily heavy dew on the banks of Lake Chad, which they thought worthy of particular mention. [\[133\]](#)

The weather conditions accompanying the beginning or end of the rains usually occur with great regularity in Bornu. In this respect the country

shows the typical Sudan climate, which is generally entirely without rain during the months of November, December, January, and February. The only noteworthy exception appears to be the actual borders of Chad. On the evening of the 22nd of January, an unusually cool day, and on the following morning, in the immediate vicinity of Chad, Foureau notes violent showers of rain, which were preceded in the afternoon by slight showers with some thunder and lightning. ^[134]

The rainfall is influenced in two different ways by the mountains on the southern border. Firstly, the volume of rain that falls here—though proper measurements of it have hitherto not been forthcoming—appears to be more intense than in the plains; and, secondly, the rains here set in earlier and do not end till later. ^[135] A greater or lesser distance from the Equator operates in a similar way. For the further south one goes, the longer is the duration of the rainy season and the greater the volume of the rainfall.

Apart from the above-mentioned exceptions, the first drops of rain do not fall in Bornu before the end of March. The downpour is heaviest in July and August. Though many of the tornadoes are extraordinarily severe and long-continued, yet the rain never lasts without intermission for a whole day or even a week, as it may do in the *primaeval* forest region of the Kameruns. ^[136]

In spite of the low rainfall as compared with that of Adamawa, which lies to the south of it, the climate of Bornu, except in the dry weather, thanks to its low-lying marshes, is distinctly unhealthy. ^[137] Besides the diseases, such as dysentery, which appear at all times of the year, malaria and its attendant evils are chiefly prevalent during the rainy and equinoctial seasons; Europeans especially are subject to it to an unusual extent, as are also the Arabs and Berbers accustomed to the healthy climate of the desert. ^[138] But even the natives are not entirely exempt from such diseases, which may assume the form of devastating epidemics. ^[139] The only part free from fever seems to be the uninhabited belt of marshy vegetation round Lake Chad, because infection from man to man is here impossible, although there are clouds of mosquitoes ready to convey the germs of the disease. On the same grounds the larger towns, which scorn all sanitary regulations, are the worst centres of disease. The irritant microbe which lives in the *anopheles*

mosquito is paralysed at a temperature below 17° centigrade, and consequently dies during the cold nights of the dry weather, in the event of its host not spending the night in the warmth of a house.

The rainy season in Bornu is dreaded by the Berbers or Tripoli Arabs not only on account of the danger to their own health, but the condition also of the pack-camels bred in the Sahara is influenced very unfavourably by it, in fact it sometimes causes great mortality amongst them, so that the caravans, whenever they can, leave the country at the beginning of the damp weather.^[140] It is not yet certain whether the camels fall victims to the same disease from which the animals bred in Bornu, horses and especially oxen, have to suffer. Inquiries into the question of whether it is spread in the form of a contagious disease like trypanosomiasis—the disease whose germ is conveyed by the tse-tse fly—have not yet been published, but at any rate myriads of flies belonging to the ‘Glossina’ species are to be found in the neighbourhood of the water-courses fringed with trees and bushes.

It is obvious that the moist heat of the rainy season is favourable to the growth of all kinds of sickness. Besides venereal disorders, malignant eye-diseases are very common, and, owing to the indolence of the natives, find every chance of being spread abroad; leprosy is by no means unknown.^[141] One of the commonest evils is that caused by the guinea-worm, a skin-disease which, though not dangerous, is protracted; the worm enters the body through the use of unboiled water from puddles, water-courses, and even from wells, and almost every native has been attacked by it at some time during his life.^[142] Cleanliness and the scrupulous observance of simple rules of health, including especially regular bodily exercise, afford, however, far-reaching protection against all these diseases, not excluding malaria itself, so that at any rate Europeans can secure themselves in some degree against sickness. All these districts are at all events far from being so unhealthy as the region of the West African primaeval forest, where the germ-destroying effect of the sun’s rays seldom penetrates.

Barth, to whom the conditions of temperature at Massawa were not yet known, even asserts that 'the average summer temperature of Kukawa exceeds that of any other place on earth'. Barth, iv. 12. [To bring degrees centigrade to degrees Fahrenheit multiply by 9 and divide by 5 and add 32.]

Cf. Barth, iv. 12, where Barth refers to 'the coldest night of his whole journey', and mentions that the temperature shortly before sunrise was only 4·5° centigrade above freezing-point. The Yola-Chad Boundary Commission repeatedly experienced similar cold nights in the winter of 1903-4, and the native carriers suffered severely from them.

Certain trees, e.g. the sappy *Candelabria-Euphorbia*, exhibit in their shade a temperature differing some three or four degrees from that of their surroundings. I can testify that this was so at Madagali, in the Mandara highlands, where my tent was pitched under a very large *Euphorbia*. The temperature in the open tent, on the contrary, did not differ from that outside.

At the end of this month Barth registered 45° centigrade in Bornu. Barth, v. 406.

In the very south of Bornu Stieber was able in March to observe temperatures which dropped from 45° centigrade at 8 p.m. to 30° centigrade. *Kolonialblatt*, XVI. Jahrgang, p. 88. Cf. also Denham, i. 150.

Barth, iii. 625. On April 12 at 1.30 p.m. the thermometer stood at 35° centigrade, at sunset it was 37·3° centigrade.

[Vid. [Appendix VIII.](#)]

Doc. sc., i. 128.

[Vid. [Appendix VIII.](#)]

Doc. sc., p. 81.

Nachtigal, ii. 766.

La Géographie, vii. 317.

[On the contrary, most people find the Harmattan tonicky and exhilarating. Cf. article by Seefried in 'M. Deutsch. Schutzgeb,' 26 (1913) 9-12].

Other electrical appearances connected with thunderstorms have not yet been observed, but Lenfant mentions seeing the Zodiacal Lights for several days in the winter. Lenfant, p. 285.

Nachtigal, i. 558; ii. 193 and 315.

Denham, i. 120; Barth, iii. 23 (German edition).

Foureau, *D'Alger au Congo*, p. 622, and *Doc. sc.*, i. 133. I can myself confirm that rain occurs on Chad at this time of the year, for on a ride from Maduari to Kukawa on the very cool morning of January 15, 1904, I travelled for a short distance in a tornado.

Cf. Rohlf, ii. 14.

[For rainfall in British Bornu vide [Appendix VIII.](#)]

[Bornu is one of the healthiest Provinces of Nigeria.]

Cf. *idem*, ii. 89.

Nachtigal, i. 733; cf. also ii. 488.

Denham notices the insalubrity which accompanies the rains with the words: 'All the quadrupeds, as well as bipeds, transplanted from the countries bordering upon the great ocean appeared to suffer alike,' i. 264; cf. also Nachtigal, i. 681.

[Lepers in British Bornu are now concentrated in two leper-camps under the supervision of the Government Medical Officers. Average numbers in Maiduguri leper-camp 950, in Geidam 120. They are housed, fed, and clothed by the Beit-el-Mal (Native Treasury). These figures do not include the pagan districts.]

[Intestinal worms are also very common, especially among the cattle-owning Shuwas.]

V

FLORA

THE flora of Bornu corresponds exactly to the geological formation and climatic conditions of the country.^[143] The usual appearance of the country is that of a tree-covered steppe, which extends south of the border-line of the regular summer rains over the vast plains of the interior of Africa from somewhat north of 8° of north latitude, from Senegal to Cape Guardafui, east of the great Central African lake and reaching southwards as far as South-East Africa. The character of this steppe is dependent on the predominance of deciduous plants, mostly belonging to the order of Leguminosae, and of many thorny trees and bushes, which very gradually give way on the borders of the country, according to local conditions, to representatives of the flora of the desert, of the damp tropical uplands or of the primaeval forest district, or else are found intermingled with them. Whilst Bornu shows in the north over a wide extent somewhat more luxuriant examples of species belonging entirely to the flora of the desert, on its own side also it spreads plants peculiar to itself pretty far into the south of the Sahara, but it lacks representatives of the above-mentioned grass-lands and primaeval forest flora owing to the predominance of evergreen trees and the appearance of epiphytic vegetation. Nowhere in Bornu can one find the dark-green forests with their hot-house humidity and tree-loving ferns, orchids,^[144] and begonias, with pandanacea and different kinds of climbing calamus, with raphia- and oil-palms, which in dense lines edge the water-courses of the grass-lands lying not much further south. Of plants which are peculiar to such districts, the banana is only to be found at one place in the country, and that curiously enough is at Zinder, in the north-west corner of the country, whither it must evidently have been introduced from outside.^[145] The raphia-palm, *Raphia vinifera*, and different varieties of paw-paw^[146] first appear on the southern edge of the Mandara mountains, which lie scattered to the south of the tenth parallel of north latitude, and only one variety of *Ceiba pentandra*, which is indeed

stunted enough in comparison with the specimens of giant *Ceiba* (*Bombax*) of the West African primaeval forest, is to be found here and there in the neighbourhood of the water-courses. There are indeed the trees of the *Ceiba* species, which, in contradistinction to the evergreens, shed their leaves in the dry weather—even in the primaeval forest. It appears rather strange that none of this kind of vegetation occurs along the course of the Lower Shari, which is always full of water, although it is plentifully represented at the sources of that river, especially the Gribingi,^[147] and although it extends for a short distance down-stream close to the banks. The reason of this phenomenon may perhaps be sought for in the fact that the long and intensive dry weather of Bornu, and also of the Lower Shari country, is not favourable to this kind of vegetation. This is analogous to the similar phenomenon on the Upper Nile, where the plant formation of the hot damp countries where the river rises extends through districts which get drier and drier as one proceeds northwards, but cannot advance at all after the latitude of Khartum.

The sole or at any rate predominant landscape as regards Bornu, so far as it depends on the flora, is just this tree-studded steppe.^[148] Even the trees are in unison with this steppe; if they do not belong to the prevailing species of acacia, they are at any rate of a habit in accordance with them. They are in general low, stunted and twisted, with deep roots and wide-spreading crowns, not higher than the trunk of a twenty-year-old oak sapling; only along the water-courses and where a deep layer of ‘humus’ soil exists do they reach a more considerable height, but even in the most favourable spots they do not much exceed 25 metres. The density with which the trees are planted depends also on the condition of the soil. Generally the trees stand in high steppe-grass at such long intervals that their crowns hardly touch, then the landscape reminds one of the extensive fruit-tree plantations of the north. The better the soil is, the closer stand the trees and thus form dense wide-spreading forests, often tangled with creepers, as for example in the riverain district of the Lower Shari from Karnak to Lake Chad.^[149] In other places, especially on the granite slopes of the hills that jut out into the Bornu plain, the trees occur so far apart, that the landscape comes to show a great similarity with the slopes of the Sierra Nevada covered with sparse ilex trees which dip down to the Rio Genil in Southern Spain. In fact, where

the true tropical trees, such as the palm and fig species, or the strange-looking 'kuka' (*Adansonia*) are wanting, the general impression of the flora is quite that of Northern Europe. Often, especially in winter when the ground is covered with fallen leaves, one can find places that at once recall the tree-bordered river valleys of the low-lying plains of North Germany in autumn. Localities which give a regularly tropical impression are at all events not in the majority, and the inspired descriptions of tropical luxuriance which Nachtigal gives on his first acquaintance with these parts, originated, as he himself has to admit, from the impression of dissimilarity between the Chad countries and the Sahara through which he had just struggled. ^[150]

By far the largest number of characteristic trees and bushes are naturally of such a stamp as are suited to the conditions of the country. The arrangements designed for the preservation of species are the thorny spikes for defence against annihilation by things that feed on them or for protection against drought during the winter months. Most of the trees shed their leaves in the dry weather, but those that do not, possess a quantity of thick leathery foliage or fine feathery leaves, succulent twigs, tendrils with tough outer fibre or thick cork-like bark. On the shea-butter trees, *Butyrospermum Parkii*, and the ilia-acacia, *Burkea africana*, and especially on the timber found in Southern Bornu, this thick cork-bark is to be observed, serving at the same time as a protection against the 'bush' fires. ^[151]

Soon after the beginning of the dry weather the high grass is quite dry, and affords ample material for the 'bush' fires that now commence. The trees and bushes, sticking out of the burnt, ash-covered soil with their bark all charred, offer then a melancholy spectacle. This may make them appear as if quite dead, but as a matter of fact the bark, of which only the outer layers have been burnt, has been able to protect the sap so well, that just before the beginning of the first rains tree and bush put forth fresh leaves and twigs or even stand dressed in full bloom. ^[152] About the beginning of January the Leguminosae begin one by one to bloom, and shortly afterwards the other trees and shrubs follow suit, so that the 'bush' is soon pervaded with an overpowering perfume.

A few species of trees form a remarkable exception, for it is just during the rains that they are leafless and seemingly dead, yet as soon as the rains are over they put forth new leaves and reach their greatest luxuriance in the middle of the dry weather.

After the first tornadoes the ash-covered earth begins to cover itself with a thick green turf, which gives the landscape a park-like appearance, and from it spring in a few days great quantities of variegated Ranunculaceae and lilies, followed soon afterwards by the flowers of such trees and shrubs as have not already come into bloom. When the flowers come into view, the season corresponds somewhat to the spring-time of Central Europe, and a similar state of things continues for the following months; by June or July the leaves are all fully out, and in October, at the end of the tornado season, the latest fruits have ripened. Among the fruit-bearing trees that grow wild are some that are made use of by the natives; as regards the greater part of those mentioned by European travellers as being useful, there are very various reports. Barth, who seems to be by no means dogmatic, praises them for the most part, but Denham expresses himself less favourably, and the greater part of this traveller's observations come pretty near the truth. ^[153] At all events the really pleasant-tasting fruits are quite rare in the 'bush'.

The flora of Northern Bornu is also strongly influenced by the Sahara, which pushes forward representatives of its own here and there far beyond its borders. In addition to the different species of thorny acacia, similar to each other in habitat, which after reaching a certain age generally have umbrella-shaped crowns, and which are so generally characteristic of all the sterile districts of Africa, the two regions have quite a number of plants in common. ^[154] The two most striking, which one meets almost everywhere, are the *Zizyphus* species and the oschar, *Calotropis procera* (called by Barth *Asclepias gigantea*). The wide distribution of the *Zizyphus* species is explained by its suitability to all kinds of soil and the usefulness to mankind of its fruit, which is mealy tasting and slightly sweet, and for the sake of its branches to form thornhedges as a protection against beasts of prey. The *Zizyphus* is found northwards as far as the Mediterranean countries ^[155] and southwards as far as the boundaries of the Sudan climate. The two varieties

principally met with in Bornu are those called by the natives ‘Kurna’ and ‘Magalia’, *Zizyphus Jujuba* and *Zizyphus spina Christi* respectively. The hardy ‘Magalia’ may take the form of magnificent shade-trees, especially in sandy soil, and constitute in uninhabited places thick thorny copses, which lead one to conclude almost with certainty on the presence of former settlements. The ‘Oshar’, far the tallest of all the Asclepiadae, ^[156] owes its wide distribution to the seeds, provided with silky floating fluff, distributed by its fruit, which is shaped like a small bladder and is larger than a man’s fist. These plants, which are in their fullest foliage towards the end of the dry weather, are to be found all over the Sahara ^[157] in favourable localities, and form on the sand-dunes of the south-west shore of Chad wide-spreading bushes, ^[158] whose grey-green leaves give to the landscape an impression of uniformity and melancholy, and extend from here even as far as Lagos. ^[159]

Of widely distributed plants belonging to the two regions in common, one of the most remarkable is the dum-palm, *Crucifera thebaica*, the sole representative of its genus, whose trunk is divided into several branches. They prefer a soil that attracts the surface water, and are to be found from the oases of the Central Sahara to far into Adamawa—though here indeed only very scattered; some splendid examples flourish especially at Ngornu, on the shore of Lake Chad. South of this region it is very rare. According to Barth the fruit may be used in large quantities for food, ^[160] but on the other hand the same traveller points out that the low-growing dum-bushes are a hindrance to agriculture. ^[161] In fact, between Kukawa and Ngornu one finds wide stretches so thickly studded with the young plants of this palm which it is so difficult to eradicate, that one is reminded of the plains of the Mediterranean littoral overgrown with ‘chamaerops’. ^[162]

The date-palm also, the most characteristic tree of the Sahara, is to be found dispersed in single examples all over Bornu, and specimens transplanted by man may even be seen in the middle of Adamawa. But it is seldom or never that it bears fruit in Bornu; the summer rains do not suit it, any more than they do the camel, and it is a proof of the desert character of certain of the sandy valleys of the neighbouring country of Kanem, that the date-palm flourishes there in full luxuriance and bears abundant fruit. ^[163]

Belonging likewise to both regions, but confined within narrower limits and generally not extending farther south than the latitude of Kukawa, there are certain plants found in Northern Bornu which are partly used for the production of salt. The most striking of these is the peculiar ‘tundub’, *Capparis aphylla* (sodata),^[164] found in the same parts as the dum-palm, the ‘ethel’, *Tamarix articulata*,^[165] the ‘retam’, *Retama spec*—according to Barth *Spartium junceum*^[166]—the *Salvadora persica*, and the *Asclepias leptadenia pyrotechnica*.^[167] Of these only the ‘tundub’ and the ‘retam’ are to be found widely distributed in Bornu. The ‘tundub’ is of importance in that the ashes obtained from it furnish the natives with salt.^[168] As a salt-loving plant it is only to be found on soil impregnated with natron, and becomes rarer as the percentage of the mineral necessary for it to thrive decreases. It does not seem to reach farther south than Kukawa. The ‘retam’, one of the European species of *Spartium* or *Ulex*, similar to the Papilionaceae, forms on the sand-dunes south of Maduari thick bushes taller than a man, which give to the places where they are found quite a northern appearance owing to its genesta-like exterior.

Other kinds of trees and shrubs which belong without any doubt to the Sudan, have nevertheless in favourable localities extended their area of distribution far into the Sahara. Among these is the ‘hadjlidj’, *Balanites aegyptiaca*, whose name, like that of the *Crucifera thebaica*, indicates a very wide distribution; with its dark-green hanging boughs it presents a very striking appearance. Its fruit, called ‘bito’ in Kanuri, plays, according to Barth, an important part as an article of diet among the natives.^[169]

It is much more difficult to define the area of distribution of the acacias than of the other well-known species of trees, because they resemble each other very much in all countries, and because it is only very insufficiently established to what species they belong owing to their great multiplicity of form. The definite classification of the various species of acacia found in these regions, which is impossible without the possession of an ample collection of material—at present not yet forthcoming—would be a task well worthy of the gratitude of botanists. But the classification is the more difficult in that each of the numerous books of travel, while multiplying the names given by the natives to these trees, apparently assign them to

different species. It is always possible that the natives, on whom botanical peculiarities are naturally lost, may assign the same name to similar species. One may therefore assume with certainty that the Arabic word ‘talha’ is a term applied to all species of acacia that exude gum arabic.^[170] By ‘talha’ are especially meant the *Acacia ferruginea* common all over Bornu with its little unmistakable reddish-brown trunk and branches and its dark golden blossoms,^[171] the *Acacia stenocarpa*,^[172] and the *Acacia tortilis*,^[173] widely distributed in North Africa. All these species, whose ball-shaped little blossoms make their appearance during the first months of the year, diffuse at that time an extraordinarily strong perfume reminding one of the violet or the heliotrope. To the finest species, which assume with especial clearness the umbrella-like form, belong the *Acacia arabica*, found far from here even in Air, and especially the ‘gawo’, *Acacia albida*. The ‘gawo’^[174] is especially remarkable for this reason, in that it strikes one from a distance by the complete bareness of its greyish-white branches during the rains; it is not till October that it puts on its feathery leaves of greenish-grey and its blossoms that remind one of willow-catkins.^[175] To be found all over Bornu and stretching far afield towards Adamawa, it reaches giant dimensions on the banks, rich in ‘humus’, of the River Yedseram, south of Bama.

The steppe-forest of Northern and Central Bornu^[176] is composed somewhat as follows. Scattered among the acacias, which predominate, are the above-mentioned species of trees—to which may be added in Central Bornu the ‘dschochan’, *Diospyrus mespiliformis*—which, however, form isolated patches in the acacia forest whenever the condition of the soil is suitable. The commonest of these is the ‘hadjlidj’ (*Balanites*). Next to this the most frequent is the ‘Kargu’ or ‘Kalgo’, *Bauhinia reticulata*, belonging to the order of Leguminosae, a shrub found as far as Munio and Zinder,^[177] which, like the *Mimosa pudica*, is remarkable for rattling at night its large leaves shaped like an inverted heart. This shrub appears mostly in the form of brushwood, prefers a sandy soil, and on the summit of the dunes along the shores of Chad replaces in some parts the *Zizyphus* and the *Calotropis procera*, with whom it also enjoys a joint tenancy. It is likewise as a brushwood shrub that the *Gardenia Thunbergia* forms sparse bushes, whose thorny branches are covered in the spring-time with big snow-white

blossoms, which first open towards evening and then diffuse a strong penetrating perfume. The branches of these trees and shrubs are often intertwined with the parasitic loranthus, whose rather large honeysuckle-like trumpet-shaped flowers probably misled Barth and made him mistake them for orchids.^[178] The loranthus with its fleshy grey-green leaves and hanging tendrils often forms great bushes, whose masses of foliage may quite obscure the green of the tree it clings to. Many other creepers attain such luxuriance that the tree they overgrow completely disappears beneath their masses. One of the most striking is the sappy cactus-like ‘digessa’, *Cissus quadrangularis*, in which the small leaves are inconspicuous as compared with the evergreen and peculiar four-edged tendrils. According to Rohlf, this plant, which is widely distributed all over the country, is one of those from which the natives prepare their arrow-poison.^[179] Besides the *Cissus* there is a kind of ‘luffa’ which is worthy of notice; its fungus-like fruit on its withered tendrils attracts the attention of the traveller in the dry weather from a long distance.

The high ant-hills in the ‘bush’ are regularly carpeted with Capparidae such as the *Cadaba farinosa* and *Capparis tomentosa*. The latter especially affords a peculiar sight in winter and early spring, when its thorny semi-creeper-like tendrils are in bloom. The blossoms, to which are attached numerous long pollen-vessels, attract—as do the acacias—through their strong honeysuckle-like perfume the whole flower-loving insect world and the metallic-coloured honey-birds.

Many trees in the ‘bush’, and especially those which prefer marshy places or the banks of streams, are inclined to isolate themselves and so form splendid clumps, solitary and circular, that give the landscape quite a park-like appearance. To this category belongs especially the tamarind, *Tamarindus indica*—in spite of its name indigenous to Africa—which does not shed its dark-green feathery leaves even in the dry weather and puts forth its orchid-shaped flowers in the winter. Its fruit serves many medicinal purposes and when pounded in water makes a refreshing beverage. Even commoner than the tamarind are the various species of evergreen fig which Bornu harbours, each in individual isolation; such isolation is grounded in their very nature. Though of great variety in the form and size of their

leaves, they all show in their habits the same characteristics. Similar to its Indian relative, the banyan, in the early stages of its growth it takes the form of a semi-parasitic creeper, which shows a distinctly astonishing inclination towards a spread both of roots and branches. Soon after it has surrounded the tree that serves it as host with a many-meshed net, the latter perishes; the inextricable tangle of wood forms a single trunk, which, as branch and bough keep on sending out new roots, finally supports of itself the giant arbour-like roof of foliage, composed of the countless roots both thick and thin that in their turn struggle earthwards. The milk-like sap of all these species exhibits in a dried-up condition more or less of the elastic properties of caoutchouc; the fruit of some species are edible. One large species, known in Bornu as 'ngabore',^[180] with poplar-like blackish-green leaves, reaches especially at Ulugo and other places on the south shore of Chad a height of over twenty metres and surpasses all other shade-trees owing to its density of foliage. Owing to its dark colouring it stands out sharply defined from everything else during the dry weather and forms a striking contrast to the uniform muddy grey of the villages. Of much less compact shape is the equally widely distributed Bignoniaceae, *Kigelia pinnata*, which is found as far as Kanem,^[181] and to which the much-travelled Hausas have assigned the name of 'nono-n-giwa' (that is to say, elephant's udders) on account of its sausage-like fruit, which is over a foot long and hangs from a long stalk. It is very commonly found in the middle of the towns. The walnut-like leaf of this tree also lasts through the dry weather, during which appear its straggling dark-red clusters of blossom.

The 'bush' in Northern and Central Bornu differs in one respect from that of the southernly districts, in that the grasses which cover the soil do not there reach the same great height; in sandy places the low-growing prickly grasses called 'ngibbi', *Cenchrus echinatus* and *Pennisetum dichotomum*, predominate, of whose notorious seed-grains Denham remarks: 'These prickles may be considered one of the pests of the country.'^[182] The unpleasant burr-like adhesive seed-grains of these grasses cling fast everywhere and are a regular plague both to man and beast—in fact, the latter's hair is often thickly matted with them. Owing to the easy way in which these noxious weeds are spread, they at present flourish all over the Sudan, wherever they are not kept under by other plants. It is

astonishing how the Bornu horses are able to eat the *Pennisetum dichotomum* without any ill results, nay, Barth goes so far as to say that they will not thrive without it. [\[183\]](#)

The Graminae flora are always predominant also in the undergrowth, for the herbaceous flowering plants only make their presence felt during the spring. Frequently the often magnificent flowers appear long before the leaves, and are either stalkless like those of a large Zingiberaceae, or else push their heads only a little above the surface of the soil like those of the strange *Amorphophallus* species and other Araceae. To one of these families perhaps belongs the bulbous plant eaten by the natives and mentioned both by Barth and Rohlfs, under the name of ‘Katakirri’ or ‘Gadagér’, as being found in Southern Bornu. [\[184\]](#)

In the districts in which the ‘bush’ consists chiefly of elements which belong also to the Southern Sahara, it experiences in places interruptions of quite foreign appearance, namely, from the flora of the shores of Lake Chad and from the desolate ‘firki’ flats.

Peculiar to the shores of Chad is the total absence of trees. This is only to be explained by the great irregularity of the water-level of the lake, which must be extremely unsuitable for steady yearly growth: during the long-enduring inundations, which put everything under water a yard deep for months at a time, those trees are bound to decay which cannot thrive except in a position which is normally dry. On the other hand, trees whose roots demand a copious supply of water, suffer from a year-long drought and find no compensation in the downpour, abundant as it is, of the short rainy season for the absence of water springing from below.

The places where trees grow—especially on the west and south shores of Lake Chad—mark the boundary of a zone of marsh vegetation poor in species but nevertheless interesting. Besides the high grasses and the *Mimosa pudica*, which is ubiquitous wherever there is water, there is usually nothing but the papyrus and the ‘ambatsch’ or ‘marria’, [\[185\]](#) *Aeschynomene (Herminiera) elaphroxylon*, both of them plants which, owing to their simultaneous appearance in the ‘sudd’ of the Upper Nile, seem to point to an earlier connexion at some time between the two

districts. The papyrus, which is also found in the brackish lakes of the Munio country,^[186] reaches in the swampy thickets of Chad the considerable height of four metres, but it is overtopped by the ‘ambatsch’. As its name (*elaphroxylon*) indicates, the latter is of unusually rapid growth—in fact, one gets the impression that the stem of this giant plant, which is as straight as an arrow and often as thick as a man’s leg, reaches its full height inside of a single year. As soon as the water recedes from any place for any length of time the ‘ambatsch’ dies, and then the shore is covered far and wide with the dead trunks overlaid with green creeping plants. The wood, when quite dry, has an elder-like pith, but is nevertheless comparatively durable and extraordinarily light—a log as thick as a man’s leg, and some seven metres long, hardly exceeds one kilogram in weight—and is put to various uses by the inhabitants of the shores and islands of Lake Chad. The ‘ambatsch’ as well as the papyrus, standing in some places a yard deep in water, is generally thickly covered with the tendrils of a luxuriant kind of *Ipomoea*, whose big purple trumpet-shaped flowers, mingled with the golden-yellow butterfly blossoms of the ‘ambatsch’, at any rate enliven what is otherwise a monotonous vegetation.^[187]

Although the marshy district on the shore of Chad is favourable to certain kinds of vegetation, yet the ‘firki’ flats appear to be without any plant growth whatever, and thus give the landscape a rather extraordinary appearance of melancholy. Although the ‘firki’ flats are, in the rainy season at any rate, a favourable soil for water-loving plants, and are in their nature fertile, yet they dry up so rapidly in the late autumn that only quickly growing plants, without being able to come to flower, can maintain themselves as long as the layer of loam retains its humidity; but after that cracks in the soil soon appear and cause the roots to dry up, so that all existing vegetation must necessarily die down. Only where there is subterraneous water or stagnant water-courses that are able to last out the dry weather are there acacias, which in some parts of this region so bare of trees furnish the only fuel, and are so stripped of their branches by the natives that with their stunted shoots they exactly resemble at first sight the pollard willows of the north. On the pools and water-courses that do last out the dry weather are found in places the flora, which appear also elsewhere in Bornu, of the smaller lakes and ponds, principally consisting of the buds

of the cosmopolitan *Pistia stratiotes*, with its lettuce-like head and two kinds of water-lilies—the *Nymphaea lotos* with yellowish white, and the *Nymphaea Zanzibariensis* with sky-blue flowers.^[188]

The farther one penetrates into the ‘bush’ towards the south-west, the south, and the south-east, the oftener do new species occur and replace by degrees the more northerly varieties. The ‘hadjlidj’ (*Balanites*) next disappears from the forest and the acacias are relegated to the sterile spots; only the *Acacia albida* remain as numerous as ever, and appear in ever finer specimens. On the other hand, the *Anona Senegalensis* makes itself noticeable amongst the undergrowth with its grey-green aromatic-smelling leaves; it is one of the few wild bushes to produce edible fruit. ‘Gonda-n-Kura’, as the wild paw-paw is called, is usually known by a Hausa name as opposed to the cultivated ‘gonda’, as the *Carica papaya* is called in the Central Sudan. Along the streams there is found a tree with dark-grey bark and leaves, not unlike though somewhat larger than the horse-chestnut, which it also resembles in the shape of its strong-smelling blossoms. It is the *Vitex cuneata*, belonging to the Verbenaceae and locally known as ‘Ngalibi’. Ink is prepared from its bark,^[189] and its black fruit is used for food. Near it are always found other kinds of trees, whose external appearance gives the impression of regular tropical growth. Next to it is the well-named silk cotton tree, *Ceiba pentandra*,^[190] though it is indeed far from reaching the size of those found in the primaeval forest; in favourable spots, however, along the river-banks where the ‘humus’ soil is plentiful, there are specimens which cause it to stand out as one of the tallest trees of the district. Large trees of this species were even used by the natives of the middle Shari as places of refuge during the slave raids.^[191] The soft silky fluff of the seed capsules is used all over Bornu as stuffing for various articles. Its relative, the *Bombax buonopozense*, is very seldom found near the *Ceiba*; the former is a tree which can never be overlooked, owing to its big dark-red blossoms, which stand out at a distance against the background of its scanty foliage.

Typical specimens of ever-increasing size now appear of the giant of the African tropical ‘bush’, the *Adansonia digitata*, or ‘Kuka’ as the Kanuris call it. It is, indeed, distributed all over the country,^[192] for the most part in

the neighbourhood of the towns, where its fibre is made into rope and its young leaves are valued as a vegetable; but it is not till one gets to Southern Bornu that one meets with examples truly characteristic of the species, the bulk of whose trunks is out of all proportion to their height; at Issege on the Yedseram there are trees measuring fifteen metres and over in circumference. The *Adansonia*, like the *Ceiba*, is perfectly bare during the dry weather, but the leaves appear with the first rains and are soon followed by the big white flowers on their long pendent stalks. Another true native of the tropics is the Deleb palm, *Borassus aethiopum*, which is almost entirely absent from Northern Bornu,^[193] but now sometimes singly, sometimes in groups, vies in height with the other giant trees of the southern districts. Its trunk, straight as an arrow but thickening to a cigar shape in the upper half and crowned with compact fan-like leaves, forms a most effective ornament to the landscape. The trunk furnishes an excellent weather-proof timber, and the big fruit, of a brilliant orange yellow, is eaten all over the country, but its somewhat acid taste does not correspond to the expectations raised by its delicious smell, which resembles that of a pineapple. While the Deleb palm is not found in certain districts—often a day's march apart—yet in other places, as in the Musgu country on the lower Logone, it appears in great numbers and in magnificent serried rows, generally, indeed, in the immediate neighbourhood of the towns.^[194] To the fig species of this district belongs the tree known as 'Ganyi', with large leathery leaves: it owes its name to the triangular incisions which are regularly found on its bark, and which are caused by the natives tapping it to obtain caoutchouc. But it is impossible that the quantities obtained of this much sought after substance can be large, for the tree is only met with here and there quite sporadically, especially in the denser parts of the 'bush' overgrown with 'vitis' creepers as thick as a man's arm along the rivers in Southern Bornu.^[195]

When the 'bush' borders on the rivers the shrubs of the undergrowth are thickly covered with the delicate green tendrils of the Sapindaceae, *Cardiospermum halicacabum*, which is ubiquitous all over the Sudan. This creeper, distinguished by its dainty leaves and bladder-like fruit, is to be met with quite regularly on the steep banks of streams. In places where it is possible for the river to overflow its banks, other kinds of plants which

prefer a marshy or, at any rate, a damp soil, have established themselves, such as various shrubs of Rubiaceae belonging to the *Sarcocephalus* species, easily distinguishable by their white globular blossoms.^[196] The swampy ground is every now and then covered with a thick aromatic carpet having a scent something like thyme, composed of one of the Labiatae belonging to the *Ocimum* species (perhaps *Ocimum viride*?); and wherever pools are able to form, the blue star-like flowers of the *Nymphaea Zanzibariensis* gleam once more together with the floating bright green buds of the inevitable *Pistia stratiotes*.

The nearer one gets to the Bornu boundary, the richer and more various is the flora. In the neighbourhood of the big rivers, indeed, the character of the landscape alters but little, although even here one or two kinds of trees appear that were hitherto unrepresented. Among these are to be reckoned the *Celtis integrifolia*, which Chevalier mentions on the middle Shari,^[197] as well as the *Khaya senegalensis*, whose towering trunks attain magnificent proportions on the upper Yedseram, and furnish the finest forest scenery of this region. The *Khaya*, which has already been rooted out over a large extent of country in West Africa, is of importance because it furnishes part of the supply of West African mahogany. One of the most botanically remarkable shrubs of this district is a true willow, *Salix*, which fringes the banks of the upper Yedseram in company with the *Mimosa pudica*, and apparently belongs to the same species as the latter, forming regular thickets on the Benue near Lau.

Very striking, however, is the change in the flora in the true 'bush', especially in places where the first traces of granite appear. Besides the Leguminosae, which have hitherto been absent, the Cisalpiniaceae become prominent, forming by far the larger number of the species of trees of this region, especially the Combretaceae, which are so characteristic of the hill country lying to the south. Everywhere one now meets the *Combretum* species, the *Anogeissus leiocarpa*, and the Terminaliae, the chief trees of the Southern Bornu 'bush'. By far the commonest of these is the *Terminalia Schweinfurthi*, the shape of whose trunk and branches might easily cause it to be mistaken for a fair-sized oak, though its big smooth-edged leaves have somehow quite a strange look. Differing from the other low-growing shrubs

owing to its maple-like leaf is the *Sterculia tomentosa*—closely related to the ‘kola’ tree, *Sterculia acuminata*,^[198] which is not found in Bornu—and also, owing to its pronounced tendency towards cork formation, the well-known *Burkea africana*. At first only appearing sporadically, but afterwards in ever-increasing numbers, one now meets two trees which play an important rôle in the Sudan, viz. the *Parkia biglobosa* of the Leguminosae and the *Butyrospermum (Bassia) Parkii*, or shea-butter tree, of the Sapotaceae. With its dainty feathery leaves and the strange shape of its blossoms, the former is an ornament of the granite hills, the foot of which is its favourite situation. In early spring the velvety dark blood-red blossoms, always in pairs, appear on long hanging stalks; in shape and size they resemble a billiard ball, and are out of all proportion to the tree’s delicate feathery leaves. From them develop, in the course of the summer, bunches of considerable-sized pods whose seeds form the ingredients of the ‘dadawa’ cakes, which are relished all over the Sudan and sold in every market; they have a nauseous smell, however, and are only digestible by a negro’s stomach. Much more important than the *Parkia* is the *Butyrospermum*. It is a moderately tall tree, its bark is full of cracks, and its foliage reminds one at a distance of that of the Spanish chestnut, or, still more, of that of the *Mangifera indica*; it bears a fruit whose large single kernel is akin to the mango, and produces a vegetable oil which keeps well and is put to various uses, and which has become of ever-increasing importance as an article of export. The fruit has a pleasant taste resembling a date.

This district also is particularly the home of the *Candelabra euphorbia*, known as ‘Garuru’, which in many respects resembles the American *Cercus* species and in shape is also like the dragon-tree of Teneriffe; on granite soil it reaches a height of ten metres and is crowned by extraordinarily thick foliage. These Succulenteae are distributed all over Bornu, and are found sporadically even in the plains, but they are never entirely absent from any of the pagan villages, which lie at the foot or half-way down the slope of the isolated hills, for they possess qualities which render them of great use to the natives. Firstly, the thorn-protected branches furnish the best imaginable live hedge, and, secondly, the milky juice that flows from the slightest incision is one of the chief ingredients of their arrow poison. The

small yellowish-green blossoms, which differ hardly at all from those of the European spurge, appear towards the end of the rainy season.

The open 'bush' of Southern Bornu produces grasses that surpass in height and luxuriance anything similar from more northern landscapes. In the great Marghi Forest, east of the Yedseram, a favourite haunt of big game, the grass—in places where the water has dried up—reaches such a height that it covers an elephant standing erect, and only the long neck of the giraffe reaches above it. At the flowering time the stalks, which can hardly be avoided when marching along a narrow path, send out at the slightest concussion clouds of yellow pollen, which is troublesome to man and beast alike. One can understand that this grass, which itself reaches the very tops of the *Terminalia*, does not allow other plants to come up. Only where in damp places short-stalked grasses form natural pastures for the game, variegated plants appear—at least, at the beginning of the rainy season—amongst them numbers of a sort of *Amaryllis* with big white and purple striped flowers and beautiful tall ground orchids, such as the lilac-blossomed *Eulophia* (*Lissochilus*) *dilecta* and *Lissochilus lindleyanus*, with similar clusters of golden-yellow blossoms.

The highlands of Mandara and the scattered granite masses, which make the landscape of Southern Bornu so attractive, form at the same time sort of islets of flora, which show unmistakable differences from the surrounding plain. Many remarkable kinds of plants only occur here—these rocks are an El Dorado for the botanist—and never stray down into the plain. It is noticeable that it is near the perennial springs, which gush out on the slopes and at the foot of these isolated hills, that one finds the only ferns in Bornu, mostly pretty varieties of *Adiantum*, which, besides, are identical with those of Adamawa. But amongst the representatives of the tree flora also many interesting species are to be found. Thus the granite rocks north of Uba are thickly overgrown with a species of fig-tree, the 'Bidjage', from which Barth says that the natives prepare their arrow poison.^[199] The 'Bidjage', *Ficus populifolia*, is similar in the shape of its leaves to the 'Ngabore' found near Lake Chad, only it has striking light grey-green leaves, but its qualities are certainly not more poisonous than those of any other species of fig.

The cultivated plants of Bornu are drawn from a wider range than are the wild ones. The majority of the former belong to species which are cultivated all over Africa south of the tropic of Cancer, or are such as are distributed over tropical countries throughout the world.

Of the Graminae cultivated in Bornu for grain by far the most important are the *Andropogon* species (*Sorghum holcus*) and the bulrush-headed millet, *Pennisetum* (*Penicillaria*), with their different varieties. In comparison with them all other kinds of grain play a quite subordinate rôle—at least at present. For the cultivation of the *Andropogon* species on the one hand, and of the *Pennisetum* species on the other, Barth assumes, and Nachtigal also lays down certain boundaries depending on the composition of the soil.^[200] But this assumption is not to be taken literally, for, especially in view of the continuous shifting of population, the area of distribution of the grain-stuffs has likewise been shifted in the last few decades.

The best known *Pennisetum* is the ‘Dukhn’, *Pennisetum spicatum* (*typhoideum*) with bulrush-like heads, which is excellently suited for cultivation in the sandy plains of Bornu. Sown at the beginning of the rainy season, the ‘Dukhn’ only takes two months to reach full maturity. The grain grown on the finer and moister land is the ‘Durrha’, *Andropogon sorghum*—called in Bornu ‘Ngafeli’—which is distributed over the whole of tropical Africa and part of southerly subtropical Africa and over wide districts in Asia; it is sown at the same time as ‘Dukhn’, but ripens about a week later than the latter.^[201] Besides the ‘Durrha’ itself two varieties are cultivated, the ‘Sabadu’,^[202] *Andropogon saccharatum*, and above all the ‘Massakwa’, *Andropogon cernuum*, a grain which is grown in water, and which, sown towards the end of the rainy season on swampy soil, is transplanted after fourteen days at slight distances apart, and after only ten or twelve weeks with the help of the heavy dews is ready to cut.^[203]

The grain which has its proper home in this region, the wild rice, *Oryza punctata*, is little valued; it grows wild in the swampy plains, exposed to floods in the rainy season, between the Shari and the Mandara highlands,^[204] forms there the chief food of the elephant, but is also gathered by the

poorer classes of the population. Rice is also always cultivated here and there in small quantities, as, for example, on the southern shores of Chad.

Somewhat more commonly one meets with plantations of maize, *Zea Mays*,^[205] which is often cultivated in well-watered parts of Bornu which have a fertile soil rich in mould;^[206] on the other hand, the European cereals, barley and wheat, are very seldom grown and then only with artificial irrigation and as a dry weather crop.^[207] The three last-named kinds of grain are regarded solely as delicacies and are scarcely to be considered as forming part of the people's food-stuffs. Similarly tubers are only partly acclimatized in Bornu and are only cultivated sporadically.

The tubers consist of the following: *Colocasia antiquorum*, *Dioscorea batatas*, *Manihot utilissima*, and *Ipomoea batatas*, all lately introduced for the first time and gradually imported into Bornu from the south. From the north the onion has been brought into the country by the Arabs. It grows excellently on the banks of Chad and reaches an extraordinary size. But there are at any rate some plants long established in Bornu which are cultivated as vegetables, such as the 'Karass', *Hibiscus esculentus*, whose pods furnish a slimy but pleasant-tasting vegetable. The mallow, allied in habits to that grown in the peasant gardens of Central Europe, and with flowers similar to those of the cotton-plant, which is likewise its relative, is to be found in almost all the villages of Bornu. Indigenous likewise to the Sudan is the giant pumpkin, *Cucurbita pepo*—or *Cucurbita maxima* according to Barth^[208]—which thrives everywhere and often covers the houses with thick tendrils and in the flooded districts of Lake Chad stretches over vast areas, which in the dry weather one can distinguish even at a distance by the big white gourds. Sesame^[209] also, *Sesamum indicum*, equally esteemed both as a vegetable and as an oil-producing plant, has long been cultivated in Bornu; its thimble-shaped white or red flowers are regularly to be found in the neighbourhood of the villages. A more debatable question is the origin of the two varieties of ground-nut which are distributed over almost the whole of tropical Africa; their scientific names are *Arachis hypogaea* and *Voandzeia subterranea*, and they are widely cultivated both as vegetables and for the production of oil; the fact that each has a name of its own in all the languages of the country, argues that they

found an entrance into the very centre of the Sudan at least a long time ago. Of true pod-bearing plants two are cultivated in many places, a bean, the *Phaseolus*, and a variety of *Dolichos*.^[210] From Asia have doubtless come the tomato, *Solanum lycopersicum*, and the cress, *Lepidium sativum*,^[211] which are both grown in the gardens of Central Bornu.

Besides these quite a number of wild plants furnish both vegetables and fruit. Bornu is extraordinarily poor in trees or bushes that are cultivated for the sake of their fruit. The only fruit-tree of this kind really worth mentioning is the paw-paw,^[212] *Carica papaya*, which, however, does not extend to the northern parts of Bornu. The Aurantiaceae fruit-trees, which are so widely distributed elsewhere in similar tropical and subtropical countries, are only represented by a few scattered lime-trees—obviously introduced by the Arabs from the North—in the gardens of the bigger towns of North and Central Bornu, such as Zinder, Kukawa, and Dikoa.^[213] It is equally seldom that one meets with true spice-plants in Bornu. It is not till one gets to the southern districts that one finds that the capsicum-pepper, imported from Adamawa, has here and there obtained a firm footing.

The predilection of the natives for narcotics, some of which are hardly ever found growing in the country, while others on the contrary are extensively cultivated, is an argument that the former were first introduced, while the latter had already long belonged to the Central Sudan, very probably even at the time of the discovery of America.^[214] The most important, in fact almost the only narcotic cultivated in Bornu, is tobacco, which is grown everywhere—especially in dried-up sandy river-beds—but above all in Musgu country.^[215] From the proofs which he adduces there is more than usual to be said for Barth's view that the species of tobacco grown there is indigenous to that region.

With much more justice might it be contested that those plants were indigenous in Bornu, for which there might be adduced the possibility of introduction from the East contemporaneously with the extension of Islam. To these belong indigo, *Indigofera tinctoria*, cultivated for the production of dye, and henna, *Lawsonia inermis*. The former is grown especially in the southern districts for the production of the indispensable dye for dyeing

blue cloths; henna, used for dyeing the finger-nails red, is grown in smaller quantities in all Mohammedan districts of Bornu.

With regard to cotton, by far the most important and almost the only cultivated plant applied in Bornu to textile uses,^[216] it is certain that it was not brought to the Central Sudan from the New World, even if one regards it as an immigrant. Rohlfs expresses himself as follows about cotton: 'It is quite certain that the cotton-shrub is to be regarded as a plant indigenous to Africa; at least, this theory is supported by its wide distribution over the African continent, as well as by the abundance of words for cotton which are to be found in all the negro kingdoms. The word 'Kalkutta', which the Kanuris use, appears to be derived from the Arabic in the same way as the German 'Kattun' (and the English 'cotton').^[217] Barth also mentions that in the time of Edriss Aloma cotton was already cultivated in the Sudan,^[218] at a time, too, when its introduction from America was at least very unlikely. The cotton cultivated in Bornu and Adamawa is the perennial variety classified by Passarge^[219] as *Gossypium arboreum*, which in many places, such as Ngornu on the shores of Chad, reaches a height of over two metres. It is cultivated in Bornu wherever the influence of Islam prevails and where the majority of the people wear clothes.^[220] Its favourite situation is the banks of rivers rich in mould and not too much exposed to floods and the vicinity of loamy low-lying ground strongly impregnated with sand.

Amongst other plants of importance which are grown for industrial purposes is the *Cucurbita* species, from which are manufactured by far the largest number of household vessels except those used for cooking. Besides the well-known *Cucurbita pepo*, which not only furnishes calabashes but also serves for the manufacture of floats, there is also the bottle-gourd, *Lagenaria vulgaris*, particularly trained as a creeper on houses or fences. Its fruit is of very many different shapes and this has led to different designations of the plant; however, it is merely one and the same species assuming various forms.

The names cited below are mostly taken from the specifications of a small collection of characteristic plants which I brought back with me from the Chad countries, and which were kindly identified for me by Professor Dr. Volkens. Other names are mostly to be found in Nachtigal, i. 554, and ii. 383; Passarge, p. 536; Foureau, *Doc. sc.*, i. 391.

In certain favourable places in Abyssinia, which is in about the same latitude as Bornu, Heuglin often met with a profusion of epiphytic orchids, the most characteristic plant of damp tropical regions. In Bornu one would seek such things in vain; they are not to be found even along the water-courses. Heuglin, *Reise nach Abessinien*, pp. 386 and 397.

Foureau, *Doc. sc.*, i. 452.

[Paw-paws have been introduced into some of the larger towns in British Bornu, but do not flourish. From the 'raphia' comes piassava fibre.]

Foureau, *D'Alger au Congo*, p. 771, and *Doc. sc.*, i. 456 and 545; and Chevalier, *La Géographie*, ix (1904), p. 351, where *Coffea excelsa* is also mentioned.

Rohlf's asserts that there is a great 'mimosa-forest', i.e. a forest of trees of the acacia species, which passes north of Kukawa and appears to 'extend like a ribbon, in many places four or five days' journey broad, right across the African continent from the West Coast to the Red Sea'. Rohlf's, ii. 285.

Lenfant, p. 152.

Nachtigal, i. 558.

Cf. Foureau, *D'Alger au Congo*, p. 576, where it is said that 'the bark at the base of the trees is always thickened somewhat as a consequence of the frequency of these fires; it becomes wrinkled and thick like cork, and forms as it were a sort of protective cuirass for the trees'.

It appears therefore that, as a matter of fact, the bush fires can do no great damage to the trees and shrubs. On the other hand, Chudeau asserts that the salt industry, which uses up a great deal of fire-wood, has absolutely denuded some parts of North-West Bornu of timber. *La Géographie*, xv (1907), p. 332.

Denham, ii. 143, where the traveller says: 'Not a fruit of any description can be found in the whole kingdom.' [This is an exaggeration. Fruit is very scarce in Bornu, but water-melons, dates, paw-paws, and limes can sometimes be obtained. Possibly some of these have been introduced since Denham's time. There are also *Kurna* and other wild fruit-trees.]

In Central Bornu I have even found during the dry season in sandy places a species of *Anastatica*, a typical desert plant.

Rohlf's mentions that a whole district in Tripoli is called after it (Rohlf's, i. 31).

Single plants reach a height of 5 metres; the stems covered with spongy bark are as thick as a man's arm. Cf. Barth, ii. 197. [Kanuri, 'kayo', Hausa, 'tumfafia'.]

Nachtigal, i. 261; Chudeau, *La Géographie*, xv, p. 327.

Cf. also Barth, ii. 319, and Nachtigal, i. 582.

[Incorrect.]

Barth, ii. 212 and 314.

Barth, iv. 46 (German edition).

Barth, moreover, mentions also a species of *Chamaerops* indigenous to Bornu (Barth, iii. 170).

Barth, iii. 96, 99, 100, 105; Nachtigal, ii. 322. Von Beurmann found in Bornu yet another unclassified species of *Phoenix*, vide *Zeitschrift für allgemeine Erdkunde*, xv, p. 287.

Barth, iii. 42-3 (Barth erroneously calls the plant 'siwak'). Nachtigal, i. 414 and 555.

Foureau, *D'Alger au Congo*, pp. 588 and 590.

Barth, iii. 26.

Cf. Nachtigal, i. 413, and Foureau, *Documents scientifiques*, i. 408.

[Called in Kanuri 'Kige' or 'Kighir'.]

Barth, ii. 314.

Cf. Foureau, *Documents scientifiques*, i. 441. In the same passage a whole number of other species of acacia are cited. [Cf. *Bulletin of the Imperial Institute*, 1908, 6. 47 and 1910, 8. 352.]

Barth, iii. 58 (there called *Mimosa ferruginea*).

Nachtigal, ii. 487.

Foureau, *Documents scientifiques*, i. 441.

[Hausa 'gawo', Kanuri 'Karage'.]

Foureau mentions the same peculiarity for the *Acacia Trentiniani*, a smaller species belonging chiefly to Air, but found also in Bornu.

Cf. also Nachtigal, ii. 383.

Barth, iv. 87-8, and Chudeau, *La Géographie*, xv, p. 333, where it is said that the northern boundary of this plant is coterminous with that of the white-ant.

Barth, iv. 118.

Petermann's Mitteilungen, Ergänzungsh., 34. p. 2, and Rohlfs, ii. p. 15. The latter says: 'The stems of this plant are, when young, four-edged and pulpy, but later become rounded. At the base they are as thick as a man's arm, and from their juice mixed with that of other plants is prepared the dreaded arrow-poison, the least drop of which, if introduced into a wound, causes almost instant death, according to native accounts.' Passarge, however, on p. 162 of his book, when mentioning this plant, only states that it is used throughout West Africa as a medicine for gonorrhoea.

Cf. Barth, ii. 371-2, and iii. 141 and 157. Nachtigal identifies this species with *Ficus sycomorus*. Nachtigal, i. 663. [For monographs on caoutchouc vid. list published by A. Challamel of Paris.]

Nachtigal, ii. 269 [*Rehaina*].

Denham, i. 76; cf. also Barth, iv. 66 (German edition).

Barth, ii. 198. [In times of famine it is even pounded and eaten by the poor.]

Barth, ii. 387; Rohlfs, ii. 30. In spite of careful inquiries during my stay in Bornu, I could not find this plant. [It is quite well known. Its root is shaped like a potato, with pulp like a radish, and milky juice.]

Barth calls this plant 'fogo'; Barth, ii. 408. ['Fogo' or 'fowo' is the Kanuri word, 'marea' the Budduma word, and 'ambach' the Arabic word. When dry the density of the wood is ten times less than that of water, and two and a half less than that of cork. It attains in two years a height of from 8 to 9 yards, with a diameter of from 12 to 15 inches. Cf. Tilho, *Journal of the Royal Geographical Society*, Sept. 1910. Barth states that canoes are made from 'fowo' wood, but this is not so. Cf. Benton, *Notes on Some Languages*, p. 51.]

Barth, iv. 67 (German edition).

Even in the lagoons and canals between the islands in the north-east of Lake Chad the vegetation has the same appearance. On the other hand the higher islands have bushes of the acacia species, *Crucifera*, *Balanites*, and *Calotropis*. Cf. M. Audoin, *La Géographie*, xii. 306.

The *Nelumbium* species, the true lotus plant, does not appear to have been yet found in Bornu.

Passarge, p. 546.

The *Ceiba*, moreover, extends pretty far north in favourable localities. Barth found trees of this species at Surrikulo in the most fruitful district of Northern Bornu. Barth, iv. 39.

Nachtigal, ii. 628.

According to Barth it is found even at Gadabuni. Barth, iv. 71.

Foureau, however, met with this tree at Zinder, where in Barth's time it was not to be found. *D'Alger au Congo*, p. 514.

Barth, iii. 187; Stieber, *Kolonialblatt*, xvi. Jahrg., p. 83; Kund, *Mitteil. a. d. deutsch. Schutzgeb.*, xix, p. 7.

But Kund speaks of patches of dense forest, which he met with at Budugar on the Logone; *loc. cit.*, p. 3.

Whilst the name of one variety mentioned by Foureau, *Sarcocephalus esculentus*, allows one to conclude that its fruit is edible (*Doc. sc.*, i. 406), other plants of this species are repeatedly referred to as being those from whose roots arrow poison is prepared. A confirmation of this is that one regularly finds on them the caterpillars of the *Daphnis Nerii*, which almost everywhere live only on strongly poisonous plants such as *Nerium*.

La Géographie, ix, p. 354.

The fruit of the *kola acuminata*, the 'goro' nut, is an indispensable luxury throughout the Sudan.

Barth, ii. 411.

Barth, ii. 458 (German edition); Nachtigal, i. 653. Cf. also Passarge, p. 463, where the view is expressed that in Mohammedan districts *Pennisetum* is the commonest, while among the pagan tribes both *Andropogon* and *Pennisetum* are found.

Nachtigal, i. 654.

Passarge gives 'Tchibe' as the Kanuri term, p. 654. ['Tchibe', or 'Chibe', simply means 'of the mouth'. 'Sabadu' is chewed by the natives like sugar-cane.]

Cf. Barth, ii. 241; Rohlf's, ii. 69.

Barth, iii. 161 ff.

According to Nachtigal (ii. 374) it is next to 'durrha' the chief grain grown on the Chad Islands.

Nachtigal, i. 654.

According to Barth they were introduced into Bornu at the same time as the onion, about 100 years prior to his (Barth's) visit. Barth, ii. 314.

Barth, ii. 638 (German edition).

[or benniseed.]

Barth also mentions the existence of the *Vicia faba* in Bornu, and even says that it is an important article of food. Barth, ii. 314. But this is a mistake.

[Kanuri name 'Lapsur'.]

It is interesting that this plant, though only lately introduced—as one must assume—has a name of its own. If the paw-paw originates from America, as is now commonly assumed, it remains the more remarkable that it has already been long naturalized in Bornu, whilst the sand-flea or 'jigger' (*Pulex penetrans*)—which we know by experience has spread over Africa the most quickly of all American products—is nowhere yet to be met with in Bornu, and in Barth's time appears not to have penetrated farther from the coast than Bagirmi. Barth, iii. 45.

The pomegranate trees in Kukawa, of which Rohlfs speaks (i. 314), were apparently destroyed when the town was sacked.

In considering the question of whether some plant or other is indigenous to the interior of the continent, much too little consideration is paid to the fact that—in spite of the presence of the Niger—the Central African countries were completely secluded from communication with the Atlantic up to the middle of the nineteenth century. It would be interesting to learn whether the adherents of the theory that such useful African plants as the ground-nut and the oil-palm were introduced from America, are aware of the fact that species of fauna, which are characteristic of South America, such as the neotropical *Eunica* species which approximates very closely to the *Crenis*, an African species of *Lepidopiera*, and the *Urania* species peculiar to both regions, are found in Africa.

Barth, iii. 229; Stieber, *loc. cit.*, p. 84; Kund, *loc. cit.*, p. 29.

Barth mentions that a stuff called 'gabaga' was manufactured from papyrus. Barth, iii. 298. ['Gabaga' is the ordinary Kanuri word for strips of cotton cloth.]

Rohlfs, ii. 10.

Barth, ii. 655. Denham even mentions cotton as growing wild; ii. 140.

p. 463.

Cf. Barth, ii. 539 and iv. 50; Kund, *loc. cit.*, p. 29.

VI

FAUNA

WHAT holds good of the flora of Bornu, as regards its geographical distribution and composition, holds good also, but in a higher degree, of its animal world. The fauna of the country is a mixture of purely Ethiopian varieties and of such as are peculiar to the Mediterranean, i.e. the Sahara region, still the boundaries of the fauna overlap each other far more than is the case with the flora, owing to the easier possibility of distribution amongst animals. This holds good especially of the birds and of the winged insects; in fact the appearance of the fauna is temporarily strongly influenced by the European birds of passage who make a winter sojourn in Bornu. Again, the Ethiopian animals are a mixture of the fauna of East and West Africa. Whilst among the vertebrates the West African species prevail, it appears that the East African predominate among the invertebrates, in fact varieties appear here whose nearest known haunt lies thousands of miles distant in East or even in South-East Africa. In general the richest animal life is always displayed in the vicinity of water, firstly because the larger animals can come here to drink at all times of the year, and secondly because at the same time the richest vegetation is found here, on the existence of which the life of the smaller animals more or less depends. According to Nachtigal, the Chad islands alone must harbour a comparatively rich fauna, even as regards mammals.^[221] The rocky country also, which almost always possesses a water-supply of one kind or another, and consequently a fresher and more varied vegetation, thus affording lurking-places of all sorts, is always preferred by the wild animals of the adjoining plain. The 'Terminalia' steppe of South Bornu appears at its poorest as regards animal life during the dry weather, under which circumstances an attentive observer may travel all day without seeing a single living creature save a few insects.

Of the mammals, the various kinds of monkeys in Bornu next deserve our particular attention: their numbers are exceedingly large and in spite of

the most impudent raids on the cornfields they are scarcely interfered with at all; they venture into the closest proximity to man, and often display a boldness of behaviour that one hardly finds in any other mammal. Thus they enter even the towns themselves, climbing the walls with the utmost unconcern, bent on deliberate mischief.^[222]

The *Cercopithecus* species, or long-tailed monkeys, are very widely distributed over Bornu. For they do not belong especially to the hiding-places among the rocks as the baboons do. They are present wherever a fairly large and dense growth of trees affords them sufficient shelter, and so they are to be found right within the Sahara, in the hill country of Aïr, the most northerly place for Ethiopian fauna, where they are even represented by two different species.^[223] They live always in troops and especially frequent the ‘bush’ in the neighbourhood of human habitations, where the millet and guinea-corn fields are their objective, in which in a short time they can cause incredible damage, in spite of all precautions by the owners. The best known long-tailed monkeys of Bornu are the *Cercopithecus viridis*, and the hussar monkey, *Cercopithecus ruber*,^[224] the former belonging more to the flat regions of the north, and the latter inhabiting rather the ‘bush’ and the granite rocks of South Bornu.

The dog-faced apes, the baboons, appear to be distributed still farther to the north than the long-tailed monkeys; but they rather prefer rocky localities and for this reason are principally to be met with in the most northerly and the most southerly districts of Bornu, though they also frequent the plain, as Nachtigal was able to observe near the town of Affade. Where, however, one finds only a fairly considerable granite formation, there one may reckon with the greatest certainty on the presence of baboons. Considering the slight protection afforded by their surroundings, which often consists only in the similarity of colouring between their skins and the rocks they inhabit, the baboons (relying on their formidable teeth) display still greater boldness than the long-tailed monkeys, and only retire from the cornfields selected for plunder if energetically pursued, barking furiously, and with their cheek-pouches filled to bursting.^[225] The only living thing that they fear is the leopard, their bitterest enemy. The commonest dog-faced monkey found in Bornu belongs

to the species *Cynocephalus babuin*,^[226] but there is another somewhat larger species in the higher mountains—whose males have a lion-like appearance owing to their thick manes—but this species is not yet definitely classified.

Analogous to the monkey tribe, one meets at Ulugo near Lake Chad and also in other places in the open acacia ‘bush’ a purely night-prowling beast, the ‘galago’, *Otolicnus galago*,^[227] which is about the size of a squirrel. Among the numerous bats is included a very large light-yellow species, which one occasionally disturbs out of the trees during the daytime, also a flying dog, *Cynonycteris stramineus*,^[228] of considerable size, which always appears in large numbers and flits round the fruit-bearing fig-trees or kuka-trees at sundown, but makes itself especially noticeable by its noisiness at the beginning of the rainy season.

Bornu is extraordinarily rich in beasts of prey, whose presence is explained by the abundance of game and the considerable flocks and herds owned by the natives. All the big feline species of Africa are found in Bornu. At the head of these stands the lion, which, owing to the smallness of the mane in the male, approaches the Senegal variety. In Bornu proper he is always rare^[229] or perhaps often overlooked, though he is certainly distributed all over the country; but he naturally prefers to frequent the thick ‘bush’ of the river banks, where the game come to water, and he is always sure of a rich and certain prey. It is here also that he is most in evidence. He is most frequently met with on the north shore of Chad^[230] and in the riverain district of the Shari and the Logone, where also other game is found in the greatest abundance. Dominik, who was able to certify to the presence of eight lions at a single spot on the Shari at the same time, relates that according to Arab reports the lions here hold regular ‘drives’ for game.^[231] According to Kund, the quantity of lions in these parts has temporarily become so great, that some smaller hamlets have been compelled to shift their quarters.^[232] The presence of the lions is so serious, that they not only endanger the herds, but even repeatedly attack human beings. Denham, for instance, relates that during his stay in Kukawa, a woman slave was seized by a lioness,^[233] and in quite recent times men have been repeatedly carried off by lions in the districts near the Shari.^[234]

The natives declare that the leopard, *Felis leopardus*, the second largest of the feline tribe found in the country, occasionally attacks human beings, and in view of the fact that similar depredations are reported from other parts of Africa, it can scarcely be doubted that they occur in Bornu. Whilst the Bornu lion prefers the plains, the leopard is rather a mountain animal, for the clefts in the rocks and the ravines thickly overgrown with vegetation furnish him with numerous lurking-places and also harbour large quantities of baboons, which form his principal prey.

Of the other feline animals the cheetah, *Cynailurus guttatus*, is the third largest; this is something like a dog and is often tamed, but it is distinctly rare. It seems to prefer very dry localities, whilst another species, the serval, *Felis serval*, with longer legs and larger ears, especially common near Lake Chad, has a predilection for swampy thickets, where it is very destructive to the wild-fowl. The last of the known species is the Summoli, *Felis caracal*, a true lynx with long prick ears; this appears to show all the characteristics of its European relative, [\[235\]](#) and like it too is only found very sporadically.

The wild dog tribe are likewise well represented in Bornu. At night-time may be heard everywhere the howl of the jackal, by far the commonest of the smaller beasts of prey. The jackal, *Canis mesomelas*, on the one hand, is useful by exterminating mice and removing carrion left lying about, but on the other hand is a nuisance, owing to the depredations which he commits in the poultry-yard, and by often dragging into the light of day corpses which have been only carelessly buried; in the latter respect he is a worse offender even than the hyena.

Remarkable for the similarity which the climate of Bornu has in many respects with that of the Southern Sahara, is the presence of so pronounced a desert animal as is the 'Fenek', *Canis cerdo*. [\[236\]](#) This was met with by Barth, not only in the northern districts round Munio, where its appearance would not cause especial astonishment, but also on the Shari. [\[237\]](#) Here also Dominik confirms the presence of that most interesting of the dog species of this region, the Hyena-dog, *Lycaon pictus*, which hunts in packs and forms the link between the dog proper and the hyena. [\[238\]](#)

As in all the ‘bush’ regions of Africa, so also in Bornu, there is no lack of hyenas; they have hitherto been classified under two species,^[239] the smaller and very common striped hyena, *Hyaena striata*, and the larger and somewhat rarer spotted variety, *Hyaena crocuta*; they make themselves useful, like the vultures and the jackals, by the removal of carrion, but like the latter they leave no buried corpse undisturbed, unless prevented from doing so by every possible means. Although primarily a carrion-eater, the hyena by no means despises a living prey, and the countless bold seizures and attacks on cattle, of which the herdsmen of Bornu have to complain, more often fall to the guilty account of the hyena—who indeed draws the greater quantity of his carrion from the neighbourhood of the shambles—than to that of the lion or the leopard.^[240] Whether they also attack sleeping men, as the natives assert, may be left undecided.

To what may be called the slinking variety of the cat tribe belongs the civet-cat, *Viverra civetta*, which is found all over Bornu, and is often kept as a domestic animal by the natives for the production of the civet, which is extracted in a pure state from the cuticle of the strong-smelling posterior gland of the *Viverra*.^[241] Three other smaller beasts of prey, relatives of the civet-cat, which live in Bornu, are also here and there kept in captivity by the natives, apparently for the extirpation of mice; these are the furze-cat, *Genetta dongolana*;^[242] the Ichneumon, *Herpestes ichneumon*; and the striped mongoose, *Herpestes taeniotus*; they are extraordinarily tame there.^[243]

The animals of the marten tribe are less numerous in Bornu. Worthy of notice also is the existence of an otter, ‘Lutra’, with very handsome jet black fur, which is caught now and then in Chad by the Kanembu, and of the African skunk, *Putorius zorilla*. In Bornu as elsewhere the rodents are numerously represented. Rats and mice are a regular plague in the low-lying districts, especially where ‘firki’ soil alternates with sand-dunes; the rats, however, are caught by the Kotokos and brought to the markets to be sold as food.^[244] A kind of burrowing mouse together with a species of hare^[245] undermine the surface of the sand-dunes studded with ‘Calotropis’ and ‘Zizyphus’ trees, thus rendering it difficult for pedestrians as well as mounted men to cross such places. Of the porcupine tribe at any rate the

Hystrix cristata is found in Bornu; the fallen quills of this large rodent, whose colouring makes them unmistakable, are often found in the 'bush'.

It is natural that in a land like Bornu, where ants and termites play so important a part, ant-eaters are not wanting.^[246] Clear traces of this animal's formidable digging-claws are to be found at practically every ant-hill, provided it is situated in the open 'bush'. The ground-pig, *Orycteropus senegalensis*,^[247] is certainly found, and very probably also one of the armadillos ('Manis') peculiar to the African 'bush' districts, though up to the present certain information of this has not been forthcoming. By far the most characteristic of the wild mammals whose tracks one meets in Bornu—one might almost say one is dogged by them—belong to the antelope family so rich in different species. The astonishing abundance of antelope in most districts of Bornu, not only of species, but also of individuals, is explained by Nachtigal on the ground that the great profusion of domestic animals suitable for the butcher renders it unnecessary to trap wild ones.^[248] The most striking of all the antelopes, if one may reckon it among that tribe, is the giraffe, *Camelopardalis girafa*,^[249] which is found, though somewhat rarely, in the great stretches of uninhabited 'bush', sheltered by the tall acacias, and which also wanders from here into the more settled districts. It belongs also to the tree-covered steppes to the north of Chad,^[250] from whence its distribution extends far into the Southern Sahara, as well as all over Bornu.^[251] The high granite hills to the south of Bornu form its first natural boundary.^[252] The antelopes proper are found in Bornu at the present day partly sporadically and partly in regular herds, separated according to species, but similar in their habits. The great majority of the species belong equally to Bornu and to the Southern Sahara.^[253] The best known or the most striking are: the widely-distributed Dorcas Gazelle, *Antilope dorcas*, often tamed as a pet—the Soemmering's gazelle, *Antilope Soemmeringii*—the *Antilope arabica*—the Mpala, *Antilope melampus*—the Bubal Hartebeest, *Bubalis bubalis*—the red Hartebeest, *Bubalis caama*—the reed-buck, *Redunca eleotragus*—the water-buck, *Cobus ellipsiprymnus*—the beautiful horse antelope, *Hippotragus niger*—the sable antelope, *Oryx leucoryx*—the addax antelope, *Addax nasomaculata*—the Kudu, *Strepsiceros Kudu*—and the harnessed antelope, *Tragelaphus scriptus*, as

well as a diminutive tufted duiker, *Cephalolophus spec.* As a rule only the addax antelope and the 'mpala' form themselves into largish herds, often a hundred strong. The former prefer dry localities; the latter, often in company with the water-buck, are especially fond of the swampy southern shore of Chad. [\[254\]](#)

Of other ruminants there is finally also found in Bornu the buffalo or bush cow, *Bos caffer*. Its favourite haunts are little-frequented places on the Chad shore and the swampy lowlands between the Logone and the Shari. [\[255\]](#)

The many-toed animals are represented in Bornu by all the giant species of Africa, as well as by the very small animals belonging to this family. Of those general in Bornu the elephant deserves by far the largest share of attention, owing not least to the tragic fate to which he is destined as a result of the ruthless pursuit to which he is subjected. Even if one always reckons that the number of these beasts is larger than it is in reality, one must add to this the fact that even travellers such as Barth were deceived as to their numbers owing to ignorance of the elephant's habits of life. Dominik rightly emphasizes the point [\[256\]](#) that the activity of three elephants is enough to leave traces that easily lead one to the false assumption that they have been made by a larger herd of these pachyderms. Only thus is it to be explained that Barth speaks of thousands of elephants, [\[257\]](#) while the number of the largest herd which he himself observed only reached 96 head. [\[258\]](#) If one takes into consideration the elephants which Barth actually saw, it constitutes a diminution already considerable in the number of these giant beasts since Denham's time, for the latter observed herds still numbering 150 or even 400. [\[259\]](#) Since then the number of elephants has steadily diminished, so that even the numbers of a herd alleged to contain 50 head, observed by the English members of the Anglo-German Boundary Commission in the winter of 1903-4, must certainly be taken as an over rather than an underestimate. Although Kund mentions at the same period the appearance of a herd 200 strong on the Shari, [\[260\]](#) his statement must certainly be grounded on greatly exaggerated reports from natives.

The favourite haunts of the elephant are the larger stretches of uninhabited 'bush' with suitable watering-places, which are to be found on the northern shore of Chad and along the rivers, especially the Shari and Logone, whose banks Barth reckons to be the places in Africa which are richest in elephants.^[261] Starting from here the animals wander farther afield even into inhabited localities—making the paths in the soft soil impassable for pedestrians owing to their deeply-trampled footprints. So there is scarcely a district in Bornu where it has not been observed, except in the extreme north-west. Where the elephant is not hunted, it does not appear to be timid,^[262] but this supposition becomes less and less tenable as the number of fire-arms increases.^[263]

The second largest many-toed animal of Africa, the rhinoceros, *Rhinoceros bicornis*, belongs likewise to the fauna of Bornu, but is much rarer and has only a restricted area of distribution. Indeed, it is only indigenous in the low-lying country on the Shari and Logone, especially in the Kung 'bush', south of about 10° north latitude, where it would appear to be more numerous and to be dreaded on account of its malicious character.^[264]

Wherever rock formations appear in Bornu and even where they are not very apparent, one is sure of finding the smallest of African creatures of this family, the widely-distributed dassie, *Hyrax abyssinicus*. Numerous wild boars inhabit the plains and must be exceedingly common on the Shari according to Nachtigal's narrative, though he does not mention their species.^[265] Where they are not hunted they appear to be far from timid, at least if one can credit Barth's account. The latter expresses himself on this subject as follows: 'A company of naked boys were splashing and playing about in the water and seemed on the best of terms with a number of wild boars; I have never seen these animals in such quantities in the Sudan as in the neighbourhood of the Shari. Calves and goats were feeding on the plain with wild boars in their midst.'^[266] It has not yet been definitely settled whether one of the bush-pigs (*Potamochoerus*) is found in Bornu; it is nearly always confused even in trustworthy reports with the formidable and strange-looking wart-hog; *Phacochoerus africanus*, with its misformed head and terrible tusches, which may even reach a foot in length. This animal

is found far inside the confines of the Sahara and is still common even in Aïr.^[267]

Wherever fairly large rivers, or collections of water that outlast the dry season, exist, the inevitable river-horse, *Hippopotamus amphibius*, is never absent. Chudeau mentions the appearance of the river-horse even in the swamps of the Manga country.^[268] Gerhard Rohlfs and his predecessors still report large herds—up to 100 head—of these pachyderms on Lake Chad,^[269] but on the other hand a contagious disease, of which Nachtigal makes mention,^[270] appears to have greatly thinned their numbers, at least temporarily, and even now one commonly finds tusks and other remains of these beasts on the shores of the lake. In the Shari-Logone region on the contrary the hippopotamus has always been equally common. Many places there ‘literally swarm’ with these animals.^[271] What holds good of the tameness of the wild boar when not hunted holds good also under similar conditions of the harmless hippo.^[272]

Finally, the most interesting of the mammals of Bornu is a water-mammal, for whose appearance, however, no definite proof appears to have been presented as yet; this is the ‘ngabara’ or *Manatus Vogelii*,^[273] which was first named by Barth on the Shari and lately has again been mentioned by Chevalier as occurring in Lake Chad;^[274] and it can scarcely be doubted that the existence of this animal, taking into consideration also the remaining water-fauna of the lake, must have some relation to the geological and hydrographical conditions of the Chad countries at an earlier epoch.

The birds of Bornu, so variegated and interesting, are as manifold and rich in species as the mammalia; not least are they interesting on account of the alternations introduced by the temporary sojourn of birds of passage. Not only the number of species but also the quantities of individual birds is remarkable. In early spring, when the acacias begin to blossom, the open ‘bush’ resembles in places a single great aviary, which re-echoes with the singing and chirping of its inmates.^[275] But even during the most comfortless period of the dry weather one can perceive everywhere in tree and bush strange birds, whose variegated plumage stands out in most

striking contrast to the dull yellowish-grey of their surroundings. The little that is known about the ornithology of Bornu allows one to perceive that, as in the case of the Mammalia, an intermingling of fauna has been brought about. A richer collection of material than is now forthcoming is necessary before a better account can be given of the birds of Bornu.

How easily even common and characteristic species may be overlooked by a non-expert traveller, even after staying a year in a country, is proved by Barth's assertion that the parrot is not found north of 8° North Latitude—and therefore not in Bornu at all.^[276] Yet at the present day not less than four varieties of parrots live in Bornu, of whom the lively emerald-green ring-necked parakeet, *Palaeornis cubicularis*,^[277] reaches as far as Lake Chad, and during the dry weather nests in the sparse acacia trees, which are the sole vegetation bordering the turbid water-courses of the 'firki' districts. In South Bornu, where along the rivers the big forest-trees and the swampy stretches of meadow-land form a pleasant park-like landscape, the conditions are also favourable for the Senegal parrot, *Poeocephalus senegalus*, which extends from Adamawa as far as here, and is remarkable for its shrill screech as well as for its variegated plumage. It and the ring-necked parakeet, both equally good on the wing, vie with the monkeys, finches, and pigeons in plundering the cornfields and thus make constant vigilance necessary on the part of the natives, who try to scare the intruders off with various things for making a noise. Two species of green parrots, or dwarf parrots, who are likewise found here, with light-grey and red heads, and who are scarcely visible and noiseless when at work, content themselves with ransacking the thick crowns of the fig-trees for the diminutive fruit.^[278]

The numerous family of the soft-billed birds is also represented in surprising and beautiful varieties throughout Bornu; it is just among them that many species new to Bornu will doubtless be discovered. Of the cuckoos, Barth mentions the honey-guide, *Indicator spec.*, which is useful to the natives, and which is called 'Schneter' by the Shuwa Arabs—probably onomatopoeically.^[279] In the low thick thorn bushes, which it scarcely ever quits, lives the brown lark-heeled cuckoo, *Centropus senegalensis*, worthy of notice owing to its splendid ruby-red eyes. In the

crowns of those acacias which are especially much overgrown with *Loranthus* and thickly intertwined with the fat tendrils of the *Cissus* lives in little flocks the long-tailed *Colius macrurus*, remarkable for the peculiar mouse-like manner which it shows. With clumsy clapping of wings and jerky flight the rhinoceros-bird of these regions, i.e. the 'Tok' or red-billed hornbill, *Buceros erythrorhynchus*,^[280] hurries from one group of trees to the next, and on the ground amongst the high 'bush' grass, the odd-looking ground hornbill, *Tmetocerus abyssinicus*^[281] sets to work to poke about like a stork.

Equally remarkable for their big beak and pleasing colouring, the kingfisher and the halcyon also appear in Bornu. The commonest is the black and white *Ceryle rudis*, a species which is also found in the Mediterranean countries, and which one often sees hovering above a stream, ere it plunges into the water to strike at a fish. A species double as large is to be found here and there in the more secluded spots of the water-courses. A relation also of the European kingfisher, only much smaller and of a darker ultra-marine blue, often darts past like an arrow, only visible for a second or two above the water, whilst the shining cobalt-blue *Halcyon semicaeruleus*, or African white-headed kingfisher, allows us to admire his splendid plumage at quite close quarters.

Where the river during the dry weather has left the steep banks free of water, those countless reedy hollows appear, close to each other, which are the haunt of the beautiful bee-eater, who has established its nesting-places here after the fashion of the sand-martin.^[282] The most beautiful and also the commonest is the *Merops nubicus*, or Nubian kingfisher, painted purple-red and turquoise-blue, numbers of whose noisy fledglings often gather thickly in the branches of the leafless trees along the banks and are visible from afar owing to their red plumage. The graceful flight of this beautiful bird cannot be fully appreciated till it has been seen in company with the kites pursuing like the swallows the swarms of locusts, chasing them through the 'bush'-fires, yet without ever being singed by the flames.^[283]

The beautiful blue long-tailed Abyssinian roller, *Coracias abyssinica*, builds in hollow trees, and does not allow itself to be disturbed in the least in its nesting arrangements by people resting for a while under the tree it

has selected. The three varieties of nightjar found in the country, belonging chiefly to the southern districts, are distinguished less by the colouring than by the peculiar shape of their feathers. When the two most remarkable species, *Cosmetornis vexillarius*, or pennant-winged nightjar, and *Macrodipteryx longipennis*, or standard-winged nightjar, appear flitting about in the dusk, the peculiar extension of the feathers of the wing gives one the impression that they do not belong to the bird but are some undefined object trailing after it. The swift, which is related to the nightjar but flies by day, has the same rapid flight as its European cousin of the 'Micropus' species.

Woodpeckers also are to be found, especially in wooded country where the trees are high; they are, however, rare and insignificant in comparison with the European varieties.

The Bornu birds of prey show many very characteristic species. The tall 'Borassus' palms swarm with hawks, *Falco chiquera*,^[284] who have selected the thick fan-shaped crowns of this tree as their eyry.^[285] The various beautiful kinds of eagle are to be found in places, but only sporadically; they are immediately noticeable owing to their colouring or to some such peculiarity. Near water fringed with thick vegetation, but especially in the 'ambach' thickets of Lake Chad, lives the black crested eagle, *Spizaetus occipitalis*,^[286] with powerful talons, which at the approach of man withdraws in short flights further and further into the sheltering thickets. The water-eagle on the contrary, *Haliaëtus vocifer*, with black, white, and chestnut-brown markings, displays like most of the fishing birds little timidity in the presence of man. For hours at a time this bird sits poised on some leafless branch overhanging the bank, watching from this look-out for its prey. Round the granite ridges of Mandara and the adjacent isolated peaks circles the useful Bataleur eagle, *Helotarsus ecaudatus*, which from a great height is able to detect the poisonous snakes in spite of their protective colouring, and which vies with the numerous varieties of kite in activity of flight. The most peculiar of the kite species is the swallow-tailed kite, *Nauclerus Riocouri*, with its deeply-forked tail. It is principally to be met with hovering over cultivated land, and is especially fond of perching on the long poles of the hydraulic contrivances by means

of which the Kanuri market-gardens are irrigated. The commonest kite is the Egyptian kite, *Milvus aegyptiacus*, which is widely distributed over North Africa, is omnipresent and ever-hungry, and is to be met with wherever meat is slaughtered or set out for sale. It takes up its fixed position near the carrion-vultures, surpasses even the monkeys in unparalleled impudence, steals bits of meat literally from off the trencher, gobbling it as it flies, yet does not disdain to pursue the insects frightened by the 'bush' fires. The majority also of the scavenger vultures frequent the same kind of haunts as the Egyptian kites, i.e. the neighbourhood of human habitation. The scavenger-vultures, *Neophron percnopterus* and *Neophron pileatus*, are semi-domesticated, and clear of refuse the streets, open spaces, and yards, where they mingle with the poultry; they not only remove carrion but also every kind of filth, and seeing that they often appear in hundreds, perform a very useful office. The walls of the town and leafless trees in the neighbourhood of the houses are where they chiefly congregate. But even carrion lying outside in the 'bush' is discovered, at any rate by the *Neophron pileatus*; often discernible only as a speck in the sky, it mounts to a considerable height, from whence no fallen wild animal escapes its sharp eyes. But the true vulture of the 'bush' is the big *Gyps Rüppelii*, or aasvogel, which loves to hover about the big 'Gawo' acacia trees along the water-courses, where the watering-places of the game assure it of the offal from the meals of the larger beasts of prey. Of the owls, which are not very common, it is worth noticing one species, which makes its nest flat on the ground among the tufts of high 'bush' grass.

In Bornu, as elsewhere in Africa, there are a very large number of different sorts of sparrows, represented by many varieties. They often appear in large flocks, and then almost always do great damage to the cornfields. The birds, which cheer the traveller with their song during the rains, belong nearly all to this family.

The song-birds are confined within narrower limits, and there is little to distinguish them in external appearance from their migratory relatives, who make a winter sojourn in Bornu, and with whom they may easily be confounded. Amongst the 'Motacilla' species which appear as migrants is included a beautiful black and white wagtail, peculiar to the Central Sudan.

The place of the European finches is taken in Bornu by the varied and many coloured family of the weaver-birds and the bright-coloured finches, the characteristic birds of Africa. They are ubiquitous even far within the oases of the Southern Sahara,^[287] and where grass-seed or corn is plentiful, for this forms practically their exclusive food, their skilfully plaited nests, shaped like a pear or a chemical retort, are never missing, and many a tree, especially those of the villages, is hung with them all over as if with fruit. Almost all species belong to the worst class of pests of the cornfields, whose yield may be considerably diminished by their formidable numbers. The measures of defence adopted by the natives against unwelcome pillage are chiefly directed against the countless swarms of these little pests. While certain species appear only in pairs, the great majority of them are found in colossal flocks. The big fig-trees near Ulugo are so crowded with weaver-birds, that their droppings fall to earth from the trees like continuous rain. The true weaver-birds are peculiar in that the males during the dry weather wear the same homely sparrow-like plumage as the hens, while very early in the spring they flaunt in gorgeous feathers of crimson and gold. By far the most beautiful of the weaver-birds is the orange weaver-bird, *Euplectes franciscanus*, or *Pyromelana*, which builds in the high grass along the water-courses and whose cock-birds during the mating season wear a velvety plumage of jet black and brilliant scarlet, which makes the bird look like a little roundabout, as indeed he is. Owing to their enormously long cock's-feather tails, the males belong to the Wida species. Although they mostly prefer the 'bush', still they are also frequently to be found in the neighbourhood of the towns; the *Vidua paradisica*, for example, builds right among the ruins of the town of Kukawa. Many of these finches, especially the little red or turquoise-blue *Astrildae*, are so tame that one can entice them near enough to catch them by scattering about a little food.

Equally gorgeous in plumage as the weaver-birds, but far surpassing them in tuneful ability, are the glossy starlings of Bornu, all distinguishable by their metallic blue or green feathers and their white eyes. The biggest representative of this family, the long-tailed *Lamprotornis aeneus*, is a never-failing phenomenon of the 'bush'.

Everywhere, but especially in the plains, is to be found the ox-pecker, *Buphaga africana*,^[288] which frequents big game or herds of cattle, and is a useful bird, for it rids the animals' hides of all sorts of vermin. An occasional though always rare attendant of the scavenger-vulture is the black and white African crow, *Corvus scapulatus*, when there is carrion about.

Connected both as regards place and time with the presence of insects is the appearance of birds that live on them. In the thorn bushes of the undergrowth, especially in the neighbourhood of the water-courses, the golden-crowned shrike, *Laniarius barbarus*, attracts our attention not only by its melodious note but also by its beautiful black, gold, and blood-red colouring. In places, especially in damp localities, one sees swallows; in the neighbourhood of rock formations where there are springs the black-billed Paradise flycatcher, *Terpsiphone melonogastra*, appears, resembling the cock's-feather-tailed Wida species, and also, if there are sufficient caterpillars, the *Oriolus bicolor*,^[289] resembling its European relative, the yellow thrush, both in note and colouring, but surpassing it in beauty. In similar places is to be heard the song of the bulbul, *Pycnonotus xanthopygus*, which reminds one in many respects of the song-birds at home.

In every flowering bush are to be found the little sun-birds, wrongly classified as *Kolibris*, or humming-birds, adorned with the most gorgeous feathers of green, red, or blue, with a metallic shimmer. The flowers of the acacia, the 'capparis', and the tamarind are especially sought after by these charming birds. One of the most important varieties is the metallic sun-bird, *Nectarinia metallica*. The acacias, when in flower and frequented by buzzing insects, are also the resort of another striking bird, the long-tailed steel-blue 'Baumhopf', with its red curved beak. It regularly flies in advance of the caravan, perhaps because this furnishes it with prey, and mocks men with its peculiar laughter.

Everywhere in the woods and fields is to be heard the cooing and laughter of the many varieties of doves and pigeons both large and small.^[290] The call of the turtle-dove, *Turtur risorius*, is one of the most commonly heard bird-notes in the Central Sudan, especially at the

beginning of the rains. The *Treron calva*, the largest of the varieties found in Bornu, with green and yellow plumage, differs from other pigeons not only in colouring but also in its whole behaviour: like the dwarf parrot, it searches the crowns of the fig-trees in true parrot fashion for their tiny fruit. While the latter bird is comparatively timid, the doves are often extraordinarily confiding; by scattering a few breadcrumbs one can entice the little *Chalcopeleia afra* just like the *Astrildae* right into one's tent.

Of the game birds, only two species^[291] have at present been recognized in Bornu, the double-spurred francolin, or 'bush-fowl', *Francolinus bicalcaratus*, and the ubiquitous guinea-fowl, *Numida meleagris*. The latter, which is widely distributed even in the Sahara, is to be found everywhere except in the rugged mountainous country, but it is especially fond of light sandy soil, where it can rake about. Perched in the big acacia trees in the neighbourhood of the dunes, one may often find in the evening large flocks of these birds, which are by no means timid.^[292]

In many districts of Bornu, especially the drier ones, is found the greatest of African birds, the ostrich, *Strutio camelus*, often in the vicinity of antelopes, whose company it is fond of seeking.^[293] It seems to appear in Bornu chiefly where the acacias predominate, and after that is scarcely known where the 'Terminalia' district begins. These birds are common also in Kanem, and Barth observed large flocks of them in the sterile country round Kukawa.^[294] At the present day the ostrich is very often domesticated, especially in the larger towns, but ostrich-farming has hitherto been so insignificant as to be quite negligible.^[295]

A country like Bornu, which exhibits vast stretches of pasture and numerous shallow water-courses with flooded meadows along the banks, offers all the conditions favourable for every kind of long-legged bird, who, as a matter of fact, are found in great variety and number. The sandy, heath-like flats around Lake Chad are inhabited by no less than three species of bustard, of which the huge *Otis Denhami*^[296] is a stately bird similar to the European bustard.

Great is the number of species of plover and snipe ('Tringa'), which meet one at every step along the water-courses; many of their varieties are

found distributed as far afield as the Mediterranean. Thus the spur-winged plover, *Hoplopterus spinosus*, well known in Egypt, is one of the commonest birds seen along the water-courses that crawl through the ‘firki’ districts. On the other hand, the very characteristic white-crowned plover, *Xiphidiopteris albiceps*, noticeable for its two yellow patches in front of the eyes, prefers the sand-banks of the river, where also it makes its breeding-ground, which it vigorously defends against nest plunderers by means of the spurs which it, as well as the before-mentioned species, carries at the bend of the wing. Wherever there are crocodiles their faithful satellite, the crocodile-bird, is to be found. Snipe, which, thanks to their protective colouring, are often overlooked, are common, especially on Lake Chad, according to Denham; this traveller speaks of them as being ‘as numerous as swarms of bees’, ^[297] so that one may conceive from this statement some idea of the great and constant abundance of bird life.

Besides the very rare white ibis, *Geronticus aethiopicus*, there is also found in Bornu the much commoner black variety, *Geronticus hagedashia*, a bird that by its dissonant, far-sounding cry, gives the game seasonable warning of the approach of man.

The stork family in Bornu are distinguished by their beauty, and to some extent by their size. Certainly the most beautiful of them is the gorgeous, rosy-red glutton, the African wood ibis, whom we often find perched in whole rows on the horizontal branches of the big trees along the banks—where also it builds its eyry—in company with the black stork, *Ciconia abdimii*. ^[298] In the reed-beds along the banks, the largest of all the storks, the gorgeous ‘Jabiru’, *Mycteria senegalensis*, ^[299] gets to work, and with its enormous bill is able to render innocuous to itself large poisonous snakes, thus proving one of the most useful birds in the country. In the neighbourhood of human habitations along the rivers the marabou, *Leptoptilus crumenifer*, is found, and where it has not yet been snared by Europeans for the sake of its feathers, is half-domesticated; it lies in wait for offal of all sorts, watching the movements of men, motionless for hours at a time, and only flapping its wings so as to intercept as many as possible of the sun’s rays, which it loves.

The more secluded and thickly-wooded spots on the river banks afford good places for the big nests of the hammer-head, *Scopus umbretta*, and here also we meet with the majority of the herons. Their largest representative, the giant heron, *Ardea goliath*, is rare, but the small snow-white *Ardea bubulcus* is an invariable attendant on the herds of cattle, whose intestinal worms form its chief food.

By far the most beautiful of all the long-legged birds living in Bornu is indisputably the crowned crane, or ‘Doboli’, *Balearica pavonina*,^[300] which is widely distributed in tropical Africa. Flocks of these birds often ransack the Kanuris’ fields after harvest,^[301] and where they have not been rendered mistrustful by European firearms, show not the least timidity of mankind. The same holds good of these birds, in so far as they have not been hunted, as Denham affirms to be the case regarding the Arcadian confidence of the water-birds of Lake Chad, when he says: ‘As I moved towards them, they only changed their places a little to the right or left.’^[302]

Apart from domestic ducks, to which belongs the beautiful blue Allen’s gallinule, *Porphyrio Alleni*, the majority of the water-fowl are composed of web-footed birds, which in many places, especially on the open stretches of Lake Chad, absolutely cover the water. It is certainly no exaggeration of Rohlfs to say that ‘there are water-fowl of all sorts in such quantities as in no other place in the world’.^[303]

Besides a large number of duck,^[304] who would reward a special study, there are chiefly three species which are always remarkable for their numbers or their characteristic form or appearance, and are scarcely ever absent from one of the larger stretches of water. These are the spur-winged goose, *Plectropterus gambensis*, the Egyptian goose, *Chenalopex aegyptiacus*, and the pelican, *Pelecanus onocrotalus*. More remarkable for their peculiar habits are the scissor-billed tern, *Rhynchops flavirostris*, and the strange African darter, *Plotus Lavaillantii*. The former is remarkable for the way and manner in which it ploughs the water with its beak as it flies for the sake of catching food, the latter for the strangeness of its movements—whether on the water or on dry land—such as is displayed by none of its relatives or in general by any other bird.

The central and south European migratory birds are certainly all to be found in Bornu, in so far as they seek these latitudes in the winter-time. One of the commonest sights during the dry weather is the ordinary stork, *Ciconia alba*, which one may meet with in whole flocks along the banks of Lake Chad, yet they do not as a rule mingle with the indigenous long-legged birds. High in the air is seen the swift, *Micropus apus*, and in the 'bush' one meets with many birds, which are well known to us in Europe. The most commonly seen of these are the wryneck, *Jynx torquilla*, the wheatear, *Saxicola oenanthe*, the wagtails, white, Ray's and blue-headed, the crested lark, *Alauda cristata*,^[305] the golden oriole, *Oriolus galbula*, and hoopoe, *Upupa epops*.

The reptiles and Ranidae of Bornu are at present only properly known in those varieties which are often seen, or which are remarkable for the benefit or damage which they cause.

Of the turtles, for whom, if only in parts, all the conditions of life are forthcoming, the only one known at present is the *Sternothaerus Adansoni*^[306] living in the bed of the River Yedseram. On the other hand, crocodiles, belonging to the species *Crocodilus vulgaris*, are to be seen everywhere that collections of water worthy of the name are to be found,^[307] but they also stay, in places right up to the foot of the mountains, in the smaller streams and pools, provided that a communication, if only temporary, exists with the larger water-courses; and it is surprising how they are able to maintain themselves here during the dry weather. But of all places the crocodile is commonest in Chad and in its affluents which contain continuously running water, especially the Shari, where indeed one may see giant specimens—as much as four metres long—sunning themselves with gaping jaws during the hottest time of the day on the high sand-banks. The great quantity of fish in the rivers prevents the mail-clad monsters as a rule from attacking men and cattle, but the contrary does sometimes take place—especially in the small but deep and dull-coloured water-courses.

In the reed-beds along the banks of the rivers and lakes often lurks the giant water-monitor, *Varanus niloticus*, with its variegated markings, while it is just the driest parts of the 'bush' that are the favourite haunts of the

thick-set dull earth-coloured land-monitor, *Varanus exanthematicus*, which has few markings. Most of the smaller lizards are quite domesticated. On every building, wall and roof, the beautiful blue and red lizard, *Agama colonorum*, sports in the hottest sunshine, in some places in company with a brown and white striped lizard with a lively turquoise-blue tail. At night-time these lizards are relieved by the confiding Geckoes, *Platydactylus*, who search the interior of the houses for every kind of vermin and thus are of great utility.

In the millet fields and amongst the ‘bush’ trees are found several varieties and large numbers of brightly-coloured chameleons, which appear on the dry stubble after the conclusion of the harvest, and are able, though not to so extensive a degree as is often supposed, to suit their colouring to that of their surroundings. [\[308\]](#)

Quite large in many places is the number of snakes, among whom several are very poisonous and of considerable size. By far the largest of all is the python, *Python Sebae*, usually to be found in the neighbourhood of the water-courses, but it is also fond of coming into the villages, whose hen-roosts it plunders, but is otherwise harmless. It reaches a considerable length; thus Barth secured a specimen, apparently belonging to this species, that measured over five and a half metres. [\[309\]](#) Distributed all over Bornu, but most commonly among the dunes undermined by field-mice and hares on the south-western shore of Chad, where it finds plenty of prey, is the puff-adder, *Vipera arietans*; [\[310\]](#) in consideration of its size—it reaches a length of over a yard and is as thick as a man’s arm—and its predilection for human habitations swarming with rats, it is a dangerous and poisonous brute, but fortunately evinces great laziness in the daytime. Commoner still is the ‘Efa’, *Echis arenicola*, which in the cool nights seeks the warm houses of the natives; its proper lurking-place is among the thick-knotted roots of the fig-tree. To it appear attributable the majority of fatal accidents from snake-bite. [\[311\]](#) The African cobra, *Naja haje*, the liveliest and most truculent of the poisonous snakes, is found indeed everywhere, but fortunately is distinctly rare. Finally, there is a very long green tree-snake, which is credited by the natives with poisonous qualities. [\[312\]](#)

Of Ranidae also there is no lack. All stagnant and sluggish flowing water harbours crowds of semi-gigantic frogs. A bluish-grey variety with a pointed head possesses the ability of making as much row as a flock of crows, and after the rainy season has set in nightly drives to desperation those travellers who are unable to accustom themselves to the nocturnal noises of the tropics. Small silvery-grey tree-frogs with a red belly—they belong to a widely-distributed species—are often found in Southern Bornu clinging to the leaves.

The connecting link between the Ranidae and the fish tribe is a peculiar animal found also in the Nile, the scaled salamander of Lake Chad, *Lepidosiren annectens*, which, moreover, is a living argument in favour of a former hydrographical connexion between the Chad countries and the Nile. The Shuwa Arabs say the salamander is a savoury dish and are keen on trapping it; the pear-shaped gelatinous capsules—generally hidden among the matted roots of a papyrus bush—in which the animal spends the dry season, are easily found in the dried-up soil on the banks of Lake Chad.

Zoologically speaking the vertebrates that have been least sufficiently investigated hitherto are the fish that live in Bornu waters. The little that is known about them again shows a similarity with the fauna of the Nile countries.^[313] A thorough investigation here would contribute to the solution of many geographical questions. Lake Chad and its affluents are beyond all conception rich in fish, sometimes as big as a man in size;^[314] the fish ascend the streams far up to their very source, and remain there even during the dry weather, provided that the water outlasts it. Among the most savoury are several species of cat-fish, which are extraordinarily abundant; at the fall of the water they make shift to do with a very small quantity of moisture, and are able to last out the temporary dryness in the humid mud. The best-known varieties are the electric cat-fish, *Malapterurus electricus*,^[315] the ‘Fahak’, or ‘puffer’, *Tetrodon fahaka*,^[316] and a species of pike, *Polypterus bicher*.^[317] All three are also found in the Nile.

Whilst amongst the vertebrates of Bornu many represent species proper to East Africa, the same is the case to an enhanced degree among the invertebrates. Many classes of insects have almost without exception East

African shapes, and only quite a few West African. Certain kinds of insects are even, considered purely from the point of view of animal geography, the most interesting kinds of fauna that Bornu has to show. The fact that numerous palearctic insects are also represented here is partly to be explained by their considerable powers of flight; but it is further noteworthy that the Sudan species, which have to undergo without exception only a short larval stage, show with a quicker period of generation a far smaller size than the corresponding European, i.e. Mediterranean species.

In a country where the distinction between the wet and dry seasons is so sharply defined as in Bornu, it is natural that the life of the legged creatures, whose whole existence is closely bound up with the constant moisture of the soil and atmosphere, is subject to considerable fluctuations as regards abundance. Thus soon after the beginning of the first rains great numbers of insects of all sorts appear, which indeed decrease somewhat in the middle of the rainy season, to reach their height during harvest time, whilst the winter is poor in insect-life. Where the species appear in successive generations following each other alike in the dry and in the rainy seasons—this holds good especially of the butterflies—they are subject in a remarkable degree to the laws of seasonal dimorphism, that is to say that the parts which remain visible in repose, show a noteworthy adaptability to the colouring of their surroundings. To the insects and such-like creatures, which are in evidence at all times of the year, belong especially those whose presence at any time is considered a nuisance, such as termites, ants, or vermin.

Of all the insect genera the beetles are the best represented, not only as regards number of species but also as regards splendour of appearance.^[318] Big camboid-beetles equipped with formidable mandibles—to this species belong the giant *Anthia venator*, found all over North Africa—gauzy-winged tiger-beetles, *Cicindela*, which scurry over the sand-banks, and big black water-beetles, *Cybister*, which people all the ponds and pools, are the most rapacious among them. In Bornu the chafer family is also rich both in appearance and species. Everywhere on dung one meets big and little sleek black or metallic tinted dung-beetles, whose giant variety is the elephant dung-beetle, *Heliocopris Sandersi*, which builds long tunnels on the ground and is the most powerful of dung-rollers. Whilst the *Dynastidae* are very scarce in the Sudan, the rose-beetles, or *Cetoniidae*, are as prominent in

Bornu as everywhere else in Africa, both as regards number of species, size, and colouring. Great quantities of these beetles, new and surprising examples of which appear every month, are first seen immediately after the beginning of the rains, when every flowering acacia swarms with them. Very many species, and quite the most beautiful ones, such as the white ant-eating *Gnathocera* species with its zebra-like markings of black, white, and yellow, or the big golden-green fork-nosed *Eudicellae* of South Bornu, seek the sap which flows from the grasses and trees; at the slightest concussion of these drinking-places they fly off with a loud buzzing.

Distinguished likewise by numerous species and brilliant colouring is the family of gala-beetles, *Buprestidae*, and the long-horned beetles, or *Cerambycidae*. Of the former there is a large and thoroughly tropical variety, with golden-yellow marking on a ground like brown leather, which is found even as far as the country of Aïr in the Sahara; it is called *Sternocera castanea*.^[319] Quantities of this giant beetle swarm about the end of the rains on the big acacia trees, on whose leaves they feed. On the dunes overgrown with broom on the banks of Chad one finds the beautiful steel-blue *Caillaudi* thickly powdered with yellow, which belong to the *Buprestidae* species, *Julodis*, characteristic of the southern Mediterranean lands. The long-horned beetles, usually with very variegated wing teguments, are remarkable for the damage they do to timber. One large and active species of an orange-yellow colour, striped with velvety black, the *Diastocera fasciata*, is noteworthy for the way in which both males and females are wont to gnaw off, like a beaver, quite large branches of certain trees, in order to find a place to lay their eggs; the stumps thus sawn through are so neatly done that they might have been produced by a turner's lathe.^[320]

Of the beetles that appear in large numbers, one should especially mention the black *Pimeliae*, a true palearctic variety, which are to be found in the sandy districts of Bornu, especially in the neighbourhood of villages amongst dirt and rubbish. Finally, the fire-flies, whose light may be observed at the beginning of the rains all the evening everywhere in tree and bush, belong to the widely-distributed species of *Lampyris*.

Besides the beetles the *Hymenopterae* in Bornu also belong to the class of insects who are of importance for the good service or the damage that they do; thus the honey-bees are certainly as numerous as the varieties belonging to this family which are hurtful to man. Wherever there are fair-sized trees with hollows in them, and especially in South Bornu, one may count with certainty on the presence of swarms of wild bees, which are distinguished in no way from those of other parts of Africa, as far as concerns viciousness and truculence, as all travellers have reason to relate. [321] Nevertheless, the natives understand very well how to get at the honey, and have also hit upon contrivances for settling the bees in places decided on by themselves. Barth and Nachtigal mention besides the tree-living bees a species which nests in holes in the ground and produces a special kind of honey. [322] Whilst the *Apis* variety, to which the stinging honey-bee belongs, only attacks one if one approaches its nest, its relative the non-stinging *Melipona* variety makes itself a regular pest by following man and beast in whole swarms in order to suck their sweat, so that many travellers, especially in view of the smaller size of the *Melipona*, have been led to regard them as flies. Especially unpleasant is the acrid smell which these *Hymenopterae* diffuse when one kills them, and this attracts still larger swarms of the intrusive insects.

The *Formicinae*, or ants, appear in Bornu in remarkable numbers though not in such masses as in the primaeval forest region. Many are apparently identified, even by travellers like Barth, with the termites, and are thus held to some extent responsible for the latter's destructive activity, [323] yet they are ubiquitous and their intrusiveness, though not always injurious, makes them in the highest degree troublesome. It is only by making a barrier of oil or water that one can protect oneself from their attentions. The driver-ant, too, *Annoma*, that unpleasant visitor from the primaeval forest, appears, according to Barth's description, to be always present in the Central Sudan. [324] At all events the ant fauna of Bornu is very large, and Gerhard Rohlfs says with perfect truth: 'No country could have so many ants and such different species of them as Bornu.' [325] Nachtigal, too, lays special stress on the number of species of ants. [326] Among the most important species of *Hymenopterae* living in Bornu is the

sand-wasp, *Pompilus*, remarkable for its size as well as for its brilliant colouring.

Bornu is poor in butterflies, if one takes into comparison the prodigious richness of the primaeval forest in such insects, at least as far as one considers number of species. In spite of this it is just the lepidopterous fauna of the country which is extraordinarily interesting. Only one single variety is West African, all the others, so far as they are not South European or cosmopolitan, or else of a general Ethiopian character, are East African.

As far as butterflies are concerned, the shimmering *Nymphalidae* species, in which the primaeval forest region is so rich, is almost entirely unrepresented. Of important forest butterflies there are only forthcoming some of the quick-flying *Charaxes* species, which are always hovering over the granite rocks of Southern Bornu, where they feed on the sap oozing from the trees or on excrement. The *Precis* species, corresponding to the European *Vanessa* species, show themselves during sharply defined periods of the year. Everywhere in the 'bush', especially feeding on elephant dung, is to be found the cosmopolitan 'painted lady', *Pyrameis cardui*. The best represented of all the butterflies are the whites, *Pieridae*, and amongst them again the most numerous are the charming *Teracolus* species, characteristic of the Sudan, which with their variegated markings, their carmine or bright blue beauty-spots flutter round the flowering *Capparis* bushes—they come out especially also in the dry weather—and so form the most beautiful living ornaments of the often desolate 'bush'.^[327] It is also remarkable how exquisitely suited to this family is the golden-red colouring of the under-side of the withered grasses during the dry weather. The *Papilionidae* species, which includes the largest sized butterflies, is extremely scantily represented in Bornu; so remarkable is this lack, that one of the species that is found, viz. the big black and white *Papilio Schultzei*, is not only a very interesting insect from the point of view of animal geography, but at the same time is also the most interesting representative of the fauna of Bornu in general, so far as it is known. This butterfly is only found in a few small places among the granite rocks north of Uba, i.e. close to the border of Adamawa. The nearest relatives of this butterfly are not found again till east of Lake Victoria Nyanza in East Africa, that is, at a distance of roughly 3,000 kilometres—the intervening countries, including the highlands of

Mandara, are apparently altogether without them; they were considered as characteristic of East Africa until the discovery of this new species. It is only geologists that may perhaps be in a position to clear up this curiosity in Natural History, for this butterfly belongs to those, who—like the European *Parnassiae*—are very unwilling to leave their usual haunts and show not the slightest migratory propensity.

The moths, which especially among the *Noctuae* exhibit many Mediterranean varieties, are somewhat more richly represented than the butterflies. The caterpillars of two variegated *Saturniidae* of the species *Epiphora*, which live on the *Zizyphus* trees, belong to the silk-cocoon producing family. These cocoons, however, are not employed by the natives. Many kinds of caterpillars, which crowd the trees and bushes soon after the first tornadoes, are eaten by the pagan tribes, as is usual in Africa; amongst such caterpillars are those of the *Saturniidae*, *Gonimbrasia osiris*, [\[328\]](#) which appear in regular swarms.

An important and sometimes fatal rôle is played by the two-winged flies, or *Dipterae*, for to them belong almost without exception in Bornu those varieties of insects, who have been ascertained to be the chief carriers of the disease germs, from which both man and beast have to suffer. Even when this does not hold good, they are for the most part perfect pests, which may become dangerous from their very numbers. All travellers are aware of the quantities of mosquitoes, often to be described as ‘in size equalling a large fly’, [\[329\]](#) which appear especially along the rivers and on Lake Chad in regular clouds, and make it impossible in many places for men and animals to stay there. Of these blood-suckers, to which belong many species of the *Culex* and *Anopheles* family—the female of the latter is the carrier of malaria—as well as the diminutive sand-fly, *Simulia*, with a very irritating bite. Denham says: ‘A chief told me near Kussery that during the last two years he had lost two children, who were literally stung to death, and from our own observation and sufferings this does not appear to be an exaggeration.’ [\[330\]](#)

There are also many other flies which are justly dreaded, such as the numerous varieties of horse-flies and the dangerous *Glossinae*, the female of which carries the tsetse-sickness among cattle and the sleeping sickness

among men,^[331] though the latter has not yet been observed in Bornu;^[332] fortunately the tsetse-fly is only found in certain tracts, but there it always causes more or less destruction among cattle.^[333] Barth as well as Nachtigal mentions the incredible number of fleas in the houses of the natives;^[334] on the other hand the 'jigger', *pulex penetrans*, which was introduced into Africa from America at the beginning of the nineteenth century, has not yet penetrated to Bornu.

The *Neuropterae* are represented by the big *Myrmeleon* species with brown-spotted wings; they are three times as large as the European ant-lion. The *Orthopterae*, as far as regards dragon-flies, are inferior to the big metallic-glistening species of the primaeval forest, but the southern districts especially excel in the family of the white ants. Everywhere, especially in the acacia forest,^[335] one meets with the considerable erections, three or more metres in height, made by these termites. The white ants bear the name of *bellicosus*, *fatalis*, or *mordax*, but only justly so in as much as their war of extermination is directed exclusively against vegetable products. Their destructive fury shows itself in the most troublesome fashion, so that nothing is safe from them; all vegetable material and objects placed on it, such as boxes, trunks, building timber, paper, or cord, is covered by these insects with the familiar crust of mud, unless preventive measures are taken against them, and underneath this the work of destruction goes forward.^[336] Many districts, in which dry branches and withered leaves or grass cover the ground, thus take on an appearance towards the end of the rainy season, which Marquardsen describes very accurately, when he says that they 'give the impression that they have been exposed to floods for a long time'.^[337] The damage done by the white ants is to some extent counterbalanced by the fact that during the rains swarms of these little creatures, which lose their wings at the slightest touch, are collected as food by the natives.^[338] Where ant-hills exist in the bush they are in a regular proportion with the number of *capparis* bushes, which thrive in unusual profusion in just such spots; and it is not impossible that this fact conceals a law of symbiosis.

The strange-looking *Mantidae* and *Phasmidae* do not leave one in any doubt as to their existence in Bornu. The number of locusts also, including the *Acridiae* with brightly coloured under-wings, is legion, they start up

wherever one goes. Rohlf observed clouds of these creatures moving from north to south and taking hours to pass a given point.^[339] At the beginning of the rains appear the *Gryllus* species, with few varieties but in large numbers. These are the insects which make themselves chiefly perceptible to one's hearing; in fact, one large species gives out such a shrill note, that one's ears ache if standing in the vicinity of this musical insect. On the other hand, the cricket, the most noisy insect in the primaeval forest region, is remarkably rare in Bornu, the *Fulgoridae* also are not common; the above-mentioned distinction between the dry and rainy seasons may not suit them.

Bugs too show a great variety of forms. Whilst many of them are deceptively like a withered leaf, others resemble the beetle species in form and markings; and the *Reduviini* administer an irritating bite when touched. A species of bed-bug is also found, and this kind of vermin is very widely distributed.

In hollow trees are to be found centipedes, *Scolopendra*, which are dreaded on account of their bite, but the giant 'Julus' species, or millepede, as thick as a man's finger, which lives in similar places, is perfectly harmless.

Of the spider tribe, there is the thick hairy *Solpuga*, which is fond of inhabited places and is shunned as poisonous. Still more dreaded on account of their sting are the various species of *Buthus*, or scorpion, which likewise exhibit a special preference for human habitations. The big black scorpion, however, as long as a man's hand, which is found on the coast, does not seem to appear in Bornu. Many varieties of the spider proper are forthcoming and are distinguished by their remarkable form and often also by their gorgeous colouring. One species with bluish-grey and golden-yellow markings and as large as a fingerbowl builds huge webs with strong yellow threads amongst the undergrowth.

Amongst the mites, there is a tick, *Ixodes*, which makes itself very noticeable as a most troublesome form of vermin; they swarm in innumerable quantities on the grass of the roads and 'bush' paths and appear even to outlast the burning of the grass in the dry weather by slipping into cracks in the soil. Cattle and sheep are often thickly covered with them and

on this account are apt to waste away from the loss of blood caused by them. Apparently the vermin mentioned by Barth under the name of 'Kari' is something similar to such a tick.^[340]

The little that is known or published about the 'Crustaceae' of Bornu leaves us in ignorance of some pretty considerable species;^[341] still the Chad and its affluents have certainly a great abundance of interesting varieties to show.

The few varieties of molluscs^[342] which are known in Bornu furnish a surprising amount of information. While among the land-snails the handsome *Achatinae*, which are found everywhere in tropical Africa, predominate both in number and size, there are varieties of molluscs in Chad and its affluents which more than any other creatures confirm the theory set forth by Professor Dr. O. Boettger, in *Nachbl. der deutsch. Mal.-Ges.*, 1905, p. 25,^[343] 'that in very early geological times the Chad district, which has now no effluent, was in connexion with the Nile water-system'. There are a large number of mussels also, whose shells cover certain spots on the banks of Chad with a stratum of the same kind; these spots were formerly under the water of the lake and appear likewise to confirm Boettger's assertions.^[344]

Among the worms the big earth-worms of Southern Bornu, which are also found in Adamawa, deserve mention; their excrement forms little piles of considerable size which are a characteristic phenomenon of the 'bush' there.^[345] Of parasitic worms there is the well-named Guinea-worm, *Filaria medinensis*, by far the best known and apparently the commonest variety.^[346]

Information as to the water animalculae of Bornu is practically non-existent, although this would perhaps be a point of cardinal importance for the geologist. There still remains a wide and fruitful field of inquiry for the scientist on this whole subject, as far as Bornu is concerned.

The principal domestic animals bred in Bornu can be enumerated in a few words. What Nachtigal said on this subject forty years ago, holds good even to-day.^[347] The most important of the animals bred there is the ox. In

Bornu their use is not confined to furnishing butcher's meat and the production of milk, but they are also constantly employed in some parts as riding and pack-animals. There are various breeds of oxen in Bornu, but they all show traces of intermixture. The most widely distributed species is the humped ox, similar to the 'zebu' of India, which is bred all over the Sudan, but which is much larger than the latter and far more varied in colour. Certain animals of this breed have the very remarkable peculiarity that their horns grow at a disproportionate rate.^[348] In the neighbourhood of Lake Chad one often sees 'Kuri' oxen with the hump scarcely formed or not at all, yet adorned with extraordinarily long and strong horns, which gives them a somewhat strange appearance.^[349] Sheep and goats cannot compete in point of numbers with the cattle. The former are of extraordinary size,^[350] while the latter, as throughout the Sudan, have very thin bodies, short legs, with short and generally brown hair.

According to Nachtigal's statement the horse bred in Bornu found its way there across the Sudan 800 years ago.^[351] It has acclimatized itself excellently, and in the course of years has adapted itself very well to the conditions of soil of the swampy low-lying plains, which are not exactly suited to a horse's anatomy. Donkeys are largely bred, but chiefly in the south-westerly districts, near the Hausa States, where they take the place of oxen as pack-animals. They are all of a very light grey colour with sharply defined zigzag stripes on the back and a single similar stripe of the same colour across the withers. Mules are not bred in Bornu.

Of the camels employed in Bornu only the kind that are bred in Kanem can remain there for long, and even they only thrive where they can find on the dunes plants to graze on such as the 'retam', which are indigenous to the Sahara.

Dogs are not often met with;^[352] they are small, mean-looking, and certainly have few of the good qualities of the European animal, and there are few varieties of breed; in Zinder they appear to live in a semi-wild state and to have undertaken there, like the vultures, the duty of scavengers.^[353] Birds are confined to cocks and hens, and here and there also pigeons and Muscovy ducks. Finally, of the wild creatures of Bornu which are

occasionally in some places tamed and made useful, the most important are the civet-cat, the guinea-fowl, and the ostrich.

Nachtigal, ii. 368.

Cf. Nachtigal, ii. 504.

Foureau, *Documents scientifiques*, ii. 1007.

[Red Patas monkey, *Erythrocebus patas*.]

Cf. Rohlfs, ii. 37.

[*Papio* sp. Some West African baboon, probably either *P. nigeriae* or *P. sphinx*.]

[One of the many West African bush-babies.]

[Small fruit bat = *Eidolon helvum*.]

Cf. Barth, v. 387.

Barth, iii. 58.

Dominik, *Vom Atlantik zum Tschadsee*, p. 186.

Kund, *loc. cit.*, p. 21.

Denham, i. 260.

Cf. Dominik, *loc. cit.*, p. 181.

Barth, iii. 168.

[Also known as *Vulpes cerda*. On Appendix, p. 182 of Denham, first edition, there is a drawing and description of this animal.]

Barth, iii. 293 (German ed.) and iv. 49 (German ed.).

Dominik, *loc. cit.*, p. 291 ff. [for illustration vid. Mecklenburg, i. 63].

Nachtigal mentions four species of hyena in the neighbouring country of Bagirmi, vid. Nachtigal, ii. 544. But it is very probable that it is only a question of varieties of the same species.

Cf. also Nachtigal, ii. 38.

Rohlfs, ii. 107.

[There is also a kind of malodorous badger, Kanuri, 'mbachamma', Hausa, 'tunku', grey with a white-tipped tail and sharp head. The Kanuris, however, consider its scent 'sweet', but their taste in perfume is peculiar. According to Koelle, they think that the odour of bugs and the odour of Heaven are of the same nature; vid. 'bermade' in Vocab.]

Cf. Barth, iii. 320; Nachtigal, i. 598 and 635.

Nachtigal, ii. 490.

Cf. Foureau, *D'Alger au Congo*, p. 646.

Cf. Nachtigal, ii. 544.

Cf. Barth, iii. 285; Rohlfs, ii. 78.

Nachtigal, i. 629.

Perhaps, however, it belongs to a species of its own.

Nachtigal, i. 561.

Cf. Rohlfs, i. 286.

The German members of the Yola-Chad Boundary Commission observed giraffe even on the northern border of Adamawa, close to the foot of the Mandara Hills.

Cf. Barth, ii. 244 and 326; iii. 45 and 357; iv. 20; Nachtigal, i. 533, 561 ff., 572; ii. 34.

[In addition to these, some of which are not found in British Bornu, there are the common red cob antelope, the white-stern gazelle, and the sittatunga. There was a specimen of the latter in captivity in Maiduguri in 1911—the only one so far observed in Bornu. It was obtained near Mulgue on the River Yedseram.]

Nachtigal, ii. 488 and 544.

Dominik, *loc. cit.*, p. 248.

Barth, ii. 190-1.

Barth, ii. 42 (German edition).

Denham, i. 75; ii. 147.

Kund, *loc. cit.*, p. 5.

Barth, iii. 162.

Nachtigal, i. 565. [The Customs Clerk when proceeding to Kauwa, inside the Chad Game Reserve, in 1908, reported that he was held up for an hour by an elephant on the road who stood and looked at him. A subordinate Customs Officer, one Shegorama, a man of imagination, assured me that once when riding inside the Reserve he had had his cap taken off by an elephant.]

[The whole of the British shore of Chad is a strict game reserve. There are a considerable number of elephants there, whose destructive habits have entirely driven the natives off the rich farming land there.]

Cf. Barth, iii. 313; Kund, *loc. cit.*, p. 4; Stieber, *loc. cit.*, p. 118; Dominik, *loc. cit.*, p. 260 ff. [The rhinoceros is also found on the Bornu-Yola border and along the Gongola. The Kanuri say there are two kinds of rhino which they call 'Kargadan' (large) and 'Kumarima' (small).]

Nachtigal, ii. 542.

Barth, iii. 311.

Foureau, *Documents scientifiques*, ii. 1008.

La Géographie, *loc. cit.*, p. 332.

Barth, ii. 326; Rohlfs, i. 331.

Nachtigal, ii. 288.

Nachtigal, ii. 573.

Nachtigal, i. 566.

[Cf. Benton, *Notes on Some Languages*, p. 286, the 'ayo' or 'manatee'. There is a photo of one on p. 208, vol. ii of Chudeau, *Missions au Sahara*.]

Barth, iii. 325; Nachtigal, i. 660; *La Géographie*, ix (1904), p. 342. A similar animal, the 'Ayu', identical with the *Manatus senegalensis*, is found in the Benue.

Cf. Rohlfs, i. 286.

Barth, iii. 240.

[*Palaeornis docilis*.]

The grey parrot is not found in Bornu. Its distribution does not extend beyond that of the oil-palm.

Barth, iii. 354.

[*Lophoceros erythrorhynchus*. Kanuri, 'zogum'; there is also a black-billed hornbill, Kanuri, 'kogoji'.]

[*Bucorvus abyssinicus*, called in Hausa, 'burtu', Kanuri, 'kagum'.]

Vid. also plate in Foureau, *D'Alger au Congo*, p. 727.

The other species, such as the green *Merops Persicus Saharæ*, are much rarer and always appear in pairs.

[Red-headed falcon—*Falco ruficollis*.]

Cf. Barth, iii. 287.

[*Lophoaetus occipitalis*.]

Cf. Nachtigal, i. 547.

Cf. Barth, iv. 169. [*Buphagus Africanus*.]

[North African golden oriole.]

Barth relates that in one place he observed thousands of turtle-doves, whose presence required special precautions to be taken for the protection of the farms. Barth, ii. 205.

[There are also the sand-grouse, Kanuri, 'fer-fer', and the little black 'rock-partridge'. The latter is about the size of a bantam; I have only seen it twice in Bornu, once at Gujba, and once near Wupti on the Maiduguri-Yola road. P. A. B.]

Denham observed flocks of guinea-fowl of as many as 100 birds. Denham, i. 70.

Nachtigal, i. 561.

Barth, ii. 238.

[Denham took home four black ostriches, which were put in the royal menagerie at Windsor, vid. Denham, Appendix, p. 199 (first edition). The Niger Company at Nafada now do a very large trade in ostrich feathers, but they nearly all come from Kanem and Wadai.]

Denham, i. 254. The *Otis senegalensis*, which is also found, is very widely distributed.

Denham, i. 120.

[White-bellied stork.]

[Saddle-billed stork—*Ephippiorhynchus*.]

[Crown-bird or demoiselle crane, Kanuri, 'gubori', Hausa, 'goraka'.]

Cf. Nachtigal, ii. 488.

Denham, i. 63.

Rohlf, i. 331 ff. Cf. also Foureau, *D'Alger au Congo*, p. 650.

[Including the very common and succulent whistling-teal, the mallard, and rarely the miniature goose, often called the cotton-teal.]

According to an oral statement by the English ornithologist Boyd Alexander, it belongs to a species of its own. Personally I cannot detect the slightest difference.

The author brought one of these home with him alive. It has not been determined to what species belonged the turtle observed by Barth in Kanem. Barth, iii. 67 (German edition).

Chudeau even met with the crocodile in the Lake of Gadabuni. *La Géographie*, xv (1907), p. 334.

Cf. Vogel in *Petermann's Mitteilungen*, 1857, p. 168.

Barth, iii. 57. One that was brought to me alive by some Shuwa Arabs at Ulugo was over four metres long.

Cf. Foureau, *D'Alger au Congo*, p. 638.

Cf. also Rohlf, i. 287.

[For a very full native account of the snakes of Bornu, vid. Koelle, *African Native Literature*, p. 189. They are enumerated there as follows:

(1) 'Kulutshi', a python, speckled, 12 ft. long, and as thick as two men.

(2) 'Abr', spotted, 9 ft. long, poisonous.

(3) 'Gangu', spotted, 6 ft. long, harmless.

(4) 'Komontugu', yellow, 6 ft. long, harmless.

(5) 'Tshibato' [jibado], black, 9 ft. long, said to spit poison and blind its victims; venom used for arrow poison; nicknamed 'Koana lageṭe' = the villainous felon. Rears itself erect and distends its head. [Adder or cobra?]

(6) 'Rokodimi', square red spots on the skin, 1 ft. long, has four legs, very poisonous, venom used for arrow poison.

(7) 'Shargo', striped black and white, 3 ft. long, nicknamed 'Kadi mallam' = the priest snake, harmless, much admired for its beauty, and sometimes domesticated.

(8) 'Kadi keḷi' = green snake, 3 ft. long, poisonous.

(9) 'Kadi tselim' = black snake, small, lives in wells, poisonous.

To these may be added from personal information:

(10) 'Wofe', light yellow, poisonous but not aggressive. Lives inside houses. About 3 ft. long.

(11) 'Fushi', striped, poisonous and aggressive. Lives in holes in the ground in dry weather, but in the grass in wet weather. About 18 inches long.

(12) 'Gergeshi', speckled, poisonous and aggressive. Lives at the bottom of trees. About 2 ft. long.

(13) 'Karua', dull green, poisonous and aggressive. Lives in the water in Chad. About 9 inches long.

(14) 'Kadi suni uguwa' = snake of the five shepherds. Green. Lives in hollows of trees. Very poisonous. It will kill five or more men if they put their hands into the hollow. About 18 inches long.

(15) 'Silla', black, harmless, lives in houses. About 6 inches long.

(16) 'Kadi digalbu' = the bed-snake. Striped. Lives at bottom of native beds. Poisonous but not aggressive. About 18 inches long.

In spite of the number of snakes found in Bornu, one seldom hears of a death from snake-bite. Can the explanation be the same as given in the old Greek epigram?

'A snake once on a Cappadocian
Its deadliest venom tried.
Was the man killed? Dismiss the notion;
The snake it was that died.'

(*Memoir of Herbert Kynaston*, by E. D. Stone).]

Cf. Foureau, *Documents scientifiques*, ii. 1054, and Hassert, p. 154. [In Chevalier, *Mission Chari-Lac Tchad*, p. 437, there is an Appendix on Fish.]

Barth, iii. 241 (?); Nachtigal, i. 660; Foureau, *D'Alger au Congo*, p. 714; Dominik, *loc. cit.*, p. 187.

Barth, iii. 36.

Foureau, *Documents scientifiques*, ii. 1054.

Idem, and Nachtigal, ii. 495.

Cf. Kolbe, *Berliner ent. Zeitschrift*, li. Jahrg., 1906, p. 334 ff., and Ch. Kerremans, 'Buprestides des environs du Lac Tchad,' in *Ann. de la Société ent. de Belge*, Brussels, 1907. [Cf. Chevalier, *Mission Chari-Lac Tchad*, p. 703, Appendix on Coleoptera.]

Foureau, *Documents scientifiques*, ii. 1021.

[These beetles are known among Europeans in Bornu as the Colorado beetle. I do not know if the term is a correct one. A small sum is spent every year in their extermination in the vicinity of Government stations.—P. A. B.]

Cf. also Barth, iii. 240.

Barth, ii. 337; Nachtigal, i. 666.

Barth, iii. 341. According to Nachtigal there is one species of ant which is the termite's chief foe. Nachtigal, ii. 556. [This is the large black travelling ant, Hausa, 'Kwalkwassa,' Kanuri, 'Kanam leirabe'.]

Barth, iii. 397. From the description there given it appears that Barth regarded ants as termites.

Rohlf's, ii. 9.

Nachtigal, ii. 556.

Cf. also Rohlf's, i. 287.

Cf. Nachtigal, i. 662 [silk-moths].

Denham, i. 119.

Denham, ii. 30; cf. also ii. 92 and Barth, iii. 251 (German edition); Nachtigal, ii. 386.

[It has been suggested that there is a mention of the tsetse-fly in the Bible. The 18th and 19th verses of the 7th chapter of Isaiah, in prophesying threatened scourges, read as follows: 'And it shall come to pass in that day that the Lord shall hiss (i.e. whistle) for the fly that is in the uttermost part of the rivers of Egypt, and for the bee that is in the land of Assyria. And they shall come and shall rest all of them in the desolate valleys, and in the holes of the rocks, and upon all thorns, and upon all bushes [in margin, or commendable trees].' The 'uttermost part of the rivers of Egypt' can hardly mean anything else than the headwaters of the Nile, viz. the Victoria Nyanza, whilst by the usual law of antithesis in Hebrew composition, the first half of the 19th verse will refer to the bee and the second half to the fly, i.e. that the flies shall rest 'upon all thorns and upon all bushes', and it is the well-known habit of the tsetse-fly to rest on the thorny ambach-shrubs near the Lake.

Arguing from this in an article in the *Royal Geographical Society's Journal* for February, 1913, Dr. Felix Oswald remarks that although sleeping sickness only entered Uganda eleven years ago from the west, it is not improbable that it may have existed in the region of the great African lakes in past ages, subsequently disappearing. In the discussion on the paper, Sir Percy Girouard gave it as his opinion that the disease had spent its force in West Africa through centuries, and had become endemic.]

The only species as yet known in Bornu, the *Glossina tachinoides*, is found in places on the bushy banks of streams. [*Glossina submorsitans*, has also been found. The only case of sleeping-sickness so far observed in Bornu was that of a prison warder who was proved to have contracted the disease which proved fatal, at Loko in Nassarawa Province three years before. The case was described by Dr. D. Alexander in the *Journal of Tropical Medicine*, for 1911.]

Barth, iii. 315 and 463 (?); Rohlf's, ii. 15; Nachtigal, ii. 573.

Barth, ii. 301; Nachtigal, ii. 486.

According to Barth termites are not found in the Musgu country, which is exposed to floods for long periods of the year (Barth, iii. 199). Here again Barth has certainly confused termites with ants. On the edges of the swampy 'firkî' patches the hills erected by the termites are to be found everywhere.

Cf. also Barth, iii. 37 and 302.

Marquardsen, *Der Niger-Benue*, p. 46.

Cf. also Nachtigal, ii. 555.

Rohlf's, ii. 16. Nachtigal also mentions a species of locust being used as an article of food, Nachtigal, i. 666. [For an account of six different kinds of locusts found in Bornu, and of the method of cooking the edible ones, vid. Koelle, *African Native Literature* p. 198.]

Barth, iii. 263. ['Kari' is the regular Kanuri word for a 'tick'.]

Foureau, *Documents scientifiques*, ii. 1060. It is certain, however, that the large land-crabs observed by me in Adamawa are also found in Bornu during the rainy season. [Cf. Chevalier, *Mission Chari-Lac Tchad*, p. 701.]

[There is a detailed study of the Chad shells in Tilho *Doc. sc.*, ii. 165 ff.]

The species adduced to confirm a correspondence between the molluscs of Lake Chad and of the Nile are:

Vivipara unicolor [the pond or river snail].

Melania tuberculata.

Corbicula fluminalis.

In addition to these, I found in a pool on the Yedseram the following varieties, which belong to the fauna of the Nile, and which were identified by Professor Dr. Boettger:

Ampullaria ovata (a very common species) [i.e. the apple snail].

Lanistes ovum.

Cleopatra bulimoides.

[Cf. [\[97\]](#).] [In Chevalier, *Mission Chari-Lac Tchad*, p. 459, there is an Appendix on Molluscs, with a plate at the end of the volume, and a Bibliography on p. 587.]

Cf. also Foureau, *Documents scientifiques*, ii. 1060 ff.

Cf. Passarge, *loc. cit.*, pp. 57, 237, 292, and 400 ff., and the views there expressed as to the rôle played by the earth-worm.

[Also several kinds of intestinal worms.]

Nachtigal, i. 672 ff.

Nachtigal, i. 687; Foureau, *Documents scientifiques*, ii. 1033 and 1039.

Cf. Nachtigal, i. 682. [Cf. [Appendix XVI](#).]

[The Bornu cattle are periodically decimated by a disease called 'fufu', generally described as 'lung-sickness', i.e. pleuropneumonia ('fufu' means 'a lung' in Kanuri). Tilho, *Doc. sc.* i. 32, mentions an epizootic liver disease, called in Budduma 'banami', as attacking Budduma cattle, and says that it is caused by the bad quality of stagnant water. There is a disease called 'gisu' in Kanuri which attacks sheep and goats, and consists of a tumour in the side.]

Nachtigal, i. 616.

[This is a curious statement. Dogs are as numerous and as noisy in Bornu as in the other countries of the Western Sudan.]

Foureau, *D'Alger au Congo*, p. 506.

VII

POPULATION

BORNU offers to the ethnologist, as well as to the botanist and zoologist, phenomena well worthy of notice. Especially he finds confirmed here the interesting fact that in a relatively short time a nation essentially uniform in character can be built up out of various races. The predominant race and the one that inhabits the largest districts in the country is the Kanuri.^[354] They are a mixed race of the kind mentioned above; their speech and character are uniform, although single individuals often show 'throw-backs' in their external appearance which point unmistakably to the bodily peculiarities of the tribes which have contributed to the creation of the new breed. We have here in front of us one of the most difficult problems of general ethnology, because all the conditions favouring the creation of a state of miscegenation from very heterogeneous elements, such as continued migrations, lack of natural boundaries, and the universal and widespread practice of slavery, coincide to a degree known scarcely anywhere else. Absolute clearness regarding the origin of the race is still more difficult to obtain to-day than it was in Barth and Nachtigal's time, because since then the campaigns of Rabeh have introduced new elements into the population of Bornu.^[355]

Roughly speaking, one will not go far wrong if one follows in the main the deductions of Nachtigal based on the philological studies of Barth, according to which the Kanuri are a mixed race of Arab tribes and Hamitic Kanembu and Tubu,^[356] which, gradually losing more and more the physical and mental peculiarities of the dwellers in the desert, became merged in the negro population of the conquered country. Of the good qualities of the desert tribes steeled by a struggle with hostile Nature, the Kanuri appear to have retained only a certain mental activity and industry, whilst the moral virtues, such as energy and chivalry, and the generous bodily characteristics, swamped by the opposite qualities of the inferior negro type, have disappeared more and more. Not only has the influence of

the intermixture of blood been made apparent, but also that of the enervating climate of the newly conquered country, which does not call into play all the powers both of body and spirit in the same degree as that of the Sahara.^[357] As regards external appearance the Kanuri in general are far inferior to many other tribes in the country owing to their heavier build and pronounced negro physiognomy. One may well describe them as of an ugly, thickset type; their women, moreover, still further disfigure themselves, at least according to European ideas of beauty, by staining the teeth red and wearing in the nostril the inevitable plug of coral or stud of metal. Their chief characteristics are untrustworthiness, love of good living, and true negro levity. Still the assertion of Monteil that Bornu is justly called by the Hausas and Fulani 'the Land of Lies',^[358] certainly suits the Kanuri in no higher degree than other tribes with a strong negro strain. When, on the other hand, the German travellers Barth, Rohlf, and Nachtigal lay eulogistic stress on the estimable qualities of such people as Shehu Umar, this is almost always a sign that in the case of these particular individuals the intermixture of negro blood has not yet been thoroughly assimilated.

The quality which shows the Kanuri in an advantageous light as compared with other peoples is their industry; in spite of their undeniable love of pleasure, they do not leave all the hard work to the women and slaves, as is so very very common in the Sudan and especially in almost all negro countries, and it is this that has rendered possible the rich cultivation of the country.^[359] In spite of this activity, much that one may consider as signs of a higher civilization appears to have been lost, such as the use mentioned above of burnt bricks in earlier times.

The influence of Islam is everywhere unmistakable; often indeed it is purely formal, but it is the predominant religion of the country, and in spite of contact—in the Eastern Sudan—with the fanatical sect of the Senussi, conducts itself throughout in tolerant fashion. On the other hand, it can scarcely be denied that Islam has favoured in every respect that deeply rooted inclination towards slave-raiding, on which the European Powers have latterly for the first time imposed some check. The ordinances of the Koran are very laxly observed, as is shown especially by the very slightly secluded life of the extremely coquettish Kanuri lady, who is thus enabled

to play a definite part in public life,^[360] and who even apart from this shows none of the reserve of her sisters in strict Mohammedan countries. This freedom goes so far that, at least in the larger towns, the women show themselves in the streets and exhibit an exceedingly wanton demeanour, to which even the ladies of royal blood form no exception.^[361] Finally, in harmony with this looseness of manners is the lack of piety towards the dead, in which the Kanuri compare very unfavourably with many pagan tribes.^[362] The practices peculiar to Islam are shown by the conscientious observance of the Mohammedan feasts; the existence of a mosque or 'masallachi'^[363] which is never lacking anywhere, though it often consists of nothing more than an enclosed rectangular space; and the instruction of a large proportion of boys in the teaching of the Koran. If the teaching generally consists only in the thoughtless repetition by rote of the first chapters of the Koran,^[364] still one meets more people in Bornu than in other countries of the Central Sudan who can reproduce the words of their native language in the letters of the Arabic alphabet. However little may be the value of the 'high schools' of the larger towns, of which Rohlfs and Nachtigal inform us, and however scanty may be the knowledge of the 'beggar-students' taught there,^[365] still it is to be observed that Barth met people at the Court in Kukawa who knew enough to be able to converse with him on the subject of Ptolemy.^[366] Moreover, the fact that a part of the history of Bornu, as we have seen above, could have been handed down in written form, is surely owing to this unmistakable impulse towards education.

It is obvious that in so purely Mohammedan a country as Bornu is, there is no pictorial art to be met with apart from the decoration of houses and utensils. On the other hand it happens that music is extensively cultivated; besides long coach-horns and the 'algeita'^[367] found all over the Sudan, which is a wind instrument evolving the most clamorous dissonances, there are also many string instruments, by means of which very pleasing melodies are produced. Music finds multifarious applications in Bornu, and even serves to spur the people to labour cheerfully in the fields.^[368]

The fact that the dress of the natives in Denham's time—apart from the fact that European influence is now predominant—was subject to certain

definite fashions, may perhaps be considered as a sign of a fairly high state of culture. In Barth and Vogel's time the women wore their hair in a sort of helmet shape, quite different from the fashion prevailing now,^[369] though the tattoo marks usual at that time, numerous parallel lines or three scars on each cheek, are still partly retained to-day.^[370] The tobe and burnous, formerly invariably worn by men, have now in many cases been replaced by the Dervish dress introduced by Rabeh. The Kanuri bestow great attention on their external appearance and show a preference for costly stuffs; besides those manufactured in the country itself, excellent cotton fabrics are imported by way of Tripoli and the Niger, and as a result of this their taste is constantly changing. This has become especially noticeable in the last decade, and is shown by the preponderance of East Sudan influence over that of the Hausa States, which was formerly predominant, in the dress of the Bornu troops, consisting of wide trousers, the fez, and the Rabeh 'gown', worn especially by the infantry. The cavalry have to a large extent retained the old equipment, which from the remotest times has been peculiar to the Central Sudan, consisting of a coat of mail worn under the clothes, made of a thickly wadded cotton quilt, which can protect both horse and rider against poisoned arrows or throwing knives, and a helmet, which is occasionally furnished with a piece of chain-mail^[371] to cover the nape of the neck. They are still armed, even to-day, with the lance, though besides this various kinds of missile weapons—such as the infantry invariably carry—are included in their equipment.

Houses and utensils have not been subject to change in the same degree as clothes, and thus they have preserved the same forms as were observed by the earliest travellers in Bornu and which were so fully described, especially by Nachtigal.^[372] In all villages or the suburbs of towns one finds round huts with conical roofs of the shape usual throughout the Sudan. The substructure consists either of mud walls or of a wooden framework covered with matting, the roof of various material usually furnished by the stalks of whatever grain is to be had locally. On the topmost point of the roof, which is generally thickly overgrown with gourd tendrils, is often fixed an ostrich egg as a symbol of fertility.^[373] In the towns there are houses usually of one rarely of two stories of rectangular design with thick walls of sun-dried brick or mud and the flat roofs customary throughout the

East. The majority of these houses stand in a compound formed by mud walls, which render it impossible to see into them from outside. The dwellings of all men of rank are built in this style, as also are the mosques which have no minarets. The bare outer walls are generally without any ornament, only the party-walls and square pillars of the houses of the great are decorated with a fresco of paint in three colours.^[374] Only such material as is found in the country is employed in house-building, viz. wood, mud—often mixed with cow-dung—and mats of various kinds.^[375] In spite of their apparent solidity the mud houses nevertheless do not afford the same protection against the ravages of the rainy season as the huts with conical roofs, which carry off the water better.

The villages, in which the round hut predominates, are as a rule open; on the other hand, places whose importance is expressed externally by the prevalence of massive buildings are surrounded by a more or less considerable mud wall, the inside of which is buttressed by sort of stairs leading upwards, and which can even be arranged for the reception of cannon.^[376] No town of any size is without a ‘dendal’, or High Street, which, apart from the market, is the centre all day long of life and activity. The internal arrangement of the houses is extraordinarily simple and is without much comfort, being generally confined to round huts filled with household furniture, among which the most prominent utensils are tastefully ornamented calabashes and carefully plaited grass dish-covers.

Amongst the employments which fill up the daily life of the Kanuri, and in which is almost universally recognized to an extensive degree the principles of the division of labour and the separation of professions, the most prominent are those connected with the operations of agriculture,^[377] whose sole implement throughout the country is the hoe, and the breeding of cattle. The Kanuri devote special attention to their market-gardens—this care is also displayed by some of the other tribes of Bornu—and thus render it possible in the dry season by means of artificial irrigation to make use of the soil and produce vegetables of all sorts. These market-gardens, of numberless small shapes, have rectangular beds lying parallel to each other, which—like the moulds of a blast-furnace—are connected by channels and are irrigated by means of hydraulic contrivances from deep wells, in

constructing which the Kanuri are past-masters,^[378] or from the surviving pools of the dried-up rivers.^[379]

The Kanuri wealth in cattle is in some places extraordinarily large, and the great attention which they pay especially to their horses is a peculiarity unmistakably inherited from their Arabic or Hamitic tribal ancestors. In harmony with this is the great love of animals displayed by the Kanuri, which forms a sympathetic trait in their character, and which is caused by their often being engaged in capturing and taming wild animals.^[380] The herds of cattle, some of which are employed as pack-animals, supply the Kanuri first and foremost with milk, which is never made use of in its fresh state, but not until it has been curdled by the admixture of cow's urine.^[381] Even the butter, which the Kanuri everywhere know how to make, always has the fatal aftertaste of this unpleasant addition to the milk.

Fishing is eagerly carried on in the rivers with small nets, the fisherman being supported above the water by two hollow calabashes connected by a stick—these are also used for the manufacture of rafts.^[382]

Whilst both men and women share in agricultural work, the manufacture of industrial products, as for example even the artistic embroidering of the short bodices worn by the women,^[383] is almost entirely confined to men. The two commonest industries are those of textile products and leather. The former obtains its raw material from the rich cotton farms of the country; the thread made from this cotton is first manufactured into strips as broad as a man's palm, in which state it is already an article of value, and these are then sewn together to form various kinds of garments. The dyeing of these garments and the embroidering of them in tasteful designs is, especially in the large towns, a common branch of industry. From the hides of slaughtered cattle^[384] the Kanuri know how to produce excellent morocco leather generally of a red colour, which is then manufactured in the country itself into durable and beautifully patterned articles, such as cushions, travelling bags, and cartridge-belts, as well as riding-boots, slippers, and so forth. A not insignificant rôle is also played by the products of the salt-works^[385]—which article is also partly obtained from the ashes of salt-producing plants^[386]—and those of the

smiths, who although obliged to obtain most of the raw material for their craft from other countries, yet have even ventured on the casting of cannon. [\[387\]](#)

The products of agriculture and industry are only sold privately to a small extent, but they are objects of the liveliest barter in the markets of the towns and the larger villages. Here the various kinds of merchandise can each show their special quarter, which at the same time serves as a workshop for many industrial products. Under the superintendence of a special functionary, the ‘Maisuk’, [\[388\]](#) and in the presence of numerous auctioneers and brokers, there is displayed in the market, especially at midday, an almost international conglomeration and an unparalleled crowd, which better than anything else shows the quick commercial instincts of the Kanuri people. They have here an opportunity to bring into play their mercantile inclinations, though these, indeed, are even to-day still united with that unreliability and faithlessness—even among the higher ranks—of which we hear complaints as early as Leo Africanus. [\[389\]](#)

On the other hand, the military spirit—as is sufficiently proved by the history of the last few years—has disappeared more and more. Already Denham was able to describe the Kanuri as an unwarlike nation. [\[390\]](#) Even the numbers of the army have steadily declined. Whilst Vogel still speaks of an army of 22,000 horse and 10,000 foot as having advanced into the Musgu country, [\[391\]](#) the numbers to-day would not amount to the half of this for the whole of the territory inhabited by Kanuri. The stimulating effect of the military spirit of Rabeh’s troops was only temporary; the only things that have remained permanent are externals, such as the ‘fantasia’ which Sanda, the Sultan of German Bornu, holds even at the present day at his capital of Dikoa on every Friday and at every Mohammedan festival; on these occasions large quantities of powder are blazed away and all the pomp at his command is displayed. Together with the disappearance of the military spirit has dwindled more and more the power and influence of the ‘Koganawa’ or higher officials, as well as that of the ‘Kachellas’ or military commanders, whose position to-day consists of little more than the title, which in long past times was far more important than at present. Moreover, with the partition of Bornu between England, France, and Germany, the old

Sultanate automatically ceased to exist. The Sultans, who formerly stood in a subject relation to Bornu, such as those of Gulfei, Kusseri, Logone, and Mandara, are to-day on a perfect equality with the Shehu of Dikoa, the descendant of the last Bornu dynasty, so that all distinction between them has disappeared.

Of the Hamitic tribes settled in Bornu Proper who have contributed to the formation of the Kanuri race, only the Kanembu, the inhabitants of Kanem,^[392] are still to be met with along a narrow strip of the south-west shore of Chad. But the traces in Bornu of an older race, which Barth asserts were distinguishable in his day, have now been lost.^[393] Even the characteristic shields of 'ambach' wood^[394] can to-day scarcely be still considered as a peculiarity of the Kanembu, as Nachtigal thought,^[395] since they are also found in use by the Kanuri and Shuwa Arabs of the villages on the south bank of Chad, which often show a very varied mixture of tribes.

To the tribes settled in Bornu before the advent of the Kanuri belong a whole string who were absorbed in the conquering people and lost their own characteristics, but there are others who retained their individuality and who can still be distinguished from the former even at the present day.

The largest and most important tribe is that of the Masa in Western and South-Western Bornu, to whom both Barth and Nachtigal devoted an exhaustive linguistic study; with them are to be reckoned the Makari or Kotoko, the islanders of Lake Chad, the Gamerghu, the Mandara or Wandala, and the Musgu.^[396]

Among the smallest of these are the Mohammedan Gamerghu—who even in Barth's time had been for the most part extirpated^[397]—whose villages are scattered along the middle Yedseram in Kanuri country, and who thus have very little chance of bringing into play their tribal peculiarities. Very important, however, on one ground or another, is the rôle that has been played in former times by all the other branches of the Masa.

The Makari or Kotoko and the Logone people, who can scarcely be distinguished from them, are the most civilized of these, although in Barth's time their Mohammedanism was not more than sixty years old.^[398] These tribes inhabit the great flooded district between the Shari, the Logone, and

the connecting system of canals somewhat north of the eleventh parallel of latitude. They are more heavily built even than the Kanuri, appear to be of a more serious and steadier nature than that frivolous people—a quality, however, which inclines them to listen to the call of witchcraft—and on that account seem to be more trustworthy and more energetic. The rest of their mode of life corresponds to their greater solidity of character. This is shown even in the building of their townships.^[399] Manifold are the forms of the massive often two-storied mud buildings, which have a pleasing appearance owing to the changing shapes of the doorways and windows and the crenellated edges of the thick walls decorated with turrets. The numerous nooks and corners of the narrow lanes and the high walls with their very narrow gateways make up a whole that reminds one, in towns such as Affade, Gulfei, Kusseri, and Logone (Karnak), of the ramifications of a castle of the Middle Ages.

Where so civilized a spirit finds expression it is natural that industries should be prominently developed; so besides excellent woven fabrics one meets all over the country with extraordinarily fine basket-work; in fact the Logone baskets, plaited in three-coloured patterns, are amongst the most beautiful industrial products of the Central Sudan. But besides this agriculture receives the most careful attention; the population is generally pretty thickly distributed and needs an unusually large quantity of food which will grow in a small space.^[400] Of the species of grain found in Bornu, maize is here especially cultivated and certainly with great care. Moreover, along the many water-courses of the country the fisheries are very important, a branch of employment that soon makes itself very noticeable by the strong smell of fish offal in the towns. The people who live along those rivers that always contain flowing water have been compelled to develop a form of navigation of the highest interest. Their roomy barges with long beak-shaped prows and flat bottoms, which Dominik compares with the dahabeas of Egypt, are cleverly built of broad planks, and are moved forward by oars as broad as a man's hand or by long poles.^[401] Besides this, smaller canoes are also employed which are manufactured from hollowed out tree-trunks. In dress the Kotoko and Logone people differ from the Kanuri principally in the fact that the tobe is hardly at all worn any longer, much more prevalent is the East Sudan

dervish dress; their weapons, on the other hand, do not appear to have any obvious differences.

Amongst the allied tribes, the Mohammedan Wandala or Mandara have to suit their habits of life to some extent to the mountains^[402] which border their country on the south, and which owe their name—not a very distinctive one—to this tribe; whilst the islanders of Lake Chad, partly intermixed with Kanembu and other tribes, are the embodiment of a people living half in and half out of the water. They, too, are for the greater part Mohammedan^[403] and fall into two great subdivisions.^[404] The Yedina^[405] or Budduma inhabit the northern and larger part of the Chad archipelago, the Kuri the smaller and southern. Whilst the former build big barges singularly resembling those of the Shari, besides possessing also smaller vessels made of ‘ambach’ wood, the means of transport used by the Kuri, though here there is a close imitation of the barges used on the Shari,^[406] are nothing more than a canoe-shaped bundle of ‘ambach’ stems.^[407] Bigger logs of this light material, which every man—Kuri as well as Budduma—carries with him on his hunting and raiding expeditions,^[408] enable them to cross in very convenient fashion even those places where the water indeed has disappeared, but where mud a yard deep forms a dangerous hindrance to traffic. This means of locomotion, excellently adapted to the constantly shifting distribution of the waters of the lake, at any time difficult of access, may have been an inducement towards the propensity for robbery of the Chad islanders, although they have sufficient resources in their own fields and herds, and the abundance of game^[409] and fish. Even if they are in friendly commercial intercourse with one or other of the villages on the banks, still the dwellers in Lake Chad as a whole correspond to what the Tuaregs are for the country to the north of the River Yo; in fact, the ‘pirates of the Chad’^[410] are scarcely less audacious than the dreaded robber bands of the Southern Sahara.^[411]

Of the tribes belonging to the Masa, the pagan Musgu people doubtless deserve our chief attention, firstly, on account of their numerically large population, and secondly, also on account of their interesting national customs, which furnish ethnographers with many hints for connecting them with the tribes of the watershed between the Shari and the Congo, and even

with the dwellers farther south. Among these characteristics belongs especially the use of the throwing-iron, a weapon which is found far within the Congo basin and takes the place of bow and arrow.^[412]

Inhabiting the flooded country of the Shari and Logone, south of the eleventh degree and the most fruitful stretches of these lowlands at that, the Musgu have been able to preserve their characteristics almost unchanged in contrast to the uniformity of those of Islam. The coarse features of the otherwise well-formed Musgu people are rendered still more unattractive by the use of lip-disks—similar to the wooden disks worn by many Indian tribes in Brazil—with which the female sex disfigure themselves, so much so that Denham mentions that the slave-dealers of Tripoli and Fezzan declined to buy Musgu women. The disks are of the size of a dollar and made out of the most varied materials, but usually from the rind of the calabash gourd;^[413] the lips are perforated and the disks forced into them, so that they project like beaks. The same or a similar foolish fashion is found in the countries on the north border of the Congo basin and even in East Africa.^[414] The character, too, of the Musgu has many features that do not exactly evoke sympathy. The worst of these is their domestic discord, which has not only prevented this people from offering a common front against the slave-raiders, but, on the contrary, has even caused them to utilize in the most shameless fashion the misfortunes of their fellow-tribesmen in order to rob and plunder.^[415] Characteristic of the Musgu, moreover, is the barbarous way in which they retain their seat on a horse—large numbers of a small, shaggy, but pretty hardy breed are raised in the country. As saddles are quite unknown, the Musgu supply the deficiency by causing an artificial sore on the place where the saddle ought to go of a horse otherwise in good condition; this sore is kept continuously open, so that the rider is as it were glued to his horse.^[416] The bridle, too, consists of nothing more than a piece of rope like a halter, which is bound round the animal's jaw.^[417] Whilst in their clothing the Musgu are also very primitive—the men only wear an apron^[418] and the women only a band of rope-like twisted grass passed between the legs—in the building of their villages one finds, on the contrary, a taste for a certain amount of household comfort, for art, and for an orderly mode of life. The most remarkable thing in these

villages—at least, as far as the Musgu country proper is concerned—are the bee-hive or tiara-shaped mud huts, which have numerous knobs on the outside. These knobs may make the houses more capable of withstanding the effects of the tornadoes, but they also make it possible to climb to the top of the hut from outside and thence to gain a view over the extremely flat tract of country.^[419] One also, indeed, meets the usual round huts of the Sudan and mud buildings, especially designed for the storage of grain. The Musgu towns, which with certain exceptions such as the capital, Musgum, are not fortified, lie extraordinarily thick, so that this district is amongst the best populated of the whole of Africa, and was formerly the favourite quarry of slave-raiders. The banks of the Logone are so thickly inhabited, that one may describe the stretch of country extending over fifty kilometres between the towns of Musgum and Mochore as one huge city.^[420]

The density of population compels the Musgu to make the most careful use of the soil, and one accordingly finds agriculture highly developed in their country, and even artificial manuring of the fields. In the neatly laid-out plantations, usually in the shade afforded by the groves of deleb palms, there hardly ever fail to be beds of tobacco, which is indispensable to the Musgu.^[421] Their commercial spirit is further shown by the rational use they make of the products of nature that grow wild. Hives, in the shape of hollowed out tree-trunks, are placed everywhere in the larger trees for the numerous swarms of bees; the surplus grass or other edible plants are collected and plaited into sort of pigtails of hay, which are likewise hung on the trees to serve as dry-weather fodder for the abundant herds of cattle; they also understand how to make ingenious contrivances for catching the fish with which nature has plentifully stocked the rivers.^[422] The Musgu and the tribes related to them fall into a whole string of subtribes, of which the Puss and the Wuliya are the best known.

The large pagan tribe of the Margi, inhabiting the southern districts of Bornu west of the Mandara mountains and a part also of those highlands, show substantial differences from the Masa. To begin with, the Margi possess great external advantages. They have finer limbs, better proportioned bodies, and above all, more pleasing features. The copper-colour mentioned by Barth^[423] may sometimes occur naturally, as in the

case of other negro tribes, but this does not exclude the possibility that the traveller was in error here, for the Margi often stain the whole body red with powdered wood. According to the study of the language which Barth engaged in, the Margi belong to the 'South African family'.^[424] In character also they differ, advantageously indeed, from the Musgu. They are easier to get into touch with, less truculent and less suspicious than the latter. But even here the national character shows the disastrous effects of the slave-raids, for the Margi, like most of the pagan tribes, replied to the persecution to which they were exposed by the Mohammedan slave-raiders, by attacks on their own part, which were rendered successful by the protection afforded by their mountain refuges and by the use of the bow and justly dreaded poisoned arrows.^[425] A visit to most of the 'kopjes' in the Margi country is even to-day not altogether without danger, although possibly their attacks on the neighbouring natives may be the result of certainly not unjustifiable suspicion.

Every Margi is easily distinguishable by his outward appearance: the men wear little leather aprons decorated with various designs, drawn between the legs and hanging down behind like a tail; the women make plentiful use of iron armlets and anklets and iron beads;^[426] they also regularly wear in the chin a little piece of stick as long as a man's finger or a little plate of wood, bone, or metal.^[427] Their villages are distinguished by great neatness and by the pleasing architecture of the round huts, where instead of the usual straw-roofs they employ an artistic kind of basket-work. In the neighbourhood of their settlements one finds as shrines large trees—usually *Adansonia* (Kuka) or fig-trees—or little copses, whose signification is indicated by a ring of stones or an earth wall. They appear to point to a species of ancestor-worship, a fact which Barth adduces as a proof of the relationship of the Margi with the South African tribes.^[428]

To the westwards the Margi adjoin the border peoples of Bornu, all of them pagan tribes within British territory, about whom even to-day we know little more than the modest amount of information furnished us by Nachtigal.^[429] They consist of the Babur, south of Gujba, and the Kerrikerri and Ngizim between Gujba and Katagum.

In the middle of the old Bornu provinces in which the Kanuri language is still predominant, along the middle course of the River Yo (or Wobe), and in the hilly district of Munio, lie the principal seats of the salt industry;^[430] here live the Manga and the Bedde, who are probably related to them. They are bounded on the west and north-west by the territory of the Hausas, who form the predominant element in Zinder.^[431]

The Hausas, a mixed race probably of Hamitic descent, but like the Kanuri with a strong infusion of negro blood, are, thanks to their enterprising spirit and commercial far-sightedness, the most active trading folk of the Central Sudan.^[432] Their travels extend from Tripoli in the north to the Bight of Biafra in the south, and occasionally also reach to the Senegal, or in the guise of pilgrims to Mecca via the Red Sea; they have thus been able to make their language the 'lingua franca' of the Central Sudan, wherever Arabic is not predominant. In all places in Bornu worth mentioning one meets with Hausa settlements of some sort, whose members have generally undertaken the rôle of merchants and money-lenders, and thus have attained prosperity or even wealth.^[433]

Finally, one must consider as border-tribes of Bornu the Tibbu or Tubu, and the Tuareg (Tuarik) or Kindin, both of whom were formerly causes of constant disturbance to the northern provinces. The 'thievish gipsy-like' Tibbu, originally closely related to the Kanuri, as has been stated above, even to-day have settlements along the lower course of the River Yo (Wobe), from whence they occasionally push their by no means purely commercial operations as far as the towns of Central Bornu. Now and then one may see men of this tribe, recognizable by the characteristic 'litham' or cloth veiling the lower part of the face, in the markets of Kukawa or Mongonu.

Much more important and fateful, however, is the rôle which the Berber Tuaregs have played in the provinces north of the Wobe. Considered by Barth as formerly 'an integral part' of the population of Bornu,^[434] they have in later times broken off and since then, while avoiding open hostilities, have continually disturbed the country by their sudden raids, which afforded them booty in the shape of slaves and cattle, and thus for a time completely blocked the roads to the Western Sudan, and even in parts

rendered impossible the cultivation of the land. The state of affairs in these districts was so insecure that the Shehus finally saw themselves compelled to appoint a special Kachella to cope with the Tuareg raids, and even recently these have necessitated the constant interposition of the French garrisons. At all events the Tuareg are by far the most turbulent and dangerous members of the whole population of Bornu. [\[435\]](#)

Besides the above-mentioned tribes, one meets at the present day in Bornu two elements in the population who are scattered in larger or smaller colonies over the country usually unconnected with each other, and though of quite different origin, yet show a certain purely external similarity, and in fact are often found in friendly intercourse with each other; these are the Semitic Arabs and the Hamitic Fulani.

First as regards the Arabs called Shuwas by the Kanuri: they form a not inconsiderable part of the population of Bornu, [\[436\]](#) and are to be carefully distinguished from the very light-coloured Tripoli merchants designated as 'Wassili', who are equally of Arabic origin, but who in most cases only make a temporary sojourn in the larger towns of Bornu, such as Kukawa, Mongonu, and Dikoa. The Shuwas immigrated into Bornu at various times, firstly with the conquerors of the country from the north, and secondly at a much later epoch, about 300 years ago, from the east, no doubt, as Barth assumes, from Nubia or Kordofan. [\[437\]](#) They are broken up into a great number of tribes, of which the Beni Hassan has remained the purest; but they have lost many of the bodily and mental peculiarities of the Semitic race, doubtless as a result of their long wanderings and also of the intermixture which has taken place with other peoples. Their language, however, which indeed through the Koran is a bond of union between most of the North African peoples, they have retained in wonderful purity. [\[438\]](#)

It is interesting to see how the Shuwas, driven by circumstances, from being nomads became a settled people. Since the camel, with which their existence had hitherto been bound up, could not stand the climate of Bornu, they turned their attention to cattle-breeding, and when later these too were annihilated by contagious disorders, the wandering herdsmen were compelled to become settled, and very capable, agriculturists, who curiously enough have established themselves in just those districts which

are the best watered.^[439] It is this totally new mode of life which—as in the case of the Kanuri—has influenced their character in an unfavourable sense. Barth and Nachtigal lay special stress on the disappearance of that chivalrous hospitality once so notable a trait among the Arabs.^[440] The Shuwas modelled their habits of life chiefly on those of the Kanuri. This is well shown in the architecture of their huts, which are similar to the Kanuri huts, only roomier and with somewhat flatter roofs,^[441] and especially in their clothing. Only the mode of dressing the hair among the women, who in spite of an often unmistakable infusion of negro blood have everywhere retained their long and not frizzled hair, differs essentially from that of the Kanuri and reminds one in many respects of that of the wandering Fulani. The style almost invariably met with—at least in German Bornu—which makes the Shuwa woman easily recognizable at a distance, consists of numerous tightly plaited braids hanging down from the temples and often also a thicker raised plait at the back of the head. Hair dressed in this style, which is alleged to have originated from Bagirmi, is kept in place, according to the usual custom in the Sudan, by a plentiful application of butter.^[442] The Arab tribes who have settled down are to be found to-day in the lowlands of Central and Eastern Bornu, but especially on the south bank of Lake Chad.

The Hamitic Fulani are less commonly seen in Bornu than the Shuwas, though in Adamawa, which borders Bornu on the south, they are the dominant race. According to Barth's investigations, they were originally herdsmen, but having penetrated in historical times from the Senegal into the Central Sudan, they finally, at the beginning of the nineteenth century, when they had become the ruling race in Sokoto, came into conflict with the Kanuri. However, in spite of their mental superiority, they were unable to secure any permanent influence in Bornu. They appear to have emigrated from Asia Minor in very ancient times by way of Morocco and Fezzan; many circumstances point to this, such as their physical build and the use by the men of the Phrygian cap.^[443] Perhaps it was they too who, in prehistoric times, brought over the humped ox from Asia. Barth says of them with absolute truth: 'There is not the slightest doubt that the Fulani are the most intelligent of all African tribes';^[444] equally just is Rohlf's dictum

that: ‘At all events they are by far the finest specimens of mankind in Central Africa.’^[445] The Borroro^[446] tribe especially, who are found even in Bornu, between the Shari and the Logone, as half-nomadic herdsmen, and who live in almost complete isolation, have retained all the peculiarities of the race—the spare, sinewy build of the dweller in the desert or the steppe, the delicately formed limbs and the refined features, which remind one of ancient Egyptian art, and which in the case of the women, with their long hair plaited into braids and their big copper earrings, even attain a certain charm. On the other hand, it is just those Fulani who formerly played so important a rôle, that have lost many of the characteristics peculiar to their race as a consequence of intermixture with the native tribes. Fulani of this kind have remained in small colonies along the middle course of the Wobe since the time of their first advance into Bornu; similar to these are the inhabitants of those Fulani settlements which have been pushed forward from Northern Adamawa into the frontier districts of Bornu.

The character of the Fulani shows many superiorities over that of the other Mohammedan tribes of the Sudan. If the doctrines of Islam have from time to time stimulated them to special fanaticism, on the other hand—whatever people may say about the degeneration of the Fulani—it is also owing to these doctrines that they have retained a stricter morality, which is shown not least by their better developed family spirit and the domestic inclinations of the Fulani woman.^[447]

Individual colonies of Kanuri extend far beyond the borders of Bornu, on the one side as far as the Sahara, on the other beyond the right bank of the Shari, and as far as Adamawa. Cf. Barth, iii. p. 293 (German edition).

Cf. Nachtigal, i. 736.

Nachtigal, ii. 417 ff.; cf. also Barth, iii. 77—8. The name ‘Berebere’ or ‘Beruere’ (Berber), by which the Kanuri are designated even at the present day by their southern neighbours, the Fulani, points to an original immigration from the Sahara. [On the subject of the Berbers, Sir H. H. Johnston has the following illuminating remarks in his recently published monograph on *The Opening Up of Africa*, p. 102: ‘The Berbers were very near to the white man of Europe in race, descent, and culture, but they were sufficiently tinged with the blood of Africa to be no longer in community of feeling with Europe. . . . They found the Romans a little too European to their liking: too dominating, too fond of method, order, tidiness, and fatiguing public works.’ It would perhaps be impossible to sum up the African’s objection to the European more succinctly.]

Cf. Barth, iv. 88-9; Nachtigal, ii. 193.

Monteil, p. 313.

Cf. Rohlfs, ii. 8.

Cf. Nachtigal, i. 723; Monteil, p. 355. Even at the present day in German Bornu the mother of the reigning Sultan, Shehu Sanda, holds an important position. [In the old Court of Bornu there were two officials called the 'Magira' and the 'Magiram', the Shehu's official mother and his official sister—not necessarily his actual mother and sister. Their duties appear to have been connected with the Royal Household, but the 'Magira' also acted as an adviser of, and interceder with, the sovereign. There is a mention of the 'Magira' in Harris, *Hausa Stories and Riddles*, p. 33 ff., where she is wrongly described as the Shehu's sister instead of his mother. These ladies are not officially recognized in British Bornu, though at one time it was proposed that they be ranked as second class chiefs and given staves of office.]

Cf. Denham, ii. 2; Barth, ii. 317; Rohlfs, i. 341; Nachtigal, i. 738; ii. 299; Dominik, *loc. cit.*, p. 155.

Cf. Nachtigal, i. 611.

['Masallachi' is the Hausa word, 'mashidi' the Kanuri.]

Cf. also Barth, ii. 543-4.

Rohlfs, i. 344; Nachtigal, i. 625.

Barth, iii. 153.

[For illustration vid. Macleod, p. 20.]

Cf. Rohlfs, ii. 46.

Polko, *loc. cit.*, p. 136. The way of dressing women's hair usual at the present day had already come into fashion in Nachtigal's time. The hair is twisted with the help of butter into countless little plaits, which radiate from the crown and lie flat on the head, being frayed out at the tips. This produces a thick crown of hair, which when seen from in front reminds one of the coiffure of European ladies. The hair is often thickly powdered with pulverized cinnamon, so that it has a chestnut-brown tint. [Cf. illustrations of p. 238 and p. 250 of Macleod, *Chiefs and Cities of Central Africa*; and for Kanembu hair-dressing and hair-ornaments, *idem*, p. 240.]

Cf. Denham, ii. 144 ff.; Rohlfs, ii. 7. Vid. also Passarge, Pl. XX, Fig. 3. In addition to this there is the tattooing mentioned above, which was forcibly introduced by Rabeh.

[There are a few suits of chain-mail in the country, said to date from Crusader times, and to have drifted across the desert. For illustrations of four different patterns of 'ngalio' or throwing-iron vid. Mecklenburg, i. 24, and for coloured illustration of 'liffedi' or quilted armour vid. *idem*, i. 53.]

Nachtigal, i. 610 ff.

Cf. Barth, ii, 308. [Instead of an ostrich egg, an inverted bottle-shaped gourd, or nowadays a glass bottle, is sometimes used, sometimes also an iron spear-head.]

Ornamentation of this sort, such as Foureau describes at Zinder (*Documents scientifiques*, ii. 934, and *D'Alger au Congo*, p. 505), I myself found in the ruined Sultan's palace in the old west town at Kukawa.

Burnt bricks are not employed at the present day. [The ruins of the palace at Gaserregomo (Kasr Koumo) of the old 'Mais' of Bornu show that it was built of bricks of extraordinary hardness. The secret of their manufacture has, however, been lost.]

Cf. Barth, iii. 128-9; Foureau, *D'Alger au Congo*, p. 510.

Nachtigal rightly considers cereals to be the staple food in Bornu, i. 652. [On the subject of agriculture cf. *The Agricultural and Forest Products of British West Africa*, by G. C. Dudgeon, and various special monographs in French published by the firm of A. Challamel, Paris.]

Monteil met with wells in Western Bornu which were 38 metres [i.e. 126 ft.] deep (Monteil, p. 317). [This is nothing unusual; wells in Gubio and elsewhere are 40 fathoms, 'nganji', i.e. 240 ft. deep.]

Cf. also Barth, iii. 116.

Cf. Denham, ii. 96; Nachtigal, i. 635 ff. [I do not think many people would agree to this statement *re* the Kanuri 'love' of animals.—P. A. B.]

This bad practice is usual even within the Sahara, according to Nachtigal, i. 557.

Cf. also Denham, ii. 88 ff.; Barth, ii. 253 (German edition) and iii. 34 (German edition). [For illustration vid. Tilho, *Doc. sc.*, i. 26. Plaited grass fish-traps are also largely used, and are formed into dams right across the rivers.]

For an illustration of one of these, vid. Nachtigal, i. 650.

[Rather from sheep and goat skins. For notes on the method of preparation, vid. *Bulletin of the Imperial Institute*, 1908, 6. 175, and 1910, 8. 402.]

Cf. Barth, iii. 44; Nachtigal, i. 544, 570; Foureau, *Documents scientifiques*, ii. 946, and *D'Alger au Congo*, p. 581. Barth mentions that salt is even obtained by burning cow-dung; vid. Barth, iii. 240 (German edition).

[For a detailed chemical study of the salt and natron of the Chad countries, vid. Tilho, *Doc. sc.*, ii. 553-600. Cf. an article by M. Cortier on 'Les Salines du Sahara soudanais' in *La Géographie* (1912), pp. 231-46.]

Barth, iii. 289; Rohlfs, i. 307. Cf. also Foureau, *D'Alger au Congo*, p. 505. According to Barth, gunpowder also is manufactured in the country, Barth, iii. 127 ff. (German edition). The powder-magazine, which blew up when the French captured Dikoa, must have been filled for the most part with locally manufactured powder.

['Suk' is the Arabic word for market, 'Kasugu' the Kanuri.]

Cf. also Nachtigal, i. 602 and 703 ff.

Denham, ii. 140.

Polko, *loc. cit.* p. 140.

Nachtigal, ii. 336 ff.

Barth, iii. 114.

For illustration vid. Pl. XXIV, Foureau, *Documents scientifiques*, ii. 954; Lenfant, p. 165.

Nachtigal, ii. 341.

Barth, ii. 341, 414, 445; iii. 161, 269 (all in German edition); Nachtigal, ii. 426, 530; Nachtigal considers Barth's designation of 'Masa' for these tribes as rather arbitrary.

Barth, ii. 363. [They are semi-pagan.]

Barth, iii. 304.

Barth, iii. 275 ff.; Nachtigal, ii. 519; cf. plate in Nachtigal, ii. 502; Bauer, *loc. cit.*, p. 101; Lenfant, p. 166.

Barth, iii. 293; Nachtigal, ii. 533.

Barth, iii. 293; Foureau, *D'Alger au Congo*, p. 684; Dominik, *loc. cit.*, p. 173. Plates in Foureau, *Documents scientifiques*, ii. 975, and *D'Alger au Congo*, pp. 685, 697, 715; Dominik, p. 189. [For illustration of Kotoko canoe, vid. Macleod, p. 194.]

They offered a very energetic and successful resistance to the hordes of Rabeh, first of all in their capital-town of Mora, and later in the hills.

[The Buddumas do not appear to have much faith in their Mohammedan teachers. One of the chiefs remarked to M. Landeroin: 'Our "mallams" are liars. They ask us for alms in order to cure us of sickness and to prevent epidemics. . . . If after a man's death his relations demand a refund of the payments made to the "mallam", he replies: "I cannot restore it, for thanks to my prayers the deceased will obtain an excellent place in Paradise, which he would lose if the 'sadaka' was refunded".' Vid. *Doc. sc.*, ii. 339.]

They are treated of in great detail by d'Huart in *La Géographie*, xi (1904), p. 167 ff.; cf. also Nachtigal, ii. 362 ff. and 373. [For notes on the Buddumas, and for Budduma and Kury vocabularies, vid. Benton, *Notes on Some Languages of the Western Sudan*, p. 38 ff. and p. 78 ff.; and for a very full study of the Buddumas, vid. Tilho, *Doc. sc.*, ii. 310-41. Cf. also Budduma and Kury vocabularies by Dr. Decorse in *Actes du XIV^e Congrès International des Orientalistes, Alger, 1905.*]

[Landeroin remarks as follows: 'The principal population of the Chad islands is generally designated under the name of Budduma, a nickname given them by their neighbours, the Kanuri, from 'budu' (= grass) and '-ma' (= belonging to). But the Buddumas' own name for themselves is 'Yedina', the etymology of which is uncertain. It may be derived from Yedi, a town on the south-west bank of Chad, whose original inhabitants, a branch of the Sos, may have taken refuge in the islands from the power of Bornu; or it may be derived from the Kanuri word Gedi, meaning East. . . . Some people say that Yedina was the name of the Kanembu tribe, to which Bulu, the mythical ancestor of the Budduma, belonged.']

Cf. also Rohlf, i. 333 ff.

[The usual Budduma canoe is made from reeds, the 'fole', a large reed with a triangular stalk and a head like guinea-corn, and the 'ngalle', a smaller reed and flat. The canoes have long curved prows like a gondola, only turned backwards. For illustration vid. Tilho, *Royal Geographical Society's Journal* for September, 1910, and Tilho, *Doc. sc.*, ii. 335 and 338; cf. Benton, *Notes on Some Languages*, p. 50. Canoes of the same kind are also made from 'ambach' branches. Cf. Tilho, *Doc. sc.*, i. 29.]

[For illustration vid. Macleod, p. 232.]

The Kuri also trap the crocodile for the sake of its meat, Nachtigal, ii. 374. The same food is also relished by the dwellers on the Shari, Barth, iii. 324.

[They, however, maintained more or less friendly relations with the sovereign of Bornu. The principal Budduma chiefs used to go every two or three years to pay him a visit of courtesy, and brought him presents which were evidence of goodwill rather than tribute, consisting of a few cows and a little butter. In return they received embroidered gowns. Shehu Ashimi offered Kachella

Koremi, chief of the Gurias, authority over all the west bank of Chad, if he would stop the Budduma raids on the Kanuris of the mainland. The Kachella accepted the country lying south of Kauwa and Ngornu, but in the following year Bornu was invaded by Rabeh, vid. Tilho, *Doc. sc.*, ii. 332 and 318.]

Denham, ii. 65; Barth, iii. 405 and 408 (German edition); Rohlfs, i. 291 ff.; Nachtigal, ii. 371, 485, 487, 489.

Barth, iii. 192; Nachtigal, ii. 531; Kund, *loc. cit.*, p. 24. Cf. also Passarge, p. 440 ff. [The 'ngalio'.]

Among the Kamerun Protectorate troops the Musgu are nicknamed the Calabashes.

Denham, i. 361; Barth, iii. 237; Rohlfs, i. 344; Lenfant, p. 145; Bauer, p. 114; Stieber, *loc. cit.*, p. 82. Cf. also Harry Alis, *Nos Africains*, p. 111 ff.; Foureau, *Documents scientifiques*, ii. 992.

Barth, iii. 201; Stieber, *loc. cit.*, p. 85 ff.

According to Stieber the sores are not produced artificially but are caused by the friction of the rider's bare limbs, Stieber, *loc. cit.*, p. 82.

Barth, iii. 198; cf. also iv. 35 and Nachtigal, ii. 584.

Here and there the chiefs wear Mohammedan clothing.

Barth, iii. 174 and 249, and illustrations on p. 208; Lenfant, p. 149; Stieber, *loc. cit.*, p. 82.

Stieber, *loc. cit.*, p. 83 ff.; Map, *Der deutsche Logone*, in *Mitteil. a. d. deutsch. Schutzgeb.*, vol. xviii.

Barth, iii. 211-12 and 229; Stieber, *loc. cit.*, p. 84; Kund, *loc. cit.*, p. 29.

Barth, iii. 190, 240, 274; Stieber, *loc. cit.*, p. 85.

Barth, ii. 383.

Barth, ii. 468 and 646 (German edition). [Cf. Benton, *Notes on Some Languages*, where Barth's Vocabulary is printed.]

The use of poisoned arrows is very widespread. Besides the Fulani and most of the pagan hill-tribes, the travelling Hausas also use them as a means of defence. Nachtigal found them even in Southern Kanem (ii. 260), and Foureau in Zinder (*D'Alger au Congo*, p. 561). It is especially as a protection against them that the cotton-wadded armour is intended. For the terrible effects of the arrow poison cf. Denham, i. 182 ff. [Cf. article on arrow-poison in *Journal of the African Society* for October, 1905, by L. W. La Chard.]

According to Barth these ornaments come from Mandara (ii. 534).

Cf. Barth, ii. 384.

Cf. Barth, ii. 380, 391, 535.

Nachtigal, ii. 431.

Foureau, *Documents scientifiques*, ii. 943 ff.

Foureau, *Documents scientifiques*, ii. 910 ff., and *D'Alger au Congo*, p. 516.

[Passarge calls them 'the African Parsees'.]

[Incorrect. With the exception of the three Government stations of Maiduguri, Geidam, and Gujba, and the Customs post of Mongonu, there are practically no Hausas in Bornu Proper.]

Barth, ii. 272; iv. 4.

Barth, ii. 203, 220, 234; iii. 42 and 48 (German edition); Nachtigal, i. 564, 572; Monteil, p. 298 ff. [The Tuareg raids have now ceased.]

Barth estimated the Shuwa Arabs in Bornu at 200,000-250,000 souls (ii. 356).

Barth, ii. 355.

Barth, ii. 356; Nachtigal, i. 687, ii. 436; Rohlfs, ii. 30. [Cf. [Appendix XIV](#) and [XIV A.](#)]

Nachtigal, i. 686; ii. 439.

Barth, iii. 284; Nachtigal, ii. 439, 511.

[The Shuwas roof their huts with corn-stalks, apparently thrown on haphazard; the Kanuris use grass.]

Cf. Nachtigal, i. 651, ii. 491; Lenfant, p. 153; Dominik, p. 197.

This has since been adopted by the other tribes of the Sudan.

Barth, iv. 143.

Rohlfs, ii. 132.

[Called in Bornu 'Abore Fellata'.] It is a proof of the intelligence of the Borroro that it is owing to this seclusion that they are able to protect their cattle against contagious disorders. Their herds provide the Borroro with everything. The milk, which amongst them is not mixed with cow's urine, is their chief food.

Cf. Barth, ii. 229, iii. 257, iv. 143; Passarge, pp. 69, 167 ff., 424 ff., 509; Harry Alis, *Nos Africains*, p. 311; Bauer, p. 133; Chudeau, *loc. cit.*, p. 331; Dominik, *loc. cit.*, p. 87. [Cf. [Appendix XIV.](#)]

VIII

COMMERCIAL CONDITIONS AND PROSPECTS

BAUER says very appropriately in his book, when treating of commerce, that 'the treasures of the Northern Kameruns consist in fertility and capacity for labour'.^[448] This statement holds good in a still higher degree for German Bornu, the northernmost part of the Northern Kameruns, and for the whole of Bornu in general; for there does not seem to be the slightest prospect of finding useful minerals here, as far as one can judge from geological conditions. All the more urgent, therefore, is the duty of the Colonial Power interested in the commercial opening-up of the country to devote the most assiduous attention to the numerous and comparatively civilized population. In a climate like this, quite unsuited to Europeans, the natives alone can extract the full market value from the natural fertility of the soil, and they are already to a large extent prepared to receive the products of European industry.

How prodigiously full of vitality the population of Bornu is, is proved by the high figure at which it has been able to maintain itself, in spite of the fact that before the beginning of the anti-slavery movement it was yearly decimated over wide districts. How destructive were the military expeditions, which generally started from Kukawa and penetrated into the heart of Adamawa,^[449] expeditions which were only undertaken because 'the treasure-chests and slave-huts were empty and must be filled',^[450] we know well enough from the descriptions of Denham, Barth, Vogel, Rohlf, and Nachtigal. Not only the Shehus, but all their vassal chiefs also, undertook razzias on their own account into the thickly populated districts. Everything that the slave-raiders had no use for—and this was the larger part—was destroyed on the spot; in spite of this, slave caravans of over 4.000 slaves were dispatched from Kukawa, of whom naturally only a very small proportion lived to reach their final goal, Murzuk and Tripoli.^[451] This inhuman, senseless, and aimless traffic, dealing in the most valuable

goods which the Central Sudan possesses, viz. its numerous and industrious population, has now ceased, since the Colonial Powers have paved the way for an orderly state of affairs. But it would not be less wrong, and like throwing the helve after the hatchet, were one now, considering the present state of the Sudan peoples, to forcibly free all so-called slaves, especially as the marriages of free-born women in the Sudan are often nothing more than a purchase on the part of the man and an entry into a kind of slavery on the part of the woman; yet this is the object actually aimed at by a false philanthropy, utterly unacquainted with the conditions prevailing in these regions. One must entirely approve, therefore, when Passarge, at present the best authority on, and the most faithful delineator of, conditions in the Central Sudan, states that: 'The abolition of slavery, even supposing it were generally successful, would mean the ruin of agriculture and the creation of a state of chaos.'^[452] People at home often take offence at the mere word 'slave', and connect it in their thoughts with such a state of things as was described in *Uncle Tom's Cabin*. The entire opposite was rather the case. The 'slave' experiences, once he has found a master, an almost invariably mild treatment on the part of his owner, who is dependent on him in many respects. The 'slaves' often enjoy a comparatively large amount of personal freedom, dwell in their own villages, and live under far better conditions than many free men, whilst in general the difference between free men and slaves is by no means obvious. Generally speaking, the term 'serfdom', as proposed by Passarge, would be much more suitable to this condition than 'slavery'; but even this term is not altogether accurate for the Chad countries, seeing that in Germany, for example, the serfs, not so very long ago, could seldom emerge from their subordinate condition. Very different was the case in the Sudan. Many slaves have risen here to wealth and great influence—such as one or other of the eunuchs of the court of Bornu or this or that kachella, who was originally a slave—and, indeed, as a foreign and therefore invigorating element, have played a not unimportant rôle in politics.^[453]

The last great events which had a definite bearing on the social condition of the population, were the conquest of Bornu by Rabeh and his final overthrow; these occurred only a decade ago. The result of these events was, owing to the facile character of the Kanuri, less a destructive

than a transforming, perhaps even a stimulating one—thanks to the new elements introduced into the country. A few years after these events, the thickly inhabited districts of Bornu—if one excepts Kukawa or Dikoa—nowhere give the impression of being countries which a short time ago were ravaged by the furies of war. Besides, one should take into consideration that the traces of devastation and destruction are soon effaced in these latitudes. What is more important is the fact that very soon after the overthrow of Rabeh, the chief hindrance to the development of prosperity in this part of the Sudan—that is to say, slave-raiding—also ceased to exist.

The districts from which originated practically all the slaves who in former days reached the markets of Mourzuk or Tripoli—that is to say, the rich Musgu country—are still at the present day the best populated part of Bornu, nay, even of the Interior of Africa, and may indeed compare in this respect with the centres of population of many European industrial districts. ^[454] But many other tracts of Bornu, with its fertile lowlands, afford the same delightful prospect. Under this category, for example, falls village after village on the right bank of the Yedseram between Isga and Malematari. In all this district there is no poverty or difficulty in supporting life; everywhere there is prosperity and the unmistakable traces of a comfortable existence. Even the hills on the border are everywhere thickly inhabited, however inhospitable and inaccessible they may appear. ^[455]

Nachtigal attempted to ascertain the population of Bornu on the same basis as Barth's estimate, and, like his predecessor, fixed the numbers at 5,000,000 for the whole of Bornu. ^[456] It is evident, however, that in this estimate the majority of the Musgu were not included in the reckoning, for the traveller's route only allowed him to perceive the small part which at that time had yet been explored. Bearing this fact in mind and, further, taking into consideration the peaceful development of the country during the last ten years, one may certainly add to-day half a million to the figures of Barth and Nachtigal. Of the numbers thus obtained, at least one-third come from the specially thickly populated portion of Bornu that belongs to Germany. ^[457]

To watch over the peaceful development of the country, to keep an eye on anything that affects it, and to suggest the policy to be followed, is the

business of the Residents appointed by the Colonial Powers; they are merely the advisers of the native chiefs, to whom must be left the direct jurisdiction over these territories according to the degree of civilization prevailing in each. It should be the prime duty of the European officials under all circumstances to prevent a return to the former state of affairs, i.e. mutual hostility between the pagans and Mohammedans—which would immediately degenerate again into slave-raiding—to make skilful use of existing divisions, but wherever possible to bridge them over. But this is all the more difficult since the pagan natives, remembering the former slave-raids, regard as an enemy any one of a different colour to themselves. Only a many years' acquaintance with the peculiarities of the country can supply a key to a political system which is full of pitfalls. Though one cannot suppose that the Europeans responsible for the administration will possess the same intimate knowledge of native languages and customs as was displayed by Barth and Nachtigal, nevertheless the mastery of at least one of the locally spoken idioms, whether Arabic, Kanuri, Hausa, or Fulani, is absolutely necessary if their efforts are to be really useful. If anywhere, this holds good of the Central Sudan, where so many threads of North African tribal life are interwoven, and where all conditions and events are under the fixed control of the particularly well-organized sect of the Senussi,^[458] which, though they hardly reach so far, are the moving spirit of the Pan-Islamic movement in North-East Africa, and whose activity was already noticed by Nachtigal.^[459] It is this last circumstance especially that prescribes rules for the other colonizing factor which brings the native into touch with European culture, i.e. for the missionaries, rules which must differ from those which hold good for the rest of Africa. In spite of unmistakable religious tolerance, at least on the part of the Kanuri, the dominant race, it must not be forgotten that Islam, to which the country owes its ancient pre-eminence in the Sudan, has lost no ground to speak of, but on the contrary is engaged in steadily advancing. Only the very greatest patience on the part of Christian missionaries, careful conduct and a systematic advance from an assured base on the coast, can enable them to conquer this country. The slightest precipitancy would not only for ever put in question the results of missionary labour, but would have a most fatal effect on the whole political state of the country. This is especially true to-

day of places where lack of suitable international arrangements has given the European Powers no opportunity of showing the great development of authority lately established in these distant regions.^[460]

The chief industry of Bornu is agriculture, for which the necessary conditions exist to a degree they scarcely do elsewhere in tropical countries. The unstinted praise which Barth assigns to the Southern Musgu country as 'the most fruitful and most richly watered tract in the world',^[461] holds equally true of many other districts of the Sultanate. No district is really bad, for even the 'firki' regions which, owing to the quickness with which they dry up, are to-day almost useless,^[462] could certainly be made productive by the employment of a more highly developed system of husbandry. There are wide tracts in the lowlands of Bornu that present a field for agriculture of unparalleled extent. What an abundance of all the necessities of life are to-day wrung from the soil by means of the primitive hoe, and what might not be produced by the mere introduction of the plough! It cannot be doubted that the intelligent and industrious natives, who exhibit such an exemplary system of horticulture with the most primitive implements, and who have long been aware of the value of artificial manuring, would soon grasp the advantages of the plough.^[463] A suitable draught animal, the humped ox, is everywhere available, and that it is suited to use with the plough has been sufficiently proved by experiments undertaken in the Southern Kameruns.^[464]

Pretty well every agricultural product of the Sudan is produced in Bornu in abundance, and certainly far cheaper than almost anywhere else in Central Africa, a circumstance to which Barth drew especial attention.^[465] Of greater significance is the fact that the plant which at the present day has the greatest importance in world-commerce, i.e. cotton, has for ages been cultivated in Bornu. The good cotton produced in the country, which is manufactured in large quantities by the natives of the Sudan, proves as a matter of fact to be far superior to the cheap cotton goods imported from Europe.^[466] And yet it is all produced by the primitive means of a native loom. Barth repeatedly emphasizes the fact that far from all of the soil of Bornu suited to the production of cotton is thus utilized.^[467] What possibilities, therefore, are disclosed for this branch of agriculture by the

utilization of all cotton soils after the introduction of better quality seed, and, above all, of a more intensive method of cultivation. Bornu, with its regular climate, is in a far more favourable position than many other cotton centres—such, for example, as the Southern States of North America, with their weather conditions which are never to be depended upon—and doubtless has a future before it so far as this product is concerned. [\[468\]](#)

Although the other vegetable products of Bornu cannot compete in importance with cotton, still the cultivation of many valuable agricultural products, which to-day play an important part in the export statistics of other African countries, could likewise experience an important increase. To this category belong especially ground-nuts and sesame.

Stock-raising has always received the same attention in Bornu as agriculture. The principal branch of this is cattle-breeding. One gets an idea of the enormous wealth of Bornu in cattle when one reads the descriptions of the Musgu country. Kund estimates that in these districts, where village joins village, a single place of some 100 farms possesses from 200 to 250 head of cattle. [\[469\]](#) Even if one takes into consideration that all districts are not suited for stock-raising, especially those where large stretches of forest make cattle-keeping difficult, still one may assume that Bornu is one of the richest countries in cattle of the whole of the Dark Continent. Flocks of sheep and goats are in places by no means inconsiderable, but they are generally met with in districts of Bornu less favoured by nature. [\[470\]](#)

From the proximity of the Sahara it is only natural that camel- and horse-breeding should have been undertaken in Bornu from early times. Although the prospects of camel-breeding have proved to be poor owing to reasons already mentioned, horse-breeding still flourishes, encouraged as it is by the possibility of always importing fresh stock from the North to reinvigorate the breed. [\[471\]](#) Nevertheless, it appears that this branch of industry has not attained the development since Denham and Barth's times that one might have expected from the descriptions of these travellers. At a review Barth saw 10,000 horses at one spot—this, indeed, comprised the entire mounted force of the country—but it is doubtful whether at the present day this number could be again assembled, though one must also

take into consideration that the occasion for such an assembly has disappeared for ever.

The existence of such a high development of agriculture, including the important occupations of cotton-growing and stock-raising, is bound to entail among a numerous population many forms of industry. It is not necessary to recapitulate here the industrial productions of the country; they have already been mentioned in the description of the various tribes. This much is certain, that everything which the industries of Bornu have accomplished—not only in the production of textile and leather goods—shows that the task of the European merchant here of creating wants and finding markets for native manufactures has already to a great extent been fulfilled. In this connexion it is to be remembered that Bornu was perhaps the first of the Mohammedan countries of Central Africa to enter into commercial relations with foreign nations, and thus came early into touch with Western products. ^[472]

A further proof of the commercial spirit of the population of Bornu is the comparatively highly developed form of currency which Barth found already in existence. Besides the very common system of barter, purchases for cash were at that time usual everywhere. The oldest medium for reckoning was a fixed weight of copper, the 'Rottl', a term which has remained even to-day as the unit of computation even after the introduction of other and commoner forms of currency. To these forms of currency belong the cotton-strips called 'gabaga', four of which go to one 'Rottl'. At that time and until quite lately the Maria Theresa dollar, ^[473] a medium of reckoning which is very widely distributed in the Mohammedan world, was also current as well as that peculiar substitute for coin, the cowry-shell; in Barth's and Nachtigal's time the exchange was 3,000 to 4,000 cowries to the dollar and 32 to the 'Rottl'. ^[474] Although the rate of exchange was fixed, still important fluctuations took place and always made cash transactions somewhat complicated. ^[475] It is only quite recently that European currency has been officially introduced and made available by the Colonial Powers, but naturally it is only able to supersede very gradually the medium of reckoning hitherto customary.

According to Rohlfs, the reason of the high development of trade in Bornu is the 'absolute freedom of trade and industry' and the freedom of all goods from tolls.^[476] Nachtigal, however, mentions a kind of river-toll on the Logone, and similar arrangements of a more local nature may even still be found.^[477]

How these trade relations—which naturally depend on the interchange of goods by the Sudan peoples among themselves—will shape in the future it is not yet possible to foresee. But this much is certain, that the chief objects of export and the lines which they take will have quite changed within a few years. Both certainly depend on the same causes.

In Barth's time slaves were still the chief export of Bornu,^[478] and even when Rohlfs visited the country things had altered little, as is proved by the figures previously quoted. Nachtigal says, not much later, that the exports of Bornu are confined 'almost entirely to slaves, ostrich feathers, and ivory'. This traveller was the first to notice a distinct falling off in the most important 'article'—a natural result of the restrictions that had begun to be enforced on the slave-trade.^[479] But it is only in quite recent years that the export of slaves across the Great Desert to the shores of the Mediterranean has ceased. Ivory as well as living merchandise has now ceased to be an important article of export, owing to the war of extermination waged for many years against the elephant. Moreover, the quantity of ostrich feathers produced in Bornu was not important enough to influence trade. But it was these three chief Bornu products alone that made remunerative the caravan trade across the Desert, inseparable as this trade was from so much risk and expense.^[480] The efforts of individual Tripoli merchants to maintain trade along the old routes in spite of quite altered conditions—by importing the goods hitherto customary and exporting the excellent Bornu leather—cannot be considered to have much vitality, especially as the condition of the Sahara has become more and more insecure.^[481] The restless robber tribes of the Desert, Tuareg and others, who were formerly dependent on the caravan trade, on which they levied voluntary or involuntary tolls, have pushed southwards, seeking new sources of subsistence, and have thus not only rendered conditions in the northern parts of Bornu increasingly precarious, but have also made trade along the old routes almost entirely

impossible.^[482] This circumstance has contributed materially to assist—partially at any rate—the development of the natural artery of commerce for the Central Sudan, the Niger-Benue route advocated long ago by Barth and Rohlfs.^[483] Thus the apprehensions of the Arab merchants, who have been intriguing for the last sixty years against European competition, have been realized.^[484] As a matter of fact the Niger-Benue route is the only natural one by which the products of the Central Sudan, especially ground-nuts, shea-butter, sesame, and gum-arabic, articles of extremely low intrinsic value, can be conveyed to the coast. Only when it was possible to bring these products to the coast in considerable quantities, and when cotton—and that, too, in large quantities—was added to them, was the existing natural water-way made use of, so far as this was possible. But there is no doubt that the importance of the Niger-Benue route, in consideration of the poverty of natural means of communication between the Central Sudan and the coast, has always been overrated. Any one who has once made the wearisome canoe journey up the Benue, lasting at least four weeks in the dry weather, when progress is often only possible by digging artificial canals, will not estimate too highly the value of this water-way. It is not till July that the water of this river, which rises very rapidly and increases many feet in depth, reaches such a height that large steamers can reach German Adamawa; but by October the water begins to fall with the same rapidity, and steamers which have not got away at the right time have to stick where they are and await the next rainy season before they can return. All merchandise that cannot be shipped on steamers during the rains must either be entrusted to the very costly and insecure canoe transport or be stored to await a more favourable opportunity for shipment. It may be imagined what an unfavourable influence these periodic possibilities of shipment have, for example, on the sale of cotton, an article which is subject to such considerable fluctuations both in supply and demand. Canoe transport would be quite out of the question for the utilization of other agricultural products, especially of course for a trade in cattle. This means of transport is so dear even for the importation of the European goods in vogue in the Sudan, that the Hamburg firm of Pagenstecher, trading in German Bornu, has had to withdraw the ‘factories’ which it had pushed forward to Dikoa and Kusseri, as they could not show a profit.^[485] As regards German Bornu

the water-ways, which for the greater part lie in British territory, have also the disadvantage that though navigation on the rivers is free in accordance with international agreements, still there is an inevitable control by the British of German trade, which under the circumstances is not at all convenient.

But the German colony of the Kameruns is in a fortunate position, for by far the shortest ideal line of connexion between the sea and Bornu, which may be regarded as the focus-point of the Sudan, runs through their territory; this line is the one from the Bight of Biafra to Lake Chad. It is the urgent duty of the German Empire to make use of this fortunate circumstance.

Barth, Rohlfs, and Nachtigal with one accord perceived many years ago the high value of the Chad countries, and any one who has personal knowledge of them will gladly join in Nachtigal's dictum that 'there may be many tropical countries in which, thanks to mighty rivers and the alternation of mountain and valley, Nature may seem more stupendous and more opulent, beauty more sublime, vegetation more luxuriant, and the soil more fruitful, but there is scarcely any country in Central Africa which, if the efforts of mankind corresponded to the powers of the soil, offers a more beneficent prospect of rapid development'.^[486]

Nevertheless, this development cannot proceed at the rate which this rich and fortunate country deserves. The fault lies with the insufficiency of its natural trade routes. Nothing but a railway can change this; it alone could render possible a full utilization of the rich cotton soil and of the other resources of the country. The conditions are very favourable, for the technical difficulties of railway building from the Bight of Biafra to Lake Chad are not too great, and the line—whose first stage from Bonaberi to the Manenguba Hills will be ready in 1910—is nowhere intended to be a trunk line, since everywhere it will pass through districts capable of development. The railway will be the first thing to disclose, as they deserve, the high value of the Chad countries; without it they have lain hitherto like the dead and useless stock on a merchant's shelves.

Bauer, p. 137.

Barth, ii. 417.

Barth, iii. 112 (German edition).

Rohlf's, i. 316; Nachtigal, i. 701.

Passarge, p. 526.

Cf. Barth, ii. 151, iii. 273; Rohlf's, ii. 2; Nachtigal, i. 715, 722; Passarge, p. 527.

Cf. Stieber, *loc. cit.*, p. 84.

As regards the density of population in the hill country, it is worthy of note that Zimmermann estimates the population of the Mandara Highlands at 250,000. Zimmermann, *loc. cit.*, p. 464.

Nachtigal, ii. 441.

This figure indeed differs widely from the official estimate given for the total population for the whole of the Kameruns, viz. 1,300,000. Passarge very rightly doubts the correctness of the official statement, and estimates that there are at least 2,700,000 inhabitants in the Kameruns (*Kamerun im Jahre 1907-1908, Koloniale Rundschau*, 1909, p. 517). But even this figure appears to me much too low. [The official figure for the British Province of Bornu on April 1, 1911, was 674,230.]

[The Senussi have proved a fruitful source of inspiration to alarmist writers, both in England and on the Continent. The influence of the Senussi in Bornu is nil, nor is it likely to spread there. The majority of the Kanuri belong to the Tojani or Tijani sect of Islam. The Tojani and the Senussi are about as likely to co-operate as a Puseyite and a Plymouth Brother, or rather as a Kensit preacher and a Broad Churchman. The following remarks, taken from Professor Margoliouth's monograph on 'Mohammedanism', recently published, bear on the point. On p. 184 he says: 'Senussi's idea was to unite Moslems against European influence, from which he failed to dissociate that of the Turks. In the heart of Africa he proposed to found an Islamic state, whither all Moslems who wished to be quit of these pernicious influences could resort for refuge. . . . His doctrine savoured strongly of Wahhabism in respect of its puritanism, e.g. prohibition of tobacco and music. . . . So far as they have interfered in disputes between tribes and governments, it appears to have been with a view to conciliation.' Again on p. 221 he says: 'The Tijanis of North Africa used to be mentioned as an exceptional case of an order which has favoured the power of France.' Cf. [p. 333](#) of the present work.]

In this connexion M. Landeroin remarks, *Doc. sc.*, ii. 528: 'It is pretty generally supposed that the tribes between the Niger and the Chad have all been Islamized, whereas in reality Islam has not penetrated very far into the heart of the population. Its propagation was favoured by the Arab merchants, and above all, by the Fulani . . . but the attachment of the natives for this religion seems never to have been more than outward, . . . and its practice seems to be decreasing more and more since the occupation of the country by France and England. Since Fulani, Arabs, and Tuaregs have had to yield to the Christians, and it no longer appears politic to flatter the masters of the country by adopting their religious beliefs, the mass of the population is quietly returning to their ancient and simple fetish practices, only retaining certain external appearances of Islam.'

If this means that Mohammedanism in Nigeria is only skin-deep, it is perfectly true. On the other hand, it is equally true that, at any rate, nominal adhesion to Islam is spreading among the pagans. For a detailed account of Islamic sects vid. Depont and Coppolani, *Les Confréries religieuses musulmanes*, Algiers, 1897; but I think they are wrong in showing on their map a Senussi 'zauwia' at Kuka. The Sarikin Mussulmi of Sokoto is, I believe, Kaderiya. In jurisprudence Bornu follows the

Hanafi school, cf. Denham, Appendix V, and foot-note thereto by A. Salame, official Arabic translator to the Foreign Office; and Benton, *Notes, &c.*, p. 200; Sokoto follows the Maliki school.]

Nachtigal, i. 192; ii. 50.

As early as 1903 two American missionaries made a temporary stay in the Central Sudan (British Adamawa), but had to retire owing to total lack of means.

Barth, iii. 236.

[On the contrary, large crops of 'masakwa' or dry weather corn are grown on the 'firki' soil. This soil is generally known as 'black cotton soil', apparently on the 'lucus a non lucendo' principle: it is not black but grey, and cotton is never grown on it.]

[The Catholic Fathers at Tshendam in Muri Province are trying to teach the natives the use of the plough, and some men and oxen have been sent from British Bornu to learn. Men who have had practical experience of farming in the Tropics have grave doubts of the success of ploughing. Land in the Tropics must be frequently allowed to lie fallow, and the labour expended in stumping the land would more than outweigh the labour saved by the use of the plough. Only experience can show whether this opinion is correct.]

Dominik, p. 219.

Barth, ii. 313. Provisions were accordingly one quarter cheaper in Kukawa than in Timbuctoo, one-third cheaper than in Kano, and half the price of those in Katsena and Sokoto. Cf. also Nachtigal, i. 692 ff.

Cf. also Nachtigal, i. 648.

Barth, iii. 282.

[With the present means of transport the export of cotton from Bornu is totally out of the question. Even if railway transport were provided, it is extremely doubtful if the export, at the present price of cotton, would pay.]

Kund, *loc. cit.*, p. 28.

[Experiments are being made in sending cattle and sheep from Bornu to Lagos via the railway at Kano.]

Cf. Rohlfs, i. 343.

Nachtigal mentions among the European products imported into Bornu, via Tripoli and Murzuk, textile fabrics, paper, perfumery, and especially cutlery from England, Solingen, and Steiermark. Nachtigal, i. 697 ff.; cf. also Dominik, p. 163 ff.

It is essential that the coin has the date 1780 to be accepted as currency. Such dollars were specially minted at Trieste until recently for use in the Sudan. [The importation of these dollars into British territory is now forbidden, but they are still imported into French and German country. Their intrinsic value is about 1s. 6d., but their exchange value in Bornu is at present 3s., though of course it fluctuates. The value of any particular dollar depends on its condition, which is decided by whether or not the brooch on the Queen's shoulder is visible. This of course gives the native a much valued opportunity for haggling. The date on them (1780) is always the same, doubtless for convenience of minting, but I have twice seen genuine Austrian dollars in circulation, one was a Maria Theresa coin of different design and date, and the other was a Francis I. The native word for a dollar—'gurus' or 'grush'—is said to be a corruption of 'groschen', vid. Decorse, *Rabah et les Arabes du Chari*, p. 56.]

Carbou, *L'Arabe parlé au Ouaday*, p. 132, states that dollars have been coined by the French Mint for the African trade.]

[Thirty-two cowries to the 'rottl' or 'rattal' still holds good.]

Barth, ii. 310; Nachtigal, ii. 690.

Rohlfs, i. 347. The dollar which the recognized broker obtains for every horse and camel sold by auction does not seem to be regarded by Rohlfs as a tax.

Nachtigal, ii. 541.

Barth, ii. 339.

Nachtigal, i. 700 ff.

Rohlfs, i. 351.

[The following trade figures of produce purchased by the Niger Company's station at Nafada on the Gongola have been kindly supplied by their agent there, Mr. C. W. Allen. The bulk of the produce comes from Bornu, though some comes from Bauchi Province. As will be seen, the figure for ostrich feathers is enormous. The war in Tripoli has diverted the feather trade to Nafada. Few of the feathers come from British Bornu, mostly from French country across the Shari. European ostrich farms are now being started there by private enterprise.

1912.

Ostrich Feathers	15,772½ lb.
Red, undyed skins	52,020 Pcs.
	T
Gum-arabic	56 11 2 2
Gutta-percha	8 1 2 8½
Beeswax	2 1 26
Cow-hides	1,638 Pcs.
Native Silk	1 5

[The Kano railway and the Tripoli War have put the last nail in the coffin of the caravan trade. Arabs are already beginning to leave Tripoli and settle in Kano.]

Barth, iii. 220, and Rohlfs, i. 351.

Barth, ii. 344.

According to a written communication from the head of the firm of Pagenstecher to the writer (July 24, 1909).

Nachtigal, ii. 387; cf. also Barth, iii. 162 ff. (German edition); Rohlfs, i. 351, and Reclus, p. 658.

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[In addition to the above-mentioned there are various books and articles which have appeared since the publication of the German original of this monograph, or which escaped its author's notice. For many titles I am indebted to Joucla's *Bibliographie de l'Afrique Occidentale Française*, and to a *Bibliography of the Sudan* printed by Dr. Karl Kumm, a copy of which was kindly sent me by the author; also to Struck's 'Linguistic Bibliography of Northern Nigeria', in the *African Society's Journal*. Items which I have not been able to examine myself are marked with an asterisk (*).—P. A. B.]

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—— vid. also under Beurmann, and under Schubert.

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MASSARI, A. La traversée de l'Afrique de la mer Rouge au golfe de Guinée, article in *Bull. de la Soc. belge de géogr.*, 1883, p. 845. [Massari was a naval officer and Matteucci a doctor. Their expedition was partly financed by Prince Borghese. They started from Suakin on the Red Sea, passed through Wadai, Kuka (where they remarked on the gaiety and looseness of morals of the inhabitants), Kano, Bida, and Egga, where they stayed at the English factory, and so down the Niger and home. They landed at Liverpool on August 5, 1881, seventeen months after disembarking at Suakin. Dr. Matteucci was attacked by fever in the train between Liverpool and London, and died the following day. I have not been able to discover the date of Massari's death, but as he lectured to the Belgian Geographical Society two years after his return to Europe, he evidently did not die so soon as is implied in note [\[38\]](#) of the present work.]

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*—— *Résultats d'un voyage dans le Soudan*, in-8, 1875. 1 fr. 25.

*—— Nachrichten von Dr. Nachtigal, in *Petermann's Mitteilungen*, ann. 1871, pp. 326, 450, 67, 201; *Zeitschrift der Gesellschaft für Erdkunde zu Berlin*, ann. 1873, p. 61; ann. 1871, p. 130; ann. 1873, pp. 249 and 311;

ann. 1874, p. 39; ann. 1874, x. 2, p. 109; *Globus*, ann. 1873, pp. 215 and 231; ann. 1873, pp. 119, 137, and 153; *Kölnische Zeitung*, July 20 and 28, 1873; *Mitteilungen der Geographischen Gesellschaft zu Hamburg*, ann. 1876, 1877, p. 305; *Tour du monde*, 2^e semestre, 1880; *Geographical Magazine*, ann. 1875, p. 178.

NELSON, REV. THOMAS. *Biographical Memoir of the late Dr. Walter Oudney and Capt. Hugh Clapperton, both of the Royal Navy, and Major Alex. Gordon Laing, all of whom died amid their active and enterprising endeavours to explore the Interior of Africa*. By Rev. Thomas Nelson, M.W.S. [i.e. Member of the Wernerian Natural History Society]. Edinburgh, Waugh & Innes; Whittaker, Treacher & Co., London, 1830. [Nelson was a personal friend of both Oudney and Clapperton, and had letters from both put at his disposal by Dr. James Kay, R.N., Professor Jameson, and Oudney's sister. He also had personal information from Clapperton's sister and other friends.]

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PRINS, P. Vers le Tchad. Une année de résidence auprès de Mohammed Abd-er-Rhaman Gaourang, Sultan de Bagirmi, Avril 1898-Mai 1899, article in *La Géographie, bulletin de la Société de Géographie*, Paris, 1900, March number.

Quarterly Review. Vid. Index vol. xl under Clapperton and Denham. [Vol. xxxiii contains a review of Denham's book; and vol. xxxix of Clapperton's second book, also two letters of Sultan Bello of Sokoto to Clapperton not printed elsewhere. On p. 112 of vol. xxxviii the *Quarterly Review*, commenting on the death of Clapperton and other explorers, remarks: 'We trust there will now be an end to the sacrifice of valuable lives in prosecuting discoveries on this wretched continent, of which we know enough to be satisfied that it contains little at all worthy of being known.']

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*REGELSPERGER. Du Niger au Tchad. La Mission Tilho, ses travaux et ses résultats, article in *Le Mois col. et marit.*, 7^e année, 1909. 1^{er} vol., pp. 97-109.

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—— *Peterm. Mitt.*, 1867, p. 333.

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SEIDEL, A. *Geschichten und Lieder der Afrikaner*. Berlin, 1896, p. 311.

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- TALBOT, P. A. Article on Lake Chad in *Journal of the Royal Geographical Society* for September 1911.
- TILHO. Délimitation franco-anglaise entre Niger et Tchad (Mission Moll), article in *La Géographie*. Paris, 1906, 13. 332-6.
- The French Mission to Lake Chad, article in the *Royal Geographical Society's Journal*, September 1910.
- *Documents scientifiques de la Mission Tilho*, 1910-11. Paris. [Of first-rate interest and value. Published by the French Ministry for the Colonies. Capt. Tilho's colleague, M. Landeroin, is responsible for the historical and ethnographical sections.]
- *TRUFFERT. Région du Tchad, le Bahr el Ghazal et l'archipel Kouri, article in *Revue de Géographie*, June-July 1903.
- TULLY, RICHARD. *Letters from Tripoli*. London, 1819, vid. [Appendix XXI](#).
- *VAN GENNEP, CH. MONTEIL ET G. DEMOMBYNES. *Principes d'enquêtes dans l'Afrique Occidentale. Ethnographie et linguistique*. [In preparation.]
- VOGEL. Nachrichten über Vogels Schicksal von Munzinger, in *Petermann's Mitteilungen*, ann. 1862, p. 346.
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- *WAUTERS, A. J. Les Voies d'accès au Lac Tchad, article in the *Mouvement géographique*, 29. (1912), 481-6.

MAPS

For the construction of the maps the following were used in addition to the authorities mentioned above:

Sketch-map in the *Deutsches Kolonialbl.*, xvii. Jahrg., p. 594.

Map 2 in the *Treaty Series*, No. 14, London, 1906.

Map of Northern Nigeria in *Colonial Reports Annual*, No. 532.

The Surveys of British Africa, London, 1907, pp. 34 ff.

The sheet marked 'Chad' in the map of the Kameruns by Moisel, 1909.

LANGUAGES

Vid. Bernhard Struck, Bibliography of Northern Nigeria, in *African Society's Journal*, 1911-12; and Benton, *Kanuri Readings and Notes on Some Languages of the Central Sudan*.

APPENDIX I

LIST OF LEPIDOPTERA

OBSERVED BY ARNOLD SCHULTZE DURING THE YOLA-CHAD BOUNDARY
COMMISSION OF 1903-4 IN BORNU AND THE ADJACENT COUNTRIES. [\[487\]](#)

FAM. DANAIDIDAE

1. *Danaida chrysippus*, L., et var. *alcippus*, Cram. This species, which is widely distributed all over Africa and Western Asia, is to be found everywhere. Caterpillars are found on all the *Asclepiadae*, especially *Calotropis procera*.

2. *D. limniace*, var. *petiverana*, D. H. In South Bornu (on the 'kopjes') sporadic and rare.

FAM. SATYRIDAE

3. *Melanitis leda*, B. South Bornu. Flights about evening time.

4. *Ypthima simplicia*, Butl. On the southern boundary at Uba.

FAM. NYMPHALIDAE

5. *Acraea neobule*, var. *seis* Feisth. In South Bornu ('kopjes').

6. *A. caecilia*, Fbr. Dile. (South Bornu.)

7. *A. pseudegina*, Westw. (South Bornu.)

8. *A. encedon*, L. On the southern boundary at Uba.

9. *Pyrameis cardui*, L. This cosmopolitan insect is found especially in the Marghi 'bush' at the end of the rains feeding on elephant dung.

10. *Precis orithya*, var. *madagascariensis*, Guen. Very sporadic.

11. *P. oenone*, var. *cebrene*, Trim. As above.
12. *P. octavia*, Cram. et f. *amestris*, Dr. In South Bornu (on the granite hills).
13. *P. antilope*, Feisth. with f. *simia*, Wallengr. As above, but somewhat commoner.
14. *P. chorimene*, Guen. As above.
15. *Catacroptera cloanthe*, var. *ligata*, Rotsch. & Jord. South Bornu. On the plains and on the granite hills.
16. *Hypolimnas misippus*, L. Sporadic in South Bornu. Granite hills.
17. *H. dubius*, Pal. Rare, on southern boundary at Uba.
18. *Byblia acheloia*, Wallengr., with f. *Crameri*, Auriv. Very sporadic.
19. *Neptis agata*, Stoll. Boundary of Southern Bornu.
20. *Hamanumida daedalus*, Fabr. Everywhere.
21. *Charaxes epijasius*, Reiche. South Bornu sporadic. Caterpillars on *Burkea africana*.
22. *Ch. achaemenes*, Fields. South Bornu ('kopjes').
23. *Ch. etheocles*, var. *viola*, Butl. Everywhere, especially on the tamarind trees. Caterpillars on various Leguminosae.
24. *Ch. candiope*, God. Rare, on southern boundary at Uba.
25. *Ch. varanes*, Cram. Everywhere that the *cardiospermum halicacabum*, on which the caterpillars feed, grows, but always sporadic.

FAM. LYCAENIDAE

26. *Dendorix caerulea*, H. Druce, at Dile, South Bornu.
27. *D. livia*, Shy. Everywhere that there are acacias in bloom.
28. *D. antalus*, Hopffer. Idem.

29. *Jolaus menas*, H. Druce. On the southern boundary at Uba. Like No. 30, especially on the *Vitex* when in bloom.

30. *J. ismenias*, Shy. As above.

31. *J. bicaudatus*, Auriv. At Dile (South Bornu).

32. *J. umbrosus*, Butl. Found hovering round scattered trees on 'firki' soil. (The caterpillars of all the *Jolaus* species are fond of the *Loranthus*.)

33. *Spindasis mozambica*, Bertoloni. Boundary of South Bornu.

34. *Cupido plinius*, Fabr. Everywhere.

35. *C. baeticus*, L. This widely-distributed species is found hovering round the 'ambach' of the zone of marshy vegetation of Chad.

36. *C. eleusis*, Demaison. South Bornu.

37. *C. cissus*, God. Everywhere.

38. *C. lysimon*, Hubn. Idem.

FAM. PIERIDAE

39. *Herpaenia eriphia*, var. *lacteipennis*, Butl. South Bornu.

40. *Pieris gidica*, God. Everywhere; especially in the flooded districts of Chad.

41. *P. creona*, Cram. Idem. Caterpillars on the *Capparis*.

42. *P. mesentina*. As above.

43. *Teracolus amatus*, var. *Calais*, Cram. In Central Bornu, especially at Dikoa on the *Capparis* bushes, on which the caterpillars feed, as do those of all the *Teracolus* species.

44. *T. chrysonome*, Shy. As above. Very common.

45. *T. vesta*, var. *amelia*, Lucas. South Bornu.

46. *T. protomedia*, Shy. Everywhere, sporadic.

47. *T. eris*, Shy. As above.

48. *T. jone*, var. *phlegas*, Butl. Everywhere, and at all times of the year.
49. *T. eupompe*, Shy, with f. *dedecora*, Feld. As above.
50. *T. evippe*, L. and f. *ocale*, Boisd. Habitat as above, but rarer.
51. *T. antigone*, Boid., with f. *phlegetonia*, Boisd. As above.
52. *T. euarne*, Shy, with f. *citreus*, Butl. As above.
53. *Eronia cleodora*, Hübn, var. *erxia*, Hübn. Everywhere sporadic, especially on the cotton plant.
54. *Catopsilia florella*, Fabr. Everywhere. Caterpillars on the Cassia.
55. *Terias brigitta*, Cram., f. *Zoe*, Hopff. Everywhere, especially in the marshy zone of Chad, where the caterpillars are fond of the 'ambach'.

FAM. PAPILIONIDAE

56. *Papilio Schultzei*, Auriv. Only on the boundary at Uba, and only at one small spot near inaccessible rocks.
57. *P. pylades*, Fabr. Everywhere on the damp sand-banks of rivers. Caterpillars on *Anona senegalensis*.
58. *P. leonidas*, Fabr. Boundary of Southern Bornu.

FAM. HESPERIDAE

59. *Hesperia Zaire*, Pl. South bank of Chad.
60. *Chapra mathias*, Fabr. Marshy zone of Chad. Caterpillars on Graminae.
61. *Cyclopides formosus*, Butl., var. *tsadicus*, Auriv. Dile. South Bornu.
62. *Tagiades flesus*, Fabr. As above.

FAM. ARCTIIDAE

63. *Utetheisa pulchella*, L. Everywhere. Common in places.

FAM. LYMANTRIIDAE

64. *Laelia euproctina*, Auriv. South bank of Chad.

FAM. SPHINGIDAE

65. *Hersi convolvuli*, L. Caterpillars common on the *Ipomoea* on the banks of Chad.

66. *Acherontia atropos*, L. Sporadic. Caterpillars on the *Vitex*.

67. *Daphnis nerii*, L. In damp places. Caterpillars on the *Sarcocephalus* species.

68. *Macroglossum trochilus*, Hübn. South Bornu.

(*Hippotion celerio*, L., and *osiris*, Dalm., are both met with in the neighbouring country of Adamawa, the former in great numbers.)

FAM. SATURNIIDAE

69. *Epiphora bauhiniae*, Guér. Everywhere that the *Zizyphus* grows, on which the caterpillars feed. (Silk-spinners.)

70. *E. Schultzei*, Auriv. As above, but more on Lake Chad.

71. *Gonimbrasia osiris*, Druce. South Bornu. Caterpillars swarm on the *Terminalia* trees, and are eaten by the pagans.

72. *Bunaea hersilia*, Westw. In South Bornu (Marghi 'bush'), caterpillars common on the elephant grass.

73. *B. licharbas*, Maas. Dile. South Bornu. Caterpillars on vetches.

FAM. LASIOCAMPIDAE

74. *Taragama diluta*, Auriv. To be met with resting on the 'retam' or broom on the banks of Chad.

FAM. SESIIDAE

75. *Sesia spec.* Marshy zone of Chad.

FAM. PLUSIIDAE

76. *Plusia chalcytes*, Esp. Boundary of Southern Bornu.

FAM. HELIOTHIDAE

77. *Heliothis armiger*, Hl. South Bornu. Caterpillars on *Sesamum indicum*.

78. *Xanthodes Graëllsii*, Feisth. Everywhere.

FAM. OPHIUSIDAE

79. *Leucanitis stolidus*, F. Everywhere in damp meadows.

80. *Grammodes bifasciata*, Petag. As above. Common in places.

81. *G. algira*, L. As above.

(The five last-named species are the commonest *Heterocerae*.)

82. *Pseudophia tirrhaea*, Cr. In Southern Bornu.

Cf. p. 128 ff. of article by Aurivillius in *Archiv für Zoologi*, vol. ii, No. 12, Stockholm, 1905. [For plates of butterflies vid. Karl Kumm, *From Hausaland to Egypt*.]

APPENDIX II

LIST OF THE BORNU KINGS

BELOW will be found a list of the kings of Bornu given to me by one Mallam Kashim. This list was shown to the late Major McClintock, who ordered the mallam to keep it until I returned from leave.

It differs considerably from Mr. Vischer's list^[488]—printed opposite it—and it also differs from the list compiled by M. Landeroin and printed in the *Documents scientifiques de la Mission Tilho*, vol. ii, p. 348. Mallam Kashim asserts that the list he gave me was copied from an old manuscript, but I do not think it has any special authenticity or correctness.^[489] It is interesting, however, for purposes of comparison. Mr. Vischer's list was compiled when he was a political officer in Bornu in 1906, and is printed with his kind permission, and by that of the government of Northern Nigeria. It is printed in the *Notes on certain Emirates and Tribes*, edited by Major Burdon and published by the N.N. Government, where also appears a table of relationships of the Shehus, compiled by Mr. Vischer. Barth's, Nachtigal's, and Landeroin's lists appear in parallel columns in vol. ii, p. 348 of *Documents scientifiques de la Mission Tilho*, and Mr. H. R. Palmer's list in the *Journal of the African Society* for October, 1912.

CHRONOLOGICAL TABLE OF THE RULERS OF BORNU

FROM SAEF BEN DHU YASAN, FOURTH CENTURY A.D. TO SHEICH ABUBAKR GERBAI
BEN IBRAHIM BEN OMAR, PRESENT DAY.

Source of information: Barth's and Nachtigal's tables of Bornu rulers; personal information. Compiled by Hanns Vischer, Geidam, December 12, 1907.

1. Saef Ben Dhu Yasan	reigned	20 years
2. Ibrahim Ben Saef	„	16 „
3. Duku Ben Ibrahim	„	250 „
4. Fune Ben Duku	„	60 „
5. Aritso Ben Fune	„	50 „
6. Katori Ben Aritso	„	250 „
7. Adyoma Ben Katori	„	20 „
8. Bulu Ben Ayoma	„	16 „
9. Arki Ben Bulu	„	44 „
10. Shu Ben Arki	„	4 „
11. Abd El Djelil Ben Shu	„	4 „
12. Hume Ben Abd El Djelil	„	from 1086-1097
13. Dunama Ben Hume	„	„ 1098-1150
14. Biri Ben Dunama	„	„ 1151-1176
15. Abd Allah Ben Bikoru	„	„ 1177-1193
16. Abd el Djelil Ben Bikoru	„	„ 1194-1220
17. Dunama Dibalami	„	„ 1221-1259
18. Abd El Kedim Ben Dunama	„	„ 1259-1288
19. Biri Ben Dunama	„	„ 1288-1306
20. Nikale Ben Biri	„	„ 1307-1326
21. Abd Allah Ben Kade	„	„ 1326-1345
22. Selma Ben Abd Allah	„	„ 1346-1349
23. Kama Gama Ben Abd Allah		1350

23. Kure Galla Ben Abd Allah	„	„	1350
24. Kure Kura Ben Abd Allah	„	„	1351
25. Mohammed Ben Abd Allah	„	„	1352
26. Edris Ben Nikale	„	„	1353-1376
27. Daud Ben Nikale	„	„	1377-1386
28. Otman Ben Daud	„	„	1387-1390
29. Otman Ben Edris	„	„	1391-1392
30. Abubakr Liyatu Ben Daud	„	„	1392
31. Omar Ben Edris	„	„	1394-1398
32. Said	„	„	1399-1400
33. Kade Afno Ben Edris	„	„	1400-1432
35. Otman Kalnama Ben Daud	„	„	1432
36. Dunama ben Omar	„	„	1433-1434
37. Abd Allah Ben Omar	„	„	1435-1442
38. Ibrahim Ben Otman	„	„	1442-1450
39. Kade Ben Otman	„	„	1450-1451
40. Dunama Ben Biri	„	„	1451-1455
41. Mohammed	„	„	1455
42. Amer	„	„	1456
43. Mohammed Ben Kade	„	„	?
44. Rhadji	„	„	1456-1461
45. Otman Ben Kade	„	„	1461-1466
46. Omar Ben Abd Allah	„	„	1467-1471
47. Mohammed Ben Mohammed	„	„	1472-1504
49. Edris Ben Ali	„	„	1504-1526
50. Moh Ben Edris	„	„	1526-1545
51. Ali Ben Edris	„	„	1545
52. Dunama Ben Mohammed	„	„	1546-1563
53. Abd Allah Ben Dunama	„	„	1564-1570
54. Edris Ben Ali Aloma	„	„	1571-1603
55. Mohammed Ben Edris	„	„	1603-1618

56. Ibrahim Ben Edris	„	„	1618-1625
57. Hadj Omar Ben Edris	„	„	1625-1645
58. Ali Ben El Hadj Omar	„	„	1645-1685
59. Edris Ben Ali	„	„	1685-1704
60. Dunama Ben Ali	„	„	1704-1722
61. Hadj Hamdun Ben Dunama	„	„	1723-1736
62. Mohammed Ben El Hadj Hamdun	„	„	1737-1751
63. Dunama Gana Ben Mohammed	„	„	1752-1755
64. Ali Ben Hadj Dunama	„	„	1755-1793
65. Ahmed Ben Ali	„	„	1793-1810
66. Dunama Ben Ahmed	„	„	1810-1817
67. Ibrahim Ben Ahmed	„	„	1818-1846
68. Sheich Omar Ben El Kanemi	„	„	1846-1880
69. Abdurrahman Ben El Kanemi	„	„	1853-1854
70. Sheich Bukr Ben Omar	„	„	1880-1884
71. Sheich Ibrahim Ben Omar	„	„	1884-1885
72. Sheich Hashem Ben Omar	„	„	1885-1893
73. Sheich Kiari Ben Bukr	„	„	1893
74. Sheich Sanda Limanambe Ben Bukr	„	„	1893
75. Rhabe the Usurper	„	„	1893-1900
76. Sheich Sanda Kwori Ben Ibrahim	„	„	1900
77. Sheich Abubakr Gerbai Ben Ibrahim Ben Omar Ben El Kanemi			

LIST OF BORNU KINGS

As given by Mallam Kashim (Maiduguri, October 12, 1912). *N.B.*—V.
= Vischer.

1. Sebu Aisami Yamanma	V. 1
2. Ibrahim Sebumi	V. 2
3. Mai Duku Brem	V. 3
4. Mai Fune Dukumi	V. 4
5. Mai Archo Funemi	V. 5
6. Mai Kaduri Archomi	V. 6
7. Mai Boiyoma Kadurimi	
8. Mai Boiyo Bulumi	
9. Mai Arigi Bulumi	V. 9
10. Mai Jil Shomi	V. 11
11. Mai Jil Rigimi	
12. Mai Ume Jilumi	V. 12
13. Mai Dunama Umemi	V. 13
14. Mai Daudu Nigalemi	
15. Mai Dalla Bikorumi	V. 15
16. Mai Tselim Bikorumi	
17. Mai Dalla Dunamami	
18. Mai Kadde Aujami	
19. Mai Dalla Aujami	
20. Mai Arri Gaji Zainami	
21. Mai Kadde Madallami	
22. Mai Usuman Kaddemi	
23. Mai Momadi Kaddemi	

24. Mai Iderisa Ashami
25. Mai Deril Arilwa
26. Mai Ume Aisa Gana
27. Mai Biri Aminami
28. Mai Kore Afuno Dalla
29. Mai Kore Goguwa
30. Mai Kore Gawua
31. Mai Kore Kura Kesa Ngilleru
32. Mai Jil Kellemi
33. Mai Momadu Baranbatama
34. Mai Momadi Jumarama Umar Iderisami
35. Mai Dagumoma Dalla Umarmi
36. Mai Kagu Umarmi
37. Mai Momadu Godimi
38. Mai Momadu Palagema
39. Mai Momadu Maza Iderisimi
40. Mai Daudu Niyalemi V. 27
41. Mai Usumanu Daudumi V. 28
42. Mai Sebu Gana
43. Mai Yusufu Sebumi
44. Mai Tselim Auwami
45. Mai Daudu Nigalemi
46. Mai Tolomaramma Kashim Biri Dunamami
47. Mai Dalla Bikurumi Birimi
48. Mai Dunama Arrimi
49. Mai Momodu Dunamami
50. Mai Duna Aminami
51. Mai Panami Umar
52. Mai Arri
53. Mai Dalla Dunamami
54. Mai Arri Gaii

55. Mai Aisa Keri Ngumaramma (F) (regent for nephew No. 56)
56. Mai Iderisi Arrimi Aloma (buried in Lake Alo) V. 54
57. Mai Momadi Panami Baranbatama V. 55
58. Mai Brem Gumsumi V. 56
59. Mai Umar Pusami Dusumami V. 57
60. Mai Arri Umarmi V. 58
61. Mai Tolomaramma Kashim Biri Umarmi (ancestor of Mallam Kashim, compiler of this list)
62. Mai Kime Mele Arrimi
63. Mai Dunama V. 60
64. Mai Aji Dunamami V. 61
65. Mai Momodu Ajimi V. 62
66. Mai Dunama V. 63
67. Mai Arri Dunamami V. 64
68. Mai Arri Panami
69. Mai Amadu Arrimi V. 65
70. Mai Dunama Lefiami (ancestor of Maina Gumsumi, present representative of old dynasty and now Ajia of Bussugua) V. 66
71. Mai Momadi Ngilleruma Gumsumi or Ngueleroma
72. Mai Ibram V. 67
73. Mai Arri Dalatumi (killed at Minarge by Shehu Umar)
74. Shehu Lamino
75. Shehu Umar
76. Shehu Abdurrahman
77. Shehu Umar
78. Shehu Bukar
79. Shehu Ibrahim
80. Shehu Ashimi
81. Shehu Kiari

82. Shehu Sanda Limannambe

83. Rabeh

84. Shehu Sanda Kura

85. Shehu Garbai

Though sometimes only in giving a man's matronymic, instead of his patronymic, e.g. No. 58 in Mallam Kashim's list is described as 'the son of Gumsu' and corresponds to No. 56 in Mr. Vischer's list 'the son of Edris'.

E.g. he introduces in No. 61 his own ancestor, who does not appear in any other list. On the other hand he includes No. 71, Nguéléroma, omitted in all other lists, except M. Landeroin's, for reasons mentioned in *Documents scientifiques de la Mission Tilho*, vol. ii, p. 346; cf. [Appendix IV](#) of the present work.

APPENDIX III

THE TUBBAS (OR SEFS) AND THE SOS

ACCORDING to the *Documents scientifiques de la Mission Tilho*, ii. 352, the dynasty of the Sefs are more generally known under the name of Tubbas. 'A young chief whose name has not been preserved but who was later known as Tubba Lowel (the first who repented) left Hindi (India) at the head of an army and went to the country of Cham (Syria), whose inhabitants paid him tribute in order to avoid war. He then directed his march to Bougdari or Goudeber (Baghdad?) then to Masr (Egypt) and Medina, exacting tribute everywhere. From Medina he went to Yemen, where he established himself without resistance. From there he went to besiege Mecca, which had refused to submit to him. He was accompanied by 4,000 mallams (priests).' To this legend M. l'officier interprète principal, Hamet Ismael, appends a note that this legend appears to be a somewhat garbled account of the history of the Tobbas or Shepherd Kings of Egypt (Hyksos): cf. *L'Arabie*, par Noël des Vergers, p. 50 ff. 'One day the king reproached the mallams that their assistance did not enable him to capture Mecca. The principal mallam replied that he ought to be content with the empire of Persia, Bougdari, Egypt, Medina, and Yemen, and should not attack the sacred city of the Kaaba, which was destined one day to give birth to a prophet. The king then repented, abandoned the siege, and declared that on the appearance of this prophet he would become his first disciple. This repentance caused him to be called Tubba Lowel (the first who repented), from the Arabic verb "itoub" = he repents. Tubba Lowel returned to Yemen, and forty years later, when Mohammed was preaching Islam, he embraced the new religion.

After the reign of Hussein, the sixth Caliph, the Mussulman Empire was divided. One of the descendants of Tubba Lowel had the north and afterwards Stambul (Constantinople); another, named Seibu Aïsami, crossed the Nile and travelled towards the Sudan. This Seibu is perhaps Barth's Sef. Later the descendants of Seibu came to Kanem, then occupied

by the Bulala, and drove back the latter to Lake Fittri. Whether the Tubbas came in from the north via Fezzan or from the east via Kordofan is not known. Barth says they came from Borku, which is north-east of Kanem, and that they were descended from the Libyan tribe of the Berdoa, who, according to Nachtigal, occupied the oases of Kufra and Tou. Having thus conquered Kanem, the Tubbas or Seibuas established their capital at Birni Njime, thirty miles east of Mao, north-east of Chad. Towards the end of the fifteenth century the Tubbas left Kanem in order to instal themselves in Bornu. According to Barth and Nachtigal the Tubbas were driven out of Kanem by the Bulala; according to our informants they left there in order to conquer Bornu, a more fertile country than Kanem.'

There is also another version of the installation of the Tubbas in Bornu (*Doc. sci.*, ii. 356). After having abandoned Kanem, about four hundred and fifty years ago, they came to Bornu, then occupied by the Sos, whose chief Guma Kandira or Dala Gumami [cf. Benton, *Kanuri Readings*, p. 27] ('Kandira' means in Kanuri 'hunter, archer') was living in straw huts at Gambaru on the River Yo. Having obtained leave to sit down near him, the Tubbas bought land in order to build their houses there. Amongst this land was a garden planted with 'kumo', which means in Kanuri 'the calabash-gourd'. When their town or 'birni' was built, it was called Birni Zerku Kumo, i.e. the town of the calabash-garden, 'zerku' meaning in Kanuri 'a garden'. [Probably the word spelt 'zerkou' by Landeroin should be 'zerge' = a dry watercourse, a natural place for a market-garden.—P. A. B.] By inversion of syllables 'zerku' became 'kuzer', hence Kuzer Kumo. Another derivation is suggested by Landeroin. Nachtigal speaks of a group of three villages to the north of Mao, whose chief bore the title of 'Kumo'. Perhaps the Tubba chiefs, who bore in Bornu the title of 'Mai' (= 'king' in Kanuri), may have formerly borne that of Kumo, hence Kasr Kumo = the palace of the Kumo. ['Kasr' is an Arabic word for a castle or citadel, sometimes used in Hausa, vid. Robinson's *Hausa Dictionary*, under the form 'kasru' to mean 'a palace'.]^[490] The Tubbas constructed their first capital at Gambaru, but finding that it was apt to be flooded by the river during the rains and was unsuited for the operations of their cavalry, they removed a few miles south to Kasr Kumo. Gambaru became the residence of the Queen Mother and her court.

In the course of time the So king got into debt to the Tubbas and finally fled to escape his creditors. The following legend is told to explain how the Tubbas got rid of the remaining Sos. The Tubbas were accustomed to stain their hands and nails with a tincture of henna. The Sos had tried to imitate them, but they were ignorant both of henna and how to use it. The Tubbas had jealously guarded their secret, but at last they agreed to reveal it on condition that all the Sos in the country were present. Accordingly on the appointed day they killed an ox and made the credulous Sos dip their hands in its blood. Their hands were then enveloped in green leaves, as the Tubbas were accustomed to do themselves, and the leaves bound round with thongs of raw hide. These thongs were not to be removed for three days. At the end of this period the thongs had shrunk so much that it was impossible to remove them without cutting them. The Tubbas then fell upon the unfortunate Sos, only one of whom escaped. This happened in the year 1452 of our era.

The Sos mentioned above were, according to the natives, the autochthonous inhabitants of Bornu. They lived chiefly by hunting and professed no religion. History has not preserved the names of their chiefs, but legend paints them as of gigantic stature and of such strength that when a So killed an elephant, he could carry it home on his shoulder [cf. Benton, *Kanuri Readings*, p. 9]; their bracelets would have been too big for our thighs and their axe-heads were as big as a man's body of the present day. But nothing in the country recalls this extraordinary strength,—no weapon, utensil or dwelling which can justify their reputation. [Since these words were written by M. Landeroin some implements have been discovered bearing on this point. During the excavations for the wall of the Maiduguri leper-camp at the end of 1911 three implements were turned up, which the Shehu at once pronounced to be So arrow and spear-heads or hoes. They are shaped like a very blunt arrow-head, and the biggest measures 14 in. by 8 in. at its greatest length and breadth, and weighs 4 lb. 12 oz. They are perhaps So hoes, and are certainly infinitely heavier than those used at the present day. They have been deposited in the Zungeru museum. No other remains were found.—P. A. B.] Another argument against the Sos being giants is that their descendants the Beddes, Ngizims, Kotokos, &c., are no taller and no stronger than individuals of other races inhabiting the country.

[But according to Miss Macleod, *Chiefs and Cities of Central Africa*, p. 196, few Kotokos are under 6 ft., and some are as tall as 6 ft. 7 in.]

Whichever derivation is accepted, it is perhaps not too fanciful to quote as parallels the Biblical phrase ‘a lodge in a garden of cucumbers’, or again the first words of the famous passage describing the accession of Queen Victoria—‘In a palace in a garden . . .’ If this sounds too poetical, the alliterative translation ‘Castle Calabash’ is easily remembered.

APPENDIX IV

SHEHU LAMINO AND HIS SUCCESSORS

THE following is a translation of pp. 358-76 of vol. ii of the *Documents scientifiques de la Mission Tilho*, published in 1911, which modify considerably Barth's estimate of Shehu Lamino and his successors. The information was obtained orally by M. Landeroin, 'officier interprète' of the French Boundary Commission. He remarks on p. 346:

'It is even probable that Barth, as well as Nachtigal, could not get impartial information regarding the period of the rise of Mohammed El Amin (more often called Sheikh Lamino), founder of the Kanemi dynasty. One may guess in fact that, owing to fear of the Sheikh, no one would have dared to say, even in a whisper, what they thought of their sovereign, for informers were held in much honour at the court of Bornu, and the gossips would soon have been denounced and executed. Barth and Nachtigal were generally escorted and guided by the spies of Shehu Umar, and they could hardly become intimate with anybody but his favourites and friends. All the information furnished to them was faithfully reported to the Shehu, who would not have failed to deal severely with those who dared to criticize his own acts or those of his father. This is doubtless why Barth was able to write, all in good faith, that Sheikh Mohammed El Amin was "a pious and God-fearing man", whereas, on the contrary he was greedy and ambitious.

To-day the Kanemi dynasty exercise no more than the shadow of power; any one, therefore, can tell fearlessly what he knows and express an opinion on those who have held the reins of power. This is why the unpublished information which we have collected will show the reigns of the last Sultans of Bornu under a new light.'

The justice of M. Landeroin's observations can hardly be disputed, and a knowledge of the present representatives of the Kanemi dynasty inclines one to accept his version of events rather than Barth's. Nguéléroma does not appear in Barth's and Nachtigal's lists, or in that compiled by Mr. Vischer in

1906. The Bornu Provincial Office copy of the latter has a note inserted in pencil by M. Landeroin himself regarding Nguéléroma.

Translated from *Documents scientifiques de la Mission Tilho* (1911), vol. ii, pp. 358-76. [\[491\]](#)

STRUGGLE AGAINST THE FULANI

Mai Ali was succeeded by his son Ahmed (Mai Amadu). During the latter's reign, Othman dan Fodio, who had just founded a Fulani empire in the Hausa states, sent one of his lieutenants, Goni Mukhtar, to attack Bornu. Mai Amadu, having become blind, had handed over the reins of power to his son, Mai Dunama. The latter was unable to repulse Goni Mukhtar, who captured Gazerregomo (Kazer Koumo) and installed himself there. Dunama fled to Maguié, one day's march to the east of Gazerregomo, where he met Sheikh Lamino (Mohammed El Lamin El Kanemi), a native of Kanem. Mai Dunama gave him handsome presents and said to him: 'You are a learned and pious "mallam", pray to God for me that He may restore me my kingdom.' This Sheikh Lamino was the son of Sheikh Langa of the Kanembu tribe and of an Arab woman from Zouïla in the Tripolitaine. After studying, like his father, with the Koiyams of Gazerregomo, he accompanied his father on a pilgrimage to Mecca. His father died at Medina, and Sheikh Lamino returned to Kanem. In answer to Mai Dunama's request, Lamino wrote three letters to Goni Mukhtar pointing out to him that he was waging an unjust war against Mussulmans, and suggesting that he should abandon Gazerregomo. At the same time he begged the Sultan of Bornu's vassals, among others the chiefs of Nguru, Wacha, and Miria, to come and join him. No reply having been received to the three letters, Sheikh Lamino said to Mai Dunama: 'Return to your capital, God will restore it to you.' Full of confidence, the Tubba Sultan marched against Goni Mukhtar and was victorious; the Fulani chief was killed in the battle inside the town itself, which he had occupied for forty days.

Mai Dunama reinstalled himself at Gazerregomo, but a year later was attacked by the Fulani chief Mallam Ali, surnamed Zaki (the lion), and

again routed. Putting his faith in the invocations of Sheikh Lamino, Dunama hastened to recall him from Kanem, gave him 100 camels, 100 oxen, 100 slaves, 100 horses, 1,000 dollars, 1,000 goats, 1,000 sheep, and 1,000 gowns, and begged him to pray to God anew that his empire might be a second time restored to him.

Sheikh Lamino and his disciples prayed continuously for seven days, at the end of which Dunama marched against his enemies. Mallam Zaki, vanquished in his turn, had to fly and took refuge at Katagum, after having been master of Gazerregomo for six months. These events happened about 1808 or 1809.

At the advice of Sheikh Lamino, Mai Dunama abandoned his capital, polluted by the Fulani, and founded Birnin Kabela, five days' march to the south-east.^[492] To Sheikh Lamino he said: 'Your prayers are always efficacious; each time that you have invoked God in my favour, He has heard you; I wish you therefore to remain near me in case the Fulani come again to attack me.'

The Sheikh agreed, but only on condition of being allowed to settle amongst the Kanembus who lived near Ngornu, eight or nine miles to the east of Birnin Kabela. The Sultan accordingly gave him authority over all the Kanembus of the district.

INTRIGUES OF SHEIKH LAMINO

The Sheikh was intriguing and ambitious. Some time afterwards, in accord with the courtiers, he had Mai Dunama deposed and imprisoned at Dikoa, according to some, at Ngornu, according to others. He was replaced by his paternal uncle, Mohammed Nguéléroma, younger brother of the preceding Sultan, Mai Amadu the Blind.

Mohammed Nguéléroma reigned for three or four years, but he soon fell out with Sheikh Lamino, who did not find him sufficiently generous towards himself. New intrigues were then started to restore Dunama, still a prisoner, to power. The Sheikh, having gained over by his presents a certain number of Nguéléroma's courtiers, had the ex-Sultan brought before him by them, and asked him: 'What would you give me if I had the kingdom

restored to you?’ Dunama replied: ‘If you have the power restored to me, you shall have half the revenues of the kingdom.’ Satisfied with this answer, Sheikh Lamino kept Dunama in his house, and the next day asked Nguéléroma to send him the principal officers of his court.

The Sultan sent them to him at once, accompanied by his son and his son-in-law. When they arrived at the gate of Lamino’s house, the latter had them introduced one after the other, with the exception of the son and the son-in-law, into a room in which had previously been placed a divan to serve as a throne and some garments of royalty. Then Dunama entered, took his place on the divan, and repeated before the assembled functionaries the promise which he had made the evening before. ‘If Lamino has the power restored to me, he shall have the half of the revenues of the kingdom’.

Those of the courtiers who had not been made privy to the plot were violently angry, but as they were in a minority, they had to restrain themselves. Already they were afraid of the Sheikh, who, supported by the Kanembu, was powerful enough to overawe them. Lamino immediately invested Dunama with the garments of royalty, and then had the son and son-in-law of Nguéléroma introduced into the chamber. At the same time he bade the trumpets sound to announce the accession, and gave orders for the public to be admitted. All could then see Mai Dunama seated on the throne, and Sheikh Lamino presented him to them as the new sovereign of Bornu.

On hearing the trumpets sound to announce the accession of Mai Dunama, the horse-boy of Nguéléroma’s son leaped on his master’s horse and galloped with loose rein to apprise the Sultan. He found the latter occupied in reading the Koran. Nguéléroma, having learnt the news, showed no signs of disturbance, but asked the horse-boy if the weather seemed overcast. The latter replied that there were no clouds. Then the Sultan shut his book, put it in a bag which he hung round his neck, and left the town on foot by the north gate.

SECOND REIGN OF MAI DUNAMA

Meanwhile Sheikh Lamino and the courtiers were bringing Mai Dunama to Birnin Kabela, which they entered by the south gate.

Nguéléroma, hearing the trumpets, stopped in the bush. Mai Dunama had search made for him, and he was found without difficulty. Brought before his nephew, the latter, without reproaching him for having formerly dethroned him, assigned as a residence a village called Yamia, situated to the south of Munio, and allowed him to take with him his four legitimate wives and forty concubines. Nguéléroma only accepted the four wives, and escorted by forty faithful horsemen, soon reached his new residence.

For three years Mai Dunama and Lamino governed jointly and divided the revenues of the state. But Lamino's authority kept increasing, and the Sultan in alarm sought in secret to get rid of him. With this view he summoned to his aid the Sultan of Baghirmi, Burgu Manda, who had been formerly the vassal of Bornu, but had profited by the embarrassments occasioned by the Fulani invasion to make himself independent of his suzerain. Glad of an opportunity of proving his power to the Kanuri, he assembled an army, crossed the Shari, and marched on Birnin Kabela.

Sheikh Lamino, informed almost immediately of the Baghirmi invasion, decided to move against the enemy with all the forces of Bornu. Mai Dunama could not avoid putting himself at the head of the army without unmasking his designs, but he wrote to Burgu Manda to explain to him the reasons of his apparent hostility. The Sultan of Baghirmi immediately replied that he approved his conduct, but suggested that in future, to avoid any misunderstanding, he and his adherents should always establish their camp to the south of that of Sheikh Lamino. The Baghirmi could thus be sure of concentrating their efforts on the army of the Sheikh and of destroying it.

But the courier who carried this letter made a mistake. Instead of giving it to Mai Dunama, he took it to Sheikh Lamino, who read it without showing any disturbance, put it in his pocket and sent the man away with a present.

Then, as the Baghirmi approached, the Sheikh's camp was struck and re-pitched some distance in advance, at Ngala; here Lamino and his Kanembu established themselves to the south of Mai Dunama's camp and waited.

Burgu Manda soon arrived. After viewing the dispositions made by the opposing army, he thought that the Sultan of Bornu had safely received his letter and had followed his advice. Accordingly he attacked the north camp with all his forces, and after a desperate battle cut to pieces the army of his friend, Mai Dunama. Before the latter could recover from his astonishment and put a stop to the misunderstanding, he was captured and slain by the Baghirmi troops (1817).

Meanwhile, Sheikh Lamino fled to Ngornu, hastily collected his most valuable possessions and sent them off northwards, and after burning the remainder, he fled towards Yo, on the River Wobe.

When Burgu Manda perceived his terrible mistake he was furious, and hastened in pursuit of Lamino. The latter, however, had a good start and the Sultan of Baghirmi could not overtake him. Burgu Manda was soon afterwards compelled to return to his own country, and after sacking Kukawa, where the Sheikh had another house, and ravaging everything as he passed, he returned to his kingdom laden with booty.

SULTAN IBRAM AND SHEIKH LAMINO REIGN JOINTLY

At Birnin Kabela the nearest relatives of Mai Dunama and Nguéléroma disputed among themselves as to who should succeed the Sultan thus killed by mistake. The Sheikh, reassured by the departure of the Baghirmi army, returned to Ngornu and again became master of the situation. Ibram (Ibrahim ben Ahmed), younger brother of Mai Dunama, wishing to assure himself of his protection, said to him: 'My brother gave you half the power and half the revenues of the kingdom; I will let you have them all, if I can obtain the title of Mai.'

Lamino immediately accepted this offer. He called for the long silver trumpets which were used to announce the accession of a Sultan, and summoned to Ngornu the chiefs of the various tribes. In front of them Ibram solemnly repeated the promise which he had just made to the Sheikh, who immediately had a seal manufactured, on which was engraved in Arabic—

Let God's will be done.
God is sufficient for his servant.
Mohammed Al Amin.

Then addressing the assembled chiefs and notables from the various tribes, he advised them to turn a deaf ear to the artful designs of the courtiers and to obey him loyally. Then, at his order, the trumpets proclaimed the accession of the new Sultan. Ibram was clothed in the royal garments, and the procession set out for Birnin Kabela.

One of the first acts of sovereignty of the Sheikh was to permit the ex-sultan Nguéléroma, exiled to Yamia, to once more approach the capital. He installed him at Belangawa, north of Geidam on the River Wobe, where he died some time afterwards.

Lamino also busied himself in organizing his household, which hitherto had resembled the school of a famous 'mallam' rather than the dwelling of a chief. His disciples were always coming and going, and allowed any one who liked to penetrate into his presence. On his demand, forty officials of the court of the Sultans of Bornu were sent to him; they soon drove away the greater part of the students, and divided amongst themselves the dignities of the new court.

This court was soon more brilliant than that of the nominal sultan Ibram. The latter, in fact, only received annually from the Sheikh 1,000 dollars (about 5,000 francs), 1,000 gowns, 1,000 cloths, 100 camels, 100 horses, 100 oxen, 100 donkeys, 100 goats, and 100 sheep, whose total value was less than 50,000 francs, which was very little to cover the cost of maintaining his entourage. The courtiers therefore gradually deserted him and went to live at the court of the Sheikh.

It is said that the latter, in order to affirm his authority, went out one day from his apartments holding in each hand a gown, which he gave as insignia of office to two slaves who were watching at his gate. The one on the right was nominated Kachella of the Kubri Kanembu (the Sheikh's own tribe), and the one on the left Kachella of the Sugurti, the best warriors of

Bornu, who fought on foot, armed with the spear, the bow, and the shield of 'ambach' wood.

At the same time he rebuilt his palace at Kukawa, which had been pillaged and polluted by the Sultan of Baghirmi, and put himself in communication with the Fulani Sultan of Sokoto, Mohammed Bello, the Sarikin Mussulmi. To the latter he wrote as follows: 'We profess the same religion, and it is not fitting that our subjects should make war on each other. Between our two kingdoms are the pagan Bedde tribes, on whom it is permissible to levy contributions; let us respect this limit; what lies to the west of their country shall be ours, what lies to the east shall be yours; as for Munio, Damagaram, and Daura, they will continue to be vassals of the Sultan of Bornu, who in return will surrender to you all his pretensions to Gober and Katsena.'

Mohammed Bello replied that he accepted Lamino's propositions, but the two Mussulman chiefs would not, or could not, abide by their word, and the Fulani incursions into Bornu continued as before.

EXPEDITION AGAINST BAGHIRMI

Sheikh Lamino never forgot a grievance and, when he could, repaid twofold any evil that had been done him. As soon, therefore, as he had reorganized and reinforced his army, his first expedition was directed against the Baghirmi in order to avenge the intervention of Burgu Manda against him some years before. So as to deceive the enemy, instead of marching to the south of the Lake, where the Shari constitutes a formidable obstacle, he marched round Chad to the north, and thus arrived in Kanem without alarming his adversaries. A rapid march brought him to Massenya before the Sultan of Baghirmi could assemble his contingents, and it was then Burgu Manda's turn to take to flight. He fled for refuge to the Sara country, whilst Sheikh Lamino razed the ramparts of his capital and destroyed his palace, the site of which was, according to our informants, systematically polluted for a week by the Bornu army. Then Lamino, having gained considerable booty and seen his enemy fly before him, satisfied with his vengeance, crossed the Shari and re-entered Bornu.

EXPEDITION AGAINST KANO

Every year in the autumn the Sheikh undertook some new expedition. Tempted by the hope of a rich booty, he directed his army against Kano, the wealthy commercial capital of the Central Sudan. Having arrived at Gorko, a little distance from the town, he learnt that a strong Fulani army, commanded by Yakubu, governor of Bauchi, was marching against him. Lamino immediately sounded the retreat, but was overtaken by Yakubu at Fake. Thanks to their fire-arms, the Kanuri repulsed the first attack of their adversaries, who were only armed with spears and bows; then in their turn they took the offensive, but without success. The two parties, having each gained a slight advantage, then returned each to their own country claiming for themselves a victory.

In the following autumn, Lamino renewed his attempt and was more fortunate, for, if he did not succeed in penetrating into Kano, he was able undisturbed to obtain possession of numerous villages, such as Kanawa, Rimi, &c. As booty and not glory was always his object, he returned to Bornu as soon as he considered his plunder sufficient. [\[493\]](#)

EXPEDITION AGAINST KATAGUM

The following year it was the turn of Katagum to be disturbed; but the town, well defended, could not be taken in spite of a two months' siege. The Bornu army had to content itself with sacking all the surrounding hamlets, and before retreating, it destroyed, out of spite, all the growing crops. A great dearth resulted for this little country.

EXPEDITION AGAINST HADEIJA

A year later Sheikh Lamino laid siege to Hadeija. He pitched his camp on the banks of the river, and summoned his vassals to aid him in his enterprise, for the town resisted vigorously. Munio, Damagaram, Machena, and Nguru obeyed; Tumbi dan Hawa, Sarikin Gummel, alone refused to join them. An envoy from the Sheikh was sent to press him, but he replied:

‘If the Sheikh and his army can do without water for two days, let them come here.’

EXPEDITION AGAINST GUMMEL

Lamino accepted the challenge. He crossed the uninhabited and waterless zone, which for forty miles separated Hadeija from Malawa, to the east of Dungass, not without losing numerous men and horses from thirst. After resting three days, he sent a messenger to the Sultan of Gummel to announce his arrival. Dan Hawa, panic-stricken, tried to obtain pardon by sending him two ambassadors, Liman Bergoma^[494] and Mallam Shetima Kinguimi, accompanied by twenty young virgins, twenty young boys, twenty camels loaded with presents, and 100 oxen. The Sheikh gave the embassy a very unfriendly reception. ‘You have given evil counsel to dan Hawa,’ said he; ‘you deserve to die, but I pardon you in memory of the father of Shetima Kinguimi, for as a young man I studied under his directions.’ Then, refusing the presents which they brought, he sent them back, saying: ‘Let your master come in person.’

Dan Hawa, little reassured by the reception accorded to his envoys, preferred to dispatch another ambassador to the Sheikh, one Shetima Allah Rhama, with much more valuable gifts.

Lamino then appeared to be softened, and accepted the presents on condition that the Sarikin Gummel came himself to ask his pardon. To give him confidence, he said in Hausa to the envoy: ‘En ka kawo shi, ka tashi da shi agaba nan,’ i.e. ‘if you bring him, you shall arise at the same time as he in my presence.’

Dan Hawa was then reassured, and accompanied by Shetima dan Allah, he repaired in person to the camp of the Sheikh. The latter made them sit down in front of him, and when the Sarikin Gummel had expressed his repentance, Lamino said to them: ‘Give each other your hands and get up.’ They obeyed. ‘Now sit down again.’ They did so. Then addressing Shetima, he said: ‘I promised you that if you brought me dan Hawa, you should arise with him in my presence; I have kept my promise.’

He then handed over the Sarikin Gummel to the Galadima of Nguru. Now the latter, who claimed hegemony over all the vassals of Bornu from the north to the west, had long hated dan Hawa, who had always defied him, so without delay he put him to death.

On learning of this summary execution, which had not been ordered by him, Lamino was, or pretended to be, violently angry, and inflicted on the Galadima a considerable fine.

He nominated as Sarikin Gummel, dan Tenoma, younger brother of dan Hawa, and at the request of the new chief, decided that he should not have to obey the Galadima of Nguru, but only Yerima Umar, his own son. He then returned to Bornu.

EXPEDITION AGAINST NGURU

The Galadima of Nguru had been profoundly humiliated, and had sworn to revenge himself. He soon assumed an independent attitude towards the Sheikh, at the same time advising the other vassal states to imitate him, but only Kosso, ex-sultan of Munio, joined him with a strong force of Mangas.

It was therefore against Wari—a place which no longer exists, but which was situated about thirty miles north-east of Kachellari—the capital of Nguru, that Lamino directed his autumn expedition of the following year. It is asserted that he camped opposite this town for a year and a half without being able to take it, or even to secure any marked success against the bands of horsemen that from time to time sallied out from the walls. He then appealed to all his vassals, and when his troops were sufficiently numerous, completely surrounded the fortifications. The Galadima Umar, feeling that all was lost, made a desperate sortie, succeeded in penetrating the besiegers' lines and fled to Hadeija. On the following day Kosso also made good his escape.

The inhabitants, thus deprived of their leaders, asked for peace. Lamino demanded from them 100 slaves, 100 horses, 100 camels, and 100 oxen, which were immediately delivered to him. He then ordered the destruction of the ramparts. The inhabitants refused to consent, and hostilities recommenced. After some time they made new proposals for peace, which

was offered on the same conditions as before. When these were fulfilled, the Sheikh demanded the demolition of the fortifications. The inhabitants again refused. There were fresh hostilities and fresh successes on the part of the Sheikh. At last the inhabitants again asked for peace, paid the tribute imposed on them, and then scattered.

Wari was delivered up to pillage, and then Lamino marched on Hadeija, where the Galadima Umar had taken refuge. Thereupon the latter fled to Sokoto and was replaced as Galadima of Nguru by his younger brother Dunama.^[495] The victorious Bornu army returned home and the Sheikh allowed it two years' repose.

A little later, Lamino, always excessively prone to suspicion, took umbrage at the popularity acquired by a 'Sufi' mallam, whose school was much frequented and who lived at Yale, to the north-east of Dikoa. He repaired in person to Yale, where he learnt that the mallam was a Koiyam of the Kullumfardo family, and quite devoid of ambition. At this juncture the mallam became blind and the Sheikh, almost at the same time, was attacked by a malady of the ear, from which he was destined never to recover.

Almost at the same time Bohari, the Sultan of Hadeija, attacked Nguru and raided the district of Selleri (Zagarari). Lamino was ill and unable to take the field in person, but his son, Yerima Umar (later Shehu Umar), took command of a strong expedition with which he advanced rapidly into the invaded territory. Bohari had no alternative between a difficult retreat and immediate submission, and having collected considerable presents he repaired in person to Umar, did homage to him, and was allowed to return to Hadeija.

Sheikh Lamino died soon afterwards, in 1835, leaving Bornu flourishing, his family aggrandized, and the nominal sultan, Ibram, without the slightest power. His son, Yerima Umar, then took the title of Sheikh (or Shehu) and succeeded him without opposition as *de facto* sovereign of the kingdom.

FIRST REIGN OF SHEHU UMAR (1835-53)

One of the first acts of the new Shehu was to cut down by half the meagre subsidy which his father had paid to the sultan Ibram. The latter protested, but in vain, for the following year the subsidy was still further reduced, and the third year he received almost nothing. Mallam Amadu Gonimi, relying on the friendship shown him by the late Sheikh, thrice made friendly representations to Umar, reminding him that it is the part of a good son to act in all circumstances as his father had acted, and requesting in consequence that he should allow Sultan Ibram the same subsidies as he had been accustomed to receive from Sheikh Lamino. Shehu Umar remained silent, but on the third occasion he replied: 'A son inherits his father's property, not his character.'

Then Ibram, spurred by necessity, wrote to the Sultan of Wadai, Sherif by name, begging him to come to his aid and rid Bornu of the Shehu and his family. The army of Wadai was set in motion, but unfortunately Ibram had confided the secret to one of his daughters, who in her turn communicated it to her lover, a young relative of the Shehu. As soon as he received the warning, the Sultan and seven of his advisers were seized and thrown into prison. The Shehu then departed with his army to arrest the invasion. The encounter took place on the banks of the Shari at Kusseri (1846) and the Bornu army was completely routed. Umar, much chagrined, returned in haste to Kukawa and had the Sultan and his advisers, who had summoned the enemy into Bornu, immediately executed. He then fled towards the north-west, taking refuge in the marshy country of Bera, near Zigagué, on the banks of the River Wobe.

Meanwhile, the Sultan of Wadai continued his march on Kukawa, and learning of the execution of Sultan Ibram, whom he had come to deliver, proposed to replace him by Ali, son of the latter, a boy of seventeen.

But a Wadai man, called Ibrahim, settled at Kukawa, pleaded the cause of Shehu Umar so well that the Sultan of Wadai dispatched the following message:

'If you consent to pay me \$1,000, I will return to Wadai with my whole army; otherwise I shall leave a strong guard with the young sultan to protect him.'

‘Where can I get \$1,000?’ replied the Shehu, for at this period silver was rare in the country, and \$1,000 constituted a very important sum.

But an Arab trader who happened to be in the country saw his opportunity to make a big profit. He brought \$600 to the Shehu, on condition that he was repaid this sum in young girls, at the rate of \$7 a head.^[496] Then Umar, having collected all the ornaments of his harem, was able to make up the balance of \$400, and sent the whole sum demanded. Sherif then returned to Wadai, leaving the young sultan, Ali, under the sole guard of the Kanembu and Sugurti.

Meanwhile, Durman (Abdurrahman, younger brother of the Shehu) happened to be with an army in the north-west of Bornu, marching towards Zinder. Ibrahim, sultan of this country, came to meet him with conciliatory intentions and peace was concluded. Durman, recalled by his brother with the idea of attacking the young sultan and regaining the reins of power, then returned towards the River Wobe. On his arrival, however, Umar, who was a bit of a poltroon, had changed his mind and suggested that they should fly to Fezzan, their mother’s country.^[497]

The younger brother replied: ‘As you seem but a womanish creature, go to Fezzan if you like. I intend remaining in Bornu.’

Umar was thus shamed into fighting. The encounter took place at Minargué, on the banks of the River Wobe. Barth calls the place Minarem. Umar undertook to hold in check the cavalry of Ali, who were reputed to be not very brave and only moderate horsemen, whilst the more valiant Durman joined battle with the Kanembu and Sugurti infantry and put them to rout. The young sultan was killed and Shehu Umar was able to re-enter Kukawa as sole master henceforward of Bornu.

Two years later, desirous of strengthening his authority over his western vassals, he proceeded to Zinder, whose sultan, Ibrahim, had disregarded his orders. Ibrahim took to flight, but his brother Tenimu repaired to the Shehu and got himself enthroned by him as Sultan of Zinder. A mosque was built to commemorate this event and the Shehu returned to Kukawa.

Five years later grave dissensions arose between him and his brother Durman. The latter reproached him warmly with the affection which he

showed for one of his counsellors, Haj Beshir, whom he consulted on every occasion and whose advice he followed in preference to that of his brother. Old men, who were living at the time and whose memories are still very clear, have informed us that the mutual affection of the Shehu and his counsellor was greater than that of two brothers and recalled that of husband and wife. Durman was much annoyed and told his brother one day in the course of a violent quarrel that he had 'the spirit of a sheep'. He then fled.

Umar and his favourite, Haj Beshir, set off in pursuit, but when they had nearly overtaken him, some of the Shehu's other brothers, who had followed the army, wrote to Durman in secret that they would not fight against him. Durman immediately faced about and by a bold stroke penetrated the camp and reached the very outskirts of the Shehu's own tents, in front of which he himself discharged several shots.

Umar and his favourite, stricken with panic, leaped on their horses and, almost unaccompanied, fled to Kukawa without stopping. Haj Beshir immediately collected all the wealth that he could carry away and hastened to take refuge at Gulfei on the banks of the Shari, leaving the Shehu at Kukawa to await events.

Meanwhile, Durman had been proclaimed Shehu by the army of Umar and re-entered Kukawa at its head. Umar, relegated to the position of a simple prince of the blood, had to leave the palace and was installed in his brother's private house.

REIGN OF DURMAN (ABDURRAHMAN), 1853-4

Durman had hardly been installed, when he charged his brother, Abba Bukar Kura, to inform Haj Beshir, of whom he was a friend, that if he would return at once to Kukawa he would grant him a pardon. At the instance of his friend and after a long hesitation, Haj Beshir consented to return. But as soon as he arrived Durman had him imprisoned, and shortly afterwards, in spite of the prayers of his brother, incensed at his ill faith, had him impaled.^[498] This did not prevent him swearing the same evening in

the presence of all the courtiers that he had never given the order for the execution of the unhappy man, but only that he should be put in bonds.

Durman reigned a bare two years: his harsh character rendered him unpopular with the courtiers, whilst he alienated the army by an unsuccessful expedition against the Musgu, a people in the south of the empire who refused to pay tribute. The malcontents resolved to replace him as soon as he returned to Kukawa by Shehu Umar, whose accommodating temper appeared all the more attractive since the autocratic character of Durman kept constantly increasing.

But Durman was warned of the plot by one of his slaves, Mustapha Mutube. He immediately assembled the courtiers and said to them: 'I know that you are plotting against me. If you want to replace me, do it at once; don't wait till I have returned to my palace.'

The courtiers exclaimed: 'It is false! you have been deceived!'

Durman called the slave: 'Who told you that a plot was being hatched against me?'

'It was Arri Fugumani, the chief of the slaves.'

Arri Fugumani, when questioned, said: 'I was joking with Mustapha Mutube; the whole thing is nothing but a jest, for which I ask pardon.'

Durman pardoned him, saying: 'From henceforward let no one else come and tell me lies.'

Two months passed, the army had returned to Kukawa, and the courtiers began to conspire afresh.

Arri Fugumani said to Mustapha Mutube: 'Yesterday's lie may become to-morrow's truth. You would do well to inform the sultan.' But Mustapha, fearing to be again contradicted, dared not say anything.

The next morning all the courtiers assembled in the western town (Bela Potebe) in front of Umar's house, and the latter's son, Yerima Bukar, brought him the sultan's drum and standard, the emblems of power. Then forming in procession they advanced towards the eastern town (Bela Gedibe), where the palace of Durman was situated. The latter, hearing the

sound of the trumpets and the cries of the mob, emerged in haste, and with a few faithful followers engaged his adversaries in unequal combat on the open space, called Gumsigine, between the eastern and the western towns. Durman had his horse killed under him. He was immediately brought the horse of one of his slaves, but he refused it, saying: 'I cannot mount a slave's horse.'

His brother, Abba Bashir, then gave him his own horse to enable him to fly; but he was unable to get far, and was soon afterwards captured and brought back to Kukawa. Shehu Umar solemnly declared that he would not harm him, and installed him in the house that he had himself just quitted.

SECOND REIGN OF SHEHU UMAR (1854-81)

Three months later, fearing doubtless lest Durman should dethrone him a second time, the Shehu assembled the principal mallams and asked them: 'Is it better to kill one man than to allow ten to be killed? If one man may cause the death of several others, may one get rid of that one? Consult your own consciences and the Book and let me know your answer.'

All guessed that the Shehu alluded to a new civil war rendered possible by a new rebellion on the part of Durman, and they replied: 'Certainly it is better to execute one man than to allow ten to be killed, especially if it is the one man who may cause the death of the others.'

Thereupon Shehu Umar gave orders to his head-slave to go and put his brother to death, [\[499\]](#) but to act discreetly so that his death should remain uncertain for some time. His orders were obeyed, and henceforward Shehu Umar enjoyed a peaceful tenure of power. But if the Shehu himself was of a pacific temper, the same was not true of his son, Yerima Bukar, a brave, bold and energetic prince, fond of war and booty.

EXPEDITION AGAINST THE NGIZIMS, 1855 (?)

Yerima Bukar's first expedition was directed against the Ngizims, a pagan tribe, said to be descended from the Sos, and living to the south of the Bedde country. Having paid tribute to the Sultan of Sokoto, they

naturally refused to pay a second time to the Sultan of Bornu. Moreover, they defended themselves courageously when Bukar entered their country. The campaign, which lasted five months, was a very arduous one for the Kanuri, but, on the other hand, they gained much booty, principally in slaves.

EXPEDITION AGAINST THE MUSGU

The Shehu succeeded in restraining for five years the bellicose temper of his son; but the Musgu, against whom Durman had already had to take severe measures during his short reign, having again refused to pay tribute, Yerima Bukar had to direct four successive expeditions against them in order to obtain their submission (1861-5).

EXPEDITION AGAINST ADAMAWA

It was next the turn of Adamawa, a very hilly country to the south of the Benue, to suffer invasion. The rich town of Balbaya was captured, but all the camels and the greater part of the horses of the army perished. The troops were soon in such a state of destitution that the Shehu had to send in haste both horses and clothes to enable his son to make an honourable entry into Kukawa (1866).

AGAINST THE BEDDES

Two years of peace followed, and then Yerima Bukar had to subdue the Beddes, who had refused to pay tribute, and had driven away the Shehu's envoys, 1869.

AGAINST FIKA AND BAUCHI

The following year he had to subdue Fika, an important group of towns, situated to the east of Gujba, whose inhabitants had driven away the Shehu's representative, Kachella Abdullahi, son of Kachella Arri Marghi, who had been appointed to this district on the death of his father. Bukar

destroyed the fortifications of the town, ^[500] imposed a heavy tribute on the inhabitants, and continued his march westwards on his way to attack the town of Misau, situated to the north of Bauchi and a tributary of the Fulani. But he was at once recalled by his father, who did not wish to involve himself in difficulties with the Sultan of Sokoto (1870).

AGAINST THE BEDDES

Bukar was not at all anxious to return. He therefore ceased to attack the Fulani, but remained on the western frontier of Bornu raiding and plundering far and wide, and gradually approaching the Beddes, whose submission had been far from satisfactory. But the rainy season was approaching, and his lieutenants feared its effects on their cavalry. They therefore advised Al Haji, the Bedde chief, to pay a certain amount of tribute to Bukar, and at the same time begged Shehu Umar to give his son formal orders to return to Kukawa (1872).

AGAINST KUSSERI

Bukar had never forgotten the help which the inhabitants of Kussери had given to the Sultan of Wadai, when, summoned by Ibram, he invaded Bornu and obliged his father to take to flight. It was they who had assisted the passage of the Wadai army by showing them a ford, and a number of them had joined the invaders of Bornu. Always on the look-out for somewhere to attack, Bukar obtained leave from his father to march against this town. The battle was just about to begin when an order arrived from Shehu Umar instructing him to make peace, as the inhabitants of Kussери had hastily sent to Kukawa a tribute of 300 slaves. The army had to return to Bornu without fighting, but not without having partly ravaged the banks of the Logone and sacked the town of Ngurkum (1879).

Two years later Bukar asked leave from his father to go on an expedition, but this time without telling him whither. The Shehu, now an old man and not daring any longer to refuse him anything, allowed him to set out, and Bukar took advantage of this to renew his designs against Kussери.

He had arrived in front of the town, when he received a message from his mother telling him that the Shehu was at his last gasp, and that, unless he returned within five days, the courtiers would nominate some one else as Shehu.

Bukar set out at once and arrived in Kukawa on the fourth day. The Shehu was still alive, but the principal eunuchs prevented Bukar from seeing him, fearing that before he died Umar would designate Bukar as his successor. Those eunuchs wished Bukar to owe the throne to themselves in order that they might make large profits. Three days afterwards, about two o'clock in the afternoon, the Shehu breathed his last without having seen his son (1881).

The chief eunuchs, Malla Kerim, Yiroma, and Mestrema, deliberated over the choice of a successor. They dreaded the accession of Bukar, whose firm and haughty character had, in his father's lifetime, more than once disappointed their schemes. Malla Kerim, who of the three commanded the most warriors, made up his mind the first. Taking the ring, rosary, and cap of Shehu Umar, and leaving Mestrema to watch the corpse, he repaired in company with Yiroma to Bukar. Giving him the ring, &c., he said: 'Your father is dead, here are the emblems of power.'

Bukar refused to accept them except in the presence of numerous witnesses, and summoned his son, Kiari, and his friends, Kassim, Bunu, Abana, Lamino, and his principal slaves. Having informed them of the death of his father he ordered them to assemble his troops. A procession of horse and foot was then formed, and they proceeded to the palace, whose four gates were guarded by the troops of Malla Kerim and could not be opened except in his presence. When they reached the corpse, the chief eunuch raised the shroud and said: 'Behold, he is dead, we resign him to you for burial.'

He then put the dead man's ring on Bukar's finger, his rosary round his neck, and his cap on his head. Outside the drums beat and announced to the people by their slow and measured roll the death of the old Shehu and the proclamation of the new. The Liman, the Legari, and the Talba came to wash the dead, and having assembled all the mallams, celebrated the obsequies that same night.

REIGN OF SHEHU BUKAR (1881-4)

The next morning very early the crowd of courtiers and notables assembled at the palace. Shetima, descendant of the ancient Tubba sultans of Bornu, gave the investiture to Bukar by clothing him in the royal gown and turban, and according to custom asked him: 'What province will you give me?'

'I grant you the Kagaburi country,' replied the new Shehu.

The Shehu's relatives then came to swear fidelity to him, and the ceremony of investiture ended with a great noise of drums, trumpets, and other instruments of music.

For seven days Bukar only resided in the palace during the daytime, passing the night in his own house. On the seventh day he distributed alms in memory of his father, and installed himself definitely in the palace, leaving his own house and all that was in it, except his women, to his son Kiari.

His first thought was to restore order in the country. 'Hitherto', said he, 'you have been plunderers of whom I was the chief. Henceforward, I shall tolerate no more plundering, and I shall punish without fail any one who disobeys me.'

During the second year of his reign he made an expedition against the Beddes, and captured the towns of Kadera, Aburraga, and Tagari.

The following year he had to repress a revolt of the people of Mulgue in the south of Bornu; but for this expedition he could only gather a few troops, for his subjects, tired of the constant wars in which he involved them, made but a poor response to his summons; some only sent a slave, others a son, others again no one at all.

The Shehu was much annoyed, and as soon as he returned to Kukawa he summoned a great meeting, where he announced that in order to punish his subjects for their half-hearted obedience he should inflict on all a fine equal to half their property. Soon afterwards long caravans of slaves and animals loaded with produce of all sorts might be seen entering the capital

every day. Never before had such wealth been accumulated in the palace of the Shehu.

But the common people and the notables murmured loudly. Everywhere in all the villages was to be heard the same malediction: 'May God cut him off and rid us of him.'

As it chanced the Shehu did die soon afterwards, long before he had seen paid in full the formidable fine which he had inflicted on his subjects, and every one agreed that God had heard the unanimous prayer of an oppressed nation (1884).

REIGN OF IBRAHIM (1884-5)

On the death of Shehu Bukar the courtiers chose as his successor his uncle, Abba Mustapha, younger brother of Shehu Umar. But, as a result of intrigues, there was a sudden change, and the next morning the high dignitaries of the Court proclaimed Ibrahim, younger brother of Shehu Bukar.

Abba Mustapha, chagrined at thus being deceived, at first refused to take the oath of allegiance and sought refuge in flight. But he soon returned on a promise from his nephew that his life would be spared. Ibrahim, however, did not keep his promise, and Mustapha was soon afterwards assassinated.^[501] He himself died after reigning one year without having been able to undertake any expedition.

REIGN OF ASHIMI (1885-93)

His younger brother, Ashimi, third son of Shehu Umar, was chosen to succeed him. After a pacific reign of eight years he was obliged to take up arms to defend his kingdom against the invasion of Rabeh, advancing from Baghirmi.

To repulse the invader he sent to meet him an army of 30,000 men, under the orders of Mohammed Tar, son of Amadu Guimi. The encounter took place a little to the south of Ngala, at Ngamagué, in April 1893.

Rabeh, who could only muster 3,000 men, took care to camp on the border of a marsh, which served to protect him against a surprise by the enemy's cavalry, who were much superior in number. Mohammed Tar, presumptuous and imprudent, got entangled in the marsh, the horses were mired, and those of their riders who could not succeed in disengaging themselves were massacred in great numbers by the enemy's gun-men.

On hearing the news of the disaster Ashimi organized a second army more numerous than the first, put himself at their head and marched against Rabeh. In spite of the support of several heavy cannon, sent him by the Sultan of Zinder, his troops lacked confidence. The Bornu army fled at the first charge and the flight was soon changed into a rout. Ashimi returned in hot haste to Kukawa in order to dispatch his most precious possessions north-westwards, and to fly himself as soon as possible.

It is said that he left Kukawa one Saturday evening and that the conqueror entered the town on Sunday morning. Rabeh immediately launched a column commanded by Abba Bukar in pursuit of the fugitive Shehu. Ashimi was overtaken at Lumburem, on the River Wobe, east of Duchi, and had barely time to escape, leaving in the hands of the enemy all his baggage, his animals, his slaves, and even his women.

The scattered Kanuri then nominated as Shehu his nephew Kiari. The latter declared his intention of marching without delay against the invader, but his partisans, alarmed at such audacity, immediately abandoned him and rallied once more round Ashimi, whose defeats had made him prudent. Kiari then accused his uncle of compromising with the enemy and had him assassinated.

REIGN OF KIARI (1893)

Three months later Rabeh advanced in person against those of the Kanuri who still maintained their independence on the banks of the River Wobe. Kiari assembled his forces and marched against him. The encounter took place at Gashagar. The Bornu army fought valiantly, but when their chief was wounded they gave ground.^[502] Kiari, who refused to fly, was

captured and taken to Rabeh, who had him executed. Thus perished the last independent Sultan of Bornu.

By kind permission of Commandant Tilho.

There is no trace of this site to-day, but it was just north-west of Ngornu and in sight of that town.

For another account of Lamino's Kano campaign, vid. [Appendix VI](#).

Bergoma was the father of Shetima, Alkali of Zinder in 1907, who confirmed this information.

Umar afterwards returned, Dunama fled, and Umar was recognized by the Shehu as Galadima.

Young girls then averaged \$40 a head in Bornu and \$200 on the Mediterranean coast.

According to some authorities, Shehu Umar was the son of a slave given to Lamino by Mai Dunama and already enceinte when she entered the Sheikh's harem.

Vogel says 'bowstrung', vid. p. 272, Benton, *Notes on Some Languages*, &c.

According to Vogel he was 'strangled', vid. p. 277, Benton, *Notes on Some Languages*, &c.

If this means Fika town itself the statement is incorrect, it was never captured either by Kanuri or Fulani.

For another account of Abba Masta's rebellion, vid. [Appendix VII](#).

[According to Gaston Dujarric, *Vie du Sultan Rabah*, the Kanuri were at first victorious but scattered to plunder the enemy's camp. Rabeh, who as usual, had kept a strong body of gun-men as a reserve, rallied his troops, gave all his generals—including his son, Fadel Allah, whose arm had been broken by a musket-ball—100 lashes apiece, and sent them forward to victory.—P. A. B.]

APPENDIX V

ACCOUNT OF RABEH

THE following translation of a German account of Rabeh, written in 1899, is not without interest, though of course it is in several instances inaccurate. Herr Lippert's chief informant appears to have been a man from Ghat. His Hausa is not always unexceptionable, and there is nothing to show that he was ever in Bornu or was actually an eyewitness of the events he describes. His account of Rabeh's campaigns should be compared with that given in Decorse et Demombyne's *Rabah et les Arabes du Shari*, and in Dujarric's *Vie de Rabah*. Rabeh did not die of poison in 1897 but was killed in battle by the French in 1900.

Mittheilungen des Seminars für Orientalische Sprachen (Berlin) for 1899.

RABAH^[503] by JULIUS LIPPERT

'History is made by personalities.' This maxim only holds good in a very limited sense for western states, but as regards the historical changes in the 'Dark Continent', and especially in the Sudan, it is absolutely true. The negro is entirely deficient in public spirit and other political virtues. In the same degree as the idea of the state is foreign to him,^[504] he lacks also the sense of patriotism and loyalty to a dynasty. It remains a matter of indifference to him whom he serves, so long as he can do so with tolerable security for life and liberty. It is only by such considerations that the success is explicable of bold adventurers, who, relying on nothing but intelligence and energy above the average and the support of a handful of devoted adherents, were able to carve out mighty empires for themselves and overthrow ancient kingdoms, who owed their hundreds of years of existence solely to a political *vis inertiae* similar to their own on the part of the neighbouring states. Thus we see, to quote only two examples, how at

the beginning of the nineteenth century Othman dan Fodio^[505] with his numerically infinitely weaker Fulani conquered the various Hausa states and welded them into the kingdom of Sokoto. The Mahdi likewise in the 'eighties subdued the Egyptian Sudan for a time in spite of European administration. Now it may indeed be urged that both these men appeared also as religious reformers; and religious fanaticism is certainly always a mighty factor in favour of success. But in the case of Samory, whose capture the newspapers announced a few weeks ago, it was solely the power of personality, which enabled him after being expelled from one territory to conquer in a very short time a new dominion about as large as Germany, and since 1885 to oppose unaided with more or less success the superior military skill of the French.^[506] It was personal ability also which enabled Rabeh, the brickmaker's son, to raise himself to the position of king of the powerful state of Bornu, and would have let him accomplish even greater things, had not the assassin's poison put a premature end to his career of conquest. Just in the last few weeks all kinds of accounts of this man have been current in the newspapers. These accounts, derived from a letter of the French traveller, Bonnel de Mezières,^[507] specially concern us Germans, inasmuch as the scene of the events therein described is situated in the hinterland of our colony of the Kameruns; but it may also be of universal interest to find briefly described here the life and death of this personality.

My sources of information are chiefly the two documents printed with a translation in the appendix to this article.

1. A biography of Rabeh in Hausa, written down for me by Mohammed Beshir-al-Ghati, Reader in the Berlin School of Oriental Languages.
2. An Arabic letter to Mohammed Beshir from Abd-el-Kader ibn Al-Haji Isa Muzi from Ghadames, concerning the death of Rabeh.

In so far as I have made use of other sources of information, I have noted them in the proper places.

Rabeh (i.e. the victor) belonged, like Zobeir Pasha, to the Arab family of the Ja'alin, who have their domicile on the upper Nile round about Khartum, but who are distributed all over the Eastern Sudan, and even as

far as Bornu,^[508] as Jellaba (i.e. importers, petty traders). His father was a poor brickmaker, who gained an honest livelihood by the work of his hands. That Rabeh has sometimes been falsely described as a slave of Zobeir's is to be attributed to the fact that he attached himself in early youth to that extraordinary man, who, prior to his appearance as a public official, carried on the business of slave-raiding. Rabeh early became a loyal friend and useful assistant in Zobeir's undertakings. It is certain that he took an active part both in the subjugation of the Bahr-el-Ghazal Province and in the conquest of Darfur. After Zobeir had gone on a mission of complaint to the Khedive in Cairo and had been detained in custody there, his son Suleiman, whom he had left behind as his representative, was driven by intrigues into rebellion, and Rabeh became one of the most prominent sub-commanders in the rebel army. The fortune of war was not favourable to the insurgents. After being totally defeated on repeated occasions, they were summoned by Gessi, the commander of the Government troops, to cease further resistance as useless and to surrender to him under promise of amnesty. In the rebel council of war which met to discuss these propositions, Suleiman and the majority of his lieutenants showed themselves inclined to capitulate on these terms, but Rabeh spoke energetically against the proposal. His advice was, either, at the price of abandoning the territories hitherto connected with their rule, to push westward with their forces, still by no means inconsiderable, where it would be easy to conquer a new dominion; or, if they were quite tired of fighting, to surrender not to Gessi, but direct to the Government authorities in Khartum. If neither of these two proposals should find acceptance, he would find himself compelled to separate from his former comrades and act on his own account. Neither alternative of Rabeh's extremely judicious counsel was attended to; Suleiman and the majority of his sub-commanders gave themselves up to Gessi,^[509] but Rabeh, after an affecting leave-taking from his old comrades-in-arms, marched away with his own troops, and those who had joined him from other detachments, to his camp in the uncertain future, accompanied by the notes of the far-sounding war-horn.^[510]

At this point it may be fitting to insert a few words as to the resources which, at the outset, Rabeh was able to dispose of. I do not think that his army was ever at any one time much over 3,000 strong. Moreover, it was

certainly only the minority of these troops that consisted of Ja'alin and Arabs of other tribes, though they formed the cavalry of the army, while the greater part must have been supplied according to requirements from the 'Basingers', i.e. negro soldiers disciplined on European lines, selected from captured slaves. One may quite fittingly compare them with the Mamelukes of Egypt or the Turkish Janissaries. From what I have heard from eyewitnesses of Rabeh's battles in Bornu, he must also have had artillery at his disposal. It is not only obvious but confirmed by the course of events, that these well-armed and well-disciplined troops, in spite of their numerical weakness, represented a formidable force when opposed to barbarous negro empires, to whom fire-arms were certainly no longer entirely unknown, but amongst whom they had not long been in exclusive use. [\[511\]](#)

II

With regard to the first decade (1880-90) of Rabeh's adventurous career, only vague and scanty accounts are forthcoming to enable us to attain an assured knowledge of events. According to Mohammed's information, he first of all invaded the country of the Dinkas (at the confluence of the Bahr-el-Ghazal and the Nile) and waged war with this tribe for three years. It is evident that this statement is based on a confusion with the before-mentioned conquest of the Bahr-el-Ghazal province by Zobeir. Ferryman states in his recently published work, [\[512\]](#) that Rabeh and his troops turned northwards, and after twice defeating the Sultan of Borku, received permission from him to establish themselves in the southern border districts of his empire. It must have been here that Rabeh, as well as the above-mentioned Sultan, received an invitation from the Mahdi, who had meanwhile seized possession of the whole Egyptian Sudan, to ally themselves with the new state, a course which they both declined with scorn. In the lengthy struggle that followed between the Mahdists and the border tribes belonging to the Sennusiya between Darfur and Wadai, Rabeh must have taken a vigorous part. So Ferryman says. I am able to refute this statement, which is offered without the slightest authority being quoted, as I consider it improbable from an impartial point of view. Borku is a small

group of oases south-east of Tibesti, whose population Nachtigal asserts to be from 10,000 to 12,000.^[513] To reach Borku from Gerra (in Southern Darfur), where Suleiman's camp was situated at the time of his surrender, Rabeh would have to traverse either Darfur, occupied throughout its extent by the Egyptian troops, or else Northern Wadai, inhabited by wild hill-tribes, and make moreover a not inconsiderable desert march. Under these circumstances I cannot conceive what could have caused him to be attracted to this small and remote group of oases, where certainly much fighting but little booty was to be expected. Slatin Pasha states briefly that he was removed to 'the remote districts of the south-west'.^[514] This statement has very great probability on its side. Even Ferryman admits that he lived in the country of the Banda and Fertit before his attack on Bagirmi.^[515] In the midst of these and other pagan Nyamnyam tribes south of Darfur and north of Upper Uele he found what he needed, viz. a suitable field for the capture of slaves, of whom he was in constant want as a medium of exchange for weapons and munitions. A full decade seems to have elapsed before the burning of towns and villages, necessarily consequent on his slave-raids, changed into a wilderness this formerly populous and flourishing district. It then became necessary to find a new field for this form of activity, which in Moslem eyes is by no means disgraceful, provided it is exercised at the expense of idolaters. He was prevented from penetrating farther south, where booty was easiest to be obtained, by considerations regarding the supply of munitions, which he procured from the north through the medium of his fellow-tribesmen, the Jellaba. For this reason he turned north-west and attempted to overrun the kingdom of Wadai. But here he must have met with energetic resistance, as was only to be expected from the close political organization of this coveted empire.^[516] So after a short time he desisted from his attempts on Wadai, and made an incursion with his troops into Bagirmi, which lies to the south-west of the latter. Ferryman is aware^[517] that during his stay in Dar-Banda and Dar-Fertit, Rabeh addressed a request to the Sultan of Bagirmi that he would promise a free passage from Kukawa, the capital of Bornu, through Bagirmi to his traders for the purpose of supplying him with munitions. Rabeh swore revenge against the Sultan for not complying with this request; he was at present, however, denied the opportunity of satisfying this feeling. The Sultan of

Bagirmi sought in vain for assistance from the Courts of Wadai and Bornu. Refusing to acquiesce, yet not in a position to make head against his dreaded opponent in open battle, he threw himself into his fortified capital of Masseña, and surrendered the open country to the enemy. Rabeh quartered himself on the country in his usual fashion, collected a rich booty in slaves, and possessed himself of several of the larger towns, such as Gulfei and Logone. But he had no intention of establishing himself permanently in the country. He was inclined to this course by misleading reports that had reached him of the natural riches of Bornu and of its internal rottenness. He seems to have prepared long beforehand for an attack on this empire, whose prestige among the neighbouring peoples was still very great in spite of its rotten internal condition. He first sent secret emissaries charged to stir up the political differences, which, thanks to the weakness of the royal authority, consisted chiefly in the mutual rivalries of the court dignitaries, and at the same time to create a disposition favourable to himself. He then allied himself with the Fulani chieftain, Mallam^[518] Hayatu, of Jamare, for the purpose of a joint invasion of Bornu. This prince, who, in contrast to the feeble successors on the throne of Sokoto of the great Othman dan Fodio, showed once more something of the warlike and statesmanlike ability of his ancestor, was at once ready for the undertaking against the hereditary enemy of the Fulani. Their united troops advanced into Bornu and pushed their way unopposed right up to the walls of Kukawa. The Sultan,^[519] Abba Ashimi, here faced them for the first time, and thanks to superior forces succeeded in repulsing the assailants. Rabeh now spent several months in Bornu territory in a difficult position, during which the provisioning of his troops especially caused him great anxiety. It was just while he was at the height of his distress that the seed sown by his envoys reached maturity. Rabeh received an invitation from one of the chief Bornu court dignitaries to return to Kukawa, where he would assist him to obtain the mastery. He naturally did not wait to be asked twice, but immediately advanced again on the capital. Another battle was fought in front of the gates of the palace, and this time Rabeh was victorious, chiefly owing to his artillery, to which the enemy could oppose nothing similar. Abba Ashimi himself remained on the field.^[520] In the campaign which followed and which led to the capture of the capital, his two nephews and

successors, Abba Kiari and Abba Sanda^[521] were slain. The conqueror entered Kukawa as the new Sultan.^[522]

III

This event forms a turning-point in the history of Rabeh. Whilst he had hitherto been more or less a mere adventurer, he had now become at one blow the ruler of one of the most powerful Sultanates of the Sudan. Remembering the practical bent of his mind with its grasp of actualities, one might expect that he would at once be conscious of the problems and duties of his new position and that he would proceed energetically towards their accomplishment. Nor did he disappoint such expectations. He sought to revive the caravan-trade with Tripoli, which had been brought to a standstill owing to the unrest of recent years, by giving the traders assurance of the most perfect security, and by other concessions. His endeavours in this respect met with success.^[523] He also entered into trade relations with the British 'Niger Company'.^[524] He removed his headquarters from Kukawa, which is extremely unhealthy,^[525] particularly in the rainy season, to the charmingly situated town of Dikoa,^[526] on the Alo river south-east of Lake Chad. For the better protection of the frontier of his newly-won territory, he left strong garrisons in Gulfei on the Shari, and in Kusseri and Logone on the River Logone. In order to found a dynasty, whose legitimacy should be acknowledged even by the rulers of the neighbouring kingdoms, he married a daughter of his ally, Mallam Hayatu, who, on his side, took to wife a daughter of Rabeh's.

His energetic never-resting nature caused this internal political activity to be accompanied by efforts to enlarge his territory. He first attempted to chastise the hill tribes of Mandara, dwelling to the south of Bornu, who had made themselves troublesome by occasional plundering expeditions into the southerly districts of the empire. The aged king of Mandara himself fell into the hands of a punitive expedition. But, while Rabeh resolved humanely enough to dismiss the old man unharmed to his home, the latter's son and successor allowed a son of Rabeh's, who had fallen into captivity, to be executed. Another expedition—to Bedde—was certainly not designed for

the occupation of this country, but for the capture of numerous slaves, who were carried off as welcome booty to Bornu. A raid into Miga^[527] country had the same successful result. On the other hand, an expedition against Katagum miscarried; the valiant stand made by the king of this country forced Rabeh to retreat hurriedly on Bornu. The King of Misau got wind beforehand of an undertaking that had been planned against his country, and hastened to conclude an offensive and defensive alliance with the neighbouring rulers of Shira and Gombe in order to meet the threatened danger. When united they felt themselves strong enough to advance and attack Rabeh. The opposing armies came into collision on the march. A battle resulted, and after a desperate struggle Rabeh's superior military skill secured him the victory. The losses of the allies were extraordinarily heavy; the King of Gombe himself was among the fallen. But even for the victor the day must have been very costly; for he returned to Bornu without further results.

So long as Rabeh confined his attacks to the small tribes living between Bornu and Sokoto—buffer-states one would call them in modern parlance—everything went very smoothly. But it is obvious that the moment he attacked the territory of a large neighbouring kingdom, complications of immeasurable import were bound to ensue. Such complications took place when he invaded Adamawa,^[528] the great south-eastern vassal state of Sokoto, whence he brought home vast booty, especially in slaves. The governors of the neighbouring Hausa states then recovered from the indolence, with which they had hitherto viewed the success of Rabeh, and entered on an energetic course of action. They had to be prepared for the fact that it might be themselves on whom he would next lay hands. Fright effected what political insight had been unable to accomplish; they formed themselves into a league for the purpose of common defence in case of an attack on one of the confederates. Especially serious for Rabeh was the fact that his father-in-law and former ally, Hayatu, joined the league. Although he had founded his kingdom of Jamare in political rivalry with the Court of Wurno,^[529] he was Fulani enough at heart to feel that the interests of his fellow tribesmen were not different from his own, in so far as the to be or not to be depended on himself. At first he attempted to intervene diplomatically. But when Rabeh, fully conscious of his power, haughtily

repulsed these attempts, Hayatu came forward as leader at the head of the league. It may be that Rabeh feared the military force of the league, or the power of England that stood behind it, or something else; at any rate it is certain that for a time he abstained from further attempts against Sokoto—though an expedition against Kano may have been contemplated—and turned his attention to a problem which awaited solution within the compass of his own empire, viz. the reconquest of the recalcitrant Governor of Zinder. Zinder was the most north-westerly province of Bornu, but its governor had frequently shown himself refractory even in Shehu Umar's time, [\[530\]](#) and had taken advantage of the confusion of recent years to free himself of even nominal dependence, and had also annexed the province of Munio which lay to the east of his own territory. After careful preparation Rabeh set out about the beginning of 1897; he was destined never to return to his capital alive. Mallam Hayatu, who, following in the lines of Oriental despots, thought cunning the better part of valour, got poison administered to him on the way to Zinder by means of his daughter, who, as mentioned above, had been married to Rabeh. Thus perished, in the midst of his activity, the still vigorous warrior at the age of fifty-eight. [\[531\]](#)

In criticizing Rabeh we must be careful not to adopt the standpoint of Christian humanity; the character of his people and of his religion must be taken into consideration, which declare as permissible many things that are held by us to be abominably cruel. Thousands of unfortunates were enslaved by him and sold either in the Sudan itself or sent to North Africa. I have myself seen in Tunis and Tripoli quite a number of individuals who were recognizable as Rabeh's slaves by the three oblique scars on each cheek. Any one who is acquainted with the conditions knows that for each of the slaves who reached their journey's end, one must add at least five other individuals who perished either in the slave raids of the foregoing campaigns, on the journey, or through sickness following on a change of climate. This sum total of misery and destruction of family happiness constitutes a terrible indictment against their originator, but we must also again recollect that these slave raids were not for Rabeh an end in themselves, but only a means to an end, and that he not only understood how to destroy, but also how to build up. When we consider that, after becoming Sultan of Bornu, he endeavoured energetically to raise the

condition of the country by a whole succession of peaceful measures, we can reconcile ourselves in some degree to his misdeeds; in fact, we may summarize our judgement of him by saying that he was—in the Shakespearian sense—a man. If one looks at the matter properly he may be described as the leaven of the Sudan; for Bornu, in particular, his mission was as much an historical necessity as the migration of nations was for the Roman Empire. As to whether his dynasty will have any permanence, who can tell? If it perishes with him, the fault lies not on him but adverse circumstances.

All that I have been able to discover as to the course of events after his death, is that one of his freedmen has undertaken the regency in the name of his son, who is still a minor, and that he has known how to guard up to the present the integrity of the empire. When M. Bonnel de Mezières,^[532] and after him the newspapers, announce that Rabeh has banished Gaurang, King of Bagirmi, and occupied his capital, Masseña, it can only refer to this regent, even assuming that the news in this form proves right at all. At the same time it is not impossible that the conqueror is the King of Wadai, who certainly stands in the position of suzerain of Bagirmi.^[533] Certainly once before, when the vassal chief, Abu Sikkin, showed himself recalcitrant, Masseña was occupied by King Ali of Wadai after a siege lasting several months.^[534] In any case it is an exquisite irony of fate to see how the actors of the drama produced here are intermingled with one another, and behave themselves as if there was no world outside their own, whilst all the time the die has long been cast against them, and they have been divided up among the European Powers.

HISTORY OF RABEH ACCORDING TO MOHAMMED BESHIR

usili rabeh. ubansa mutum jalaba shi da basha alzubair mutane jalaba. uban rabeh shina yi gini kasa, talaka ne. babu shina da komi. rabeh ya tafi wurin basha alzubair ya zama galadimansa. suna tafiya darfur suna yi yaki dasu shekaru biyar. su ji labarin mutane masr su da ingliz suka tafi achikin darfur. su yi masa hila wayo suka che ka tafo. ya zo basha alzubair, su kama shi. galadimansa rabeh ya gudu ya tafi agarin janke (dinka) ya tafi shi da

dakaransa shina yi yaki achikin janke shekaransa oku. ya tashi ya tafi agarin wadai shina yi yaki kwanansa goma sha bokkoi achikin kauye. ya tafi agarin bagirmi shina yi yaki da sarikin bagirmi shekaransa biyu. achikin bagirmi shina aikewa achikin barnuh^[535] da samo shina yi magani dakeau domi bakinsu shi banban suna yi fada. ya rubutu ga malam hayatu laya ya che masa ina sonka da abuta. mallam hayatu ya che kua, nima ina sonka amma ka zo mu yi shawara zamu chi barnuh. rabeh ya che to mu gamu achikin hanya. ya tashi su gamu suka che mu je achikin barnuh mu chi barnuh. sarikin barnuh ya ji labari suna taruwa dayawa ya che ku yi shiri dakeau. rabeh ya zo barnuh achikin kofan gari, sarikin barnuh ya fita suna yi fada, sarikin barnuh ya kore rabeh. ya tafi akauye ya zamna, shina yi yungwa watansa tara achikin ankalaiwa suna zamne wuri daya. da su samau chimaka su komo achikin barnuh sun yi yaki. sarikin barnuh dakaransa su gudu suna barshi shina yi fada su kasheshi; aba hashim dafari aba kiari da aba sanda dan uwan aba kiari, su mutu. ya shiga barnuh ya zamna shekaransa oku da shashi. ya tafi agarin sarikin mandara ya kai yaki achikin mandara. ya kama sarikin mandara zofo ne ya kawoshi achikin barnuh. yaro sarikin mandara ya kama yaron rabeh. rabeh ya che da mutaninsa kada ku kasheshi wanan zofo ne. ya che masa yaron sarikin mandara aiko mani da ubana. ya che masa zan kasheshi zofo ya che, kadan na mutu da. safi kai kua ka mutu da azafari.^[536] ya sakeshi ya tafi agarinsa mandara.

saura maganan rabeh. basha malla karim na chikin barnuh ya aike^[537] wurin rabeh ya che, ka zo ka shiga achikin garin barnuh. ya che ina da yungwa babu abinchi. sun tafo su shiga achikin gari ya zama sarikinsu. ya zauna watansa fudu ya tashi ya tafi achikin garin bade shina yi yaki da su, allah ya bashi nasara. ya kama mutane bade ya damresu achikin sasari ya kawosu achikin barnuh. ya koma ya tafi agarin mika suna yi yaki da sarikin mika. allah ya bashi nasara ya kashesu. ya tafi agarin katagum suna yi fada dakeau. sarikin katagum ya koreshi ya gudu ya komo achikin garinsa barnuh. ya zamna watansa biyar ya tashi ya tafi agarin miso. sarikin miso ya fita suna yi yaki shi da rabeh. sarikin miso ya aike wurin sarikin shira da wurin sarikin gombe su taru achikin garin miso. su tashi su tareshi achikkin hanya. suna yi yaki dakeau ya kashe masu mutane dayawa rabeh ya yi nasara dakeau. sarikin gombe ya mutu achikin yaki. rabeh ya komo achikin barnuh, ya zamna watansa tara. ya che ga mutaninsa ku yi shiri ku tafi

achikin adamawa da yaki. su tafi su yi yaki achikin adamawa su kama mutane dayawa su damresu achikin sasari su kawosu daga barnuh. malam hayatu sarikin jamare ya ji labari ya che masa domi ka tafi achikin kasan kakanina? rabeh ya amsa masa na tafi. malam hayatu ya che, dakeau. ya aike masa da magani wurin matatasa yariniya malam hayatu shika aureta. ama rabeh shekaransa tokwos da hamsin ya mutu.

TRANSLATION

Rabeh's origin. His father was a Jellaba man as also was Zobeir Pasha. Rabeh's father was a poor man and a clay digger (for making bricks); he had no property. Rabeh went to Zobeir Pasha and became his lieutenant. They went to Darfur and made war there for five years. They heard news that the Egyptians and the English had come to Darfur. They (the English) laid a cunning trap for him (Zobeir) and said, 'Come to us.' They caught Zobeir Pasha. His lieutenant Rabeh fled and came to the Dinka country. He and his soldiers went and made war in the Dinka country for three years. He got up and went to Wadai and made war in the surrounding country for seventeen days. He went to Bagirmi and made war with the King of Bagirmi for two years. Whilst he was in Bagirmi he sent poison to Bornu and made powerful medicine because they were at strife. They made war. He wrote a letter to Mallam Hayatu and said to him: 'I want to be your friend.' Mallam Hayatu likewise said: 'I like you, but come and let us consult how we shall conquer Bornu.' Rabeh said: 'Very well, let us meet on the road.' He got up. They met and said: 'Let us enter Bornu and conquer it.' The King of Bornu heard news that they were assembling in large numbers and said (to his people): 'Make careful preparations.' Rabeh came to Bornu and reached the gate of the town [*i.e.* Kuka]. The King of Bornu came out and they fought. The King of Bornu repulsed Rabeh, who went into the surrounding villages where he sat down. He suffered from hunger for nine months in Ngellewa where they remained concentrated. When they got provisions they re-entered Bornu and made war. The King of Bornu's soldiers fled and left him on the battlefield and he was killed, first of all Abba Ashimi died, then Abba Kiari and Abba Sanda, brother of Abba Kiari. He (Rabeh) entered Bornu and sat down there for three years and a

half. He went to the King of Mandara and brought an army to Mandara. He caught the King of Mandara, who was an old man, and brought him to Bornu. The son of the King of Mandara caught a son of Rabeh's. Rabeh said to his people, 'Do not kill him, he is an old man.' The son of the King of Mandara said to him, 'Send me my father.' He said to him, 'I am going to kill him.' The old man said: 'If I die in the morning, you will die in the afternoon.' He let him go and he went to his country of Mandara.

The story of Rabeh continued. Pasha Malla Karim of Bornu sent to Rabeh and said, 'Come, enter the town of Bornu.' He said: 'I am hungry, I have no food.' They met, they entered the town, he became their king. He sat down for four months. He got up and went to Bedde country. He made war with them. God gave him the victory. He caught the Bedde people and bound them in chains and brought them to Bornu. He came back, he went to Mika^[538] country. They made war with the King of Mika. God gave him the victory. He killed them. He went to Katagum. There was a stiff fight. The King of Katagum repulsed him. He fled and came back to his own country of Bornu, where he sat down for five months. He got up and went to Misau. The King of Misau came out and he and Rabeh fought. The King of Misau sent to the King of Shira and to the King of Gombe (telling them) to assemble inside the town of Misau. They got up and met on the road. There was a stiff fight, but Rabeh killed many of their men and gained a brilliant victory. The King of Gombe died in the battle. Rabeh returned to Bornu, where he sat down nine months. He said to his people: 'Prepare yourselves and go with an army to Adamawa.' They went and made war in Adamawa. They caught many men and bound them in chains and brought them to Bornu. Mallam Hayatu, the King of Jamare, heard the news and said to him: 'Why have you entered the country of my fathers?' Rabeh answered him: 'I have entered it. (What about it?)' Mallam Hayatu said: 'Good!' He sent him poison through his wife. Rabeh had married Mallam Hayatu's daughter. Now Rabeh was fifty-eight years old when he died.

FROM A LETTER OF ABD-EL-KADIR IBN AL-HAJI ISA MUZI OF GHADAMES TO
MOHAMMED BESHIR, dated 17 Rejeb, 1315 (*i.e.* 12 December, 1897).^[539]

. . . And if you inquire as to the state of affairs in the Sudan, know that everything is in order. Only a letter has reached Ghadames from the Sudan, in which they have informed us that Rabeh has of a truth been killed by poison. The Sultan of Jamare said: ‘Assemble yourselves, ye Sultans of the Sudan.’ They said: ‘Let us prepare ourselves and fight against Rabeh.’ And the Sultan of Mandara and the Sultan of Bagirmi and the Sultan of Kano and the Sultan of Katsena and the Sultan of Daura and the Sultan of Kazaure and the Sultan of Zaria and the Sultan of Bauchi and the Sultan of Bedde and the Sultan of Damerghu and the Sultan of Tanamari and the Sultan of Gagiduna and the Sultan of Asben they all assembled themselves (and said): ‘Let us march against Rabeh and kill him.’ His father-in-law, Mallam Hayatu, was he who killed him by poison. He sent the poison to his daughter with the message: ‘Give it thy husband.’ After that^[540] he said to his son-in-law: ‘Why have you taken the land of my fathers?’ Rabeh answered and said unto him: ‘I have taken it.’ Then he went away, prepared the poison and said to her: ‘Give it to thy husband.’ And now Rabeh is dead. It is six months ago since I read it in the letter from the Sudan.

[p. 263] FROM A LETTER OF HERR HANS RESENER, Editor of the *Aegyptischer Kurier*, to the Editor of the *Mittheilungen*.

. . . Herr Julius Lippert in his article on Rabeh, the notorious ruler of the Kingdoms of the Central Sudan, which is shortly to appear in your valuable periodical, states: ‘his father was a poor brickmaker, who gained an honest livelihood by the work of his hands. That Rabeh has sometimes been falsely described as a slave of Zobeir’s is to be attributed to the fact that he attached himself in early youth to that extraordinary man, who prior to his appearance as a public official carried on the business of slave-raiding. Rabeh early became a loyal friend and useful assistant in Zobeir’s undertakings.’

I may be permitted to observe on the contrary that Zobeir Pasha has more than once represented to me that his relationship to Rabeh was as

follows—Rabeh’s father was a slave of Zobeir’s father. The lot of a slave in Africa who happens to be in good hands is by no means hard, he eats out of the same dish as his master, &c., &c. Rabeh was brought up in familiar intercourse with Zobeir without any distinction or preference being made; the fact that Rabeh was a slave of Zobeir’s father did not prevent this. At the same time Rabeh was Zobeir’s foster-brother; Zobeir always had a great affection for him and when his (Zobeir’s) father died, he gave Rabeh his freedom.

[Translation published by kind permission of Dr. Sachau, Director of the Seminary. The name is spelt Rabah by the French and Germans, Rabeh by the English. The latter form gives the Bornu pronunciation, but perhaps the former is the correct Arabic form. The town of Calatrava in Spain, the head-quarters of a famous order of religious knights, is properly Kalat Rabah = Rabah’s citadel, vid. article on Calatrava in *Encyclopaedia of Islam*.]

This certainly holds good in general for Mohammedan nations, in so much as with them the conception of the state is identified with that of the ruler. Any one who requires proof of this, let him read any twenty pages whatever of Aug. Müller’s *Islam*, vol. ii; he will then be convinced.

‘dan’ in Hausa is the Arabic ‘ibn’ (son).

Cf. G. A. Krause’s ‘Samori’, in No. 483 of the *Vossische Zeitung* for October 15, 1898.

Vid. *Le Mouvement géographique*, 1898, No. 47 (Nov. 20).

Cf. Nachtigal, *Sahara und Sudan*, ii. 234.

They had to repent of not having followed Rabeh’s advice; for in spite of the assurances made to them they were massacred in an altogether assassin-like fashion on July 15, 1879, owing to calumny and intrigue (cf. Slatin Pasha, *Fire and Sword in the Sudan*, p. 28, German edition). [‘I caused Suleiman and nine of the principal rebel chiefs to be shot, for Suleiman, after a futile attempt to make my troop revolt, had tried to escape,’ vid. Romolo Gessi, *Seven Years in the Sudan*, p. 329.]

Cf. Slatin Pasha, *idem*, pp. 7-28.

[The Bornu troops had included gun-men since the sixteenth century (vid. [p. 17](#)). The Kanuri word for gun is ‘bendege’, which has usually been derived from Venediga = Venice, the original source of supply. Huart, however, in the *Encyclopaedia of Islam* under ‘bundukiya’ derives it from Latin ‘(nux) pontica’ = a hazel-nut, hence a bullet. He gives ‘bunduki’ = a Venetian sequin, as coming from ‘Bundukiya, the Arabic name of Venice’.]

Ferryman, *Imperial Africa*, London, 1898, p. 354.

Sahara und Sudan, ii. 141.

Fire and Sword in the Sudan, p. 25.

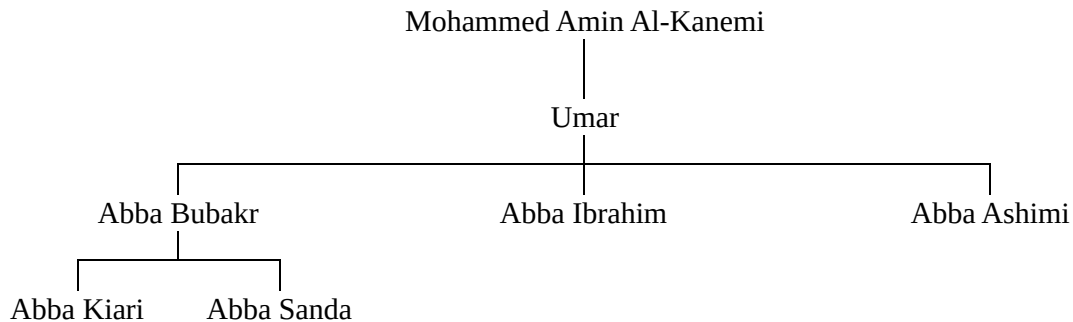
Imperial Africa, p. 355.

Cf. Nachtigal, *Sahara und Sudan*, iii. 51 ff.

Imperial Africa, p. 356.

The Hausa word Mallam (or Mallami) is, of course, the Arabic ‘mumalim’ = teacher. It is employed like the Arabic ‘Imam’ (in Moroccan Arabic ‘Faki’) for the ‘scribes’.

Sultan is here used in the broader sense; the rulers of the latest (or Kanem) dynasty in Bornu made use of the official title of Sheikh. Their names in succession are: Mohammed Amin Al-Kanemi, Umar, Abba Bubakr, Abba Ibrahim, Abba Ashimi, Abba Kiari, and Abba Sanda. Their relationships are shown in the following table:



‘Abba’ corresponds in this connexion to the Arabic ‘Saïd’. Cf. Nachtigal, *Sahara und Sudan*, i. 582. [Abba is a Bornu title meaning ‘Prince’.]

According to Ferryman, *Imperial Africa*, p. 356, Abba Ashimi was not killed, but after the loss of the battle fled to Zinder. [As a matter of fact he was murdered by his nephew, Abba Kiari.—P. A. B.]

‘Sanda’ is the Arabic ‘Umar’.

According to my calculation the conquest of Bornu took place in the year 1894.

Cf. Grothe, *Tripoli and the Caravan-trade with the Sudan*, Leipzig, 1898, p. 22.

Ferryman, *Imperial Africa*, p. 356. [A Niger Company mission under M^CIntosh (‘Sarikin Charlie’) visited Shehu Ashimi in 1890, starting from Yola, but were unable to conclude the commercial arrangements which formed their object.—P. A. B.]

Cf. Nachtigal, *Sahara und Sudan*, i. 732 ff.

Dikoa, which in earlier days was once a place of residence of the Bornu kings, lies in the German sphere of influence. With regard to the favourable situation of the town cf. Barth, *Travels and Discoveries*, iii. 122 ff. German edition. The natives in their superstitious fashion give as a ground for the change of residence that the ghosts of the Sultans murdered there appeared to Rabeh by night in the royal palace at Kukawa. This motive is not new; thus Mohammed Sherif, Sultan of Wadai, removed his residence from Wara to Abesher, because, as popular report had it, ‘bad ghosts had made the old royal town uninhabitable’ (Nachtigal, *Sahara und Sudan*, iii. 77). Of course this is not to be taken seriously. New ruling families are frequently accustomed to select new places of residence in order to wipe out as far as possible remembrances of the old dynasty, just as Kukawa itself was first founded by the Kanem dynasty at the beginning of the nineteenth century (Nachtigal, i. 586).

[Fika?]

Belonging for the greater part to the German sphere of influence.

Hayatu was the son of Saïdu, grandson of Sultan Mohammed Bello, and consequently great-grandson of Othman dan Fodio, the founder of the empire. (Wurno, formerly the residence of the

Sultan of Sokoto, is a town a few miles north of Sokoto.)

Cf. Nachtigal, *Sahara und Sudan*, i. 730.

[This is incorrect.—P. A. B.]

Vid. Introduction to this article.

Cf. Nachtigal, *Sahara und Sudan*, ii. 712 ff., iii. 281.

Idem, ii. 726.

This spelling is common and refers to the etymology of Bornu = the land of Noah.

Corruption of 'azuhur'.

Misprint in the original of t for i.

[Fika?]

This letter, written in the Tripolitan dialect, is printed with all its errors. My conception of the meaning is set out in the accompanying translation. [As I know no Arabic, I can only give a translation of the German translation.—P. A. B.]

The sequence is confused here.

APPENDIX VI

THE BAUCHI ACCOUNT OF SHEHU LAMINO'S KANO CAMPAIGN^[541]

Extract from a *History of the Bauchi Emirs*, translated from the manuscript of Mallam Mustapha, tutor and adviser to the sons of Yakubu, first Emir of Bauchi, by A. Holdsworth Groom, Assistant Resident, Central (Bauchi) Province, Northern Nigeria.

AFTER this he (Yakubu) made a war and went to Dass. He remained there five years, and fought with them and conquered half of them. He was fighting with the other half when the news of Kanumbu (Shehu Lamino of Bornu) reached him, and he started and went to Fake. They met Kanumbu, whose horsemen numbered 44,444. He had driven away the village chiefs of Kano, the chiefs of Kano city, and the Emir of Kano, Dabo. All had fled. Yakubu came and found him encamped. When Kanumbu saw him he said: 'This slave of the Fulani is a liar, what does he think he will take from off me?' Yakubu said to his mallams: 'What is your advice? Ye behold the multitude of his horses. Ye behold also the number of my horses, 2,500. What say ye?' They replied, 'The king's word is ours.' Yakubu said, 'I know not how to defeat him, neither do I know how he may be killed, but one thing I do know, I know that he has no power to raise the dead, that he has no power, if rain be lacking, to cause it to fall, that he has no power, if the grass does not spring up, to cause it to do so.' The followers of Yakubu said, 'As for these things Allah alone is able to do them.' Yakubu replied, 'Since ye know this, that no one is able to do anything save Allah, let us take courage and fight with him, for we shall defeat him and kill him, because no one is powerful save Allah.' Then they joined battle and slaughtered them, and scattered them.^[542] Kanumbu fled, leaving behind him his flag, his tents and drums, and went away to Bornu. Yakubu started and went to Sokoto and brought to the Sarikin Mussulmin, Bello, son of Shefu dan Fodio, Kanumbu's flag, his tents, and his drums. Bello rejoiced greatly and

made a prayer for Ibraihim, son of Yakubu, that Allah would give him the kingdom of his father, that Allah would give the children of Ibraihim the kingdom of Ibraihim;^{[\[543\]](#)} that Allah would establish the seed of Ibraihim in the house of Yakubu. Yakubu returned and came home.

Cf. *Documents scientifiques de la Mission Tilho*, vol. ii, p. 363.

The Habe Archers that followed Yakubu, famous for their shooting and their poisoned arrows, are said to have contributed largely to the victory.

The present Emir of Bauchi is the grandson of Ibraihim.

APPENDIX VII

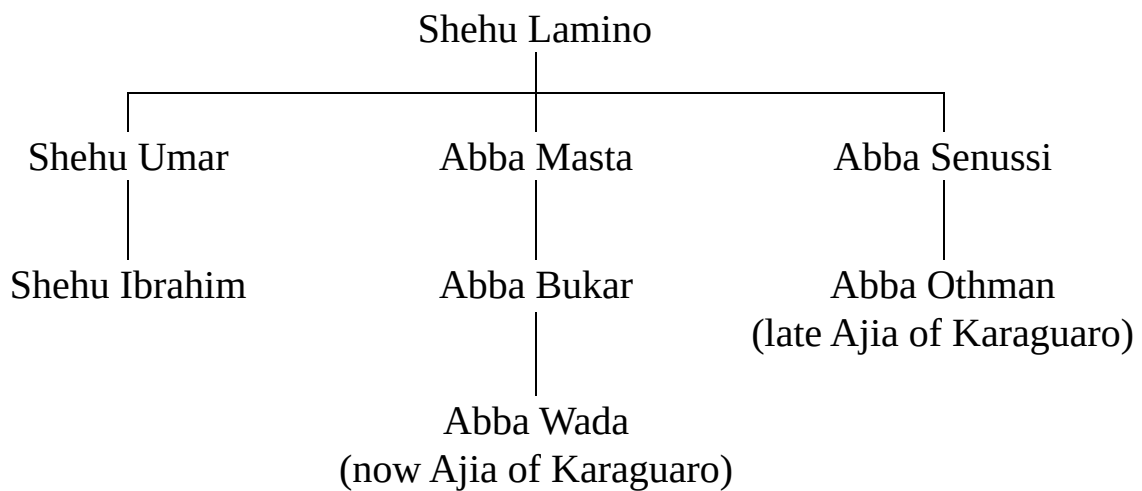
REBELLION OF ABBA MASTA

Extract from 'Travels of Haji Mahomet, a Fulani of Kano', translated from *Hausa Stories and Riddles*, by H. G. Harris, p. 33. [\[544\]](#)

WE reached the King of Bornu. There was a civil war going on. One of the Bornu princes was fighting with the King of Bornu, and trying to take the kingdom from him. This prince, whose name was Abba Masta, had 100,000 men and 40,000 horses. The King of Bornu's name was Shehu Ibrahim, he had 100,000 soldiers and 100,000 slaves called 'Kachellas'. Abba Masta refused to obey him. The Shehu said: 'Why do you refuse, you are my younger brother? Is it a question of politics? If you want a title, I will give it you.' But his courtiers said: 'No, don't give it him, consider your own interests. Is Abba Masta your elder brother, will you obey him?' But the Shehu would not listen to them, he was angry, he rose up, shook out his clothes, went to his house, and beat his drum, proclaiming, 'He that is my friend, let him follow me'. . . . 40,000 horsemen left the town and the Shehu was preparing to follow them, when his official sister, the Magira, came and caught hold of his gown, saying: 'Do not go out now, wait a while, let them go. Do you wish to injure your own interests? When they have gone, we will follow on behind.' So the army went on, and the Shehu stayed two days at home. Then he beat his drum, and said: 'Let all my friends rally round me'. They assembled, and followed him. He met Abba Masta, and sent four 'mallams' with a message to him saying, 'Let us not quarrel over politics, which are but transitory things, let us not fall out over them.' Abba Masta replied that he intended to fight, so the 'mallams' returned to the Shehu and said: 'Abba Masta is evilly disposed; he intends fighting.' The Shehu sent back the 'mallams' with another message, saying, 'Let us not fight, let not Mussulman blood be shed.' But Abba Masta continued to advance, and was close on their heels. The alarm was given in the Shehu's camp, and they opened fire. Thereupon, a certain young man

came to the Shehu and said: 'I will kill him.' The young man took his spear and pierced Abba Masta through the ribs. Abba Masta fell from his horse; he said to his followers, 'Fly, the battle is lost.' Thirty thousand of his followers were slain, some fled to Mandara, some were captured and brought to the Shehu, and the rest scattered. People said, 'Abba Masta has failed to seize the kingdom.' The Shehu said, 'Kill the prisoners.' They were killed. There was a pursuit. The defeated army fled screaming. The Shehu had the victory. He returned home, assembled his people, and said, 'Give judgement between me and Abba Masta, which of us was in the wrong.' They said, 'He was in the wrong. He refused to obey you, you tried to make peace, but he refused. He behaved like a heathen.' Then the Shehu beat his drum, and ordered Abba Masta's men to be brought before him. A court of justice was held, and some were killed and some spared, in accordance with the Shehu's orders. Five thousand were executed, and the people said, 'They have met with their deserts.'

The following genealogical table explains the relationships of Abba Masta:



Translation published by kind permission of Hermann G. Harris, Esq.

APPENDIX VIII

METEOROLOGICAL OBSERVATIONS IN BRITISH BORNU

THERMOMETRICAL and barometrical observations have been taken in Bornu for some years past by the Government Medical Officers; at Maiduguri in Southern Bornu from 1905, and at Geidam in Northern Bornu from 1907. The observations in all cases are taken at 9 a.m. in the shade of an open grass-roofed shed, with maximum and minimum thermometers.

Average monthly temperature, ascertained by taking the monthly averages for the years recorded.

<i>Month.</i>	<i>Geidam. Maiduguri.</i>	
	°	°
January	70·65	74·28
February	75·6	76·9
March	84·13	84·78
April	89·41	91·77
May	90·58	90·62
June	89·81	86·42
July	84·6	82·31
August	80·29	78·76
September	83·77	81·46
October	84·48	83·51
November	77·66	79·17
December	71·16	73·88

Average yearly temperature.

<i>Year.</i>	<i>Maiduguri.</i>	<i>Geidam.</i>
	°	°
1905	80·9	—
1906	82·14	—
1907	82·68	81·01
1908	83·02	81·35
1909	82·83	81·57
1910	78·56	82·23

The record maximum temperature of 120° was obtained at Maiduguri on April 6, 1907; the record minimum temperature of 41° at Maiduguri on March 30, 1910; the record fluctuation of 61° at Maiduguri on March 10, 1910.

The record maximum temperature at Geidam was 119° F. on March 28, 1911, and the record minimum 44° F. on January 8, 1912; the record fluctuation at Geidam was 51° F. on May 1, 1911, and January 24, 1912.

RAINFALL

<i>Year.</i>	<i>Rainfall at Maiduguri.</i>	<i>Rainfall at Geidam.</i>
	<i>Inches.</i>	<i>Inches.</i>
1905	26·04	—
1906	31·65	20·80
1907	23·63	18·15
1908	23·30	22·39
1909	31·99	21·22
1910	19·53	16·78
1911	30·00	10·46

The record rainfall for 24 hours was 3·87" at Maiduguri on August 27, 1911.

[I am indebted for these figures to the kindness of Drs. Ellis, McLeay, Moiser, and Cobb of the West African Medical staff.—P. A. B.]

DIRECTION OF PREVAILING WIND

According to the observations taken at Maiduguri, which is just below 12° North latitude, the prevailing winds are as follows:

January, ENE.
February, NE.
March, E.
April, SW.
May, SW.
June, SW.
July, SW.
August, SW.
September, SW.
October, NE.
November, NE.
December, E.

APPENDIX IX

LAKE CHAD AND THE BAHR EL GHAZAL

THE French Mission under Captain Tilho devoted a good deal of attention to the problem of the Bahr el Ghazal. Their conclusions are summed up on p. 61, vol. ii of the *Documents scientifiques*: 'The Bahr el Ghazal, as far as we examined it (i.e. to Fantrassou), is neither an affluent nor an effluent but simply a prolongation of Chad. . . . This conclusion does not solve the problem of the mythical river, which, according to old maps, formerly connected Chad with the Nile.'^[545] Twelve varieties of shells are common to Chad and the Nile, and indicate that there must at one time have been a connexion by water between the lake and the river (*Doc. sc.*, ii. 170); cf. also note ^[343] of the present work.

Again, on p. 601 ff., vol. ii: 'Topography and hypsometry on the one hand, chemistry, geology, and malacology on the other, agree in declaring that Lake Chad is not the remains of an ancient sea, that it is hardly a lake, in spite of its appearance of being an alluvial lake, and that it should be regarded as the expansion of a powerful sheet of fluvial water flowing north-eastwards, rather than a true lake. . . .

It appears legitimate, *a priori*, to suppose that this sheet of fluvial water, after having expanded to form Chad, may continue its course subterraneously, protected by a mantle of sand against the intense evaporation occasioned by the burning sun of the tropics and the parching winds of the desert. Besides, have we not in the Sahara many other examples of water-courses prolonged subterraneously very far from the place where they seem to lose themselves in the sands? . . .

Moreover, if topography and hypsometry can only show us the possibility of the waters of Chad being prolonged subterraneously, chemistry demonstrates the necessity. The amount of chloride in the waters of Chad is in fact so small, that one is bound to admit that there is an outlet somewhere. . . . Chemical analysis has shown that the water of Chad almost

everywhere contains less chloride than the drinking water of Copenhagen for example. Consequently, it is absolutely impossible that it can be a self-contained lake. . . .

It is impossible to definitely trace the course of the connecting link between the Chad and the Nile, assuming that it existed, . . . until French Borku and English Darfur are really brought under control.'

Cf. the article on the connexion between Lake Chad and the Nile in the *Journal of the Royal Geographical Society*, vol. xxxviii, p. 463. [Commandant Tilho is now in charge of the Mao district to the north of Lake Chad and is making a special study of the problem of the Bahr el Ghazal. Chudeau, *Missions au Sahara*, vol. ii, p. 232, discusses the problem of the Bahr el Ghazal.]

APPENDIX X

SOME KANURI DERIVATIONS (MOSTLY PROPER NAMES)

ABOUT half of these derivations were given me by the late Major A. M^cClintock, D.S.O., Acting Resident, Bornu, with a request that I would revise them. I have added a few notes in square brackets [] and some derivations of other proper names, also in square brackets. Some of the derivations are obviously ‘aetiological’, but others are, I think, correct. Major M^cClintock heads his collection—‘Lies and Otherwise’.

P. A. B.

[**Afuno** (the Kanuri name for the Hausas). King Iderisa Aloma of Bornu went to war with Kano and found all the people there wearing nothing but arse-cloths (‘funo’). He therefore called them ‘am-funo’ = arse-cloth people. Cf. Benton, *Kanuri Readings*, p. 19. For other derivations vid. Merrick, *Hausa Proverbs*, p. 92.]

[**Allanjurori** = the place of Lowan Allanjuro.]

Alo. Mai Iderisa, a Bornu king, was at war in Bagirmi; during a fight he was standing under a tree; up in the tree hidden by the branches was one of the enemy, who threw his knife at the king and wounded him very badly. The Bornu army returned by easy stages with their sick king via Dikoa to the lake now known as Alo. Here the king died from his wounds and was buried. After his death there was much lamentation, until the king’s son said, ‘It is the work of Allah’ [‘Allah kido’ = God did it]. All the people then chanted the word Allah and the lake has since been called Alo.

[Another explanation suggests that the word is derived from the Tubu ‘alo’, meaning a water-melon. When Mai Iderisa died, the melons with which the place was covered burst asunder from grief and formed a lake.]

[**Ardimini.** A town in Margawa District. This word means in the Koiyam language the same as the Kanuri ‘arditsani’ (3rd pers. plur. negative indefinite from ‘ardingin’) = they do not agree. The Koiyams are a pastoral people who do not care to settle down.]

Argum = millet. Birds were responsible for carrying this seed and dropping it in Kanem, at Birnin N’gebe, where the Bornu kings lived prior to their coming to Gashergomo. The seed these birds brought was taken to the king, who ordered horsemen to follow their flight and find out where it grew. It was found in the Gongola valley, and they brought some back to the king, who had it prepared and ate it. He then said, ‘Hold this safe and plant it.’

The word ‘argum’ or ‘arkum’ is a corruption of the Kanuri ‘arre kemne’ = come, hold it fast. [‘Arre’ is imperative from ‘iseskin’, and ‘kemne’ or ‘kumne’ is imperative from ‘kemgin’ = I close the fist.]

[**Arriaskuri.** Name of a village opposite the Government station at Maiduguri, said to have been founded by one Arri Yasku-ma, i.e. Arri from Yasku, a town near Lake Alo now abandoned.]

[**Bonigaral.** This was a slave village exempted from tax: its inhabitants could therefore take their ease. ‘Bone’ is 2nd pers. sing. imperative from ‘bongin’ = I lie down; ‘ngaral’ is an emphatic adverbial suffix, cf. Koelle, *Kanuri Grammar*, p. 283.]

[**Bornu.** Or, as the natives write and pronounce it, Barnu. The usual etymology given by the natives is that it is a corruption of the Arabic ‘Bahr Nuh’ = the water of Noah, because the ark was supposed to have rested on the rocks of Hadjer Hamis on the south bank of Chad.^[546] Another is that it is derived from Kanuri ‘burum nui’ = the lake is dead, has disappeared, pointing to a shrinkage of Lake Chad; cf. [Appendix XIV A.](#)]

[**Borsari** = the place of Kachella Mbursa, grandfather of Kachella Mahmut, ex-Ajia of this district.]

Bussugua = the fig-trees [parasitic *figus*]. There is one in Ngubala rest-camp. [This is where the Major died, June 24, 1912:

Νιγηρία γὰρ οὔποτ' οὐδέν' ἄνδρ' ἐκὼν
αἶρεϊ πονηρόν, ἀλλὰ τοὺς χρηστοὺς αἰεί.

P. A. B. and H. B. H.-H. with apologies to Sophocles' *Philoctetes*, line 436.]

[**Damasak**. Short for 'mbarena damasak' = we are tired, worn out. 'Mbarena' is 1st pers. plur. present participle of 'mbareskin'; 'damasak' is an intensive adverbial suffix, said to be only used with the verb 'mbareskin' and only with the 1st pers. plur. of that verb. The town of Damasak was founded by pilgrims returning from Mecca.]

[**Debira**. For 'dabbi-ri' = the place of the long-handled hoe. It is a blacksmith's village.]

Dikoa (vid. **Alo**). When Mai Iderisa was returning to Birnin Gashermo via Dikwa he left some of his people there. The ruling potentate, an ally of Bornu, said, 'Yes, you can stay here, but I mark a boundary for you.' He then marked out the site of the present town of Dikwa, which in the Magari [or Mackeri] language means 'a marked out place, a boundary'. [Another suggested explanation connects it with the Kanuri word 'digo-wa' = grandsons.]

Galadima, from 'galti', meaning to finish his job properly, i.e. if in war a man was told to capture a village and did so properly the verb 'galti' applies.

Hence Galadima, the man who does things properly, who can be trusted to carry out affairs.

[For another derivation vid. Benton, *Kanuri Readings*, p. 25.]

[**Gazerregomo** (Kasr Kumo). Vid. note in [Appendix III](#), p. 248.]

[**Geidam**. A certain man came and sat down by the river. People asked him, 'Why do you sit down here all alone?' He replied, 'Ngai-dam namgin,' literally, 'thus exactly I am going to sit down.' 'Dam' appears to be a semi-enclitic specific adverb of the kind described by Koelle, *Kanuri Grammar*, p. 283. An American might translate it 'right here'.]

Grema. From Kanuri [root 'gere'] = to eat. The story is that one of the old kings of Bornu insisted when riding about the country on tasting all wild fruits and everything that could be eaten. He had one 'boy', who was particularly good at bringing him these things. This boy died, the king felt his loss, he therefore appointed a man who was to do this work. Hence the name 'Grema'.

[**Gujiba.** A certain king's son went and sat down on this site. The king summoned him but he refused to come. The king said, 'Nguji-ba?' = has he rebelled?

'Guji' or 'nguji' is an impersonal verb meaning he rebels, he is obstinate; '-ba' is the interrogative particle.]

[**Gusamalla.** At the time of the capture of Gazerregomo by the Fulani, a 'mairam' (princess) came and sat down where Gusamalla now is. Fugitives ('nguzama' from 'nguzaniskin') joined her and formed a town. The '-la' is for the postpositive '-lan', which is here to be translated as a genitive, cf. Koelle, *Kanuri Grammar*, p. 300. The name therefore means '(the town) of the fugitives'.]

Hausa, from the Hausa word 'haushi': change the *h* to *f* you get 'fushi', which means to be angry; vid. Robinson's *Dictionary*, p. 88, 'He feels angry' = 'yana yin haushi'.

The story is that on the first invasion of this country by the Hausas they met with opposition they did not expect, and were naturally angry about it. They said, 'These people (original inhabitants) make me angry' = 'Ba-Fushi'. *F* comes roughly off the tongue, *H* is the accepted difference. 'Haushi,' between the *f* and the *u* a vowel must come or no black man could say the word.

I now say further that the much disputed derivation of this word Hausa is now solved for any one who knows anything about the matter. [!]

[The Major was always very hot on this being the correct derivation. For other explanations of Hausa vid. Merrick, *Hausa Proverbs*, pp. 91-3, and Robinson's *Hausa Dictionary*, second edition, p. 88 a, and Tremearne, *Niger and West Sudan*, pp. 51-64.]

[**Kaba.** A Marghi town, said to have been founded by a brother of the Marghi king, Mai Ibrahim, who broke away from him and founded a town of his own. ‘Ka-ba’ = no stick, i.e. without proper authority.]

Kachella, from Katsaga = a spear [and ‘-la’ or ‘-lan’ = on. ‘Kachella’ is a military title, they are those who sleep on their spears.]

Kagaburi, the name of the country between Mobber and Geidam, i.e. between Dutchi and Geidam. ‘The country of the fools’. ‘Kagabu’ in Kanuri meaning a fool.

[**Kaiuri**, the place of the ‘Kayo’ (Hausa, ‘tumfafia’ = *Asclepias gigantea*).]

[**Kanembu**, the people of Kanem, ‘-bu’ being the plural form of the adjective in Kanuri; cf. Koelle, *Kanuri Grammar*, p. 34.^[547]]

[**Kano.** The Sau who marked out Kano said to the people, ‘Is it enough for you?’ They said, ‘Yes.’ ‘Well then,’ said he, ‘pass along (“Konogo”) and enter in.’ Cf. Benton, *Kanuri Readings*, p. 26.]

Kanuri, made from the old Sau, now Beri-Beri [Kanuri] words, ‘Kam’ = a man, ‘nur’ = light-coloured, ‘-ri’ = place of or possession of. This was the name given by the Saus to the first arrivals from the North, meaning literally, ‘the light-coloured men.’ These men, the first Mohammedans in Bornu, were called Beri-Beri owing to their long sojourn in Berber before crossing the desert to the fertile Chad and its neighbouring rivers. The language of these people was the language of Berber and Arabia. On obtaining possession and settling down to Islamize the country, they adopted the language of the Saus, the now more or less Kanuri language.

[Another derivation connects Kanuri with Kanem, the country via which the Kanuri entered Bornu. Neither derivation explains why a tribal name has the place termination ‘-ri’ in it. I do not know what authority the Major had for saying that ‘-ri’^[547] could imply ‘possession of’.^[547]]

[**Karaguaro.** This district is ‘bush’. ‘Karaga’ = bush, ‘Karaga-wa’ = belonging to bush, ‘Karaga-wa-ro’ = to the bush (place).]

Karda, people living in small hamlets outside proper Kanuri villages, so called from the Kanuri word ‘Karta’ [‘Karngin’] = to separate. They were

originally goat and sheep-herds, and therefore of necessity lived in the 'bush'.

Konduga. There is close to Konduga a village named Salalari. In this village lived a hunter and his son. The son, knowing that the marsh close to our present Konduga was a good place for game, told his father that he wanted to go and live there and hunt. At this the father was angry and told the boy, 'If you do, you don't come back.' That is the meaning of the name. 'Kondi' or 'Qwondi' in Kanuri meaning 'to sit apart' or 'sit down by yourself or on your own'.

[Konduga or Kondega is short for 'keḷa-nem kong dega', literally, 'by yourself abide.' 'Kong' is an emphatic suffix or specific adverb; cf. Koelle, *Kanuri Grammar*, p. 283; 'dega' is imperative from 'degaskin' = to enter in, to abide.]

[**Kwalme Shuwas.** Vid. [Appendix XIV.](#)]

[**Lagarete.** Lagari are a Koiyam tribe, '-te' is the demonstrative pronoun used as a definite article; vid. Koelle, *Kanuri Grammar*, p. 185.]

[**Luskuri.** 'Lasku' or 'lusku' means in Kanuri the hole made by a 'bush-cat' or fox; '-ri' = the place of.]

Magumeri. 'Magume' is the old tribal Kanuri mark, that is, the two long marks down the arm and leg supposed to be made by a sword; two is for princes, &c., three marks any Kanuri. '-ri' = place of. Therefore, Magumeri means the place of the Kanuri princes or chiefs—or a properly bred old stock Kanuri.

[Magumi is the name of one of the chief Kanuri clans.]

Maiduguri. 'Mai' = a king, 'dugu' a corruption of 'dugo', which means 'presently'. Maidugu means the son of a ['maina'] prince or the grandson of a king (really 'presently a king'). '-ri' = the place of. Hence Maiduguri means 'the dwelling place of a king's grandson'.

[Cf. 'madugu' in Hausa = headman of a caravan. The place is frequently misspelt Maidugari, because people think it has something to do with the Hausa word 'gari' = a town.]

Maifoni. ‘Mai’ = a king, ‘foni’ [or ‘funi’] = a turban. A slave of a Bornu king introduced this place to his master’s notice on account of its running water and pleasant site. The king rewarded the slave by giving him a turban, which is a sign of office; hence the name.

[There is an ancient connexion between Maifoni and the kings of Marghi. According to one account the Marghi town of Kaba (*q.v.*) was originally also called Maifoni, a corruption of Mafundi, which means ‘a tall, stout man, a man in his prime’, because it was founded by a powerful man, who broke away from the authority of the Marghi king, Mai Ibrahim. A later Marghi king, Mai Jalo, complained to Shehu Lamino, that in paying his annual visit of homage to Kukawa, he found the journey a fatiguing one, as it was all ‘bush’ between Kaba and the centre of Uje. Lamino therefore gave him permission to select a site as a ‘rest-camp’, and he chose Maifoni (now Maiduguri station) as his half-way house. The following is a list of the Marghi kings of Maifoni.

Connexion between Maifoni and Marghi.

The founder of Maifoni was a man called Mele Gumsumi, and his title was Kaloma. He left Birni and sat down at Kasachiya (near Kaba) to farm. He was succeeded by Kaloma Umar Kumaisimi, he by Mai Isa Mairemmi. Mai Isa reigned for forty years.

He was Mai Momodu Mairemmi.
succeeded by

- „ „ Mai Duguribe Gumsumi (he belonged to the ruling family of Birni).
- „ „ Zuri Gumsumi.
- „ „ Mai Nassar.
- „ „ Mele Nassarmi.
- „ „ Usuman Kumaisimi.
- „ „ Usuman Gumtimi (he went and made war with the Sau town of Almis).
- „ „ Mai Mele Hawami.

- „ „ Mai Momodu Panami.
- „ „ Mai Jalo Bawomi.
- „ „ Mai Tahir.
- „ „ Mai Jalo Aisami.
- „ „ Mai Momodu Maungusuma.
- „ „ Mai Dalla.
- „ „ Mai Momodu Kowas.
- „ „ Mai Mallam Gorumba.
- „ „ Mai Zogoma.
- „ „ Mai Mele Balumi.
- „ „ Mai Sunoma.
- „ „ Mai Momodu Kabumi (son of Sunoma).
- „ „ Mai Budum.
- „ „ Mai Ibram (contemporary of Shehu Lamino).
- „ „ Mai Jalo.
- „ „ Mai Momodu Gajimi.
- „ „ Mai Arri.
- „ „ Mai Mele.
- „ „ Mai Momodu Biyemi.
- „ „ Mai Dogum.
- „ „ Mai Yagudima.
- „ „ Mai Mele.
- „ „ Mai Arri (now Lowan of Kaba in West Marghi, Kaba and Maifoni having been separated under the British administration).]

[**Margawa**, the ‘marga’ trees. The pounded wood of this tree is boiled and used to sweeten certain native dishes.]

[**Marte**, said to be a Sau word meaning a blanket or thick cloth.]

Masagua or **Masakwa**, the dry-season corn. The first of this grain was found in elephants’ droppings. The King of Kasa (now in German

territory) was the first man who noticed this grain in elephant-droppings. He picked it out and planted it. When he watered it, it grew freely and was the beginning of the seed afterwards sown widely in waterlogged ground, and bearing a grain similar to guinea-corn. The idea is that the elephant ate guinea-corn at some distant place and brought it as described to Chad shore. 'Musu' in Kanuri is anything rotten [from 'meṣeṅgin']. 'Musukwa' = something that comes from a rotten thing.

[**Masu.** A certain man came and sat down here and said, 'If they want me, let them seek me' ('matsa', 3rd pers. plural, 2nd indefinite, from 'mangin').]

Mbarma or **Mbaruma**, the man in a village who collects from the villagers any money due by them. 'Mbaro' meaning in Kanuri 'to pay rent, to pay on behalf of ———'.

['Barungin' = to redeem from slavery, to pay 'pansa', so 'mbaru-ma' is the man who is responsible for some one else. 'Mbarma' is a small village or hamlet headman, as opposed to 'bullama' or 'beḷa-ma', headman of a large village.]

[**Mongonu.** The King of Bornu, at that time settled at Kabela, sent his brother to sit down on this site. When the people saw that the country was fertile, they said, 'Mai (shiga) ngurno gono' = the king has shown him kindness. This became corrupted into Mongonu or Maungono.]

[**Ngomati.** Nguma is the name of a Kanuri tribe, '-ti' is the demonstrative pronoun used as a definite article; vid. Koelle, *Kanuri Grammar*, p. 185.]

Ngunse [Nganzei], which in Kanuri means 'to milk'. This used to be a rich district and a source of wealth to the fiefholders, who made the most of it and milked it of its wealth.

['Ngangin' = to milk; 'nganzei', 3rd pers. plural, 1st indefinite or perfect. The district is very rich in cattle.]

[**Nguru.** Some hunters killed an hippopotamus ('ngurutu') and made bracelets out of the skin. Such bracelets are called 'nguru'.]

Shetima, from the word 'sheida' = a witness. 'Shetima' means 'the tried man, the proved man'.

Shuwa. The word ‘ashe’ in Kanembu and Kanuri means ‘the strong-headed man, wilful or disobedient man’ [‘ashi’ = obstinacy], ‘-wa’ = the people of. Hence the word ‘ashiwa’, now ‘Shuwa’.

They are originally supposed to have been so perverse that they would not follow the Prophet, thence they became a wandering tribe. [Cf. [p. 328.](#)]

[**Uje**, said to be a corruption of the Shuwa-Arabic word ‘wajēhu’ = a face. Koelle, in the *Polyglotta Africana*, gives the form ‘wush’ = face, in the Arabic dialect of Beran. The district known as Uje occupies a central position in Bornu, and was therefore called the ‘face’ of the country.]

[**Yajua.** A Fulani ‘ardo’ called Yaji settled at this spot. The Kanuri, therefore, called it ‘beḷa Yaji-wa’ = Yaji’s town, ‘-wa’ being the adjectival termination.]

Yarima, from ‘ya’ = mother, ‘yari’ = the mother’s place, ‘yarima’ = he who is in the mother’s house. Hence in a king’s family the son who takes charge of the mother’s compound is called the Yarima. This title seems to go from father to son, not through sons of the ruling house. Thus now in Bornu, Yarima Kassim [ex-Ajia of Mufio] has the appointment, Bukar Yarimami [now Ajia of Gusamalla-Ngunse] refusing it on his father’s death; both these men are descendants of old Bornu ‘Yarimas’.

[**Yedseram**, or **Yadseram**. The river forming the Anglo-German boundary. A small girl called Yachua was bathing in it and was drowned. The people said, ‘Komadugu Yachua tshetshina’ = the river has killed Yachua, so the river was named ‘Yachua tshetshina-ram’ = the river which killed Yachua (‘-ram’ is an adjectival suffix); this became corrupted into Yadseram. Another suggested derivation connects it with the village of Yadza near Issege.]

Said to mean in Arabic, ‘the rock without a rival, the solitary rock.’ There is a photo in Macleod, *Chiefs and Cities*, p. 206. *Doc. sc. de la Mission Tilho*, vol. ii, p. 355, quoting Nachtigal, speaks of a rock known as Hadjer Teous on the left bank of Chad, and a rock of this name is shown in Nachtigal’s map (reproduced on p. 14, vol. i of Tilho) on the south of Chad. Possibly Hadjer Hamis is sometimes called Hadjer Teous, but according to my information Hadjer Teous means ‘the he-goat rock’, and is situated north of Chad between Kanem and Wara. I am inclined to think, however, that they are the same. In the first edition of Denham there is a plate facing p. 261 of ‘Hager Teous called by the Natives the Foot Stool of Noah’. It is not shown on the map, but from the letterpress must

have been on the south or south-east of the Lake; cf. also Chevalier, *Mission Chari-Lac Tchad*, p. 410, and [p. 45](#) of the present work.

Cf. Appendix XIV A, [p. 335](#).

APPENDIX XI

FESTIVALS IN BORNU

<i>Date.</i>	<i>Kanuri Name.</i>	<i>Arabic Name.</i>	<i>Notes.</i>
10th Moharrem	Ngumori Tsuru- mbulo-be	Ashura	Day of creation of Adam and Eve, and of the first fall of rain. Every one in Bornu kills a fowl. 'The feast of the belly-filling.' In Kanuri 'tsuro' = belly, and 'yimbuluskin' = I fill. The day is sacred to Shiah Moslems as the anniversary of the death of the martyr Hussein. According to Michell, <i>An Egyptian Calendar</i> , p. 74, a particular dish called 'ashura' is made on this day. [548]
Last Wednesday in Safar	Moddua Laraban-ganí-be	Arbaa Mayidur	Said to be the day of the last bath of the prophet. He died fourteen days after it. On this day prayer is offered for averting trouble during the year. The Kanuri name means literally 'The telling of beads of Wednesday-not', i.e. no more Wednesdays, the last Wednesday of the month.
12th Rabi-el-owwal	Ngumori Lebbi	Mulid en-Nebi	Birthday of Mohamet. This is the chief festival in Bornu.

15th Shaaban	Lowal-be Ngumori Nussufu Shaaban- be	Leylet en- Nusf min Shaaban (called in India ‘Shab i Barat’)	On this night Allah registers the good and evil deeds of mankind. The Kanuri name means literally ‘the festival of half Shaaban’, i.e. the Mid- Shaaban festival.
1st Showwal	Ngumori Asham-be	Id el-Fitr, Turkish ‘Ramazan Bairam’	First day after the Ramadan fast. This is the chief festival among the Hausas. In Kanuri ‘asham’ = fast.
10th Zu’l Heggeh	Ngumori Laya-be	Id el-Kebir, Turkish ‘Kurban Bairam’	Ram festival. Commemorates Abraham’s attempted sacrifice of Ishmael (not of Isaac as according to the Old Testament).

It consists of wheat boiled and sweetened with dates, nuts, and other dried fruits. Presents of this dish, sometimes with small gold coins sprinkled upon it, are sent to friends and relatives from the harems of the wealthy. The common name for it is ‘hobub’. This dish, or one similar to it, was prepared at the Persian New Year. It probably celebrated the coming in of the harvest. All the old fruits, &c., that had been kept during the winter were made use of for the dish, and the season of fresh fruits was thus inaugurated.

APPENDIX XII

THE CALENDAR

THE names of the Arabic months, as given in Michell's *An Egyptian Calendar*, and as transliterated from Kanuri mallams, differ somewhat.

<i>Michell.</i>	<i>Kanuri.</i>
1. Moharrem.	Muharam.
2. Safar.	Shafur.
3. Rabia el-owwal.	Rabi-il-lawal.
4. Rabia et-tani.	Rabi-il-lahir
	or
	Rabi-it-tsani.
5. Gumad el-owwal.	Jimad-al-lawal.
6. Gumad et-tani.	Jimad-al-lahir
	or
	Jimad-at-tsani.
7. Regeb.	Rajab.
8. Shaaban.	Shaaban.
9. Ramadan.	Ramadan.
10. Showwal.	Shawal.
11. Zu'l-kaadeh.	Zulgada.
12. Zu'l-heggeh.	Zulhaji.

With regard to the year, the Liman says that A.D. 1912 is A.H. 1329 with a few days of 1330. Other mallams say that it is A.H. 1330 with a few days of 1331, and this reckoning agrees with Michell and with the *Sudan Almanac* published by the Egyptian Government. This discrepancy is explained by the following extract from Major Burdon's notes on Sokoto in *Historical Notes on Certain Emirates and Tribes*, p. 61:

‘An extraordinary point in the reckoning of these Nigerian Mohammedans is that, while right in the day of the week, they are almost invariably wrong by a year and a day in the remaining part of any date. . . . The explanation given is that neither the year nor the day is counted until each is completed. It is a perfectly intelligible system, but the logical sequence to it would appear to be its application also to the days of the week.’

The names of the English months are transliterated as follows:

Yunair.
Fabrair.
Maris.
Ibril.
Mayu.
Yunihi.
Yulihi.
Awgustus.
Sabtumbir.
Aktumbir.
Nufambir.
Duyusambir.

To convert approximately a date of the Hejira to that of the Christian era, add to the former 622 and subtract from the sum three years for every century of the Mohammedan date; e.g. to convert A.H. 1318 to A.D., $1318 + 622 = 1940$; $-(13 \times 3) = 1901$ (and part of 1900). And inversely, A.D. 1900 $= 1900 - 622 = 1278$; $+ 39 = 1317$ A.H. (and part of 1318).

MEANING OF THE NAMES OF THE ARABIC MONTHS (from Michell).

1. *Moharrem* = the Forbidden, called Moharrem el-haram (the sacred). One of the four months of truce, in which all acts of hostility were strictly forbidden among the Arab tribes. It is considered unlucky to make a marriage contract in Moharrem.

In Bornu 'Moharrem' villages were those which were forbidden to be taxed, i.e. excused from the general tax. They were squeezed by a Court favourite instead.

2. *Safar*, derivation, according to Fresnel quoted by Lane, from *Sifar* = empty, either because their granaries being empty, the Arabs used to travel to procure grain; or because they went on predatory expeditions, leaving their homes empty; or because they left Mecca empty. The fairs in Yemen used to be called Safarieh (vid. Masudi and Lane's *Arabic Dictionary*). This month is called Safar el-Muzaffer (the auspicious); and also sometimes Nezlet el-Hagg (the descent or alighting of the pilgrims), because the Mecca pilgrims begin to return to Egypt towards the end of Safar.

(Mallam Zakaria of Maiduguri derives it from an Arabic word 'safar', I travel, because it is the travelling month; cf. Swahili 'safari' = a hunting expedition?)

3. *Rabia el-owwal* = the first Rabia. 'Rabi' expresses verdure and spring rains.

4. *Rabia et-tani* or *el-akher* = the second or last Rabia.

5. *Gumad el-owwal* = the first Gumad. Generally derived from 'Gamada', dryness or hardness, as applied to the earth after the cessation of the rains of the preceding spring months (cf. Kanuri 'ngamdu' = dry, lean).

6. *Gumad et-tani* or *el-akher* = the second or last Gumad.

7. *Regeb*, one of the four months of truce. The Prophet commended prayer and fasting in this month.

8. *Shaaban*, probably so called because the Arabs were wont, after the peace of Regeb, to separate ('Shaab') on marauding expeditions, and also to seek water, this month originally falling in the great heat of June and July (vid. Lane's *Dictionary*).

9. *Ramadan* = intense heat. The fasting month. The eve of Ramadan is called 'Leylet er-Ruyeh', the night of observation, because men are appointed to watch for the new moon, and then give evidence at the Court of the Kadi.

10. *Showwal*, so called, not as Lane tells us (*Arab. Dict.*), because it is the breeding season of camels (from ‘showwal’, to raise the tail), but the season when the she-camels, being seven or eight months gone with young, raise their tails (camels generally couple in the winter). Or, possibly, the word refers to a deficiency of the camel’s milk in the season of great heat.

11. *Zu’l-kaadeh* = the possessor or holder of truce or abstention. One of the four months of truce. Lane explains it as the month in which the Arabs broke in their young camels (‘el-kaadat’) for riding.

12. *Zu’l-heggeh*. The month of pilgrimage. One of the four months of truce.

APPENDIX XIII

LIST OF BORNU TRIBES

(1) Babur, (2) Bedde, (3) Burra, (4) Chibuk, practically Marghi, (5) Fika, including Bolawa (M.) and Gamawa (P.), (6) Filani, (7) Gamergu (P.), (8) Kanembu, (9) Kanuri, (10) Kerikeri, (11) Kwoyam, (12) Manga, (13) Marghi, (14) Mobber, (15) Ngizim, (16) Ngussur, semi-pagan (Gujba), (17) Shuwa, (18) Tera, (19) Kanawa, spread over all three divisions (practically none—but Filani), (20) Shira, sub. to Fika (semi-pagan).

Clans or tribes comprised in Kanuri: Kanembu, Mobber, Manga, Koyam, Karda, Kagama, Ngussur, Magumi (the original Kanuri), and several minor clans, the Lere, &c.

Filani clans: Kitije (P.), Jafun (Kanawa?). Abore: Sankara, Girije, Daiyi, Daneji, Warumonde, Bikerke, Bagaji, Mamaji, Bijingel, Uda.

In Tera: Hinna, Maga, some Tangali, and Nimaltu (south of Wuyo and Bima).

APPENDIX XIV

ETHNOLOGICAL NOTES ON BORNU PROPER

THERE are eight chief races in Bornu Proper:

(*a*) Kanuri, (*b*) Kanembu, (*c*) Shuwa, (*d*) Fulani, (*e*) Koyam, (*f*) Manga, (*g*) Gamerghu, (*h*) Mobber.

(*a*) The Kanuri or Beriberi are supposed to be a mixed race of Arab tribes and Hamitic Kanembu and Tubu, which, gradually losing more and more the physical and mental peculiarities of the dwellers in the desert, became merged in the negro population of the conquered country, the half-mythical ‘So’, a giant race, whose descendants are supposed to be represented at the present day by the Beddes, Buddumas, Kotokos, &c. One of the chief Kanuri clans is the Magumi, to whom belongs Maina Gumsumi, now Ajia of Kaiuri district, the representative of the old ‘Mais’ of Bornu.

(*b*) The Kanembu, i.e. the inhabitants of Kanem, are an Hamitic race, whose relationship to the Kanuri may be roughly described as similar to that between the Lowland Scots and the English. Their chief clans are the Sugurti and the Kubri (the Shehu’s clan).

(*c*) According to Schultze, quoting Barth, ii. 355, the Shuwa Arabs immigrated into Bornu at various times, firstly, with the conquerors of the country from the north, and secondly, at a much later epoch, about 300 years ago, from Nubia or Kordofan. ^[549] They are broken up into a great number of tribes, of which the Beni Hassan has remained the purest, but they have lost many of the bodily and mental peculiarities of the Semitic race, doubtless as a result of their long wanderings, and also of the intermixture which has taken place with other peoples. Their language, however, they have retained in wonderful purity.

There is a vocabulary of Shuwa Arabic in Koelle’s *Polyglotta Africana*, and some notes on the language appeared in Decorse and Demombyne’s

Rabah et les Arabes du Chari. A book on Arabic spoken in Wadai and to the east of Chad by H. Carbou has recently been published, vid. [List of Authorities](#).

From the best information available in British Bornu the Shuwas may be divided into two classes, (A) those who entered Bornu before Lamino's time, (B) those who entered with or shortly after Lamino.

To (A) belong (1) the Joama, originally settled north of Mongonu, but migrated to Karaguaro and Magumeri after the Fulani invasion; (2) the Maiyin or Meyin, originally settled at Marguba; (3) the Saraje, originally settled in Uje district. All these three tribes entered Bornu about 300 years ago, ten or twenty years before Birni Gazerregomo was built. They came from Darfur. There were no Shuwas in Bornu prior to this. This statement differs slightly from Schultze. None of these tribes are now numerically important.

To (B) belong (1) the large and wealthy tribe of the Kwalme Shuwas, (2) the Kurata, (3) the Beni Hassan, mostly in Mandara and Musgu, of very pure descent.

The Kwalme are divided into many clans, of which the following may be enumerated:[\[550\]](#)

- (1) Wulad Sarar.
- (2) Wulad Salim.
- (3) Wulad Himet.
- (4) Wulad Kanem.
- (5) Beni Badder.
- (6) Dagana.
- (7) Beni Muharab.
- (8) Asali (to whom belonged Lowan Adam, murdered in 1906).
- (9) Salamat (murderers of Lowan Adam).
- (10) Wulad Amir.

(11) Beni Wail.

(12) Ajeni.

(13) Wulad Hodir.

(14) Wulad Abu Isa.

N.B.—‘Wulad’ in Shuwa means sons, ‘beni’, daughters. The derivation of ‘Kwalme’ is variously given, but its real origin is probably unknown. Some connect it with ‘kworram’ = amber (Kanuri) and ‘mi’ = son of, from the fondness of the Shuwa women for amber ornaments. Others derive it from a Shuwa word ‘galib’ = it is too difficult, they are too difficult for me (Kanuri ‘tegeri’, Hausa ‘gargare’); ‘galib’ became corrupted into ‘galim’, and by transposition to ‘galmi’. This term is said to have been applied to the Shuwas on account of their obstinate character by the Arabic-speaking peoples east of Chad. Some support is lent to this derivation by that given for the word ‘Shuwa’. The word ‘ashe’ in Kanembu and Kanuri means ‘the strong-headed man, wilful or disobedient man’ [‘ashi’ = obstinacy, ‘-wa’ = the people of]. Hence the word ‘Ashiwa’ now ‘Shuwa’. They are originally supposed to have been so perverse that they would not follow the Prophet, thence they became a wandering tribe [cf. [p. 318](#)].

(d) The Fulani or Fellata of Bornu belong chiefly to three clans, the Abore Fellata (Hausa, ‘Borroro’), the Mare Fellata (the term ‘Mare’ is said to be a Fulani word for a tree resembling the ‘dorowa’ or locust-tree, whose pods are used in making the floors of native huts), and the Kitiji Fulani (only found in Gujba and among the pagan Babur and Burra tribes; they are themselves pagan). The origin of the Fulani has been, and still is, hotly disputed.^[551] They have not coalesced to any extent with the bulk of the Bornu population and remain aliens and, to a certain extent, pariahs. The Kanuri have many proverbs and sayings embodying their contempt of them as ‘bushmen’.

(e)^[552] The Koyams seem to have had the same origin as the Tubba or Sef dynasty of Bornu. They are said to have reached Kanem from Yemen, travelling slowly and living on the alms offered them as wandering students. With the consent of Mai Arri ben Haj Umar (1645-84), Sheikh Abdullahi and thirty-nine companions established themselves at Belbelec,

sometimes called 'the town of the Kullumfardos', where they founded a mosque and a school. The origin of the term 'Kullumfardo' is said to be as follows. When Abdullahi and his companions, all Koyams but not of the same family, presented themselves to Mai Arri, the latter asked them who they were. They replied, 'Koyam', adding 'Kullum fardun', Arabic words meaning 'all, separately', wishing to explain that they were all of the Koyam tribe, but of different families. At the present day it is only the descendants of Sheikh Abdullahi, their first 'Mokaddem' or religious chief, that call themselves 'Kullumfardo'; the descendants of the thirty-nine others are called Koyams. Sheikh Abdullahi was succeeded by his son, Sheikh Umar. Owing to Tuareg raids and to famine Belbelec was abandoned and the Koyams scattered. Some went south and, abandoning their rôle of students, became herdsmen and acquired wealth. They took the name of Kel Etti, a term whose meaning and origin are unknown, and became subdivided into seven fractions, each with its chief.

A small number of disciples followed Sheikh Umar to Nupe, where he remained some time, but he afterwards returned to Bornu and re-established himself at Gaskeru, situated, like Belbelec, north of the River Yo.

Umar was succeeded by his brother, Mustapha, who was the first to join political functions to the religious ones, to which his family had hitherto confined themselves. He appointed his nephew, Mahmut, as chief of the warriors and said to him: 'You shall be the Sheikh's auxiliary or lieutenant' (El Auan, Lowan). Lowan was thus originally a military title, and later became applied to the head of a district or a town.

At the end of the eighteenth century Gaskeru was destroyed by the Tuaregs and the Koyam 'Mokaddem' established himself at the pool called Sandaram, inside the walls of the Bornu capital of Gazerregomo (Kasr Kumo).

When the Fulani captured Gazerregomo the Koyam 'Mokaddem' followed the varying fortunes of the exiled king. When Shehu Lamino became head of the state, the Koyams applied to him to be dispensed from payment of tax as they had always been hitherto. Lamino, who had studied among the Koyams in his youth, consented and established the Koyam 'Mokaddem' at Zigaba on the north bank of the Yo.

In the last year of Lamino's reign, however, a great famine occurred and the Koyams again dispersed. The incursions of Rabeh scattered them still further, and the present representatives of the Koyam 'mokaddems' now live at the village of Kullumfardo in Munio.

The Koyams in British Bornu are semi-nomadic cattle-owning people living chiefly in the districts of Bussugua, Ngunse, and Gusumalla.

(f)^[553] The Mangas are mostly bilingual, speaking both their own dialect (said to be related to Bedde) and Kanuri. M. Landeroin inclines to the opinion that they are an indigenous tribe conquered by the Kanuri. Manga is said not to be their original tribal name, but to have been given them by the Kanuri.

It is said to be a corruption of 'Madinga' which signifies according to some 'gens difficiles', according to others 'the cunning people'.

The Mangas inhabit the country both north and south of the River Yo.

(g) The Gamerghu are a branch of the Masa family, but are now almost extinct. Their villages are scattered along the course of the River Yedseram. Their chief industry is horse and cattle stealing. Barth gives a detailed study of Masa in his *Central African Vocabularies*, and there is a vocabulary of the Gamerghu dialect, collected by Barth, in *Notes on Some Languages, &c.* (Benton). They are semi-pagan.

(h)^[554] The Mobbers, like all the other tribes, assert that they originated from Yemen. Serfs of the Magumis, who were themselves the subjects of the Tubbas or Sef dynasty, they were installed by the latter along the course of the River Yo, from which they never strayed far, as they had no knowledge of well-digging.

Their chief towns at the present day are Bosso, on the northern bank of the Yo where it debouches into Chad, and Yo on the opposite south bank. Nachtigal says of the Mobbers: 'This tribe is according to some a remnant of the Beddes, according to others a mixed race of Kanembus and Beddes or of Sos and Kanembus.' Like the Mangas, they are bilingual, speaking their own dialect as well as Kanuri.

Little reliable information is yet available regarding the pagan and semi-pagan tribes in the British Province of Bornu. A certain amount about the Bolewa of Fika, including a short sketch of the language, is printed in Benton, *Notes on Some Languages of the Western Sudan*.^[555] There is a short account of the Beddes in Koelle, *African Native Literature in Kanuri*, p. 210, and a vocabulary in Koelle, *Polyglotta Africana*. Overweg visited Babur in 1852, cf. Benton, *Notes on Some Languages*, p. 222; for Barth's notes and vocabulary of Marghi cf. *idem*, pp. 78, 134, and 144. For list of Marghi kings cf. [Appendix X](#) under Maifoni. There is a vocabulary of Kerrikerri in Koelle, *Polyglotta Africana*.

Cf. [Appendix XIV A](#).

Cf. Appendix XIV A, [p. 334](#).

The latest account of the Fulani is contained in Mr. H. R. Palmer's articles now in course of publication in the *African Mail*. The first article appeared in the issue of December 29, 1911. Cf. also 'Notes on the Origin of the Filani', by Capt. A. J. N. Tremearne, in the *Journal of the Royal Society of Arts* for June 17, 1910. Cf. also article by T. von Stephani in *Der Islam*, iii, 1912, 352-7.

Account mostly taken from *Documents scientifiques de la Mission Tilho*, ii. 396 ff.

Account mostly taken from *Documents scientifiques de la Mission Tilho*, ii. 421.

Account mostly taken from *Documents scientifiques de la Mission Tilho*, ii. 391.

In addition to authorities there quoted, there are also articles by the late Major George Merrick in the *Journal of the African Society* for July 1905 and October 1905. I was not hitherto aware of these.

APPENDIX XIV A
DERIVATION OF SHUWA AND KANURI

ACCORDING TO M. CARBOU

SOME writers, especially d'Escayrac de Lauture and Vivien de Saint-Martin, assumed that the Shuwas were descended from the Koreishites, pagan Arabs driven from Arabia by Mahomet. M. René Basset, however, declares that the Koreishites were not driven out but became converted, and pertinently asks, if they were driven out as pagans, at what period were they converted? There is no record or tradition of the Shuwa Arabs ever having been anything but Mussulmans. Cf. Carbou, *La Région du Tchad et du Ouadaï*, vol. ii, p. 11.

M. René Basset and MM. Hartmann and Becker incline to derive 'Shuwa' from the Arab word *shaua* = sheep, and to make it signify nomad shepherds as opposed to a sedentary people. Vid. Carbou, *La Région du Tchad et du Ouadaï* vol. ii, p. 20.

'The Arabs of Bornu and of the "Territoire Militaire du Tchad" belong to the religious order of the Tojani. This brotherhood, which preaches toleration, is the one which counts most adherents throughout this region: the Senussi are not found, except in Wadai and the northern countries (Borku, &c.). A very simple way of recognizing the Senussi is that they pray with the arms crossed, instead of keeping them parallel with the body, as all the other Maliki sects do.' Carbou, *La Région du Tchad et du Ouadaï*, vol. ii, p. 18. Cf. note [\[458\]](#) of the present work.

H. Carbou, vol. ii, p. 36 ff., enumerates the tribes of the Kwalme Shuwas as follows (cf. [p. 328](#) *supra*):

- (1) Oulad Mehareb [my No. 7].
- (2) Oulad Serrar [my No. 1].

- (3) Oulad Salem, Oulad 'Amer [my Nos. 2 and 10].
- (4) Assalé [my No. 8]. They are the descendants of Ali el Esselé (Ali the Bald).
- (5) Dagana [my No. 6], said to be descended from Othman Abou Diguen (i.e. the long-bearded). They are mentioned both by Nachtigal and Matteucci.
- (6) Oulad el 'Aouan [not mentioned in my list, but apparently a subdivision of the Dagana].
- (7) Oulad Mansour [not in my list].
- (8) Oulad Amiré, Oulad Ghanem, El Haouarti, Oulad Saïl, Oulad Mehoï [not in my list].
- (9) Beni Ouail [my No. 11].
- (10) Oulad Abou Issé [my No. 14].
- (11) Oulad abou Ghader [not in my list].

Carbou classifies the Wulad Himet [my No. 3], Salamat [No. 9], Badder [No. 5], and Beni Hassan under the Djoheïna Arabs, not the Kwalme.

Two derivations are given of Salamat. One, that their ancestor was named Salam. Another, that they are descended from a pagan slave, who, when reproached for not fulfilling his religious duties, replied 'nadem salla mat' (praying is killing work); hence he was nicknamed Sallamat. As M. Carbou remarks, p. 56, this explanation is a little far fetched.

M. Carbou, vol. i, p. 4, says:

'Kanem is a corruption of Keunoum or Konoum. This latter word has been explained by means of the Tubu term "eunoum" or "onoum" which means south, and of the letter "K" which serves to form the substantive. But the Kanembu use the same word to signify the south, and it is especially in their language that "K" serves to form the substantive.'

For all practical purposes the Kanuri and Kanembu dialects are the same. 'Anum' or 'Anem' is the Kanuri for south, but I know of no use of 'K' for forming the substantive, unless one assumes it is an abbreviation of

the Kanuri ‘Kam’ meaning a man. Kanem-bu would then mean ‘the southerners’. Cf. [p. 314](#) of the present work.

M. Carbou, vol. i, p. 20:

‘This name, Kanuri, is explained in different ways. Some say that it is composed of the Arab word “nur” (light) and the letter “K” which serves to form the substantive. Kanuri would then mean “the people of the light”, and this name would have been given to the inhabitants of Bornu because they were the propagators of Islam in the country. Others say that Kanuri is from the Kanuri “kannu” = (fire), and means “the people of the fire”, a nickname given them by the Fulani, who as fanatical Mussulmans considered the inhabitants of Bornu as destined to the flames of hell as a reward for their sins.’

M. Carbou himself says that the Arab derivation has little value. The Fulani one has less still: if the Fulani or any other foreigners wished to nickname the Kanuri, would they not use their own language to do so? Cf. [p. 314](#) of the present work.

M. Carbou, vol. i, p. 21, has an elaborate argument to prove that the word Bornu is derived from ‘birni’ meaning in Kanuri ‘a capital town’. The Sef dynasty were driven out of their capital Njime by the rival dynasty of the Bulala and went and founded another capital at Gazerregomo. M. Carbou states that M. René Basset considers this etymology of Bornu from ‘birni’ as very probable. In spite of this high authority, the etymology is not convincing. ‘Birni’ or ‘Bɛrni’ might be corrupted in Arabic writing into ‘Bornu’, or ‘Burnu’, but it is important to remember that the natives never write ‘Bornu’ always ‘Barnu’. The Arabic vowel signs which we represent by i, e, and o or u, are often interchangeable in Kanuri writing, e.g. I have seen the word for ‘horse’ spelt ‘fir’, ‘fɛr’, and less commonly ‘fur’, it could not conceivably be spelt ‘far’. Cf. [p. 311](#) of the present work.

APPENDIX XV

SELECTIONS FROM UNPUBLISHED CORRESPONDENCE

REGARDING OUDNEY, DENHAM, AND CLAPPERTON'S MISSION TO BORNU IN 1821-

4

THE following extracts are taken from the Tripoli Consular Correspondence, now in the Record Office, and from papers in the archives of the Royal Geographical Society. The latter are distinguished by the letters [R. G. S.].

The following 'précis' from the accounts given of the travellers in the *Dictionary of National Biography* and in Nelson (vid. [List of Authorities](#)) will help to elucidate the extracts. ^[556]

Walter Oudney (1790-1824), born of humble parents in Edinburgh, where he picked up sufficient knowledge of medicine to become a surgeon's mate on board a man-of-war. 1814 promoted surgeon. At the Peace he returned to Edinburgh, graduated M.D., when his inaugural dissertation was 'de Dysenteria Orientali', and set up in private practice. Studied Chemistry and Natural History, and had hopes of being appointed University lecturer on Botany. He contributed a paper to the *Edinburgh Medical and Surgical Journal* for July, 1820, entitled 'Cases of "Ileus" from a twist of the Colon'. Appointed to the African Mission 1821, died at Murmur near Katagum on January 12, 1824. Mrs. Larymore, in *A Resident's Wife in Nigeria*, p. 97, states that his grave is still pointed out by the natives. Nelson says that Oudney asked Clapperton to hand his papers to Barrow, i.e. Sir John Barrow, Secretary to the Admiralty, who edited the account of Clapperton's Second Expedition. In the Introduction to the account of the First Expedition, however, Denham states that Oudney practically left no papers. Nelson fears they may have been lost, but remembering Oudney's constant ill health, I think it is quite possible that there were none beyond the few rough notes, half illegible and of little interest, which are now among the Denham papers in the R. G. S. Oudney

is described as of middle stature and slight build, with a pale, grave face, pleasing manners, and possessed of much enterprise and perseverance. He was never married. In a letter to a friend from Mourzouk, quoted by Nelson, Clapperton says that 'Oudney is much admired by the local ladies for the blackness of his beard, and myself for the strength of my moustachoes'. Oudney, in a postscript on the same sheet, says, 'Clapperton is a strange-looking figure with his long sandy-coloured beard and moustachoes'.

Dixon Denham (1786-1828), educated at Merchant Taylors School. Articled to a solicitor, but in 1811 joined the army in the Peninsula as a volunteer with the 23rd Royal Welsh Fusiliers. In May, 1812, was appointed a Second Lieutenant in that corps; promoted to First Lieutenant in 1813. Distinguished himself at the battle of Toulouse by carrying Sir James Douglas, commanding a Portuguese brigade, out of fire when that officer had lost his leg. The following copy of a letter from General Douglas to Col. Arbuthnot is among the Denham papers in the R. G. S.:

MY DEAR ARBUTHNOT,

I am very sensible of the Marshal's^[557] kindness in desiring me to recommend any officers of my Brigade to him who may have distinguished themselves in the late affair. I avail myself of his permission to mention for the third time Brigade-Major Maher and Lieutenant Denham of the 8th Regt., my aide-de-camp. If His Excellency can in any way be useful to Lt. Denham it will be a circumstance peculiarly grateful to me. He is an officer I have lived in habits of the strictest intimacy with—and whose conduct and attentions on the 10th^[558] I am totally at a loss how to repay.

Signed J. DOUGLAS,

Commg. 7th Brigade.^[559]

Toulouse, 13 April, 1814.

To

Col. Arbuthnot,

Mily. Secretary,

&c., &c., &c.

Transferred to the 54th Foot,^[560] who were in reserve at Huy during the battle of Waterloo. In 1819 he entered the senior department of the Royal Military College. In Oct. 1821 he purchased a company in the 3rd Buffs. In Nov. 1821 he was put on half-pay, given the local rank of Major in Africa, and sent to join the African Mission. Amongst other things, he explored part of the shores of Lake Chad, which he named Lake Waterloo. He returned to England in June, 1825, and was promoted to a majority in the 17th Foot.^[561] In Nov. 1826 he was given an unattached lieutenant-colonelcy, and sent to Sierra Leone as superintendent of liberated Africans^[562] on the West Coast. In 1828 he was appointed Lieutenant-Governor of the colony of Sierra Leone, but died there of fever in May of the same year.

It is perhaps of some interest to note that Denham appears to have been very modern in his ideas of the proper way of keeping in health in the tropics. In letters to his brother [R. G. S.], written before they left Mourzouk, he mentions that he always rises before sunrise, takes horse-exercise, is very careful about where he gets his drinking and even his bathing water from, and also says that

he has a daily shower-bath, made by pouring water through a wooden box with holes bored in it.

Hugh Clapperton (1788-1827), born at Annan in Dumfriesshire, son of George Clapperton, surgeon.^[563] At thirteen he was apprenticed as a cabin-boy in a ship trading between Liverpool and America. He showed his spirit by refusing to black the captain's shoes. He was charged with a petty act of smuggling at Liverpool, and sent on board the naval tender, which carried him to Plymouth, where he was made cook's mate. Eventually, through the interest of his uncle in the Marines,^[564] he was made a midshipman, and served off the coast of Spain and in the East Indies. On his way out to the East Indies he was ordered to take charge of a boat which was sent to the relief of a ship in distress. The boat was capsized and all the occupants drowned, except Clapperton and one other. As he was being hoisted on board again, 'he had his feelings strongly excited, on hearing the wives of the Scottish soldiers on board exclaiming, "Thank Heaven, it is na our ain kintryman, the bonny muckle midshipman that's drowned after a'!—"'

In the East Indies he joined the *Clorinde*, one of whose officers was Mr. Mackenzie, youngest son of Lord Seaforth. Clapperton formed a great friendship with this young man and nursed him through a dangerous illness. Mackenzie eventually went home and died, but particularly asked his mother, Lady Seaforth, to treat Clapperton as her own son. She did all in her power to help Clapperton when he returned to England, but, according to Nelson, Clapperton was too proud and touchy to accept much. He had been deeply wounded by an insinuation of his messmates that he was a tuft-hunter.

At the storming of Port Louis, Isle of France, in Nov. 1810, he was the first in the breach and hauled down the French colours. He was next sent to Canada and was cast away off the coast of Labrador, and lost the use of the first joint of his left thumb from frost-bite; the thumb became crooked and Clapperton got the nickname of 'Hooky'. This is Nelson's account, but in the memoir by Clapperton's uncle prefixed to the account of his second expedition, it is stated that he lost his thumb when trying to carry a boy during a retreat across the Canadian Lakes over the ice.

At one time Clapperton thought of marrying an Indian princess and settling in Canada. He used to give great feasts to the Indians, and indeed this led to a deficiency in his accounts to the victualling department. This deficiency was

afterwards deducted from his half-pay and involved him in financial difficulties, from which he was not altogether relieved until his return from his first African expedition.

Whilst serving on the Canadian Lakes he used to have a hole broken in the ice for him to bathe every morning, but could not persuade any of his companions to join him.

He used to decorate his log-book with sketches of the sails carried by his ship day by day, together with sketches of the coast, harbours, &c. The officers who examined him for his promotion to lieutenant were so struck with this log-book that they sent it to the Admiralty, where it still remains.^[565] He was promoted lieutenant in 1816.

In 1817 he returned to England on half-pay and became acquainted with Dr. Oudney. Nelson tells an amusing story of how Clapperton was taken in by a swindler. This gentleman, who represented himself to be a minister of religion anxious to pay his tailor's bill, borrowed £10 from Clapperton, and in return asked him to dinner at Barclay's Hotel in Edinburgh.

'An excellent dinner was set upon the table and discussed. Madeira, champaign and other expensive wines were called for, and the two got cheerful, joyous, happy, glorious. At length the swindler made a pretence for going out a little, and Clapperton found he had to pay £2 or £3 for the dinner as well as the original loan, which he never saw again.

He was appointed to the African Mission in 1821, returned to England in June, 1825, and was promoted Commander. In the same year he was sent out on a second expedition, whose object was to reach Sokoto from the West Coast. Clapperton reached Sokoto, but died at Chungary or Jungavie, near Sokoto, in April, 1827.^[566] The account of his expedition was published by his servant, Richard Lander. Clapperton had a noble figure; he was six feet high and broad-chested.^[567] He never married.

It will probably have been fairly obvious to any one who reads between the lines of Denham and Clapperton's account of their travels, that their relations were not altogether harmonious. That they were not so is abundantly clear from the Tripoli Consular Correspondence and the Denham papers in the R. G. S., in fact, for a great part of the time they seem hardly to have been on speaking terms. Doubtless there were faults on both sides, though on the whole the correspondence leads one to side with Denham. Clapperton himself admits in

his journal on the road to Sokoto during his second expedition that he was not a mild-tempered man.^[568] Oudney, his bosom friend, calls him ‘a rough diamond’, and he appears to have been clearly in the wrong in refusing to obey Denham’s orders, though the latter was very likely not tactful. It was not unnatural, though it may sound snobbish, that the ex-A.D.C. to one of Wellington’s brigadiers should find himself little in sympathy with the ex-cabin-boy^[569] and the self-taught surgeon’s mate, both of them Scotchmen, a fact little likely to prejudice an Englishman of those days in their favour. In a private letter to his brother Charles, now in the R. G. S., Denham remarks of Oudney: ‘Neither had his professional or pleasurable pursuits ever placed him in any other situations than the gun-room of a man-of-war and his rooms in Edinburgh, twice on a horse in his life, and except by water I think he had never travelled 30 miles from Edinburgh.’

In another private letter to his brother Charles he calls him ‘Dominie Sampson with more cunning’. In another, apropos of Clapperton, he laments that a ‘gentleman’ had not been chosen as his assistant.

On Feb. 1, 1822, before they left Tripoli, Denham proposed to his companions in writing that Europeans—except Dr. Oudney, who would be busy with scientific work—should take it in turn to be ‘on guard’, or, as we might say, ‘officer of the day’, for twenty-four hours. Duties—to see sentries posted, stores issued, &c., &c. Dr. Oudney objected and the scheme was not adopted. Oudney and Clapperton considered the proposal of too ‘martial a nature’, and seem to have thought Denham was trying to ‘boss’ them.

Denham was, indeed, far from conciliatory, and there is a most offensive letter from him to Clapperton, written some time after their arrival at Kuka, in which he plainly states that he considers Clapperton’s conduct throughout to have been unworthy of an officer and a gentleman, and suggests that he shall mend his ways. The breach was very much widened when Clapperton discovered that Denham had not informed him of a charge of the grossest immorality that had been made against him by some of the Arabs of the caravan. This is the charge referred to in the Sheikh’s^[570] letter to Clapperton printed on p. 430 of vol. ii of the *Travels*. Denham afterwards wrote a formal statement disclaiming any belief whatever in the charge, and explaining that the strained relations existing between them had made him unwilling to mention the subject.

The real fault of the matter lay with the Home Government, who had not made the powers and position of the various members of the expedition sufficiently clear; hence the jealousy and distrust, without which the materials brought back by the Mission would probably have been more ample than was actually the case. Denham and Clapperton were both fine specimens of their respective Services, and deserve equal honour and credit. The extracts from correspondence will now be intelligible.

The travelling names of the various members of the Mission were as follows:

Major Denham.	Rais Khaleel.
Dr. Oudney.	Tibeeb.
Clapperton.	Abdulla.
Toole.	Rais Ali.
Tyrwhitt.	Tair.
Hillman.	Ali.

(a) Disagreements

Extract from official letter of Colonel Hanmer Warrington, British Consul-General at Tripoli, to Robert Wilmot, Esq., M.P., Under Secretary of State for the Colonies, dated Tripoli, July 4, 1822.

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PS.—Much difficulty has already arisen and much undoubtedly will arise unless one Gentleman is appointed as the Head of the Expedition, as opinions, Interest, and private Feeling will always clash and prove detrimental to the Public Service. Pray excuse the observation.

H. W.

Extract from letter of Capt. Clapperton to Consul Warrington, dated Morzuk, September 10, 1822.

. . . A word for the Major, he has gone^[571] without ever having communicated to us his intentions directly or indirectly, and left me without any

instructions. I shall be silent upon his Conduct any further than his loss will be a gain to his country. . . .

Extract from letter of Major Denham to Consul Warrington, dated Sockna, October 11, 1822.

. . . Dr. Oudney's natural disposition leads him prematurely to suspect a want of good Faith in those around him; the feeling may be pardonable while it continues to be an unexpressed one, and the Slight disagreements that have existed between us, I say slight because on my part I am sure there was nothing personal in our differences, hardly deserve such insinuations as are thrown in the letter he writes to you.

It is a feeling however that has been rankling in his Breast ever since the day of his appointment, when they wrote him from the Colonial Office to say that Clapperton might accompany the Expedition as my assistant, while he was to consider himself as Consul at Bornou.^[572]

This feeling is anything but in unison with the mildness of the D^r. exterior, and unjust towards me, as I was not the framer of my own Instructions. . . .

On our joining the party at Memoon I found that Clapperton had commenced Sketching the route, and from delicacy to him I abstained from interfering in any way, as by so doing I must have taken some of the Credit which I was willing exclusively should attach to himself.

I have differed in opinion from Dr. Oudney and do so still on some points, but this arises, I may say entirely, from my taking the Instructions of Lord Bathurst as the Basis upon which I act, and his, to use his own expression, endeavouring to forget that he ever received any. . . .

Extract from letter of Dr. Oudney to Consul Warrington, dated Mourzouk, October 12, 1822.

. . . My worthy friend Clapperton is quite disgusted with his [i.e. Denham's] returning and threatens to return, but I hope I will be able to prevail. . . . Clapperton's loss would be that of a rough diamond which could not be found in a country such as this. . . .

Letter from Consul Warrington to Robert Wilmot, Esq.

Private.

Tripoli, Nov. 4, 1822.

SIR,

I have abstained for some time past acquainting you with a Hostile Disposition existing in the Southern Mission. I think it would be culpable in me, if I were to delay any longer offering my opinion, and which I conceive ought to be done by the medium of a private Letter.

I have therefore the honour to refer you to the accompanying copies of Letters which will speak for themselves.

I fear that Rancorous feeling may militate against the Public Interest, and I am inclined to think, that in the present, as well as in similar cases, blame may attach to each Party.

The Bone of Contention appears to be the great jealousy which has from the first manifested itself between Dr. Oudney and Major Denham, and probably the Breach has been widened by various paragraphs in the News Papers, saying that the Mission is under the Immediate direction and auspices of the latter.

For my own part I conceive their Instructions are clear enough, and that the Orders of those Gentlemen are perfectly distinct, and ought not to cause any division of feeling, or friendly understanding.

I am certain it is impossible to reconcile these Gentlemen, and I should strongly recommend that Lt. Clapperton should be attached to Dr. Oudney. They are countrymen, very Old Friends, and Dr. Oudney has undoubtedly the most commanding Influence over Lieut. Clapperton.

They are prosecuting their Duty with the most zealous Exertion. . . .

It was impossible to have selected two People so well suited to the Service. . . .

Major Denham is of a different cast of character, and is more the Man of the World, and He both thinks and acts widely different, but at the same time he evinces every Disposition and Inclination to perform His part of the Interesting Research. . . .

Extract from letter of Consul Warrington to Robert Wilmot, Esq., dated Tripoli, December 31, 1822.

. . . In Quarrels Blame may attach to each Party, and I think that it is the case in the Present Instance. The Great Bone of Contention appears to me a jealousy as to whom the Mission is confided to. I think the Orders and Instructions speak for themselves, which clearly show that Oudney and Denham are distinct and separate, it being the duty of the Former from his Official appointment to afford every facility and to give every assistance to the latter. . . .

Extract from letter of Dr. Oudney to Consul Warrington, dated Bilma, January 14, 1823.

. . . My worthy friend Clapperton and the Major have had a little quarrel on the subject of duty. The latter was pleased to issue orders, rather too much of the field, directing Clapperton to give him from time to time, when he might require, the Latitude of the different places and the different courses. He refused to obey, and here the matter rests. He (C.) came out as his companion, not to be subjected to the whim of any man. He had been abandoned in Mourzouk, but notwithstanding that and the martial tenor of the orders, he in his spirited reply told him he would assist when he called upon him, but would in no respect perform what it was the other's to execute. I allowed my friend to settle the business himself, and I am well pleased with the manner he has conducted the affair. . . .

Extract from letter of Capt. Clapperton to Consul Warrington, dated Bilma, January 14, 1823.

. . . The Major has taken upon himself a very extraordinary authority in writing to me on service to give up to him copies of my courses and obs^{ns} since we left Morzuk which I have very politely refused to do . . . I allowed him to copy all my obs^{ns} up to Morzuk before he left that place for Tripoli, as he is not able to take any himself. After the cruel and un-Englishlike abandonment of me at that place without leaving me his instructions, writing to say he was gone, or how I was to act, can this man expect that I will allow him to take the credit of what I do? No, I engaged to come out as his companion and assistant, not to do the whole of the duty for which he was sent out to do, and which he would claim as his own. Were I to do myself and my country so much injustice as to give him a copy—he has been educated at his country's expense, let them see what he can do, and I will let them see what I can do who never cost them

sixpence for my education^[573] and am anxious to let the world know who that Lt. Clapperton is whom he mentioned in the papers along with his own as if by accident . . . I cannot say when you will see me again, for though the Major may and will return, as his zeal is all wind, I shall do or die if they will let us into the country. . . .

Extract from letter of Major Denham to Consul Warrington, dated Bilma, January 14, 1823.

. . . Mr. Clapperton by that consequential stupidity, for I know not what else to call it, made himself so unpopular that Boo Khaloom^[574] has long since ceased to notice him even by a common salute, and the Arab Sheikh scarcely speaks to him; except with the lowest Arabs and the servants of the Kafle he converses with no one. He has thrown off all controul and refused to act under my orders by Letters which I shall send you copies of the first opportunity. I told him for his own sake I hoped he would alter his determination, that for me it was a matter of little importance, he nevertheless persisted and the consequences must fall on his own head. The Doctor, who says but little since his illness, took no part in the business—before the curtain—but I verily believe they both thought I should be sadly distressed at this declaration of my assistant. . . .

Extract from letter of Consul Warrington to Robert Wilmot, Esq., dated Tripoli, March 31, 1823.

I have just received Letters from the Travellers which are generally speaking of a very satisfactory nature, altho' that Hostile Feeling between Major Denham and Lt. Clapperton appears so much to have increased that I feel myself fully authorized to avail myself of the Conditional Power Invested in me, in your Dispatch of the 12th Janry. last.

I send copies of the Letters I have received with my answers. . . .

Extract from letter of Consul Warrington to Dr. Oudney, dated Tripoli, March 31, 1823.

. . . With regret I perceive it is hopeless to anticipate a favourable termination to the misunderstanding between Major Denham and Lt.

Clapperton. It is not for me to give my opinion of right or wrong, but it is my duty to endeavour to prevent the public service suffering, and rendering abortive the views of His Majesty's Government, and the most effectual way I conceive to be to remove Lt. Clapperton from Major Denham and exclusively attach him to you. He will therefore consider himself justified in acting with you independent of Major Denham, and I hereby attach the whole responsibility on myself by the assumption of such power, and I feel persuaded that the Earl Bathurst will approve of the decision. . . .

I shall write both Major Denham and Lt. Clapperton on the subject of the latter being removed from the former, and to prevent mistakes you will be pleased to consider this as official. I think you acted wisely not to become a party in the embroil. . . .

Extract from letter of Consul Warrington to Major Denham, dated Tripoli, March 31, 1823.

. . . It will afford you infinite pleasure to learn that the Bone of Contention is removed, His Majesty's Govt. having most handsomely accommodated matters by placing Lieut. Clapperton under the exclusive orders of Dr. Oudney, and I am commissioned by Earl Bathurst to signify that His Lordship deems it advisable on Public Grounds at the same time that Mr. Tyrwhitt is authorized to join you in the place of Mr. Clapperton.

Extract from letter of Major Denham to Consul Warrington, dated 'Kouka in the Kingdom of Bornou, 4 April 1823, Therm^r 104 in the Hut'.

Mr. Clapperton has made great havock amongst the Gazelles, but it is meat I love not, neither do I the sport, for you are obliged to lay about for hours under the trees in order to catch these poor animals asleep, when you get a shot; this appears to me not giving the animal a fair chance. I have hit but one and he gave me such a look with his soft black eye—which has a wonderful brilliancy—that I vowed to sin no more in that way at least. . . .

Extract from letter of Consul Warrington to Major Denham, dated Tripoli, July 11, 1823.

. . . I trust in God the very circumstance of Clapperton being removed to Dr. Oudney will be the means of restoring Friendship and good will. I am ordered to convey to you both the Displeasure and animadversion of His Majesty's Govt. and you may both depend if this Hostile disposition still continues it will be fatal to the future Prospect of you both, and there is a handsome delicacy in making me the medium of conveying this Disapprobation, probably not wishing to wound your feelings, in such a remote situation. Let me beg and entreat you will meet each other halfway and accommodate this affair. . . .

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PS.—Probably my expressions not being sufficiently strong as alluding to the misunderstanding between you and Lt. Clapperton, I beg leave to intimate to you both the strong sense which is entertained of the Impropriety of your Conduct by His Majesty's Government.

[A letter in similar terms was sent by Warrington to Clapperton.]

Extract from letter of Consul Warrington to Dr. Oudney, dated Tripoli, August 1, 1823.

Three days since I received your Despatch of the 15th May. . . .

I trust Denham and Clapperton are reconciled, as I do not see any symptoms of hostility in this communication. . . .

Extract from letter of Consul Warrington to Dr. Oudney, dated August 25, 1823.

SIR,

This letter will be delivered into your Hands by Ensign Toole of the 80th Regiment, a most Gentlemanly Young Man and agreeable Companion. . . . He is expressly sent as an assistant to Major Denham instead of Lt. Clapperton, who is attached to you. . . .

I send Major Denham 350\$ by Mr. Toole as it must be unpleasant to Him to apply to you on every trifling occasion, and I have given Mr. Toole 139\$ for the same purpose. . . .

Extract from letter of Clapperton to Warrington, dated Kuka, Dec. 10, 1823.

. . . I had the pleasure of receiving your Letter when I was laid up, which added much to my Recovery, and Denham's civilities to me when sick has made matters as well as they can be in this country. . . .

Extract from letter of Major Denham to Consul Warrington, dated Mourzouk, Nov. 25, 1824.

. . . It would be the extreme of Injustice were I not now to state that after the recovery of Mr. Clapperton from his very severe sickness until his departure for Soudan, so great an improvement took place in the understanding between us, that our Evenings were passed in front of poor Dr. Oudney's hut, who was unable to sit in the Air, with a degree of sociability which had not existed since my return to Fezzan with Bookhaloom.^[575] Subsequent to Mr. Clapperton's return from Soudan the same cordiality has been preserved, he has indeed been uniformly attentive and willing to afford me every assistance during the tedious and fatiguing journey we have made together. . . .

(b) Tyrwhitt and Toole

The following extracts concern the two young men who were sent up at a later date to strengthen the Mission.

Letter from Dr. Oudney to Consul Warrington.

Tripoli, Dec. 10, 1821.

SIR,

You already know my sentiments on the importance of having a British Resident at Mourzouk, and Capt. Denham agrees with me in stating that the Greatest Benefit may be expected by us from such an Agent. By such an arrangement our communication with you will be kept up, and as you judiciously remark, should Sickness or any unforeseen occurrence compel us to fall back, a safe Asylum shall be found at Mourzouk. We cannot but commend your choice of Mr. John Tyrwhitt. He appears to be a Gentleman well calculated for such a situation, and he will undoubtedly support with Credit the name and character of the British Nation.

I have the honour to be,

Sir,
Your most obed^t
H^{ble} Serv^t,

WALTER OUDNEY.

H^r Warrington, Esq.,
Consul General,
Tripoli.

Letter from Consul Warrington to Earl Bathurst.

Tripoli, 12 December, 1821.

MY LORD,

I have already had the honour to recommend to your Lordship's consideration the appointment of a resident Vice-Consul at Mourzouk.

The good arising to the expedition under the direction of Dr. Oudney and Capt. Denham is so obvious from that appointment that I shall only refer your Lordship to the accompanying papers Nos. 1 and 2, and I trust the nomination of Mr. John Tyrwhitt to that situation will meet with your approbation.

He is cousin to Sir Thos. Tyrwhitt and Nephew to the Champion and a Gentlemanly Quiet young Man, and by not Trading in any shape He is not likely to excite suspicion and jealousy.

I mention Mr. Tyrwhitt's pedigree, my Lord, from a thorough conviction that the higher a Man's connections are, the less He thinks of dangers and difficulties and can endure Privations better than other Men. With every sentiment of the highest consideration,

I have the honour

to be, My Lord,

Your Faithful &

ob^t serv^t,

HANMER WARRINGTON.

The Right Hon^{ble}
Earl Bathurst, K.G.,

His Majesty's Secretary of State,
&c., &c., &c.,
Colonial Office.

[The appointment of a Vice-Consul at Mourzouk was negatived by a letter from the Treasury to the Colonial Office dated January 17, 1822, on the ground that the probable advantages would not justify the grant of the salary.]

The next extract—though of a previous date—shows who Mr. John Tyrwhitt was.

Extract from letter of Mr. Tyrwhitt, Senr., to Consul Warrington, dated Cook's, April 12, 1818.

MY DEAR WARRINGTON,

As it is an object of the highest importance to me to get my son John abroad for some time . . . I know not any one so proper to make an application to on this occasion as yourself. . . . He has been very wild and extravagant. . . . He is now out of the Navy. . . . I wish him to have the least possible pocket money as he has a great propensity to extravagance. . . .

[It appears that Col. Warrington owed Mr. Tyrwhitt, Senr., £500, on which he had to pay interest at the rate of 14% per annum. Mr. Tyrwhitt asks Warrington to take his son into his family and keep an eye on him: in return he is willing to waive the 14% per annum, i.e. about £70. The son remained with Col. Warrington for three years, to whom he appears to have been a source of considerable expense.]

Extract of letter from Tyrwhitt to Warrington, dated Mourzouk, July 25, 1822.

I am not much disappointed at not being confirmed. . . . However, to make the best of it, it is a long way to come for nothing, not even thanks. I suppose the Doctor will pay my expenses down. I must not return to Tripoli empty-handed, for there are some fine ostrich feathers here, and I had promised both my Mother and Sister to send them some. . . .

Extract of letter from Consul Warrington to Robert Wilmot, Esq., M.P., Under Secretary of State for the Colonies, dated Tripoli, August 26, 1822.

. . . I conclude Dr. Oudney will take Mr. Tyrwhitt along with him as a volunteer, as from his easy good-natured Disposition, ever alert to make Himself useless [*sic*], I feel persuaded Dr. Oudney could not have a better person. . . .

Extract from letter of Consul Warrington to Robert Wilmot, Esq., dated Tripoli, Nov. 4, 1822.

. . . Two more Gentlemen would be of very important advantage to Dr. Oudney, as he has a wide Field indeed to act upon. . . . I can send them from Tripoli to Bornou for 100£ provided they travel as Mr. Tyrwhitt did from Mourzouk. I believe that Gentleman would be happy to make one of such a Party, and Dr. Oudney wishes to have him with the Mission. . . .

[Tyrwhitt, however, returned to Tripoli and went for a trip to Italy.]

Extract from letter of Consul Warrington to Lieut. Clapperton, dated Tripoli, July 11, 1823.

. . . I hope to send Tyrwhitt off immediately, as I am in daily expectation of seeing him, unless Spider-like he may possibly entangle himself in his own Webb, as it is said a lady of that name has a little influence. My patience is near exhausted, and if he does not come, I shall be much displeased with him. . . .

Extract from letter of Consul Warrington to Robert Wilmot Horton, Esq., [\[576\]](#)
Under Secretary of State, Colonial Office.

Tripoli, 18 July, 1823.

SIR,

I have the honor to refer you to No. 1, being a copy of a Letter from Mr. Tyrwhitt, who from sickness is prevented repairing to Bornou to join the Mission.

I consider myself fully authorized by the Earl Bathurst to send one Person as an Assistant to Major Denham in lieu of Lieutenant Clapperton, who is now

attached to Dr. Oudney, and conceiving it is perfectly immaterial who is sent provided he is equal to the respective duty required of Him, I shall therefore send this dispatch to the Government of Malta and solicit His Honor the Lieutenant Governor will be pleased to nominate a Gentleman for the appointment. . . .

Extract from letter of Major Denham to Consul Warrington, dated Kouka, Jan. 20, 1824.

. . . I am sorry for the cause of Tyrwhitt's declining the appointment, but he has certainly acted the wisest part in so doing. Strictures and gout, or indeed either of them, would afford strong reasons why a man should not wish to embark in a service of this nature.

Malta, August 10, 1823.

DEAR TOOLE,

I have received to-day a letter from Mr. Consul Warrington at Tripoli, stating that Mr. Tyrwhitt, who was to have proceeded to Bornow to join Major Denham as his Assistant on the African Mission, is prevented doing so by illness, and requesting that some young Gentleman, who would volunteer in so arduous an undertaking, may be sent over from this to take Mr. Tyrwhitt's place.

Colonel Warrington mentions that the Person sent should possess some knowledge of the sextant and of the Artificial Horizon, and further must be prepared to encounter some dangers, much fatigue and many privations. He must be a young man of strong constitution and inured to fatigue and fasting, and above all must possess great ardor and zeal in the undertaking.

Being well aware of the many qualifications you possess for a Mission of this kind, viz. that you are an excellent linguist—speaking French, Italian and German fluently, and that you understand the Maltese perfectly which is nearly Arabic, that you are a good Draughtsman—can survey, understand the use of the sextant, and also that you are healthy and stout and inured to fatigue at all times, and having often heard you express the greatest thirst and ardor for travelling in remote countries, and besides being fully convinced that there is no one in Malta so well qualified as you are for the object proposed by Colonel Warrington, I have written you this, that you may volunteer to go, if you should

feel inclined to embrace an undertaking, which I must impress upon you necessarily involves in it many dangers and immense fatigue.

I must also state to you that Colonel Warrington in appointing an Assistant to Major Denham acts under the instructions of Earl Bathurst, but he does not mention in his letter that he is authorized to hold out any specified prospects of future advantage, but you may depend upon this that my Lord Bathurst can never fail to give due consideration to your services, provided you make yourself useful to the Mission.

Your necessary Expenses will all be defrayed, and in these you must pay every attention to the strictest economy, and I understand also that a small personal allowance is given, but what it is I do not know.

If you accept this proposal, you must be ready to start for Tripoli in 48 hours, as there is a Vessel on the point of sailing, and it is absolutely necessary that you proceed to Tripoli without delay,—and in that case you had better lose no time in applying for leave of absence for a twelve-month in a regular way through your Commanding Officer to the Major General Commanding.

I am, &c.

H. GREIG. [\[577\]](#)

Malta, August 10, 1823.

2 p.m.

DEAR GREIG,

In answer to your note of this morning, I hasten to assure you that I embrace the proposal of going as an Assistant to the African Mission with the greatest delight, and I shall be ready to start for Tripoli in 48 hours, having obtained the necessary leave of absence from my Commanding Officer and Major General Sir Manley Power.

I do not mean to say, that I accept this situation without the prospect of encountering both danger and fatigue, but I have no fear of either, and I look forward to the journey with real pleasure.

The opinion you are good enough to express of my qualifications for the Mission is much too partial, but I promise you, that neither zeal, industry, nor a Constant desire to make myself useful will be wanting.

It is far from my intention to allude to any remuneration for my Services, but I trust, that if I make myself useful to the Mission, that my Conduct will be favourably considered by Earl Bathurst.

I have now been six years an Ensign in the 80th Regiment,^[578] and I may mention that my Ensigncy was given to me by H.R. Highness the Duke of York at the request of my revered friend and Patron, Sir John Stewart, by whom I was brought up and educated. And I trust it is not asking too much, that if sickness or other causes should prevent my return from this expedition, that my Lord Bathurst may be entreated to interest himself to procure my Ensigncy for one of my younger Brothers, and you are aware that I have four, all of whom are under the age of 18 and quite unprovided for.^[579]

I am, &c.

E. TOOLE.

Extract from letter of J. Maclean, Major Commanding 80th Regt. to Hector Greig, Esq., Acting Chief Secretary to the Government of Malta, dated August 12, 1823.

. . . Major Denham (whom I have the pleasure of knowing) will, I am persuaded, consider himself fortunate in the selection you have made, as he will find in Mr. Toole a zealous and able Assistant as well as a most Gentlemanly Companion.

Did I not consider the present Undertaking as likely to prove of advantage to Toole, I should particularly regret his departure from the Regiment at this time, as, in consequence of the intended departure of Lieutenant Penny for the Continent, it was my intention to have appointed him to do the duty of Adjutant during the absence of that Officer.

Extract from letter of Consul Warrington to Robert Wilmot Horton, Esq., dated Tripoli, August 25, 1823.

SIR,

I have had already the honor to inform you that in consequence of Mr. Tyrwhitt's Health not allowing him again to join the Mission, that I had applied to the Lt. Governor of Malta to send a Gentleman instead of Him, and He has

accordingly been pleased to approve of Ensign Toole of the 80th Regiment joining the Mission as Assistant to Major Denham.

I have no hesitation to declare that the Public Service will be the gainer by the Exchange, as I can safely say I never saw a finer young man, or one more calculated for the undertaking. . . .

Extract from letter of Consul Warrington to R. Wilmot Horton, Esq., dated Tripoli, December 20, 1823.

. . . This Morning I have been informed Mr. Tyrwhitt is in Malta on his way here. I am most happy to hear it, as I am confident there is full Employment in the Interior for the whole Mission. The Road is as free as I always said it was, and the Inst. Mr. Tyrwhitt arrives I will send Him off. . . .

Extract from letter of Consul Warrington to Dr. Oudney, dated January 17, 1824.

MY DEAR SIR,

The arrival of Mr. Tyrwhitt and his appointment to the Mission will afford you all, without doubt, much satisfaction. As it will be about three weeks before he leaves this, I forward a Courier with copies of Dispatches for you and Major Denham, which I have received from the Colonial Office. . . .

Your little godson, Walter Bornow,^[580] is now under the Vaccine Inoculation, from whose Arm I send you the Matter, and Buttabel^[581] writes to state its introduction here and the efficacy of it. I know no one whose Natural Disposition and extensive Ability is so devoted to the benefit of our Fellow Creatures as your own, and I think you will evince it by endeavouring to introduce the Vaccine. . . .

Extract from letter of Consul Warrington to Mr. Toole, dated Tripoli, January 21, 1824.

. . . His Lordship, I am glad to say, approves of your appointment, and has ordered £100 per annum salary being allowed you as well as Mr. Tyrwhitt. . . .

Extract from letter of Consul Warrington to R. Wilmot Horton, Esq., dated Tripoli, February 24, 1824.

. . . 28 Feb. I took leave of Mr. Tyrwhitt at Melra yesterday morning, fifty miles from this. . . .

Extract from letter of Major Denham to Consul Warrington, dated Kouka, April 16, 1824.

. . . My amiable and clever companion and friend, Ernest Stuart Toole, who really deserved all that his Friends at Malta wrote about him, has fallen a victim to that Fever of this Country with which I have seen so many attacked.^[582] He died at Angala, a town about 60 miles South East of where I am now writing, on the 26th of last February. . . .

He was most generally beloved here, particularly by the Sheikh, with whom we used sometimes to pass an hour together in his Garden; and indeed I believe we all might have died without creating the sensation that my poor friend's death has occasioned. . . .

Assisted by the Sultan's slaves, Columbus^[583] and myself laid him in a deep grave to the N.W. of the Town.^[584] . . .

A few more details of Toole's death are given below.

Extract from letter of Denham to H. Greig, Esq. [who from an endorsement on the back we learn was Toole's brother-in-law], dated Kouka, March 10, 1824. [R. G. S.]

. . . He swallowed a little tea and seemed revived; about 6 he attempted to speak, but could articulate intelligibly nothing but the word Mother two or three times, and from that time until half-past 11 on the same day, when he breathed his last, he scarcely moved, and expired without even a struggle or a groan. . . . Almost the last words he uttered were requests that his papers . . . should in case of any accident befalling him be delivered by me into your hands. . . . His request I shall, please God, comply with. . . . His sketches will, I think, be valuable, very valuable, and should I not be permitted to return myself, I do hope that with any notes of mine that may be considered worthy of publishing, they may be given to the world, for any literary fame that my exertions may entitle me to will lose half its value in my estimation if the name of my amiable colleague is not coupled with my own. . . .

[I have not been able to trace these papers or sketches, in spite of kind assistance from E. L. Bonavia, Esq., of the Chief Secretary's Office, Malta, and from the Crown Agents. Mr. A. C. Maberly of Eccles, whose wife was a great-niece of Greig, informs me that none of Greig's papers have been preserved.]

Extract from letter of Major Denham to Consul Warrington, dated Kouka, May 23, 1824.

MY DEAR SIR,

Mr. Tyrwhitt arrived here on the 20th in good health, after performing the journey with great Expedition. Nothing can be more satisfactory than the Letters of which he was the Bearer, and the Presents for the Sheikh were everything that could be wished.

I send you, according to an Old Promise, a Bottle of the Niger Water. . . .

Extract from letter of Consul Warrington to R. Wilmot Horton, Esq., dated Tripoli, August 18, 1824.

. . . I shall send four Spears and a Shield and a Bottle of Niger Water (which I had the whim to request might be sent me) to you, and I shall feel flattered if you accept. Of no value but as curiosities. I have another bottle, and in drinking success to the Mission I hope I shall find its efficacy in inspiring me with additional zeal for the Promotion of Discovery. . . .

[There seems to have been a craze for Niger water. Barth promises to drink a friend's health in Niger water, vid. Benton, *Notes on Some Languages, &c.*, p. 298.]

Extract from letter of Denham to Lord Bathurst, dated Kouka, May 23, 1824.
[R. G. S.]

. . . I have filled and secured in the best way that I am able, a stone bottle with the Niger water; it was taken from a spot about two miles from the embouchure of that River.

[In the same letter he suggests that the R. Yo or Wobe is the continuation of the Niger. This water was probably therefore bottled somewhere near where the

Yo flows into Lake Chad.

Even at the present day it is a common saying that if you have once drunk from the River Niger, you will always return to the Coast.]

Extract from letter of Mr. Tyrwhitt to Consul Warrington, dated Woodi, Sept. 6, 1824.

MY DEAR CONSUL,

The Travellers having at last determined on returning to England, I have accompanied them thus far on their road, it having been settled that I should remain at Kouka as Vice-Consul in compliance of the wishes of the Sheik till Lord Bathurst's pleasure shall be known. . . .

I must confess that the remaining behind does not afford a very pleasant or very cheering Prospect, but I hope under all circumstances I have acted for the best, and that the step I have taken may meet with approbation. . . .

[Tyrwhitt died at Kuka on October 22, 1824, and his death was reported to Warrington by the Sheikh, who also sent an inventory of his effects (which included 52 books), cf. Denham, vol. ii, p. 434. It does not appear what eventually became of his property. It was to be handed over to Capt. Pearce, who accompanied Clapperton on his second expedition, and who seems to have been intended to be Consul in Bornu. Pearce, however, died soon after landing on the Coast. War had broken out between Sokoto and Bornu, and Clapperton was unable to revisit the latter country. No attempt to re-establish the consulship appears to have been made until the expedition of Richardson and Barth in the 'fifties.]

(c) **Hillman**

Another member of the expedition was a naval shipwright named William Hillman, a native of Somersetshire. He appears to have been a typical old salt, but in spite of constant ill health reached England again in safety. His pay was £120 per annum (vid. Denham, Introductory Chapter).

Extract from letter of Major Denham to Consul Warrington, dated 'Kouka in the Kingdom of Bornou, 4 April 1823, Therm^r 104 in the Hut'.

. . . Old Chips sometimes causes me to laugh, which does my heart good, he has such a knack of rallying from Death's door almost that his illness seems to have no danger in it. Drink Grog he will whenever he can get it, tho' it kills him. He told me the other evening that he had made up his mind not to learn 'Harribik'. 'I can't make no hand on it,' said he, 'so I get Columbo to teach me a little Hitalian, for I tell you, B—— the Navy Board. I hope the Colonial Board will provide for me, who knows but what I may be a King's Messenger, and the King, God Bless him, may say, Hillman or Mr. Hillman, here take this letter to Hitaly, why I must know Hitalian for that. . . .

.

Sunday, April 6, 1823.

Oh for a glass of the Bashaw's cold punch. We had yesterday a most suffocating day with Therm. at 104. This bids fair to be just such another. The water here is worse than at many places on the Road, altho' here it is a little Cooler, for there we drank it about the heat of tea. I found but about three bottles of Rum on my return to Mourzouk, and Hillman, when I complained, tho' Groggy at the time, said, 'Why I'm not the man to say a word against another, No, No, but that are little Tyrwhitt did drink, by G—— like Winkins—for myself I can't touch liquor now.'

Extract from letter of Consul Warrington to R. Wilmot Horton, Esq., dated Tripoli, June 13, 1824.

. . . In one of Earl Bathurst's Dispatches he suggested the propriety of Mr. Hillman's constructing a Boat to explore the outlet of the Great Lake.

The idea was so good and so obviously beneficial for the object of the Research, that I followed up His Lordship's suggestion by requesting the Boat may be built at Malta in that Portable way to be carried on two Camels, and taken to Pieces and put together without difficulty. The Government of Malta kindly acceded to my wishes, and Mr. Simonds, a most scientific gentleman, is now engaged in constructing one. . . .

[We hear no more of this boat, but it is interesting in view of Barth and Overweg's famous boat. From a letter of Columbus^[585] to a friend in Tripoli it appears that some of the members of the expedition did embark in boats (presumably native canoes) and navigate the waters of Chad for a short distance.]

Denham's views are given in the following:

Extract from fragment of a letter from Denham to Lord Bathurst, dated June 18, 1824. [R. G. S.]

. . . Without two boats could be built to carry fifty men each with fire-arms they would stand but a bad chance with the Biddoomy on the Lake. Hillman says that one shipwright alone could not build a boat of any kind, and he has certainly never since our arrival here enjoyed a sufficiently good state of health for him to make the attempt had it been requisite.

The following is a description of the Sheikh's victory over the Baghirmi invaders:

Extract from letter of Major Denham to Consul Warrington, dated Kouka, April 16, 1824.

DEAR SIR,

Our minds have been greatly relieved within the last ten days by a victory obtained by the Sheikh's people over a very large force of the Sultan of Baghermi, headed by two hundred of the principal persons in his Kingdom, including nine of his sons, seven of whom are amongst the killed and one a prisoner. . . .

The two Guns^[586] for which Hillman had made Carriages and myself Cartridges with Canisters holding 16 Musket balls each, appear to have done great execution, altho' only fired once, and have greatly alarmed the people, even in whose defence they were levelled. . . .

A fuller account of the battle is contained in the following:

Extract from letter of Denham to his brother Charles, dated Kouka, April 18, 1824. [R. G. S.]

. . . The Sheikh advanced to within 200 yards before he unmasked his little guns, supported by about fifty slaves and Arabs with fire-arms, who were formed to the right and left, while immediately in the rear he appeared himself at the head of his Kanemboo infantry with the double barrelled English gun in his hand and an ammunition belt on his shoulders dressed in a simple barracan like an Arab. He had told the Chiefs previously that it was his intention to fight on foot, that he expected all the Arabs to follow his example, and to encourage the slaves who were but young at the use of the firelock. . . . He placed his two Chiefs, Barca Gana and Ali Gana, on each flank with the Bornou and Shouaa horse, with orders if attacked, to make the best of it, but on no account to assist him. The Baghermi, who made for his Green Flag in one solid mass, were not a little alarmed at the effects of the guns at such a distance, . . . and a volley from the Arabs added to the confusion. They now fell back on the Bornou horse, who as usual gave way, and the attack of the Kanemboo with the Sheik at their head alone decided the fate of the day. . . . The little stream, called Gambalarou, near which the battle was fought . . . is said to have run red.

Again, Denham, vol. ii, p. 415, it is recorded that Hillman made a covered cart, to be used as a carriage or conveyance for the Sheikh's wives; the wheels were hooped with iron, and it was extremely strong, though neither light nor handsome.

Extract from letter of Hillman to Consul Warrington, dated Tripoli, Feb. 11, 1825.

HON^D. SIR.

After many difficulties I have got safe and sound once more in Tripoli, thank God. . . . I have taken the Liberty of sending a Lion's Skin, which I trust you will be pleased to accept . . .

With every due respect

Your most obliged

& very humble

serv^t.

W. HILLMAN.

Hillman was sent home in charge of the animals and baggage, via the long sea route (vid. Nelson, p. 94).

(d) **Sheikh El Kanemy** [*Shehu Lamino*]

The enlightened character of this ruler is made very clear in Denham's book and in the Correspondence. The two following extracts are only quoted to show the style in which he was addressed by the British Consul at Tripoli, and the type of articles he valued as presents. Bornu is very deficient in fruit and vegetables, and even the efforts of the British Administration have not been very successful in inducing the present native authorities to take serious steps to remedy the deficiency.^[587]

Letter from Consul Warrington to Sheikh Lamino [El Kanemy].

MOST EXCELLENT AND NOBLE SIR,

May the Great God who directs the Prosperity and happiness of all Mankind reward you for the Kindness and attention which you have been pleased to show my Countrymen, and that you will continue the same and extend your Powerful Protection to any place they may wish to go is my fervent Prayer.

Your three Sons, I am most happy to say, are about to return to their most respected Parent, and I sincerely hope they may arrive safe and find you in good health.^[588]

My August King and Master has been pleased to order some Presents to be sent you as a small token of that respect He feels towards you for your great liberality and kindness towards his Servants . . .

I send you by this opportunity a Saddle, Bridle, &c., as a Personal Mark of my great esteem and respect towards you.^[589]

I beg you will at all times command me whenever you may send any Person or Persons down here and that you may enjoy Health, Strength, and Happiness is the ardent wish of your Sincere Friend and Servant

HANMER WARRINGTON.

Extract from letter of Major Denham to Consul Warrington, dated Kouka, April 26, 1824.

. . . The Sheikh begs you will send him by the earliest opportunity the following articles, and for which he says he shall pay me.

Seeds, both flower and vegetable.

Fifty locks of guns.

Tea.

Winter melon seeds.

A set of small Cairo coffee cups.

Two or three Glass cups with handles, holding a pint or thereabouts.

4 Turkish Coffee pots.

[Two boxes were sent to the Sheikh by Denham after his return to Tripoli; the extract printed below concerns them. The presents brought by Tyrwhitt are mentioned in Denham and included swords, pistols, gold watches, and rockets.]

Extract from a letter of Denham to the Sheikh, dated December 4, 1824, apparently from somewhere in Fezzan.

. . . Ibrahim will give you a box in which is Tea, Sugar, Coffee, the glass cups you wrote for, two Coffee Pots and some Flints, and my friend Said will give you 50 Gun Locks and a small box containing coffee cups and seeds. You will also find two Musical snuff-boxes which my sister begs you will accept for the kindness you have shewn her brother . . .

[The Sheikh sent presents to the King of England.]

Extract from letter of Denham to Mr. Wilmot Horton, dated 44 Lower Brook Street, June 1, 1825.

. . . A horse from about the 10th Parallel of North Latitude with the 'Libida' or cloath covering wadded with cotton, which they use in battle as a defence from the poisoned arrows; four parrots, one of which died on the journey across the desert, two ostrich skins, a large box of 'zibet' or musk from the civet cat,

and twelve specimens of the manufactures of Soudan and the towns in the interior of the Kingdom of Bornou, to which he added one of his largest tents entirely of Bornou manufacture. These were all embarked on board the Brig *Britannia* previous to my leaving Leghorn, with several animals and birds, with all the minerals, plants, and other articles we had collected in the country.’^[590]

(e) Cost of the Mission.

Extract from letter of Consul Warrington to Earl Bathurst, dated Tripoli, September 11, 1825.

. . . In my office the sums drawn for amount to £8575 17s. 7d. The £5000 to the Bashaw^[591] reduces this sum to £3790, out of which must be deducted the 2500\$,^[592] say £530, will leave the expenditure at £3260 17s. 7d. Say the mission existed for three years and a half, which would make £931 13s. 7d. per annum . . .

Obituary notices appear in *Georgian Era*, iii. 75, 82; *Annual Register* for 1828, pp. 210, 495; and *Gentleman's Magazine* for 1828, pt. i, p. 568.

i.e. Marshal Beresford, in command of the Portuguese armies.

The battle of Toulouse was fought on April 10, 1814. The 8th is now The King's (Liverpool Regt.). I have not been able to trace at what period Denham was transferred to it. Perhaps it is a clerical error of the General's. The *Army List* never shows him as in the 8th Regt.

7th Portuguese Brigade, attached to 6th British Division. Vid. Oman, *Wellington's Army*, p. 372.

Now 2nd Batt. Dorsetshire Regt. Denham's diary during the Waterloo campaign and afterwards in Paris is in the R. G. S.

Now the Leicestershire Regt. His original commissions, now among the Denham papers in the possession of the Royal Geographical Society, show him as Captain in the 17th Foot and Major in the Army.

For an interesting account of the duties of this post and for the evidence given by Mr. Reffell, Denham's predecessor, vid. *Report of the Commissioners of Inquiry into the state of Sierra Leone*, 2 parts, 1827. A few sentences are printed on p. 14 of H. C. Lukach, *Bibliography of Sierra Leone*, Oxford, 1910.

In the four-volume edition of Denham, vol. iv, p. 256, it is mentioned that Clapperton's grandmother was a daughter of Col. Campbell of Glenlyon, the officer in command of the troops who committed the massacre of Glencoe.

Thus Nelson, but, according to M'Diarmid, Clapperton wrote to Mr. Scott, a banker of Annan, who applied to General Dirom [i.e. Lt.-Gen. Alexander Dirom, vid. *Dict. Nat. Biog.*], whose wife applied to her cousin, Capt. Briggs of the *Clorinde*.

Thus Nelson, but I have been unable to trace it either at the Admiralty or the Record Office, in spite of the kind assistance of the Admiralty Librarian, W. G. Perrin, Esq. Cf. [Appendix XX](#).

For details of his death vid. p. 277 of the account of his expedition. His grave has never been identified, but a brass memorial tablet has been put up in the Sokoto Provincial Office by Major Alder Burdon, C.M.G., first British Resident of Sokoto Province.

There is a very fine portrait of him in the quarto edition of the account of his expedition to Sokoto. From this and frequent references in the Tripoli correspondence, it is obvious that he was an exceptionally handsome and powerful officer. Cf. [Appendix XVII](#).

p. 176: 'At no time am I possessed of a sweet and passive temper.'

Sir John Barrow, who edited Clapperton's journal of his second expedition, remarks on p. xviii that Clapperton was evidently a man of no education. This is too severe. His letters are not ill-expressed, though occasionally somewhat incoherent.

i.e. Lamino.

Denham came to the conclusion that the Bashaw of Tripoli was wilfully delaying the Mission. He therefore returned from Mourzouk to Tripoli and embarked for Europe, in order to get the Home Government to expedite matters. He was overtaken, however, at Marseilles, by an urgent message from the Bashaw, returned to Tripoli, and was soon able to get matters settled to his satisfaction. His action was both prompt and successful, but it would have been more tactful to have informed his companions of his plans.

When Mr. Hanns Vischer was in Tripoli in 1906, he was presented by the heir of Col. Warrington with a book on Geometry, bearing on the fly-leaf the inscription, 'Walter Oudney, M.D., Surgeon R.N. and H.B.M.'s Col. at Bornou'. The book is by Leslie, published by Constable in 1817. Mr. Vischer presented the book to the Bornu Provincial Library.

In a letter to Warrington, dated Kuka, March 31, 1823, Clapperton says: 'He has been at the Royal Military College and Greenwich Observatory, I never was.'

The Arab merchant who was head of the 'Kaffila' or caravan in whose company the travellers journeyed from Tripoli to Bornu.

Vid. note on [p. 345](#).

He appears to have assumed the additional surname of Horton about this time.

Mr. Greig was Acting Chief Secretary to the Government of Malta. He retired in 1846, and died in 1873.

Now the 2nd Batt. South Staffordshire Regt.

This was also his dying request (vid. Denham, vol. ii, p. 26), and it was granted by Earl Bathurst.

I do not know what became of little Walter Bornow. I fear he must have died young, as he does not appear among the numerous Warringtons in the Tripoli Consular Correspondence in the 'fifties.

Head Marabout of the Bashaw of Tripoli.

He arrived in Kuka on Dec. 23, 1823—'a robust, healthy-looking young man, with a double-barrelled gun slung at his back' (vid. Denham, vol. i, p. 461).

Alias Adolphus Simkins, a native of St. Vincent, and a sort of dragoman to the Mission. He signs himself A. A. Simkins, but Denham, Introductory Chapter, spells it Sympkins. He accompanied Clapperton on his second expedition. There was also a Gibraltar Jew named Jacob attached to the Mission as 'store-keeper'.

Cf. Denham, vol. ii, pp. 23-6. Toole was only twenty-two when he died.

The letter is in dog-Italian, which I am unfortunately unable to read.

Vid. Denham, vol. i, p. 429. They were four-pounders, presented to the Sheikh by the Sultan of Fezzan. Denham also relates that Hillman made a large chair, 'which pleased the Sheikh excessively'. The Sheikh sent him some 'gabaga' [i.e. cotton strips, used at that time as currency in Bornu], but Hillman returned them, saying: 'No! the King of England pays me—I don't want that: but I am much obliged to the Sheikh, nevertheless.'

John Belford, Lyon's shipwright, made gun-carriages and a 'coach' for the Sultan of Mourzuk. Vid. Lyon, p. 179.

For presents sent to Shehu Omar vid. Benton, *Notes, &c.*, pp. 195 and 210.

They had been detained in Mourzouk as hostages by the Governor of Fezzan.

This was a present to Warrington from the Bashaw of Tripoli and was worth \$1,000. It appears to have been rather a white elephant, for in one letter Warrington hints to Horton that he would like to present it to him. Presumably Horton intimated he had no use for it.

In charge of Hillman.

Paid to secure and retain the goodwill of the ruler of Tripoli. Without this the Mission could not have succeeded.

Drawn but not expended. Warrington's arithmetic is obscure.

APPENDIX XVI

BREEDS AND VARIETIES OF CATTLE, SHEEP AND GOATS [\[593\]](#)

COLOURS

Cattle.

White ox or cow	bida
Coal-black	balle (<i>Koiyam</i> , kolliu)
Black and white	yāmi (<i>Shuwa</i> , yabbe)
Red	hamra
Yellow	ole
Strawberry roan	boni
White belly	saje
Deep red	mera
Dapple grey	bage

Goats.

White	bullam, keṛi
Coal-black	tselimma
Lightish black	keara
Black and white	kari
White with black neck	mallam
White and red	bururu, tosuno
Dapple grey	kurguma

Sheep

White	bida
-------	------

Coal-black	billa
White with black ears	batinge
White bellied	kordi
Speckled	ngangala
Half white and half red or black	wuda
Big ears and head	bellani
White with black neck	balo
Chestnut	kengar
White tail and stern	dubulu

BREEDS

Cattle.

Bare—Big-horned Bornu cattle, no hump; derived from Arabic Bahr = water, because these cattle came from Chad.

Kuri—Ditto: name of tribe living in Chad.

Wadare—Small (wada = dwarf) humped, short-horned.

Jawai—Very small, short-legged, from Mandara.

Abore—Spreading horned, humped Fulani cattle.

Sheep.

Wagare—Large Bornu sheep.

Bellani—Big-headed, big-eared.

Wuda—Short-haired Asben sheep.

Yeriram—Long-haired sheep from the North: wool is manufactured.

Goats.

Kani Kenji—Common Bornu goat.

Kani Tukshi—Long-haired goat.

Kani Musgu—Small, short-legged pagan goat.

NOTE. On p. 193 of the Appendix to Denham's book, first edition, it is recorded that he brought home a pair of ox-horns measuring 42½ inches in length and 23¼ inches in circumference.

Cf. Note [\[349\]](#), and vid. 'List of Authorities' under [Dechambre](#).

APPENDIX XVII

NOTE ON PORTRAITS OF TRAVELLERS AND AUTHORS

WHO HAVE WRITTEN ABOUT BORNU

BARTH (in a frock-coat) in Petermann's *Account of the Progress of the Expedition to Central Africa*, reproduced in Joseph Chavanne, *Die Sahara*, Wien, 1879, and in E. Schauenburg, *Reisen in Central-Afrika*, There is a later photo of Barth in Gustav von Schubert's *Heinrich Barth*, 1897, and also one in Cust's *Modern Languages of Africa*. The Royal Geographical Society of London have a photograph of him, full face, wearing his orders.

BEURMANN. I have not been able to trace any portrait.

CLAPPERTON. The frontispiece to the account of his second expedition is a reproduction of a very fine portrait, painted by Gildon Manton, engraved by Edward Finden. The same portrait is reproduced in the four-volume edition of Denham, in Chavanne, and in Schauenburg.

Sir H. H. Johnston in his *Pioneers in West Africa* states that the original is now in the National Portrait Gallery. This is a mistake. The original appears to have been painted in duplicate. One is in the possession of John Murray, Esq., of 50 A Albemarle Street, whose firm published Clapperton's book, and the other belongs to Col. J. K. Stewart-Mackenzie of Seaforth, Brahan Castle, Ross-shire. Col. Stewart-Mackenzie is the present representative of the Seaforth family, and from a note under a woodcut, on p. 7 of the *London Literary Gazette and Journal of Belles-Lettres* for 1829, it appears that the portrait was painted for Lady Seaforth, mother of Clapperton's friend Mackenzie, vid. [p. 340](#) of the present work. A portrait of this Mackenzie by Raeburn is also in the possession of Col. Stewart-Mackenzie.

Mr. Murray's portrait shows Clapperton as sandy-haired and of a pallid, almost ghastly, complexion—perhaps a result of his African experiences.

This same portrait was reproduced in *Scribner's Magazine* for 1891, vol. ix, p. 186. A small copy of it in oils on wood, about 12 inches square, is in the possession of the Royal Geographical Society. A label on the back of it states

that it was exhibited in 1905 at the Naval, Shipping, and Fisheries Exhibition, but I have not been able to trace how it came into the Society's possession or when it was executed.

DENHAM. A portrait painted by T. Phillips and engraved by E. Finden (exhibited at the Royal Academy in 1826) appears in the four-volume edition (1831) of Denham's book, but not in the earlier editions; this portrait shows him bare-headed and wearing a burnous. There is a print of it in the Reference Department of the National Portrait Gallery and a woodcut in *Scribner's Magazine*, 1891, vol. ix, p. 187.

The original is in the possession of John Murray, Esq., whose firm published Denham's book. I was allowed to see both the Denham and Clapperton portraits by the courtesy of Mr. John Murray, jun. A copy was made some years ago and is in the possession of Dr. W. H. Denham Rouse of the Perse School, Cambridge, who is a collateral descendant.

There is also a miniature of Denham, in uniform wearing the Waterloo medal and carrying a sabre presented to him by the Bashaw of Tripoli. This was painted by Newton and is reproduced on p. 7 of the *London Literary Gazette and Journal of Belles-Lettres* for 1829. The original is in the possession of Capt. H. A. Denham of 16 Hallam Street, W., as well as the Waterloo medal and the sabre. The originals of the water-colours reproduced in Denham's book, a plated bowl used by the explorer in Africa, and a throne or curved wooden stool presented to him by the Sultan of Bornu (probably the same as or similar to the one mentioned in the account of the traveller's first interview with the Sultan), are in the possession of Capt. L. S. Denham of Taywell, Goudhurst, Kent. Capt. Denham also informs me that the explorer brought home a Bornouese cloak, which he presented to George IV at Windsor.

KOELLE. Photo in Cust, *Modern Languages of Africa*.

MATTEUCCI. I have not been able to trace any portrait.

MASSARI. I have not been able to trace any portrait.

NACHTIGAL. There is a photo of him in a tobe and fez in his book. Another photo in European dress appears in J. Chavanne, *Die Sahara*, and another in D. Berlin's *Erinnerungen an Gustav Nachtigal*, 1887. There is also a photo in Cust. A photo of him in evening dress is in the possession of the Royal Geographical Society of London.

NORRIS. Photo in Cust, *Modern Languages of Africa*.

LOUDNEY. I have not been able to trace any portrait.

OVERWEG. There is a portrait in Petermann's *Account, &c.* This is reproduced in Chavanne and in Schauenburg, and in von Schubert.

RICHARDSON. There is a portrait of him in a burnous with the hood drawn over his head in his book *Travels in the Desert of the Sahara*, 2 vol., London, 1848. This is reproduced in Chavanne. There is a print of it in the Royal Geographical Society's collection, and also in the Reference Department of the National Portrait Gallery, where also is a small photo of a bust of him which was in existence in 1884. There is another portrait of him in a double-breasted frock-coat, buttoned up to the neck in Petermann's *Account, &c.* [\[594\]](#)

ROHLFS. There is a portrait in Chavanne and numerous ones, taken at various periods of his life, in Guenther, *Gerhard Rohlfs*. The Royal Geographical Society of London have a very faded photo of him in gown, turban, and scimitar.

SCHULTZE. Photo in Mecklenburg, *From the Congo to the Niger and the Nile*.

TOOLE. I have not been able to trace any portrait.

TYRWHITT. I have not been able to trace any portrait.

VOGEL. There is a portrait in Petermann's *Account, &c.*, reproduced in Schauenburg. There is a portrait in native dress and turban in Chavanne.

James Richardson was born in Lincolnshire in 1806, was educated for the evangelical ministry, attached himself to the English Antislavery Society and under its auspices went out to Malta, where he took part in the editing of a newspaper and also engaged in the study of the Arabic language. For account of his published works vid. Benton, *Notes, &c.*, p. 150 ff.

APPENDIX XVIII

(VID. NOTE [\[62\]](#))

FADEL ALLAH AND THE FRENCH

A PORTION of Rabeh's force was not present at Kusseri as they were in garrison at Karnak (Logone) under Fadel Allah. The latter retreated upon Dikoa, pursued by the French [vid. [Guilleux](#), 'List of Authorities']. He was routed on the River Yedseram, but escaped and rallied his forces at Mugelbu in Southern Mandara. From here he communicated in the autumn of 1900 with the British Resident at Ibi on the River Benue. From Mugelbu Fadel Allah moved back to the Kilba hills on the Yola-Bornu border, and later traversed Marghi, via Chibuk, where his camp can still be seen, then on to Marguba on the Maiduguri-Gujba road. From here he made a raid on Maiduguri and defeated the Shehu's general, Mestrema Musa, chief eunuch, just outside the town. Mestrema Musa (Mallam Musa) is now Ajia of Borsari. Fadel Allah then continued his raid to Dikoa, which he found abandoned and burnt. Shehu Garbai fled from Dikoa to Ngala where he gave battle to F. Allah, but was defeated and fled to Kanem.

The French collected their forces and Fadel Allah had to retire hurriedly on Marguba, whence with his brother Moman Niebé in command of the advance guard with the women and baggage, he continued his retreat south-westwards; the pursuing French shelling the town of Billaraba, south of Gujba, under the mistaken impression that Fadel Allah was there, before returning to Dikoa.

Fadel Allah camped for some months in the spring of 1901 at Gwani and Bima on the River Gongola and again communicated with the British Resident of Ibi. Fadel Allah pressing for an interview with a representative of the British Government, Major M^cClintock was sent up by the Acting High Commissioner.

This officer met Fadel Allah at Burguma on the Maiduguri-Gujba road, and, counselling him to remain in camp pending the decision of the British Government (vid. note [\[67\]](#)), marched back through the Marghi district to Yola.

Fadel Allah disregarded this instruction and sent one of his lieutenants, named It, to dig up a gun which had been buried near Dikoa. The French attacked and killed It, and again Fadel Allah retired to Gujba. Here, on the Ansei

swamp, the final skirmish took place on August 27, 1901. Fadel Allah was shot through the body on horseback, but was got away to Mutue, where he was buried in the swamp. The French dug him up, cut off his head and took it back to Dikoa. The name of the French commander was Dangeville. Moman Niebé surrendered and returned with the French to the French Congo.

APPENDIX XIX

HEAD-QUARTERS OF BRITISH BORNU

WHEN Bornu was occupied in the spring of 1902 by the British force under Colonel Morland, and Shehu Abubakr Garbai invited in from Dikoa to become Head of the greater bulk of the land of his fathers, now stripped of districts apportioned by treaty to Germany and France, it was promised that the Shehu should restore and repopulate ruined Kuka if he could: in the meantime he made his temporary capital at the ancient site and large market village of Mongonu, some seventeen miles southward on the Maiduguri road.

On the appointment a few months later of a Civil Resident, this officer found it expedient to reside himself at Mongonu, leaving the eastern garrison still at Maiduguri, where Colonel Morland had placed it, with a fort. Next year, when the Shehu made his experimental move to Kuka, the Resident decided that the institution of internal caravan tolls, the paucity of the administrative staff and other considerations required his residence at the more central locality of Magumeri, where the Residency continued until the end of 1904, when Kuka became the joint capital of the British and the native administration.

At the end of 1906 it became evident that no efforts of the Shehu could attract a population or trade approaching half of its volume or importance prior to the sack of the city thirteen years previously, and that mere sentiment did not justify the retention and rebuilding of a site unhealthy and depressing, and no longer central or convenient. The Resident and the Shehu accordingly moved then down to Maiduguri, the capital of British Bornu to-day.

The government station of Maiduguri, civil and military, is built upon the sandy rise, some 1,000 feet above mean sea level, on the left bank of the little River Alo, upon which the small village of Maifoni used to stand. This is the place wrongly termed Mabani by Barth, whose map also shows the stream flowing the wrong way. The situation is healthy, commanding, and with a good water supply. Maiduguri, the large market village, is some 3½ miles away; and between this and the government station, upon the sites of the little hamlets of Kalua, is built the Shehu's quarter, known at first as Yeriwa, but now as Shehuri. For general convenience, the absorbed hamlets of Kalua, the village of Maifoni

(now moved into 'Shehuri'), Shehuri and Maiduguri, themselves—comprising in all a population of some 10,000, together with the government station—are known as Maiduguri, likely soon to develop into a thriving commercial centre, on the highway to the Shari, Darfur, and the Nile.

APPENDIX XX (vid. [p. 341](#))

CLAPPERTON'S PROMOTION TO LIEUTENANT

CLAPPERTON passed for promotion to the rank of Lieutenant in 1813. He was given the acting rank of Lieutenant whilst in command of the *Confiance* schooner on the Canadian lakes, but the Admiralty refused to confirm this. He was finally promoted in 1816 and ante-dated to 20th March, 1815. For the latter fact and for the subjoined certificate I am indebted to the kindness of W. G. Perrin, Esq., Admiralty Librarian:—

Pursuant to an Order from Sir Samuel Hood, Bart., K.B., Vice Admiral of the Blue and Commander in Chief of H.M. Ships and Vessels employed and to be employed in the East Indies and Seas adjacent, &c., dated 5th February 1813.

We have examined Mr. Hugh Clapperton, who appears by Certificate to be more than Nineteen Years of age and has been to sea more than six years in the Ships and qualities undermentioned, viz.:—

Ship.	Bounty.	Place Where Born.	Age.	Entry.	Quality.	Discharge.	Cause.	Time.			
						Time.		Y.	M.	W.	D.
Renommie	—	Scotland	—	25 Novr. 1806	Ab	24 Octr. 1807	—	0	11	3	5
Do.	—	—	—	25 Octr. 1807	Midn.	10 June 1808	—	0	8	5	0
Venerable	—	—	—	11 June 1808	Midn.	27 Feb. 1809	—	0	9	1	5
Passenger to join H.M. Ship Clorinde as pr certificates	None	—	—	28 Feb. 1809	Supy. Midn.	14 March 1810	—	1	0	2	1
Clorinde	—	—	—	15 Mar. 1810	Midn.	20 June 1811	—	1	2	2	0
Do.	—	—	—	21 June 1811	M. Mate	5 Feby. 1813	—	1	8	1	2
								6	2	3	6

He produceth Logs kept by himself in His Majesty's Ships Renommie, Venerable, and Clorinde, and Certificates from Captains Livingstone, King, Pascoe, Edgecombe, and Briggs of his diligence and sobriety and obedience to commands. He can splice, knot, reef a sail, work a ship in sailing, shift his tides, keep a reckoning of a ship's way, by plain sailing and Mercators, observe by the sun or star, find the variation of the compass, and is qualified to do his duty as an able Seaman and Midshipman.

Given under our hands on board H.M.S. Clorinde, Bombay Harbour, 7th
February 1813.

(Sgd) THOS. BRIGGS, Captain H.M.S. Clorinde.

WM. H. WEBLEY, Captain H.M.S. Illustrious.

GEO. ELLIOT, Captain H.M.S. Hussar.

APPENDIX XXI

A BORNU PRINCE AT TRIPOLI

THERE is a book the full title of which is 'Letters written during a ten years' residence at the Court of Tripoli, published from the originals in the possession of the family of the late Richard Tully, Esq. the British Consul'. London, 1816, quarto. Third edition, 2 vols., octavo, 1819 [five coloured plates].

Vol. ii, pp. 51-5, under date of July 28, 1789, there is a short description of the visit to Tripoli of a 'black prince of Bornu' returning to his own country from Tripoli, who was accompanied by three wives, one of whom had learnt enough Italian to express herself in 'Lingua Franca'.

According to Tully, the prince was very well informed and wore 'pearls of uncommon size' as well as 'large gold ear-rings set with the most valuable jewels'.

The prince gave a highly coloured description of Bornu, which he declared abounded with grapes, apricots, and pomegranates, whilst the 'enormity' of slave-dealing was not indulged in by his countrymen, who left such things to the Christians and Pagans!

Tully further states that the people of Tripoli had such an idea of the numbers of the Bornu army, that they say when the army leaves the town a large date-tree is laid before the gate, on which each man steps as he passes, and 'as the foot soldiers go through the gate they wear out the body of this tree'.

[For visit of a Bornu prince to Tripoli in 1853 vid. Benton, *Notes, &c.*, p. 188.]

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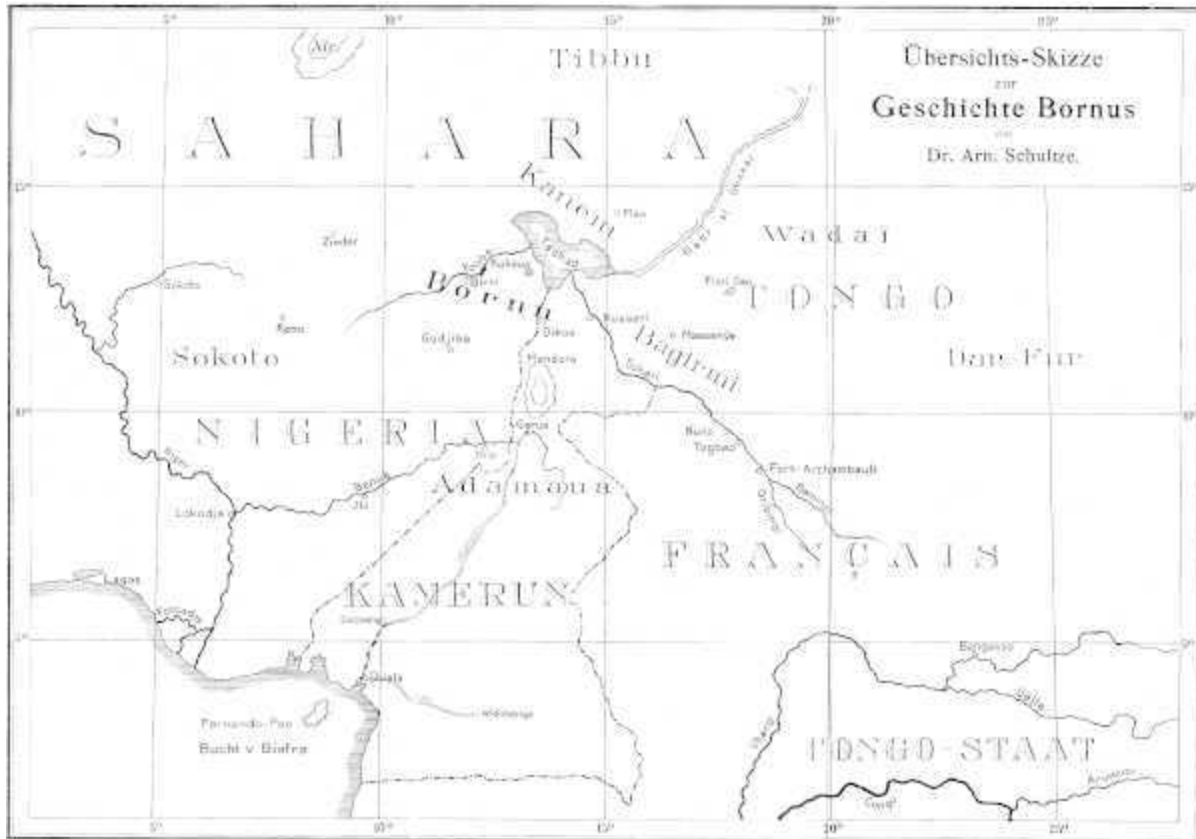


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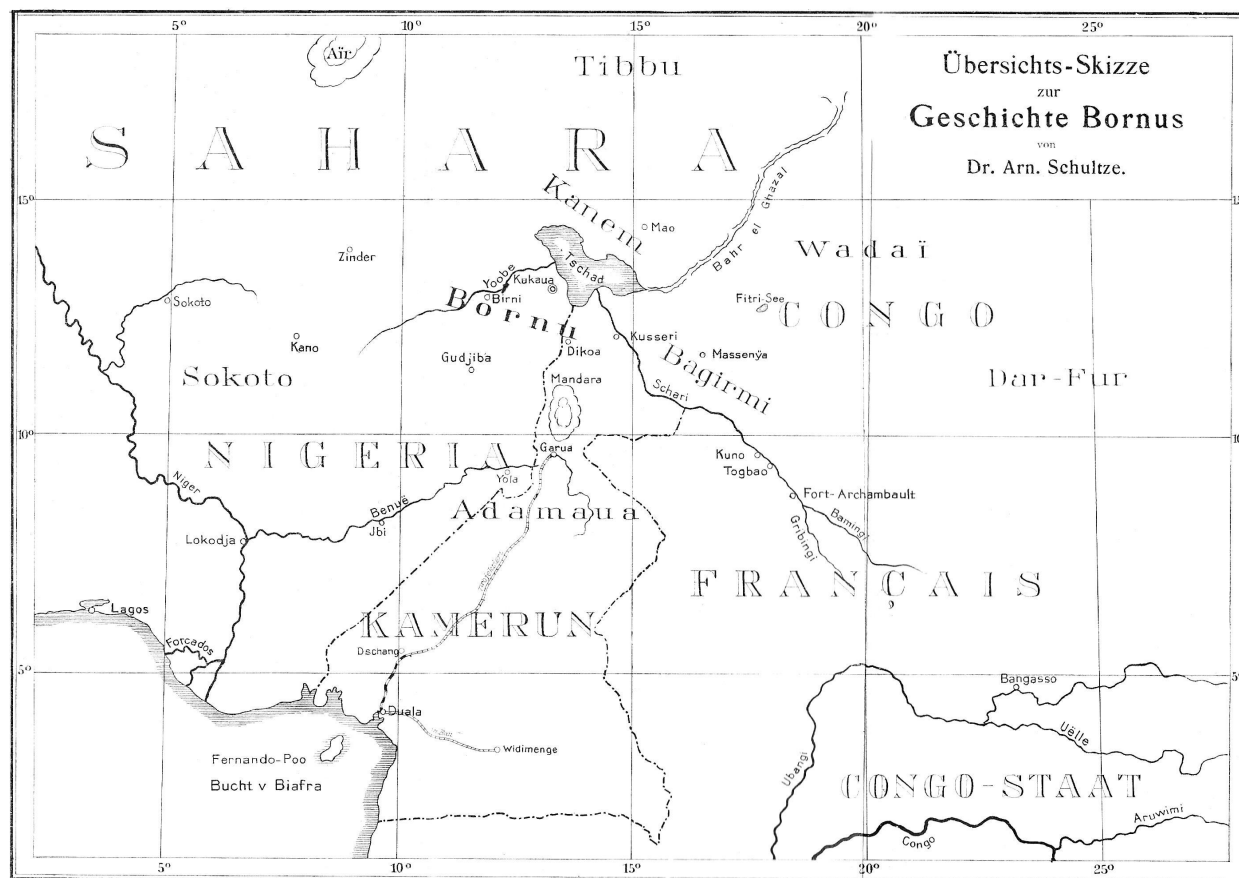
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